1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

X665 MLB DVT

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
6			2017-08-14

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DRAWING TITLE: SCHEM,MLB,X665

Apple Inc.

DRAWING NUMBER: 051-01649 SIZE: D

REVISION: 6.21.0

BRANCH

	DESCRIPTION	SCHEMATIC	BOARD	MCO	PANEL
MLB	POR FORM FACTOR MLB	051-01649	820-00967	056-04348	057-00903
DEV_MLB	ACRYLIC ONLY DEVELOPMENT MLB	951-04366	920-03304	056-04350	N/A
FLEX_PWR_BTN	FOR POR PWR BUTTON	051-02379	821-01249	056-04296	N/A
GS5_REDHEAD	SSD	051-02311	820-00899	056-04271	057-00906
FLEX_CAMERA_IA	CAMERA MODULE FLEX	051-01861	821-00978	056-04578	N/A
FLEX_INTERCONNECT_CAM_MIC	HUB FLEX	051-02325	821-01216	056-04194	N/A
FLEX_CG_MIC_LEFT	FLEX FROM HUB TO LEFT CG MIC	051-02369	821-01245	056-04245	N/A
FLEX_CG_MIC_RIGHT	FLEX FROM HUB TO RIGHT CG MIC	051-02370	821-01216	056-04246	N/A
FAN FLEX LEFT	FLEX FROM L FAN MODULE TO MLB	N/A	N/A	099-09233	N/A
FAN FLEX RIGHT	FLEX FROM R FAN MODULE TO MLB	N/A	N/A	099-09234	N/A

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE
SYSTEM PCB SUMMARY

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D
REVISION 6.21.0
BRANCH

PROGRAMMABLES

TBT T ROM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00133	1	IC,SPI SERIAL FLASH,8MBITS,3.0V,USON8	U2890	CRITICAL	TBT_T_ROM:BLANK
341S00738	1	IC,T29,AR1,VTBD,P0 DEV,X665	U2890	CRITICAL	TBT_T_ROM:POC
341S00795	1	IC,T29,AR,V15.4,PROTO 0,X665	U2890	CRITICAL	TBT_T_ROM:P0_V15P4
341S00846	1	IC,T29,AR1-T,V15.10,PROTO1,X665	U2890	CRITICAL	TBT_T_ROM:P1A_V15P10
341S00902	1	IC,T29,AR1-T,V15.10,EVT,X665	U2890	CRITICAL	TBT_T_ROM:EVT_V15P10

TBT X ROM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00133	1	IC,SPI SERIAL FLASH,8MBITS,3.0V,USON8	U3490	CRITICAL	TBT_X_ROM:BLANK
341S00738	1	IC,T29,AR1,VTBD,P0 DEV,X665	U3490	CRITICAL	TBT_X_ROM:POC
341S00795	1	IC,T29,AR,V15.4,PROTO 0,X665	U3490	CRITICAL	TBT_X_ROM:P0_V15P4
341S00847	1	IC,T290,AR2-X,V15.10,PROTO1,X665	U3490	CRITICAL	TBT_X_ROM:P1A_V15P10
341S00903	1	IC,T29,AR2-X,V15.10,EVT,X665	U3490	CRITICAL	TBT_X_ROM:EVT_V15P10

SD READER ROM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00024	1	IC,SERIAL FLASH,2MBIT,3V,8-USON,2X3X,6MM	U6230	CRITICAL	SD_ROM:BLANK_WINBOUND
335S00282	1	IC,SERIAL FLASH,2MBIT,3V,8-USON,2X3X,6MM	U6230	CRITICAL	SD_ROM:BLANK_MACRONIX
341S00733	1	IC,ROM,SDCARD CTLR GL3227A,V001,POC,X665	U6230	CRITICAL	SD_ROM:POC
341S00783	1	FW,SDCARD CTRL GL3277A,V002,POC,X665	U6230	CRITICAL	SD_ROM:P0_V002
341S00897	1	IC,ROM,SDCARD CTL GL3227A,VAP03,EVT,X665	U6230	CRITICAL	SD_ROM:EVT_VAP03

ETHERNET ROM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00269	1	IC,SPI SERIAL FLASH,32MBIT,3V,SOIC8	U6090	CRITICAL	ETHERNET_ROM:BLANK_ADESTO
341S00751	1	IC,ETHERNET,VTBD,PROTO 0,X665	U6090	CRITICAL	ETHERNET_ROM:POC
341S00793	1	IC,ETHERNET,VTBD,PROTO 0,X665	U6090	CRITICAL	ETHERNET_ROM:P0
341S00817	1	IC,ETHERNET,V2.9.24,PROTO1,X665	U6090	CRITICAL	ETHERNET_ROM:P1_V2P9P24
341S00864	1	IC,ETHERNET,V2.7.36,MDI NORMAL,P1B,X665	U6090	CRITICAL	ETHERNET_ROM:P1B_MDI_NORMAL
341S00901	1	IC,ETHERNET,V2.7.64,MDI NORMAL,EVT,X665	U6090	CRITICAL	ETHERNET_ROM:EVT_MDI_NORMAL

BT SERIAL FLASH

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00255	1	IC,SPI SERIAL FLASH,2MBIT,1.8V,DFN8	UG450	CRITICAL	BT_FLASH:BLANK_ADESTO
335S00256	1	IC,SPI SERIAL FLASH,2MBIT,1.8V,DFN8	UG450	CRITICAL	BT_FLASH:BLANK_WINBOUND
341S00726	1	IC,BT,VTBD,POC,X665	UG450	CRITICAL	BT_FLASH:POC
341S00844	1	IC,BT,VTBD,POC,X665	UG450	CRITICAL	BT_FLASH:P1B
341S00892	1	IC,BT SFLASH,VTBD,EVT,X665	UG450	CRITICAL	BT_FLASH:EVT

WLAN SERIAL EEPROM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00214	1	IC,EEPROM,SER,UWIRE,16K,1.8V,DFN8	UG410	CRITICAL	WLAN_EEPROM:BLANK_ONSEMI
335S00216	1	IC,EEPROM,SER,UWIRE,16K BITS,1.7V,DFN8	UG410	CRITICAL	WLAN_EEPROM:BLANK_ROHM
341S00725	1	IC,WIFI ROM,VTBD,POC,X665	UG410	CRITICAL	WLAN_EEPROM:POC

GPU BOOT ROM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S0399	1	IC,FLASH,SERIAL,SPI,1M X 8,3.3V,MLP8,LF	UA920	CRITICAL	GPU_BOOT_ROM:BLANK
341S00729	1	VBIOS,ROM,VEGA10 GPU,V001,POC,X665	UA920	CRITICAL	GPU_BOOT_ROM:POC
341S00792	1	IC,VBIOS,ROM,VEGA10 GPU,V002,P0,X665	UA920	CRITICAL	GPU_BOOT_ROM:P0_V002
341S00863	1	IC,VBIOS,ROM,VEGA10 A0-XLA GPU,P1B,X665	UA920	CRITICAL	GPU_BOOT_ROM:P1B_A0_XLA
341S00862	1	IC,VBIOS,ROM,VEGA10 A0-XTA GPU,P1B,X665	UA920	CRITICAL	GPU_BOOT_ROM:P1B_A0_XTA
341S00888	1	IC,VBIOS,ROM,VEGA10 XLA GPU,A1,P1C,X665	UA920	CRITICAL	GPU_BOOT_ROM:P1C_A1_XLA
341S00887	1	IC,VBIOS,ROM,VEGA10 XTA GPU,A1,P1C,X665	UA920	CRITICAL	GPU_BOOT_ROM:P1C_A1_XTA
341S00899	1	IC,VBIOS,ROM,VEGA10 XLA GPU,A1,EVT,X665	UA920	CRITICAL	GPU_BOOT_ROM:EVT_A1_XLA
341S00898	1	IC,VBIOS,ROM,VEGA10 XTA GPU,A1,EVT,X665	UA920	CRITICAL	GPU_BOOT_ROM:EVT_A1_XTA
341S00943	1	IC,VBIOS,ROM,VEGA10,XTA GPU(V15)DVT,X665	UA920	CRITICAL	GPU_BOOT_ROM:DVT_A1_XTA
341S00944	1	IC,VBIOS,ROM,VEGA10,XLA GPU(V15)DVT,X665	UA920	CRITICAL	GPU_BOOT_ROM:DVT_A1_XLA

SILICON

CPU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S00415	1	CPU,SKYW,AM9S,L0,ESP,2.4,8C,130W,LGA2066	CPU	CRITICAL	CPU_X86:08C_EVT1
337S00360	1	CPU,SKYW,QLRX,L0,ES,2.4,10C,140W,LGA2066	CPU	CRITICAL	CPU_X86:10C_EVT1
337S00435	1	CPU,SKYW,QMVZ,QS,8C,3.2,120W,BGA2066	CPU	CRITICAL	CPU_X86:08C_3P2GHZ_QMVZ
337S00434	1	CPU,SKYW,QN57,QS,10C,3.0,120W,BGA2066	CPU	CRITICAL	CPU_X86:10C_3P0GHZ_QN57
337S00453	1	CPU,SKYW,QNH8,QS,2.3,18C,140W,LGA2066	CPU	CRITICAL	CPU_X86:18C_2P3GHZ_QNH8

CPU SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
998-06198	1	CONN,CPU,SOCKET,R4,LGA2066,SKYLAKE W	U1000	CRITICAL	CPU_SOCKET:POC
511S00004	1	CONN,SOCKET R4 LGA2066	U1000	CRITICAL	CPU_SOCKET:POR_P0

PCH

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S00359	1	IC,PCH-H,KBL,SSKU02,QLL9,ES2,A0	U7000	CRITICAL	PCH:ES2_MAF_QLL9
337S00390	1	IC,PCH-H,KBL,SSKU02,QM36,SAF,ES2,A0	U7000	CRITICAL	PCH:ES2_SAF_QM36
337S00436	1	IC,PCH-H,KBL,SSKU02,QML1,SAF,QS,BGA885	U7000	CRITICAL	PCH:QS_SAF_QML1

GPU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
998-11669	1	VEGA10 SDLE ADAPTER (VDDC)	UA000	CRITICAL	GPU:SDLE_VDDC
998-11670	1	VEGA10 SDLE ADAPTER (VDDCI/MVDD)	UA000	CRITICAL	GPU:SDLE_VDDCI_MVDD
337S00452	1	IC,GPU,AMD,VEGA10,XLA,A1,LF,PS,BGA2013	UA000	CRITICAL	GPU:XLA_08_SEC
337S00451	1	IC,GPU,AMD,VEGA10,XTA,A1,LF,PS,BGA2013	UA000	CRITICAL	GPU:XTA_16_SEC

TBT X,T

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S00254	2	IC,TBT,ALPINE RIDGE,SLLSM,PRQ,C1,CSP337	U2800,U3400	CRITICAL	TBT_AR:C1

ACE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S00961	4	IC,CD3215,ACE,C00,USB PWR SW,BLNK,BGA96	U3100,U3200,U3700,U3800	CRITICAL	ACE:C0

SOC VR

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S00268	1	IC,PMU,CALPE,D2249A0,OTP-BG,CSP324	U7800	CRITICAL	PMU:A0_B

SOC

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
339S00372	1	POP,SOC,GIBRALTAR+2G 20NM,M,B0,CSP1406	U3900	CRITICAL	SOC:B0_2G_MI_SCK

SILICON

SD CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
998-07938	1	IC,GL3227A,USB3 3.0 TO SD CARD READER 4.0	U6200	CRITICAL	SD:POC
338S00312	1	IC,GL3227A,USB3 3.0 TO SD CARD READER,QFN48	U6200	CRITICAL	SD:P0_A2

ETHERNET CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
998-08198	1	ETHERNET,10GBASE-T,PCIE,MAC+PHY	U5800	CRITICAL	ENET:A0
998-09963	1	IC,ETHERNET CONTROLLER,AQC107,B0,BGA224	U5800	CRITICAL	ENET:B0
338S00332	1	IC,ETHERNET CONTROLLER,AQC107,B1,BGA224	U5800	CRITICAL	ENET:B1

CPU CORE VR CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01192	1	IC,PXE1610-PM001,5+1,CTRL,QFN48	U9000	CRITICAL	VR_CPU_CORE:PM001
353S01239	1	IC,PXE1610-PM002,5+1,CTRL,QFN48	U9000	CRITICAL	VR_CPU_CORE:PM002
353S01376	1	IC,PXE1610-PM003,5+1,CTRL,QFN48	U9000	CRITICAL	VR_CPU_CORE:PM003

CPU VCCIO VR CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01193	1	IC,PXE1110-PMA001,1+0,VCCIO,CTRL,QFN40	U9400	CRITICAL	VR_CPU_VCCIO:PMA001
353S01240	1	IC,PXE1110-PMA002,1+0,VCCIO,CTRL,QFN40	U9400	CRITICAL	VR_CPU_VCCIO:PMA002
353S01377	1	IC,PXE1110-PMA003,1+0,VCCIO,CTRL,QFN40	U9400	CRITICAL	VR_CPU_VCCIO:PMA003

CPU VDDQ 01 VR CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01194	1	IC,PXE1110-PMB001,1+0,VDDQ_01,CTRL,QFN40	U9800	CRITICAL	VR_CPU_VDDQ01:PMB001
353S01241	1	IC,PXE1110-PMB002,1+0,VDDQ_01,CTRL,QFN40	U9800	CRITICAL	VR_CPU_VDDQ01:PMB002
353S01309	1	IC,PXE1110-PMB003,1+0,VDDQ_01,CTRL,QFN40	U9800	CRITICAL	VR_CPU_VDDQ01:PMB003
353S01378	1	IC,PXE1110-PMB004,1+0,VDDQ_01,CTRL,QFN40	U9800	CRITICAL	VR_CPU_VDDQ01:PMB004

CPU VDDQ 23 VR CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01195	1	IC,PXE1110-PMC001,1+0,VDDQ_23,CTRL,QFN40	U9900	CRITICAL	VR_CPU_VDDQ23:PMC001
353S01242	1	IC,PXE1110-PMC002,1+0,VDDQ_23,CTRL,QFN40	U9900	CRITICAL	VR_CPU_VDDQ23:PMC002
353S01310	1	IC,PXE1110-PMC003,1+0,VDDQ_23,CTRL,QFN40	U9900	CRITICAL	VR_CPU_VDDQ23:PMC003
353S01379	1	IC,PXE1110-PMC004,1+0,VDDQ_23,CTRL,QFN40	U9900	CRITICAL	VR_CPU_VDDQ23:PMC004

GPU CORE VR CONTROLLER

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S00997	1	IC,IR35217-4199,BUCK CTRL,QFN56,7X7MM	UD000	CRITICAL	GPU_VR:P0_V4199
353S01246	1	IC,IR35217-4207,BUCK CTRL,QFN56,7X7MM	UD000	CRITICAL	GPU_VR:P1C_V4207
353S01380	1	IC,IR35217-4210,BUCK CTRL,QFN56,7X7MM	UD000	CRITICAL	GPU_VR:EVT_V4210
353S01469	1	IC,IR35217-3078,BUCK CTRL,QFN56,7X7MM	UD000	CRITICAL	GPU_VR:DVT_V3078

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BOM CONFIGURATION

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BOM VARIANTS

BOM NUMBER	BOM NAME	BOM OPTIONS	
685-00136	COMMON PARTS,MLB,X665	X665_COMMON_PARTS	
639-04744	PCBA,MLB,08SS,08C,X665	X665_COMMON,X665_XLA_08_SEC,X665_08C	DVT
639-04745	PCBA,MLB,08SS,10C,X665	X665_COMMON,X665_XLA_08_SEC,X665_10C	DVT
639-05079	PCBA,MLB,08SS,18C,X665	X665_COMMON,X665_XLA_08_SEC,X665_18C	DVT
639-04746	PCBA,MLB,16SS,08C,X665	X665_COMMON,X665_XTA_16_SEC,X665_08C	DVT
639-04747	PCBA,MLB,16SS,10C,X665	X665_COMMON,X665_XTA_16_SEC,X665_10C	DVT
639-05080	PCBA,MLB,16SS,18C,X665	X665_COMMON,X665_XTA_16_SEC,X665_18C	DVT
939-05214	DEV,PCBA,MLB,16SS,18C,X665	X665_COMMON,X665_XTA_16_SEC,X665_08C,X665_DBG_EE	DVT DEV
939-04922	DEV,PCBA,MLB,GFX VDDC	X665_COMMON,X665_ADAPT_VDDC,X665_00C	DVT DEV
939-04923	DEV,PCBA,MLB,GFX VDDCI MVDD	X665_COMMON,X665_ADAPT_OTHR,X665_00C	DVT DEV

SYSTEM PARTS

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-01649	1	SCH,MLB,X665	SCH	CRITICAL	
820-00967	1	PCBF,MLB,X665	MLB	CRITICAL	
685-00136	1	COMMON PARTS,MLB,X665	CBOM	CRITICAL	X665_COMMON

COMMON BOM GROUPS

BOM GROUP	BOM OPTIONS
X665_COMMON_PARTS	COMMON,ALTERNATE:PART,ALT_CMN,X665_USBC,X665_NETWORK,X665_T290,X665_PWR,X665_SD,X665_X86_SUPPORT,X665_GPU,X665_DBG_MP
X665_USBC	ACE:C0,TBT_AR:C1,TBT_T_ROM:EVT_V15P10,TBT_X_ROM:EVT_V15P10
X665_NETWORK	BT_FLASH:EVT,WLAN_EEPROM:POC,ENET:B1,ETHERNET_ROM:EVT_MDI_NORMAL,ENET_REFCLK:OSC
X665_T290	PMU:A0_B,SOC:B0_2G_MI_SCK
X665_PWR	PWR_SEQUENCE:SMC,VR_CPU_CORE:PM003,VR_CPU_VCCIO:PMA003,VR_CPU_VDDQ01:PMB004,VR_CPU_VDDQ23:PMC004,GPU_VR:DVT_V3078
X665_SD	SD:P0_A2,SD_ROM:EVT_VAP03
X665_X86_SUPPORT	PCH:QS_SAF_QML1,CPU_SOCKET:POR_P0
X665_GPU	GPUCLK:XTAL
X665_DBG_MP	XDP,XDP_CONNECTOR,ESPI_DBG,DBG_FANTACH

GRAPHICS BOM GROUPS

BOM GROUP	BOM OPTIONS
X665_ADAPT_VDDC	GPU:SDLE_VDDC,GPU_BOOT_ROM:EVT_A1_XLA
X665_ADAPT_OTHR	GPU:SDLE_VDDCI_MVDD,GPU_BOOT_ROM:EVT_A1_XLA
X665_XLA_08_SEC	GPU:XLA_08_SEC,GPU_BOOT_ROM:DVT_A1_XLA
X665_XTA_16_SEC	GPU:XTA_16_SEC,GPU_BOOT_ROM:DVT_A1_XTA

CPU BOM GROUPS

BOM GROUP	BOM OPTIONS
X665_00C	CPU_X86:OFF
X665_08C	CPU_X86:08C_3P2GHZ_QMVZ
X665_10C	CPU_X86:10C_3P0GHZ_QN57
X665_18C	CPU_X86:18C_2P3GHZ_QNH8

EE ONLY DEBUG

BOM GROUP	BOM OPTIONS
X665_DBG_EE	DEBUG_UART,DEBUG_LEDS,DEBUG_LEDS_GFX

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COMPONENT ALTERNATES TRACKED IN <RDAR://24198640>

SYSTEM EE

197S0588	197S0591	ALT_CMN	ALL	J135 <RDAR://30128206>	XTAL,24.000MHZ,20PPM,9.5PF,60OHM,1.6X1.2
353S4376	353S3384	ALT_CMN	ALL	J135 <RDAR://30128375>	IC.LMV331,COMP,OPEN COL,2.7V,60UA,1SC70
376S0972	376S00075	ALT_CMN	ALL	J135 <RDAR://30128436>	MOSFET,N-CH,30V,100MA,13.0OHM,VMT3,HF
107S00112	107S00006	ALT_CMN	ALL	J135 <RDAR://30128488>	THERMISTOR,NTC,47K OHM,1%,4073K,0603
311S00112	311S0437	ALT_CMN	ALL	DAVID <RDAR://30156705>	IC,74LVC2G126,DUAL,3-STATE BUFFER,DFN8
353S00711	353S2073	ALT_CMN	ALL	DAVID <RDAR://30174231>	IC,NCS210,CURRENT SENSE,200V/V,SC70-6
371S00089	371S00085	ALT_CMN	ALL	DAVID <RDAR://30174773>	DIODE,SCHOTTKY,40V,2A,LOW VF,0603
371S0684	371S0495	ALT_CMN	ALL	DAVID <RDAR://30174773>	DIODE,DUAL SCHOTTKY,30V,200MA,SOT-363
372S0186	372S0185	ALT_CMN	ALL	DAVID <RDAR://30174982>	XSTR,NPN,65V,100MA,3P SOT883B
376S00015	376S0610	ALT_CMN	ALL	DAVID <RDAR://30195640>	XSTR,FET,DUAL N-CH,60V,3.3OHM,SOT-363
376S00001	376S0659	ALT_CMN	ALL	DAVID <RDAR://30196068>	MOSFET,P-CH,20V,4.2A,60MOHM@4.5V,SOT23
376S00146	376S1061	ALT_CMN	ALL	DAVID <RDAR://30197441>	FET,P-CH,20V,0.1A,3.8MOHM,DFN1006
311S00116	311S0324	ALT_CMN	ALL	DAVID <RDAR://30244248>	IC,74LVC2G34,DUAL BUFFER GATE,DFN6
311S00133	311S00130	ALT_CMN	ALL	DAVID <RDAR://30273650>	IC,74AVC1T45,XCVR,DL SUPPLY,1BIT,DFN-6
353S01041	353S01042	ALT_CMN	ALL	DAVID <RDAR://30275632>	IC,TS3312,V-REF,1.25V,0.15%,5MA,QFN8
371S00095	371S0567	ALT_CMN	ALL	DAVID <RDAR://30318051>	DIODE,SWITCH,100V,500MA,SOD323,2P
372S00008	372S0173	ALT_CMN	ALL	DAVID <RDAR://30352970>	XSTR,PNP,50V,100MA,100MVW,HFE=200,CST3
376S00224	376S1128	ALT_CMN	ALL	DAVID <RDAR://30430125>	NFET,30V,590MA,670MOHM,SOT883B
311S00141	311S0653	ALT_CMN	ALL	LEE <RDAR://330543421>	IC,PI6ULS5V9,I2C LEVEL TRANSLATOR,DFN-8
138S00013	138S0772	ALT_CMN	ALL	DAVID <RDAR://30703933>	CAP,CER,2.2UF,20%,10V,X6S,0402
311S0649	311S0541	ALT_CMN	ALL	DAVID <RDAR://30735093>	IC,SINGLE,2-INPUT AND GATE, 5P SOT553
138S0714	138S0713	ALT_CMN	ALL	DAVID <RDAR://30812224>	CAP,CER,10UF,6.3V,X5R0.7MM,0402,SAMSUNG
152S0098	155S0137	ALT_CMN	ALL	DAVID <RDAR://30812705>	IND,CHIP EMIFIL,1000OHM,200MA,1OHM DCR
116S00007	116S00006	ALT_CMN	ALL	DAVID <RDAR://30812830>	RES,MTL FILM,0 OHM,TIGHT TOL,3A,0402
311S0426	311S00007	ALT_CMN	ALL	DAVID <RDAR://30904120>	IC,74LVC1G07,SNGL BUFFER W/OD OUT,SOT891
107S00015	107S00011	ALT_CMN	ALL	DAVID <RDAR://31047426>	RES,MTL FLM,0.00075OHM,1%,4-L,0612,BLK
107S00101	107S00005	ALT_CMN	ALL	DAVID <RDAR://31047611>	RES,SENSE,0.005 OHM,1/3W,1%,100PPM,0306
107S0250	107S0248	ALT_CMN	ALL	DAVID <RDAR://31047727>	RES,SENSE,0.003OHM,1W,4-TERM,1%,0612,CYN
107S0240	107S0255	ALT_CMN	ALL	DAVID <RDAR://31048025>	RES,MTL FLM,.001OHM,1%,1W,0612,4-TERM
107S00041	107S00039	ALT_CMN	ALL	DAVID <RDAR://31048140>	RES,MF,.0005 OHM,1%,200PPM,1W,0612-4T
132S00064	132S0409	ALT_CMN	ALL	DAVID <RDAR://31048516>	CAP,X6S,0.1UF,20%,16V,0201
138S00006	138S0835	ALT_CMN	ALL	DAVID <RDAR://31048596>	CAP,CER,3-TERM,4.3UF,20%,4V,X5R,0402
138S00084	138S00060	ALT_CMN	ALL	DAVID <RDAR://31050082>	CAP,X5R,47UF,20%,6.3V,0603
138S0641	138S0700	ALT_CMN	ALL	DAVID <RDAR://31073310>	CAP,CER,X5R,2.2UF,20%,10V,0402
138S0703	138S0648	ALT_CMN	ALL	DAVID <RDAR://31073310>	CAP,CER,X5R,4.7UF,20%,6.3V,H=0.6MM,0402
138S0759	138S0762	ALT_CMN	ALL	DAVID <RDAR://31073517>	CAP,CER,1UF,20%,4V,X6S,0201
138S0739	138S0706	ALT_CMN	ALL	DAVID <RDAR://31077229>	CAP,CER,X5R,1UF,20%,10V,0201,SAMSUNG
138S0945	138S0706	ALT_CMN	ALL	DAVID <RDAR://31077229>	CAP,CER,1UF,20%,10V,X5R,0201
138S0831	138S00049	ALT_CMN	ALL	DAVID <RDAR://32926922>	CAP,CER,X5R,2.2UF,20%,6.3V,0201
155S0588	155S0367	ALT_CMN	ALL	DAVID <RDAR://31153563>	FLTR,EMI BEAD,120OHM,25%,3A,0603
155S00155	155S0441	ALT_CMN	ALL	DAVID <RDAR://31153579>	FERR BD,220 OHM,25%,1.5A,0.1 OHM,0603
155S00007	155S0667	ALT_CMN	ALL	DAVID <RDAR://31153776>	FLTR,COMMON MODE,90 OHM,100 MA,0202
197S00048	197S00036	ALT_CMN	ALL	DAVID <RDAR://31153886>	XTAL,25MHZ,30PPM,20PF,50OHM,500UW,2016
740S00035	740S0115	ALT_CMN	ALL	DAVID <RDAR://31183655>	FUSE,FAST-ACTING,2A,35V,26MOHM,0603
128S00042	128S0329	ALT_CMN	ALL	DAVID <RDAR://31203639>	CAP,TA,POLY,220UF,20%,6.3V,25MO,1.8A,B2
155S0731	155S0699	ALT_CMN	ALL	DAVID <RDAR://31824320>	FERR BD,120 OHM,25%,1.8A,60MOHM DCR,0402
311S0489	311S00118	ALT_CMN	ALL	DAVID <RDAR://32143855>	IC,SNGL,2-INPUT AND GATE, 5P SOT353-1
131S00246	131S0741	ALT_CMN	ALL	DAVID <RDAR://32160407>	CAP,COG,270PF,2%,50V,0402
152S00358	152S00208	ALT_CMN	ALL	DAVID <RDAR://31205760>	IND,MLD,1.0UH,20%,2.0A,120 MOHM,1608
152S00552	152S00002	ALT_CMN	ALL	DAVID <RDAR://31225584>	IND,PWR,2.2UH,20%,3.3A,82MOHM,2520
152S0835	152S00203	ALT_CMN	ALL	DAVID <RDAR://31226192>	IND,WW,2.2UH,20%,240MA,0.221 OHM,0603
152S1499	152S0864	ALT_CMN	ALL	DAVID <RDAR://31227799>	IND,47NH,2%,0.58OHM,0402,LF,HF
155S0830	155S0316	ALT_CMN	ALL	DAVID <RDAR://31244976>	FLTR,EMI,FER BD,600 OHM,100MH
197S0481	197S0343	ALT_CMN	ALL	DAVID <RDAR://31249918>	XTAL,25MHZ,20PPM,12PF,3.2X2.5X.6MM,85C
132S00153	132S0198	ALT_CMN	ALL	DAVID <RDAR://31284382>	CAP,X7R,1000PF,5%,2KV,1206
197S00096	197S00104	ALT_CMN	ALL	DAVID <RDAR://31249998>	XTAL,50MHZ,10PPM,8PF,40OHM,2016

SYSTEM EE

377S00103	377S0057	ALT_CMN	ALL	DAVID <RDAR://31309975>	SUPPR,TRANS,VARISTOR,6.8V,100PF,0402
377S00104	377S0068	ALT_CMN	ALL	DAVID <RDAR://31312281>	SUPPR,TRANS,VARISTOR,27V,47PF,0402
371S00127	371S00096	ALT_CMN	ALL	DAVID <RDAR://30359683>	DIODE,ZENER,3.0V,5%,250MW,SOD882-2
353S00636	353S4037	ALT_CMN	ALL	DAVID <RDAR://30432514>	IC,NX3DV42,DIFF USB SW,DPDT,QFN10,0.33MM
311S00138	311S0436	ALT_CMN	ALL	DAVID <RDAR://30474773>	IC,74AVC4T245,XCVR,4BIT,DUALSPPLY,QFN16
378S00074	378S00072	ALT_CMN	ALL	DAVID <RDAR://31561515>	LED,GREEN,568NM,4.3MCD@1MA,0402
378S00073	378S00067	ALT_CMN	ALL	DAVID <RDAR://31561520>	LED,RED,621NM,2.5MCD@1MA,1X0.6X0.25MM
311S0595	311S00155	ALT_CMN	ALL	DAVID <RDAR://31204779>	IX,XCVR,DL-BIT/DL-SUPPLY,2ST-OUT-QFN-10
371S00046	371S00082	ALT_CMN	ALL	DAVID <RDAR://31824517>	DIODE SCHOTTKY,30V,2A,LOW VF,0603
155S00067	155S0581	ALT_CMN	ALL	DAVID <RDAR://31230560>	FERR BD,240 OHM,25%,420MA,0.31 DCR,0201
311S0596	311S0593	ALT_CMN	ALL	DAVID <RDAR://32226608>	IC,VOLT-LEVEL TRANS,W/CONFIG GATE,SOT886
377S0178	377S00031	ALT_CMN	ALL	DAVID <RDAR://30735965>	SUPPR,TRANS,BIDIR,5.5V,0.2PF,0201
197S00129	197S00130	ALT_CMN	ALL	DAVID <RDAR://32396224>	OSC,XTAL,50MHZ,50PPM,3.3V,5032
376S00246	376S0865	ALT_CMN	ALL	LEWIS <RDAR://32512513>	NFET,20V,4A,0.038OHM,SOT23
197S00118	197S00120	ALT_CMN	ALL	LEWIS <RDAR://32818477>	XTAL,24MHZ,30PPM,9.5PF,60 OHM MAX,1612
155S00363	155S00367	ALT_CMN	ALL	JOHN <RDAR://32886387>	CHOKE,COMMONE MODE,67OHM,20%,400MA,0805-4
155S00364	155S00367	ALT_CMN	ALL	JOHN <RDAR://32886387>	CHOKE,COMMONE MODE,90OHM,50%,300MA,0805-4
157S00024	157S00026	ALT_CMN	ALL	JOHN <RDAR://32886430>	XFMR,LAN,10G,1CH,120UH,4532-8
197S00143	197S00142	ALT_CMN	ALL	ALAN <RDAR://33418271>	XTAL,27MHZ,10PPM,12PF,40OHM,2520
102S0880	102S0879	ALT_CMN	ALL	DAVID <RDAR://33305301>	RES,0.010OHM,1%,100PPM,1/2W,1206,BLK,CYN

DC-DC

152S00573	152S1778	ALT_CMN	ALL	SURESH <RDAR://29604818>	IND,0.105UH,20%,46A,0.35MOHM,6.8X6.8X5MM
152S00685	152S1778	ALT_CMN	ALL	LEWIS <RDAR://30977597>	IND,0.105UH,20%,46A,0.35MOHM,6.7X6.5X5MM
152S1684	152S1870	ALT_CMN	ALL	SURESH <RDAR://29190171>	IND,PWR,MOLD,2.2UH,20%,105MO,2.0A,2520
152S00408	152S00329	ALT_CMN	ALL	SURESH <RDAR://29078048>	IND,MLD,1.5UH,20%,13A,10.45MO,8.44X8X2.4
128S0364	128S0264	ALT_CMN	ALL	VINCENT MARKUP 01/20/2017	CAP,TANT,POLY,68UF,20%,16V,50MOHM,D2E,BLK
138S0775	138S0860	ALT_CMN	ALL	SURESH EMAIL 09/20/16	CAP,CER,X6S,1UF,10%,25V,0402
152S00398	152S00204	ALT_CMN	ALL	SURESH <RDAR://29078168>	IND,PWR,0.22UF,20%,6.7A,23MOHM,2012
152S00741	152S00699	ALT_CMN	ALL	LEWIS <RDAR://30499425>	IND,MLD,0.15UH,20%,90A,0.65MO,11.2X10X4
128S0444	128S0323	ALT_CMN	ALL	LEWIS <RDAR://29189353>	CAP,TANT,POLY,330UF,6.3V,25MOHM,2.4A,D2E
152S00747	152S00691	ALT_CMN	ALL	LEWIS <RDAR://30637603>	IND,MLD,1UH,20%,13.5A,6.6MOHM,7X6MOHM,7X6.6X3MM
152S00744	152S00692	ALT_CMN	ALL	LEWIS <RDAR://30638646>	IND,MLD,1.5UH,20%,9.5MOHM,12A,7.0X6.6X3
152S00743	152S00694	ALT_CMN	ALL	LEWIS <RDAR://30567068>	IND,MLD,3.3UH,20%,5A,32MOHM,4.85MOHM,4.85X4.7X3MM
152S00738	152S1685	ALT_CMN	ALL	LEWIS <RDAR://30568336>	IND,MLD,0.1UH,20%,21A,6MOHM,4.4X4.2X1.2
152S00748	152S00697	ALT_CMN	ALL	LEWIS <RDAR://30645234>	IND,MLD,0.36UH,20%,40A,0.8MOHM,11.2X10X4
128S0412	128S0237	ALT_CMN	ALL	LEWIS <RDAR://31254336>	CAP,TANT,PLY,BK,330UF,20%,6.3V,9MOHM,D3L
152S00608	152S00696	ALT_CMN	ALL	LEWIS <RDAR://29077951>	IND,MLD,0.22UH,65A,0.64MOHM,11.15X10.1X4
152S00774	152S00692	ALT_CMN	ALL	LEWIS <RDAR://31693715>	IND,MLD,1.5UH,20%,8.6MOHM,12A,7.0X6.6X3
353S01283	353S01013	ALT_CMN	ALL	LEWIS <RDAR://32001361>	IC,IR3575-4206,DR MOS,60A,PQFN31,6X6MM
138S00047	138S00073	ALT_CMN	ALL	LEWIS <RDAR://32229710>	CAP,CER,10UF,20%,6.3V,X6S,HRZTL,TY,0402
152S00802	152S00698	ALT_CMN	ALL	LEWIS <RDAR://32231069>	IND,MLD,4.7UH,20%,8A,40MOHM,7X6.6X3MM
152S00803	152S00693	ALT_CMN	ALL	LEWIS <RDAR://32232571>	IND,MLD,2.2UH,20%,8A,35MOHM,5.0X4.7X3MM
152S00801	152S00695	ALT_CMN	ALL	LEWIS <RDAR://32229059>	IND,MLD,1UH,20%,23A,2.9MOHM,11.2X10X3.5
128S00038	128S00039	ALT_CMN	ALL	LEWIS <RDAR://32254335>	CAP,TA,POLY,150UF,20%,6.3V,18MO,2A,B2
128S00050	128S0351	ALT_CMN	ALL	LEWIS <RDAR://32261377>	CAP,TA,POLY,330UF,20%,2.5V,13MOHM,2A,B2
128S00008	128S0380	ALT_CMN	ALL	LEWIS <RDAR://32678648>	CAP,TA,POLY,220UF,20%,0.009 OHM,2.5V,B2
128S0631	128S0352	ALT_CMN	ALL	LEWIS <RDAR://32678456>	CAP,TANT,POLY,330UF,2.5V,9MOHM,B2
128S0375	128S0378	ALT_CMN	ALL	LEWIS <RDAR://32678872>	CAP,AL POLY,39UF,16V,27MOHM,2.35A,BLK

BLC

126-00012	126-0306	ALT_CMN	ALL	LINDA KAMAS 02/08/2017 EMAIL	CAP,AL EL,100UF,20%,35V,0.3A,6.3X11,BLK
152S00740	152S1668	ALT_CMN	ALL	LINDA KAMAS 02/08/2017 EMAIL	IND,MOD,33UH,9.4A TYP,39MOHM,17X17X7MM
138S1078	138S0810	ALT_CMN	ALL	LINDA KAMAS 02/08/2017 EMAIL	CAP,CER,2.2UF,10%,100V,X7R,1206,SFT,MRTA
124-00003	126-00013	ALT_CMN	ALL	BLC OUTPUT CAPS	CAP,AL-PLY,18UF,20,100V,105C,3A,10X13,BK
371S0748	371S0731	ALT_CMN	ALL	BLC INPUT DIODE	DIODE,SCHOTTKY,30V,1A,SOD-323
371S0782	371S0783	ALT_CMN	ALL	BLC OUTPUT DIODE	DIODE,RECT,STPS8H100G,100V,8A,TO263
376S0845	376S1067	ALT_CMN	ALL	BLC POWER FET + 12V POWER FET	MOSFET,P-CH,30V,9.4MOHM,23.7A,5X6 DFN
376S1204	376S1106	ALT_CMN	ALL	BLC LED SINK FET	MOSFET,NCH,100V,12.5A,24MOHM,3X3 DFN
740S0145	740S0146	ALT_CMN	ALL	BLC FUSE	FUSE 6A 32V SMD 0603
107S0359	107S0229	ALT_CMN	ALL	BLC BOOST SENSE RES	RES,FILM,2W,.03 OHM,1%,2512,SMD
376S1184	376S1183	ALT_CMN	ALL	BLC BOOST FET	MOSFET,NCH,100V,30A,31.5MOHM,3X3 MLP
376S1256	376S1073	ALT_CMN	ALL	BLC SHORT PROTECTION FETS	XSTR,FET,N-CH,150V,1.4A,425MOHM,SSOT-6L
155S0831	155S0797	ALT_CMN	ALL	BLC FERRITE BEADS	FLTR,EMI,FER BD,600 OHM,100MHZ,0603
138S0805	138S0834	ALT_CMN	ALL	LINDA KAMAS 02/24/2017 EMAIL	CAP,CER,10UF,10%,25V,X7R,1206,SOFT

N&V

138S00077	138S00035	ALT_CMN	ALL	AKASH BUILD MATRIX REV 3.0	CAP,X6S,20UF,20%,2.5V,HZTL,TY,0402
138S00169	138S00168	ALT_CMN	ALL	AKASH BUILD MATRIX REV 3.0	CAP,X6S,43UF,20%,2.5V,NU,0603
138S00111	138S00036	ALT_CMN	ALL	AKASH BUILD MATRIX REV 3.0	CAP,X6S,20UF,20%,2.5V,NU,0402
138S00188	138S00043	ALT_CMN	ALL	DAVID <RDAR://31632406>	CAP,X6S,10UF,20%,6.3V,0402,NU

WIRELESS

339S00411	339S00404	ALT_CMN	ALL	METE EMAIL ON 05/16/2017	MODULE,WIFI/BT,HARPOON,USI,ES7.1,LGA324
339S00425	339S00404	ALT_CMN	ALL	METE EMAIL ON 05/16/2017	MODULE,WIFI/BT,HARPOON,M,ES7.3,LGA385
339S00427	339S00404	ALT_CMN	ALL	METE EMAIL ON 05/16/2017	MODULE,WIFI/BT,HARPOON,USI,ES7.3,LGA385
339S00426	339S00404	ALT_CMN	ALL	METE EMAIL ON 05/16/2017	MODULE,WIFI/BT,HARPOON,M,ES7.5,LGA385
339S00428	339S00404	ALT_CMN	ALL	METE EMAIL ON 05/16/2017	MODULE,WIFI/BT/HARPOON/USI/ES7.5,LGA385

SOC

339S00373	339S00372	ALT_CMN	ALL	LYNN 03/21/17 SILICON SYNC	POP,SOC,GIBRALTAR+2G 21NM,H,B0,CSP1406
339S00377	339S00372	ALT_CMN	ALL	LYNN 03/21/17 SILICON SYNC	POP,GIBRALTAR+2G 20NM,M,B0,ATK,CSP1406
339S00378	339S00372	ALT_CMN	ALL	LYNN 03/21/17 SILICON SYNC	POP,GIBRALTAR+2G 21NM,H,B0,ATK,CSP1406

AUDIO

155S0660	155S0513	ALT_CMN	ALL	DAVID<RDAR://31153630>	FERR BD,22OHM,25%,1.8A,40MOHM DCR,0201
138S1103	138S0719	ALT_CMN	ALL	DAVID<RDAR://31153099>	CAP,CER,4.7UF,20%,10V,X5R,0402
155S00034	155S0706	ALT_CMN	ALL	DAVID<RDAR://31153821>	FERR BD,120 OHM,25%,1.3A,0.09 OHM,0402
138S0841	138S0638	ALT_CMN	ALL	DAVID <RDAR://31205208>	CAP,CER,10UF,10%,25V,X5R,0805,MURATA
155S00203	155S0894	ALT_CMN	ALL	DAVID <RDAR://31153858>	FLTR,MLD,180 OHMZ,3.4A,0.025OHM,2016
353S01341	353S00920	ALT_CMN	ALL	DAVID <RDAR://30173238>	IC,LDO REG,1.8V,250MA,LOW NOISE,STO23-5

PD

099-09276	870-3024	ALT_CMN	ALL	JANET H IN 05/18/2017 EMAIL	SPRING FINGER,P5S-20X30X31,J28
099-09277	870-1415	ALT_CMN	ALL	JANET H IN 05/18/2017 EMAIL	SPRING FINGER,3.2 X 1.2 X 0.8MM,SMD

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BOM ALTERNATES

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SHIELD FENCE AR

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-11526	1	SHIELD FENCE,MLB,AR,X664	SHIELD_FENCE_AR	CRITICAL	

SHIELD CAN RJ45

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-10607	1	SHIELD,CAN,RJ45,X664	SHIELD_CAN_RJ45	CRITICAL	

SHIELD CAN HARPOON

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-10608	1	SHIELD CAN,HARPOON,X664	SHIELD_HPN_CAN	CRITICAL	

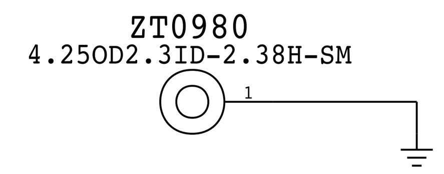
SHIELD FENCE HARPOON

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-12956	1	SHIELD,FENCE,WIFI,X664	SHIELD_HPN_FENCE	CRITICAL	

SOLDERED PARTS

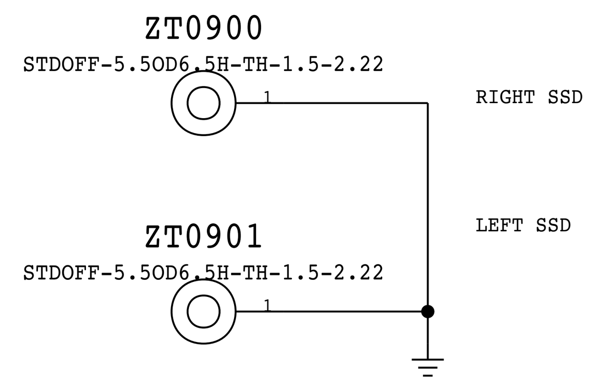
AUDIO BOSS

(860-00773)



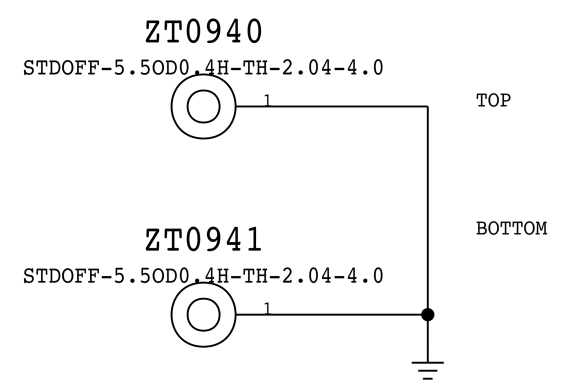
GS5 SSD STANDOFF

2.35MM HOLE, 5.8MM PAD, WITH BOSS INCLUDED (860-00739)



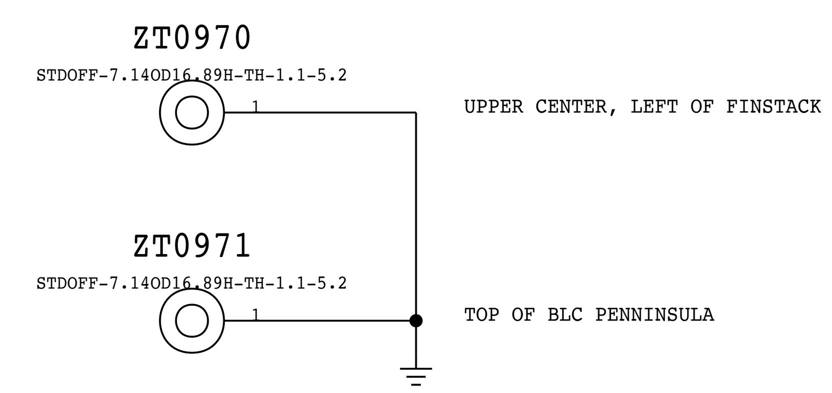
WIRELESS STRAIN RELIEF STANDOFFS

4.11MM HOLE, 6.00MM PAD, WITH BOSS INCLUDED (860-00713)



MLB STANDOFFS

5.38MM HOLE, 8.5MM PAD, WITH BOSS INCLUDED (860-00708)



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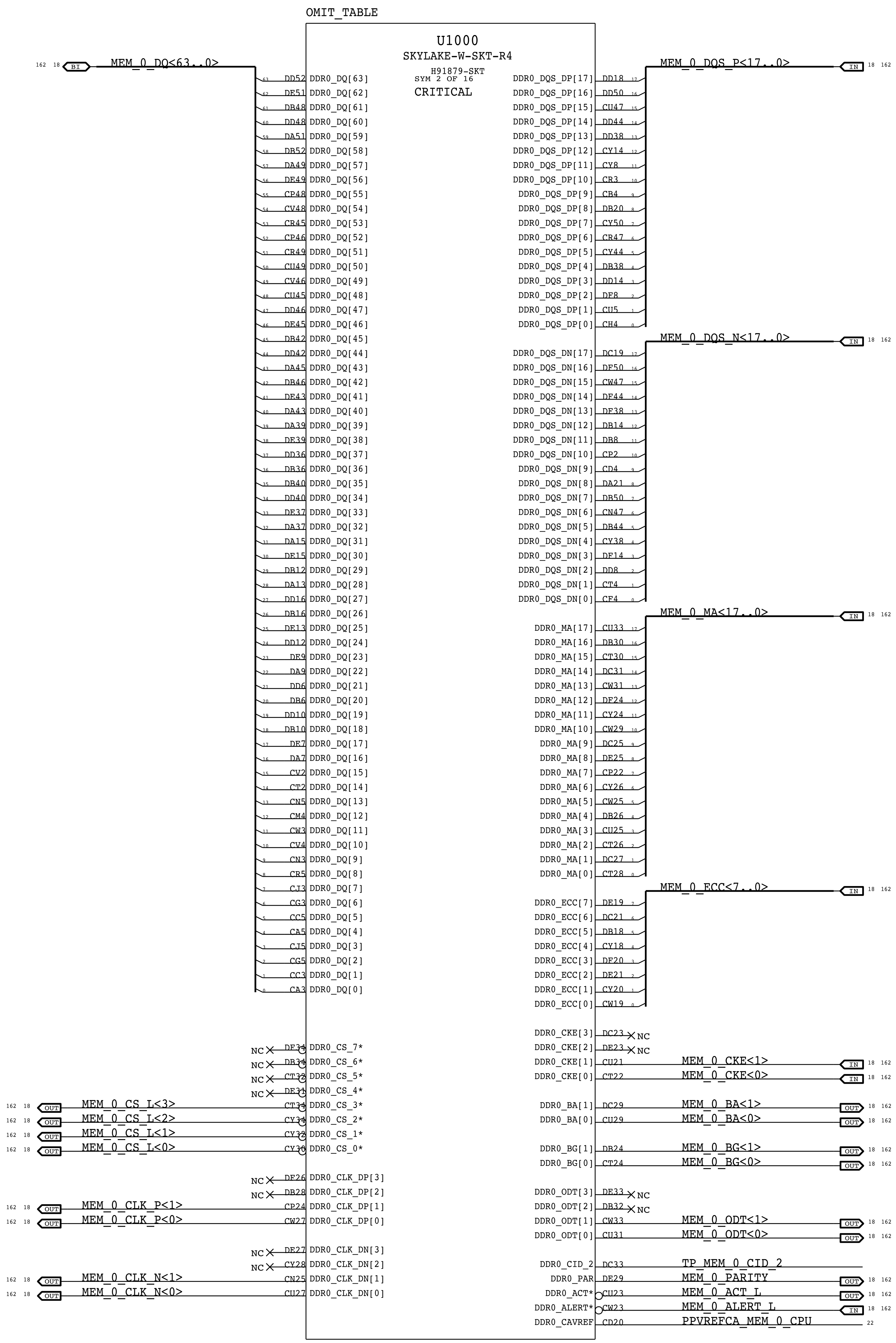

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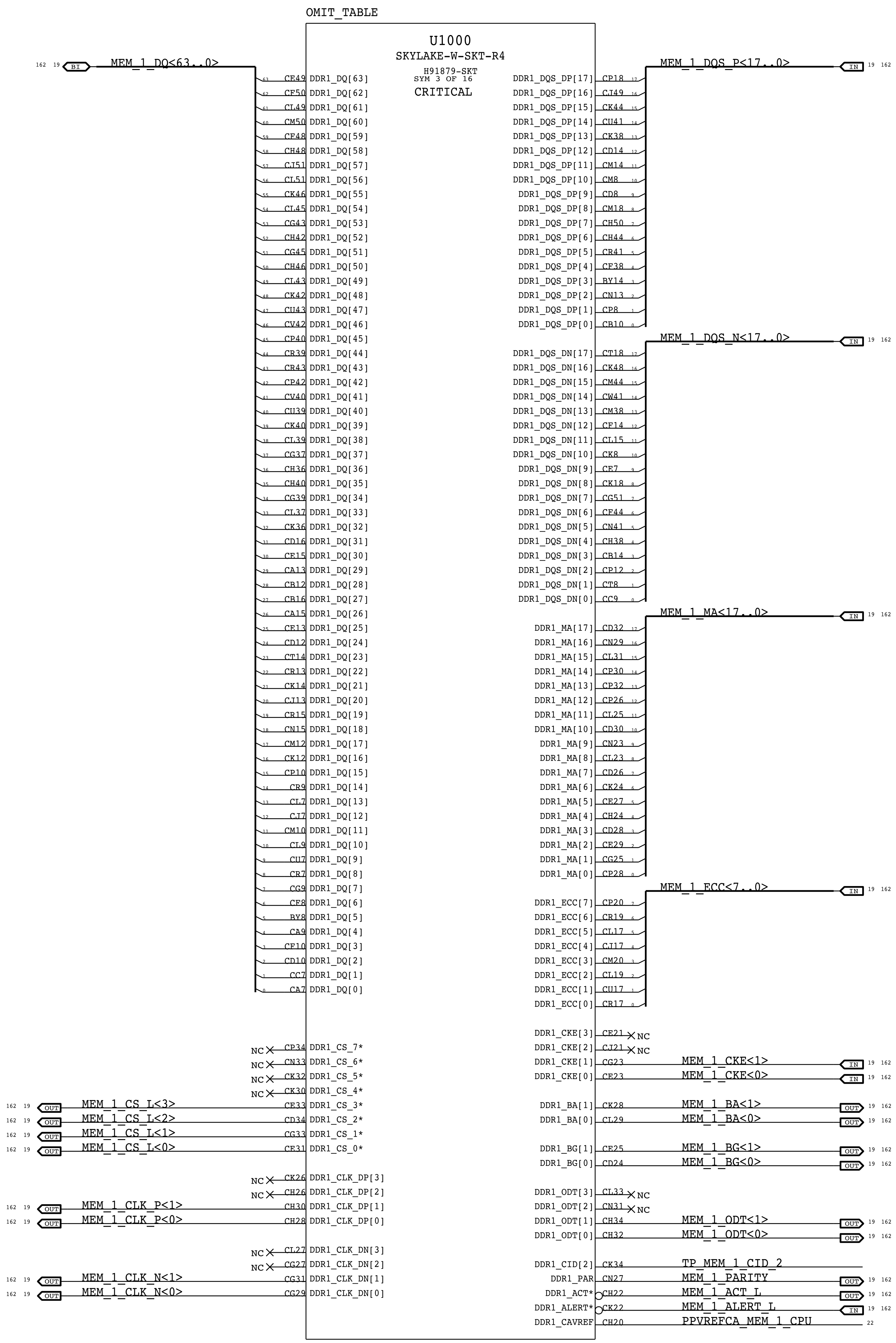

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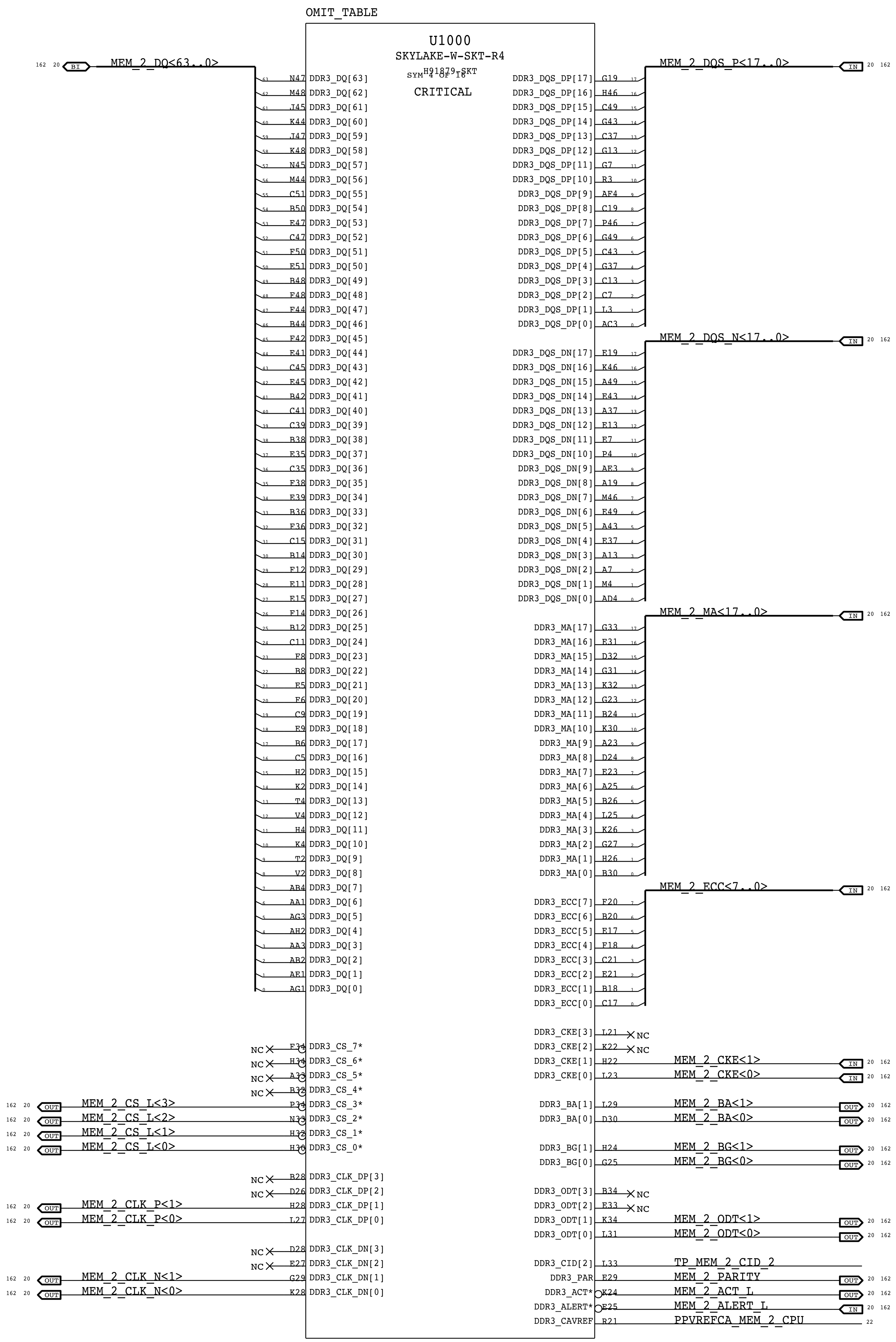

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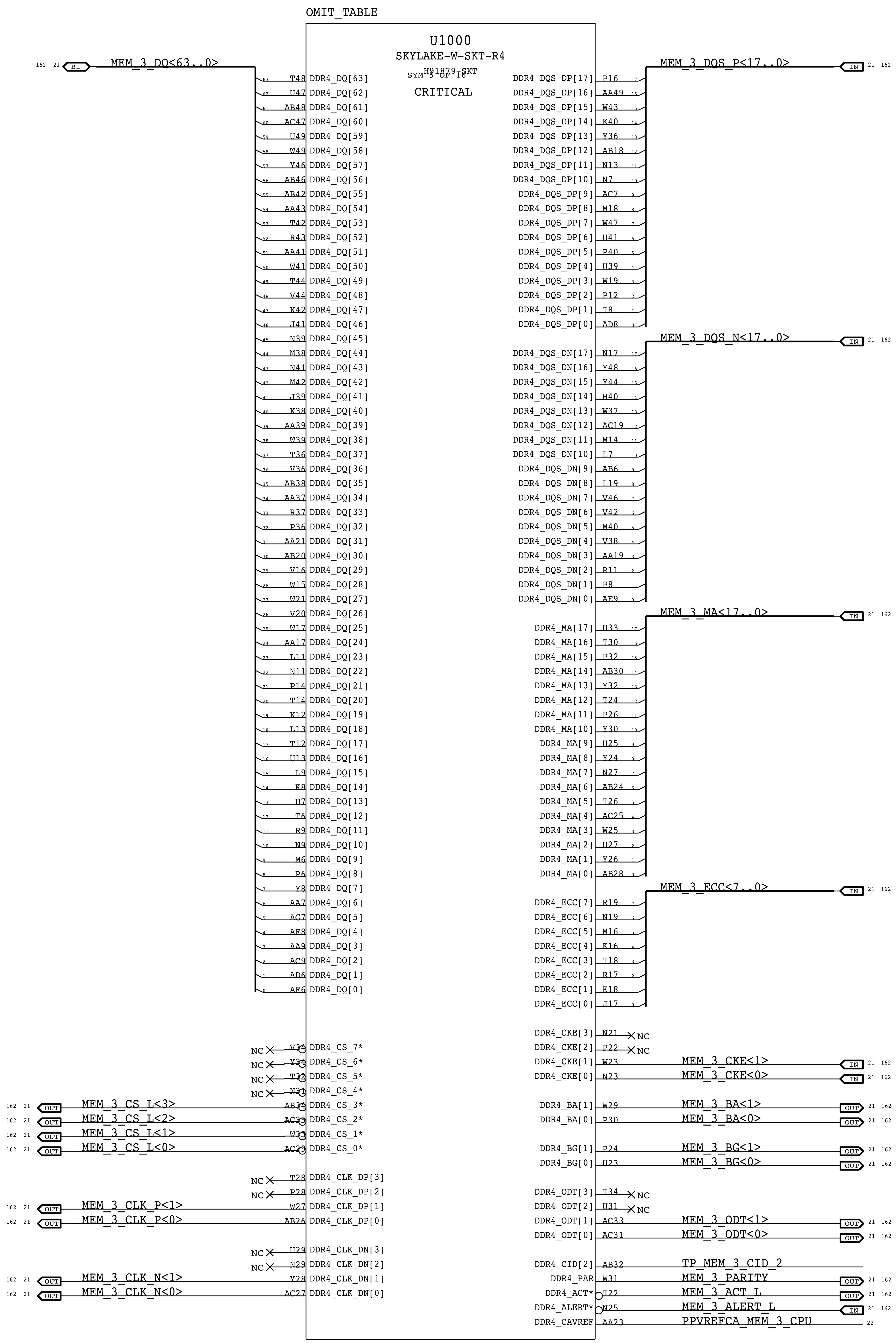

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PAGE TITLE

CPU PCIE

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SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

CPU DMI, MISC

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SYNC_MASTER=DEV_MLB SYNC_DATE=11/07/2016

PAGE TITLE: CPU SVID, BCLK, JTAG

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DRAWING NUMBER: 051-01649 SIZE: D

REVISION: 6.21.0

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PAGE: 16 OF 255

SHEET: 13 OF 171

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

CPU VCCIO, VDDQ, VCCSA

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CPU VCORE DECOUPLING

20UF X130

PLACE UNDER CPU

BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

CPU VCCIN

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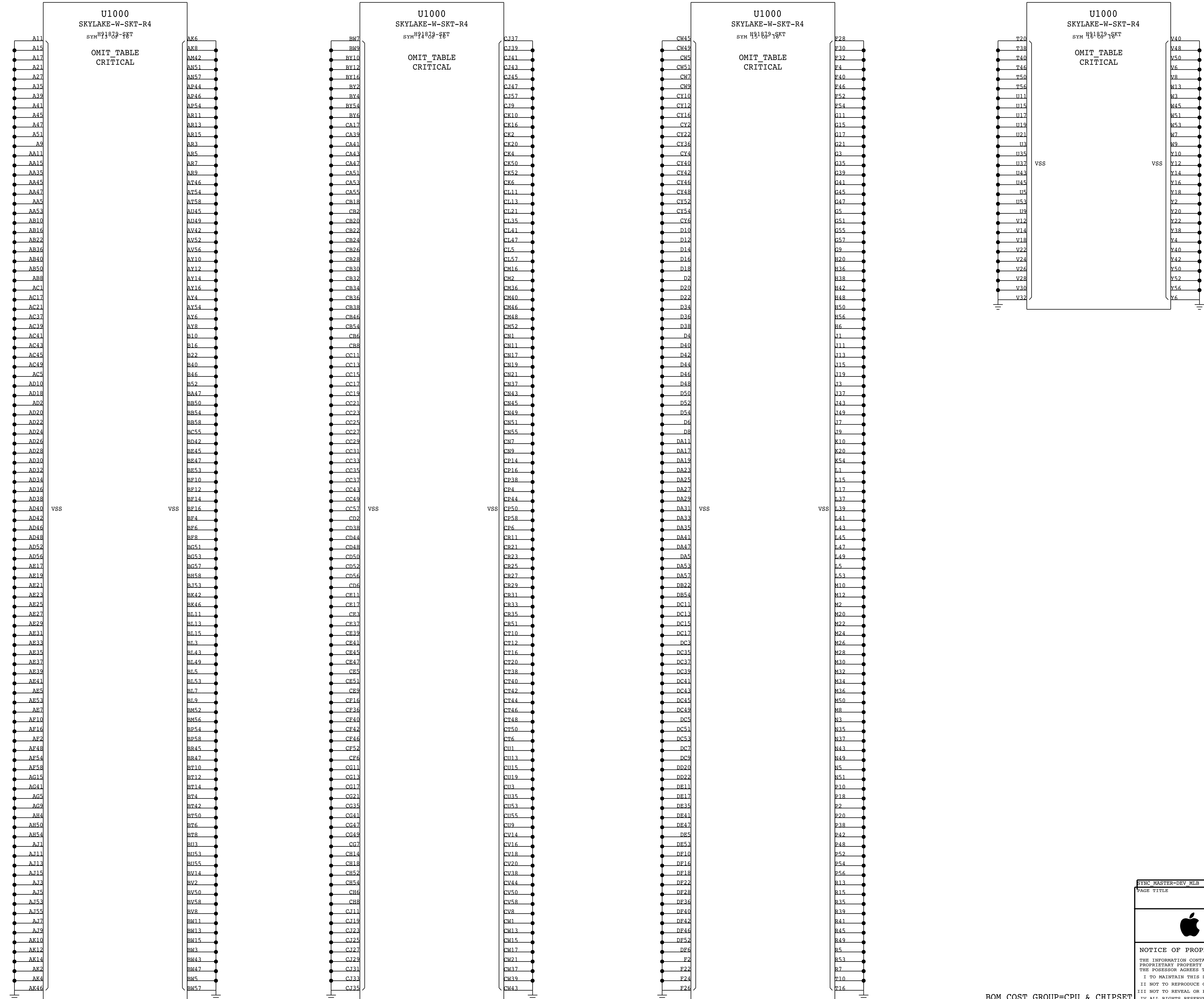

BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

CPU GND

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SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

CPU SUPPORT

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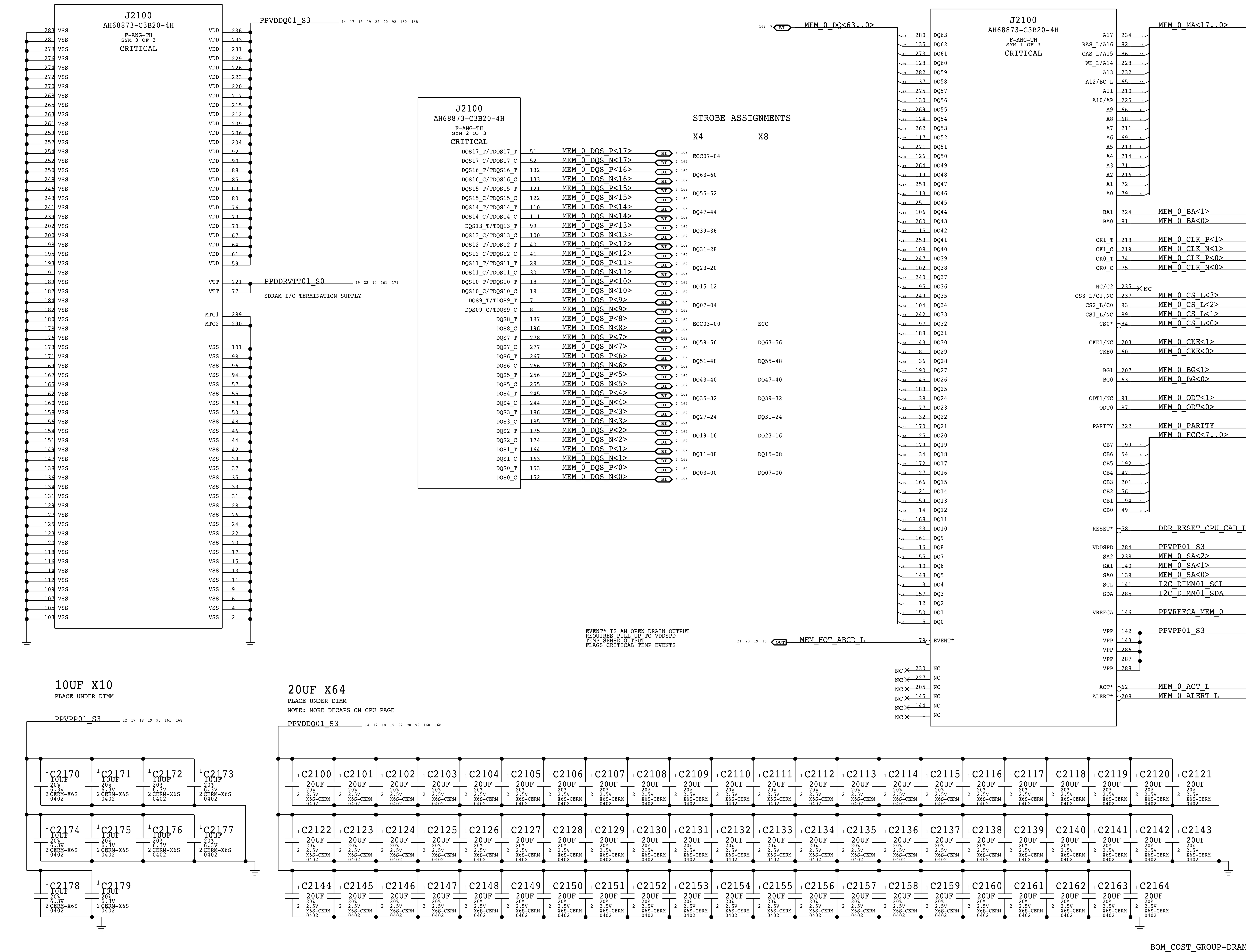

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PAGE TITLE

DDR4 DIMM CONN CH0

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BOM_COST_GROUP=DRAM

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

DDR4 DIMM CONN CH1

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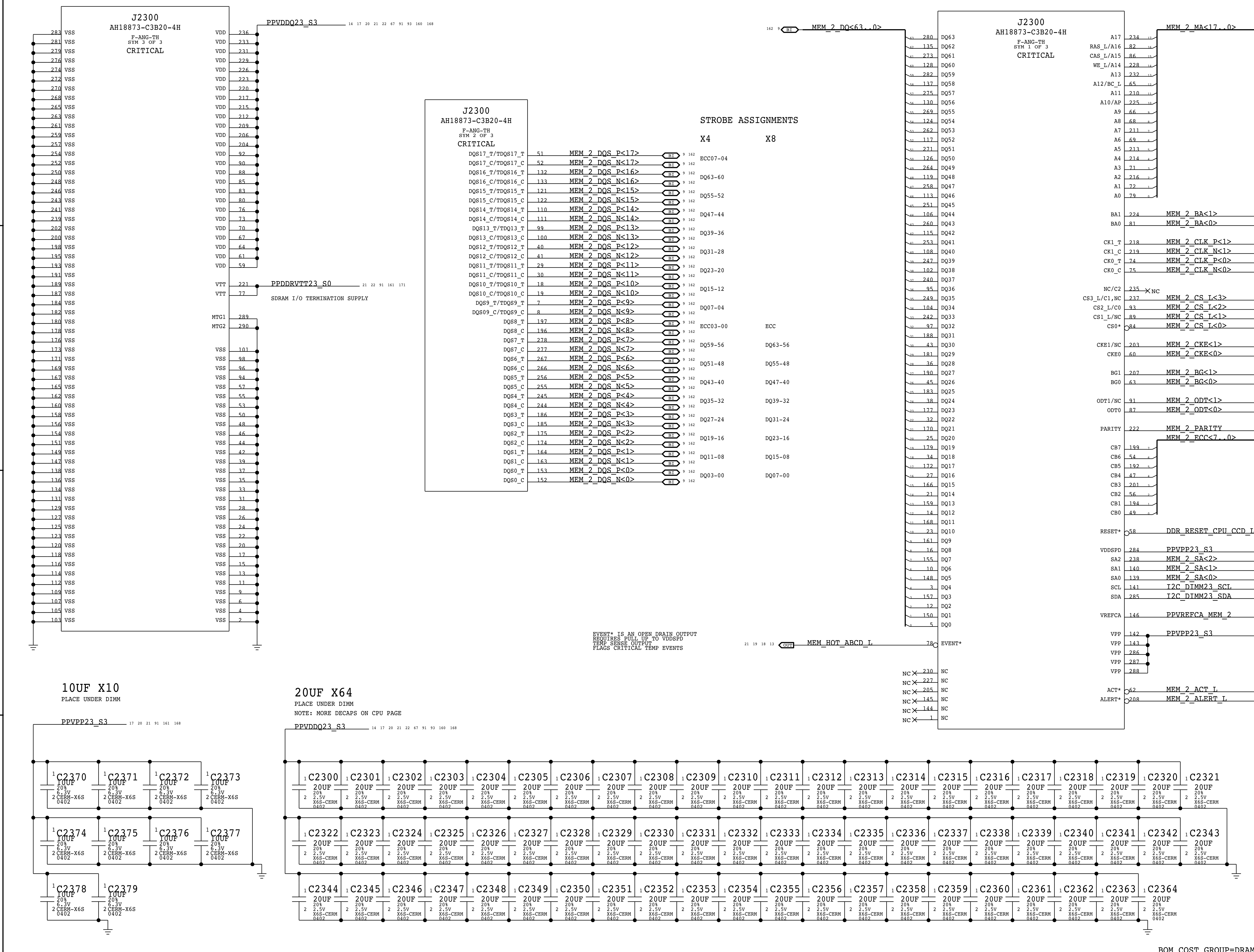

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PAGE TITLE

DDR4 DIMM CONN CH2

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SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

DDR4 DIMM CONN CH3

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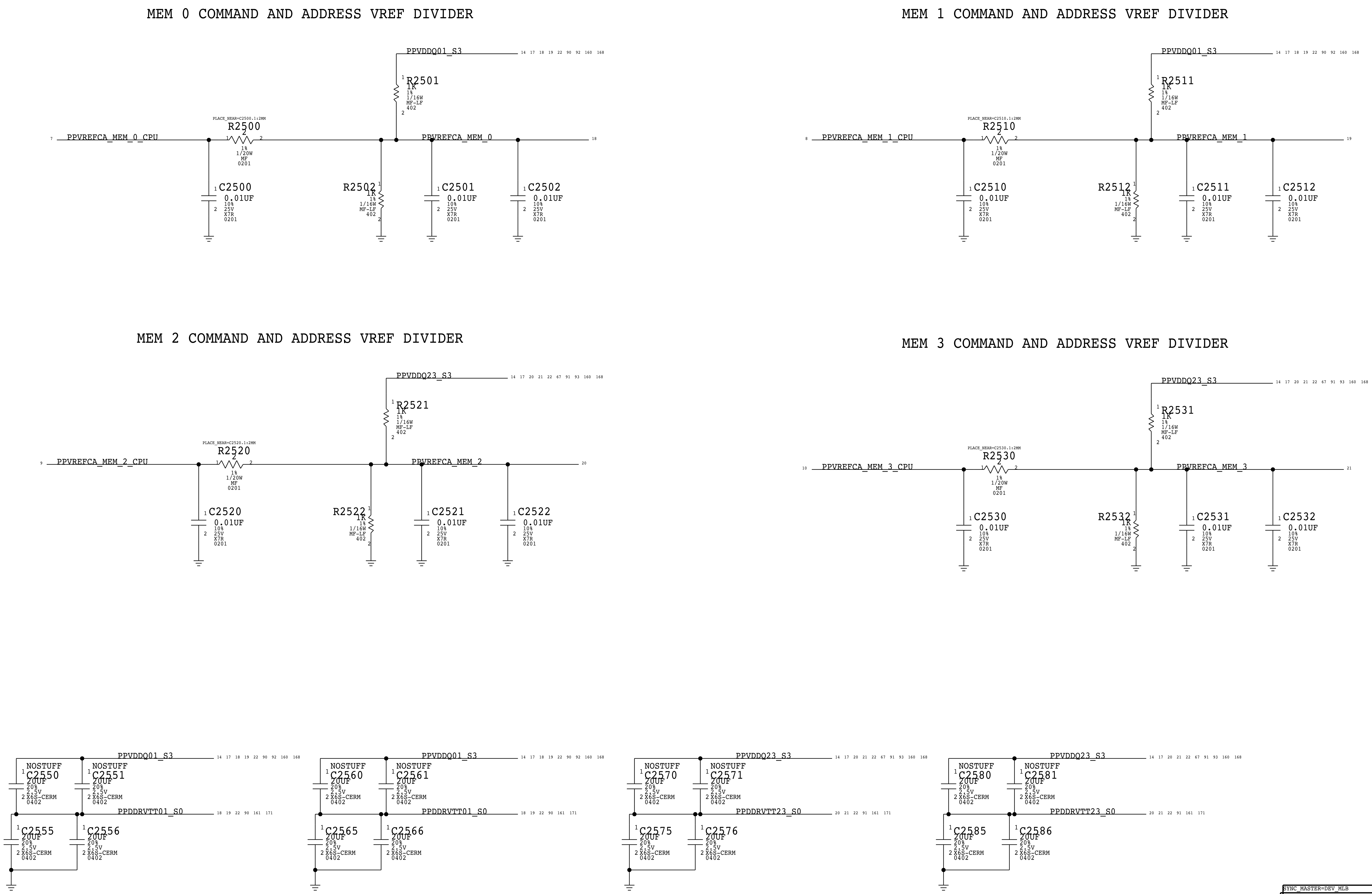

BOM_COST_GROUP=DRAM

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

DDR4 CA VREF, VTT

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SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

USB-C (T) CONNECTORS

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PAGE TITLE

USB-C (T) HIGH SPEED 1

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SYNC_MASTER=DEV_MLB   SYNC_DATE=11/06/2016

PAGE TITLE

USB-C (T) HIGH SPEED 2

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PAGE TITLE

USB-C (T) SUPPORT

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T PRIMARY ACE USB-C PORT CONTROLLER (UPC)

BOM_COST_GROUP=USB-C

SYNC_MASTER=DEV_MLB SYNC_DATE=11/07/2016

PAGE TITLE

USB-C (T) PORT CONTROLLER A

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T SECONDARY ACE USB-C PORT CONTROLLER (UPC)

BOM_COST_GROUP=USB-C

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE: USB-C (T) PORT CONTROLLER B

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SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

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USB-C (X) CONNECTORS

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USB-C (X) HIGH SPEED 1

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BOM_COST_GROUP=TBT

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USB-C (X) HIGH SPEED 2

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PAGE TITLE: USB-C (X) SUPPORT

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X PRIMARY ACE USB-C PORT CONTROLLER (UPC)

J137 USBC SUPPORTS 5V @ 3A
PP12V IS FOR PROGRAMMING ACE ONLY

NOTE:
DEBUG MUX IS CONNECTED TO
PRIMARY ACE OF 1ST RIDGE (XA ACE)
R3752-R3755, R3790-R3792 ARE OPTIONAL

BOM_COST_GROUP=USB-C

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE
USB-C (X) PORT CONTROLLER A

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X SECONDARY ACE USB-C PORT CONTROLLER (UPC)

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE
USB-C (X) PORT CONTROLLER B

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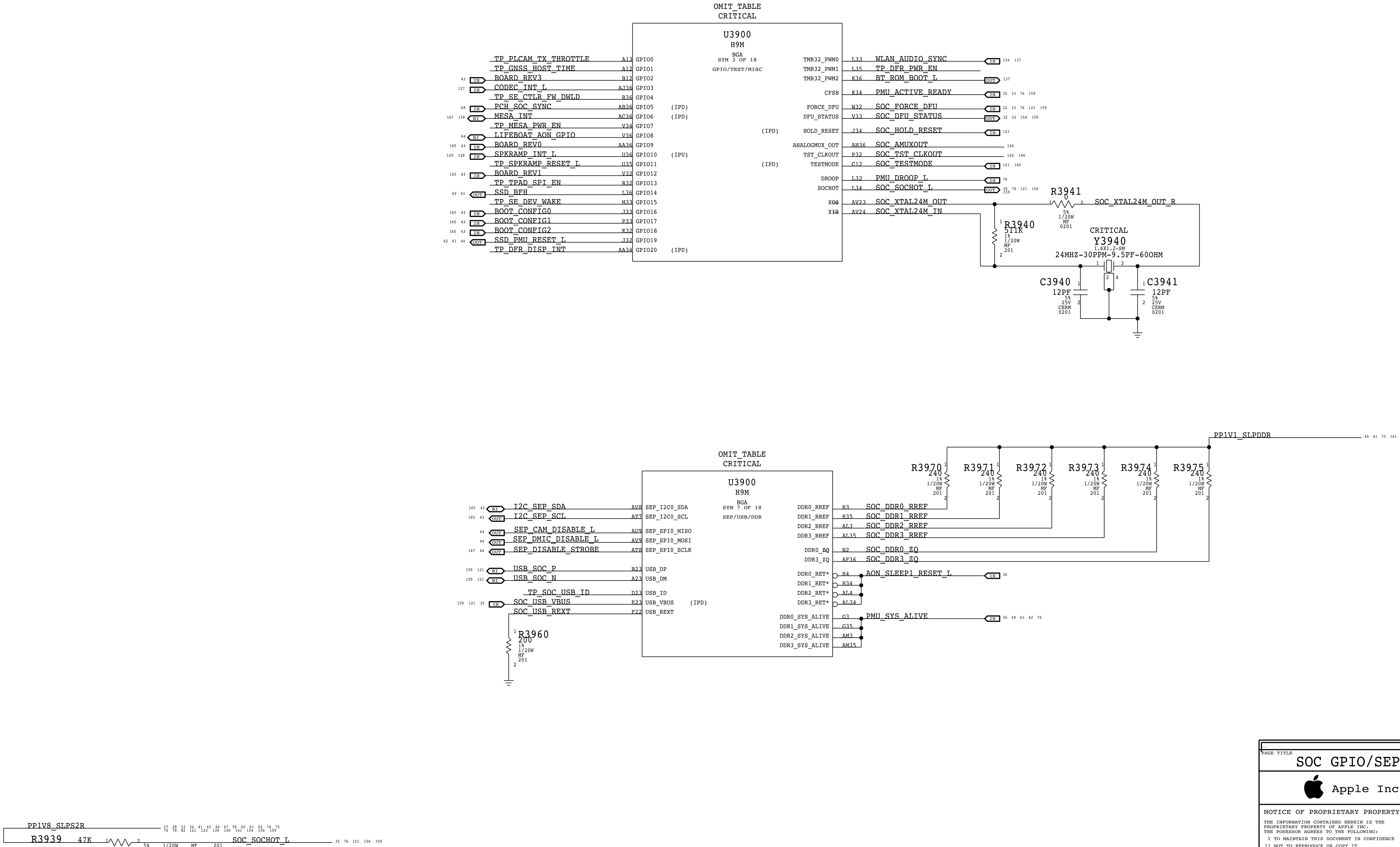
Note 1) IPU represents SW configured state, not HW default
Note 2) Internal and external pulls are preliminary assumptions. To be verified.



BOM_COST_GROUP=SOC

PAGE TITLE
SOC GPIO/SEP/USB/DDR/TEST

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BOM_COST_GROUP=SOC

PAGE TITLE

SOC AOP/AON/SMC

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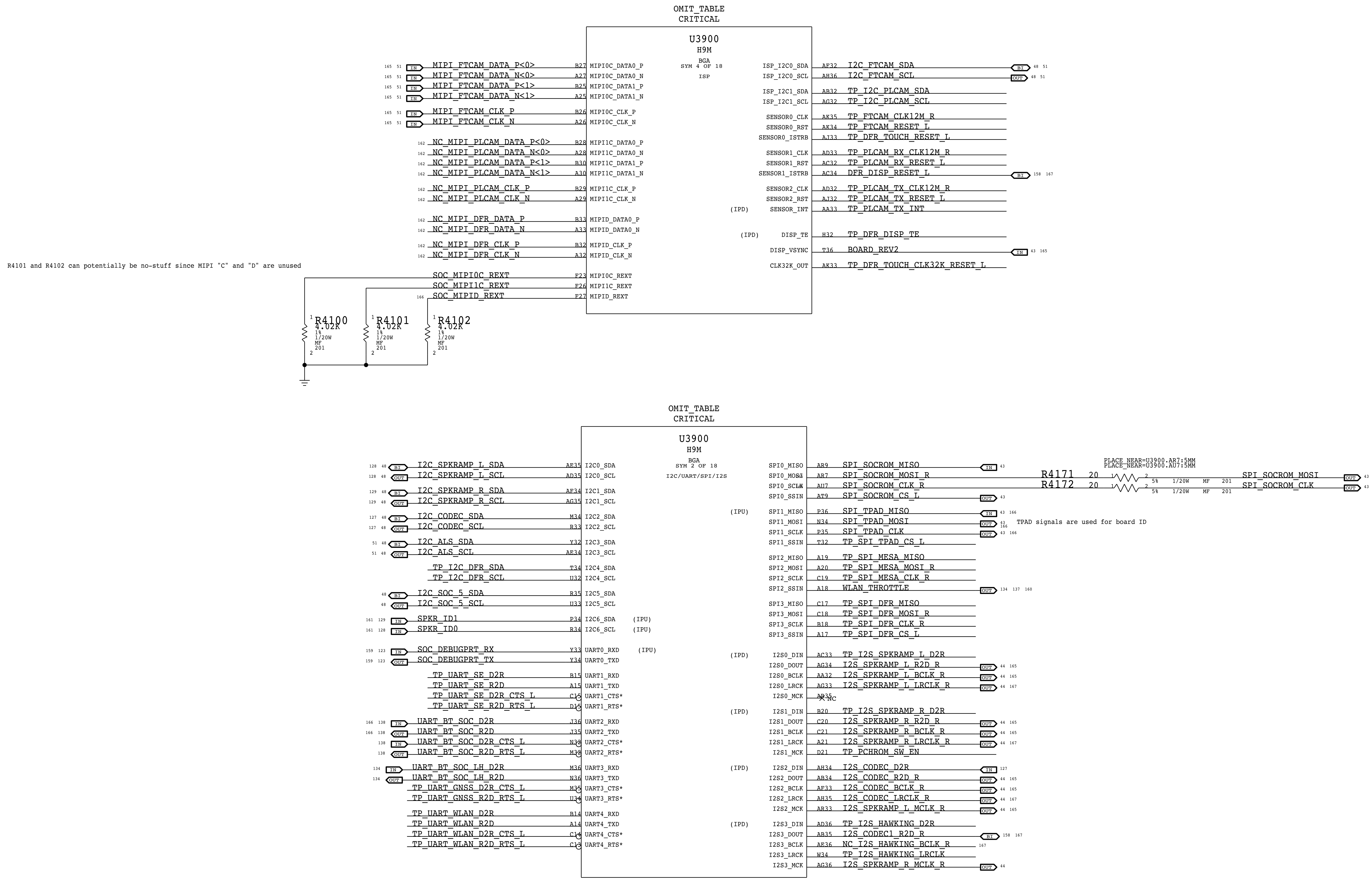

BOM_COST_GROUP=SOC

PAGE TITLE
SOC ISP/I2C/UART/SPI/I2S

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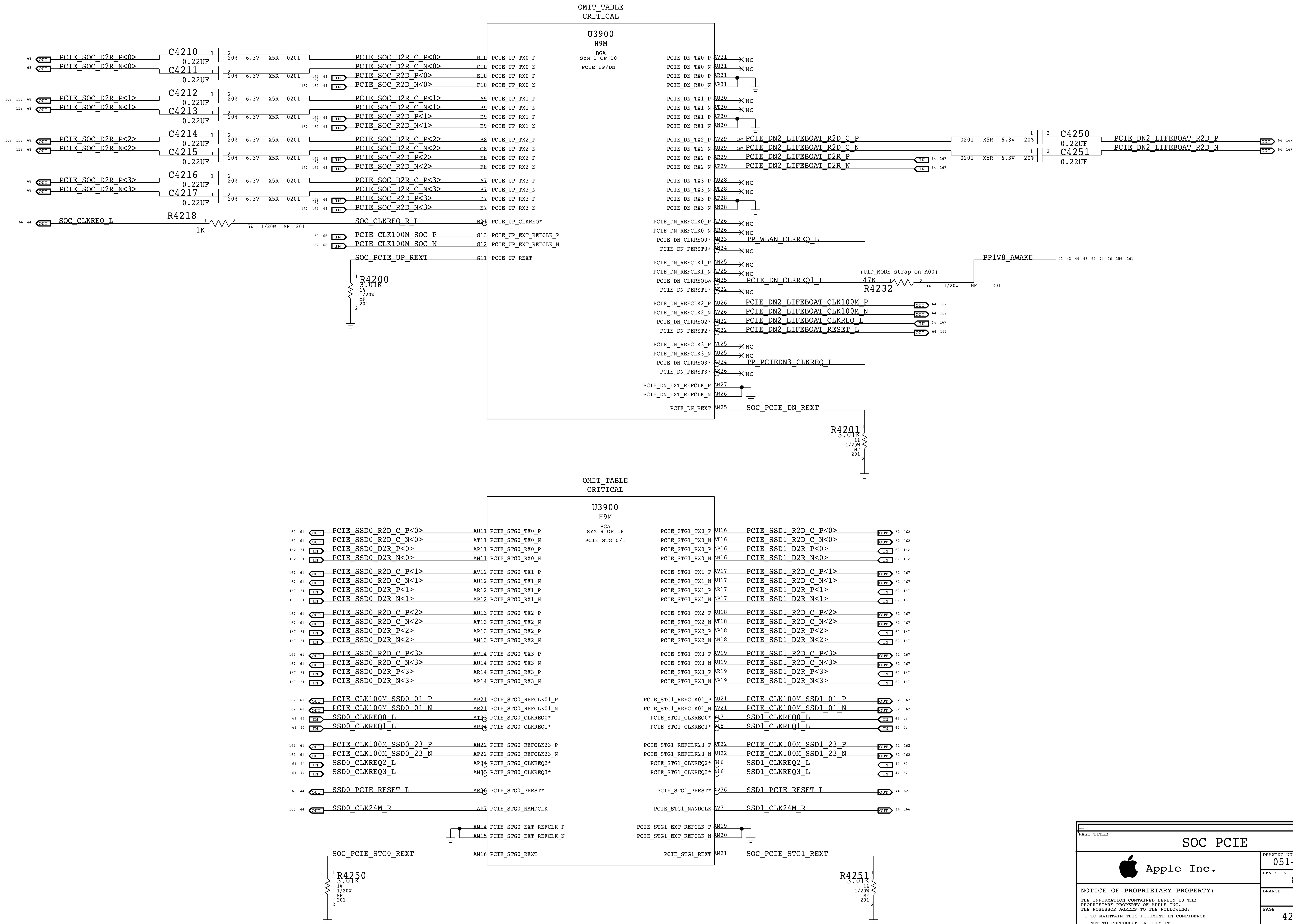

BOM_COST_GROUP=SOC

PAGE TITLE

SOC PCIE

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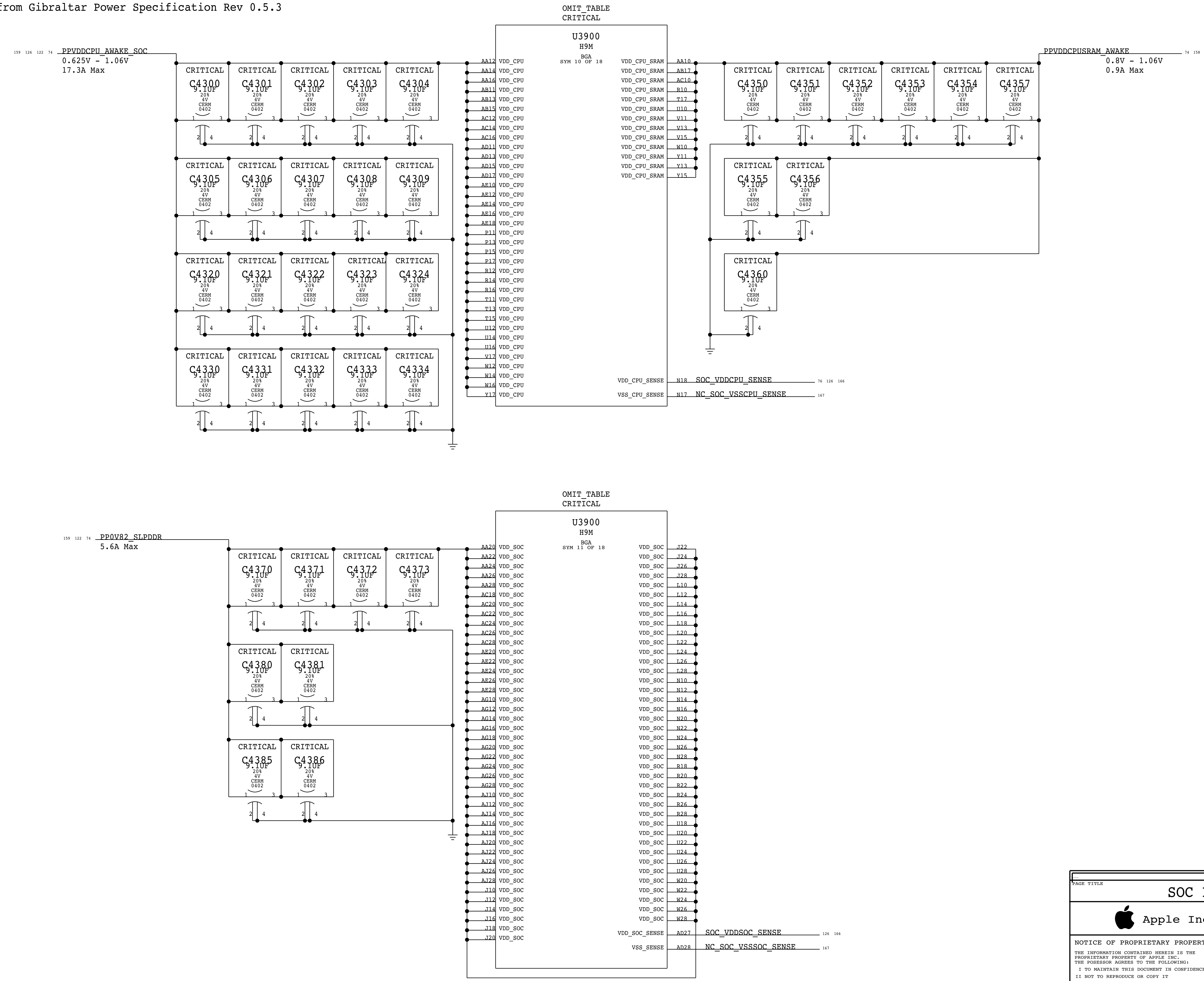
Current estimates @ 105C & 2GB from Gibraltar Power Specification Rev 0.5.3



BOM_COST_GROUP=SOC

PAGE TITLE
SOC POWER 1

Apple Inc.

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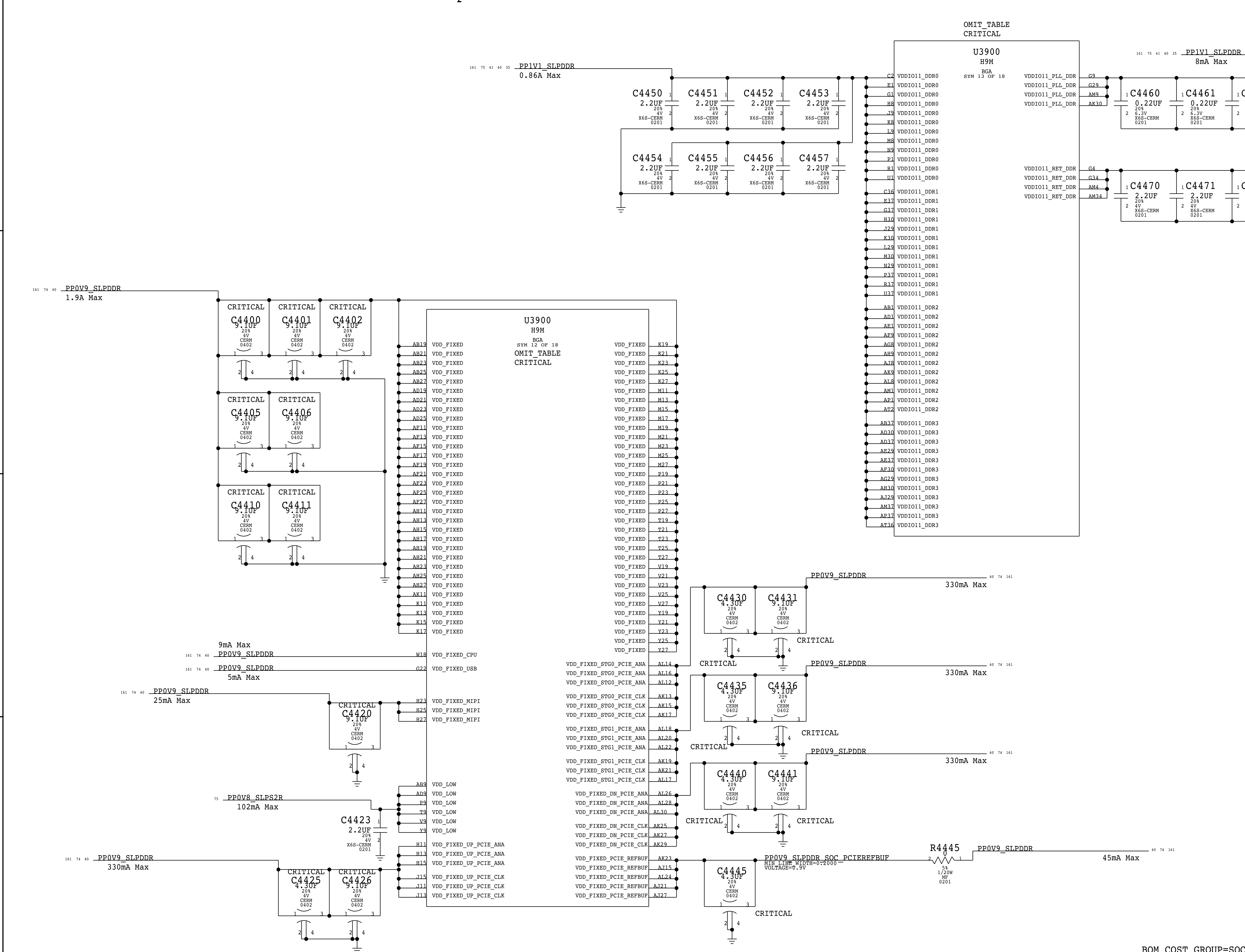
Current estimates @ 105C & 2GB from Gibraltar Power Specification Rev 0.5.3



PAGE TITLE
SOC POWER 2

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Current estimates @ 105C & 2GB from Gibraltar Power Specification Rev 0.5.3

BOM_COST_GROUP=SOC

PAGE TITLE
SOC POWER 3

Apple Inc.

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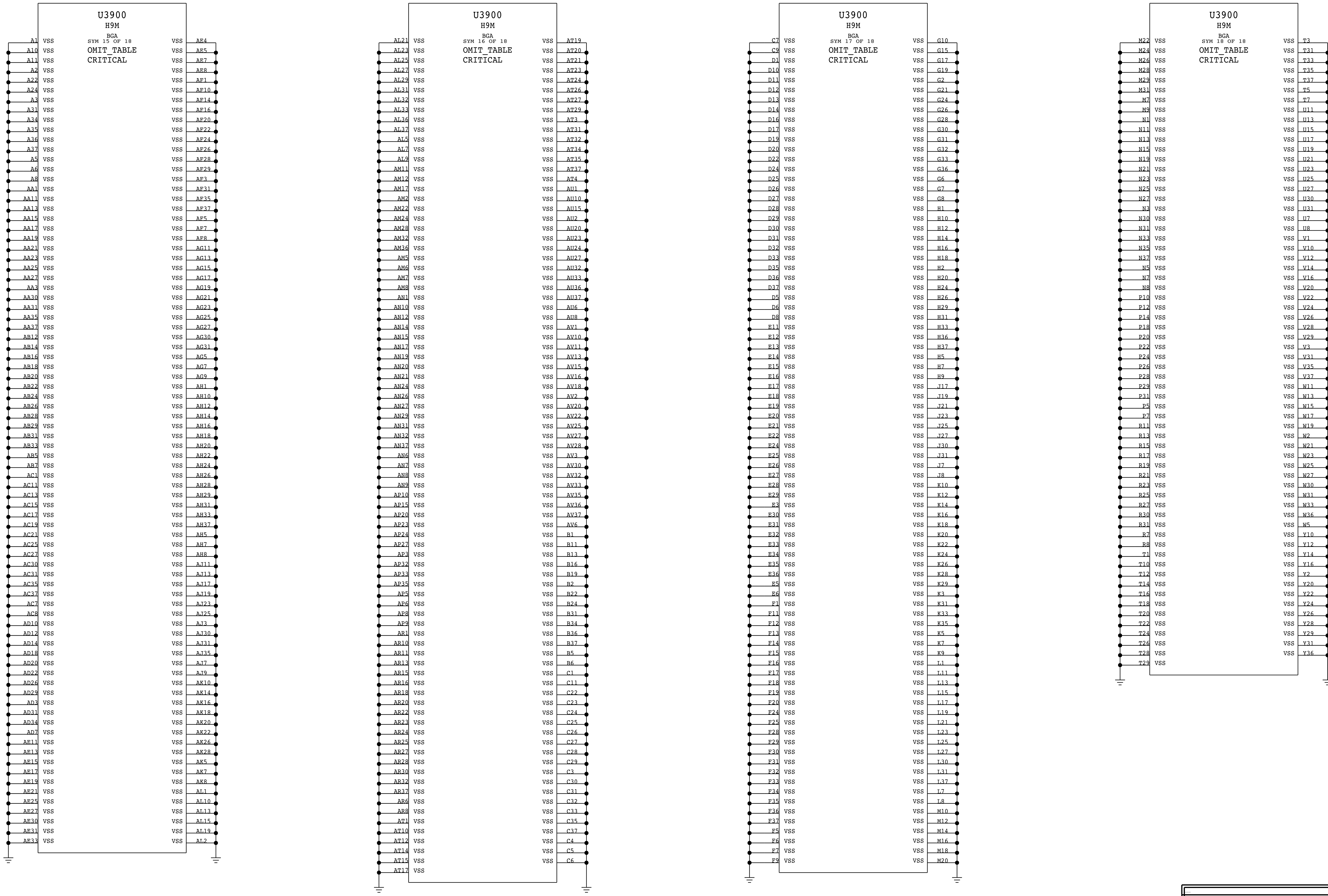

BOM_COST_GROUP=SOC

PAGE TITLE
SOC GROUND

Apple Inc.

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BOOTCFG0	Test Mode
0	Disabled
1	Enabled

BOOTCFG2	BOOTCFG1	Frequency
0	0	40 MHz
0	1	6 MHz
1	0	24 MHz
1	1	Invalid

Platform	BOARD_ID[5:0]
X589 KBL-U	000010
X589 CNL-Y	000011
J137 AP	001010

BOARD_REV[3:0]	
P0_DEV:	0000
P0:	0001
P1A:	0010
P1B:	0011
P1_DOE:	0100
EVT1:	0101
EVT2:	0110
DVT:	0111

PAGE TITLE: SOC SHARED SUPPORT

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SOC PROJECT SUPPORT 1

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PLATFORM THERMTRIP SUPPORT

INTENTION IS PMU GETS CPU/GPU THERMTRIP SO THAT IN THE CASE SMC HAS ISSUES, CALPE CAN STILL BRING DOWN THE SYSTEM GRACEFULLY

CALPE HAS AN INTERNAL DEBOUNCE, SMC DOESN'T, SO SMC VERSIONS OF THERMTRIP ARE MASKED

PCH GETS A COMBINED CPU/GPU THERMTRIP, SO IF EITHER ASSERTS, PCH SHOULD BRING DOWN THE SYSTEM GRACEFULLY

BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE	PLATFORM THERMTRIP SUPPORT
Apple Inc.	DRAWING NUMBER 051-01649 SIZE D
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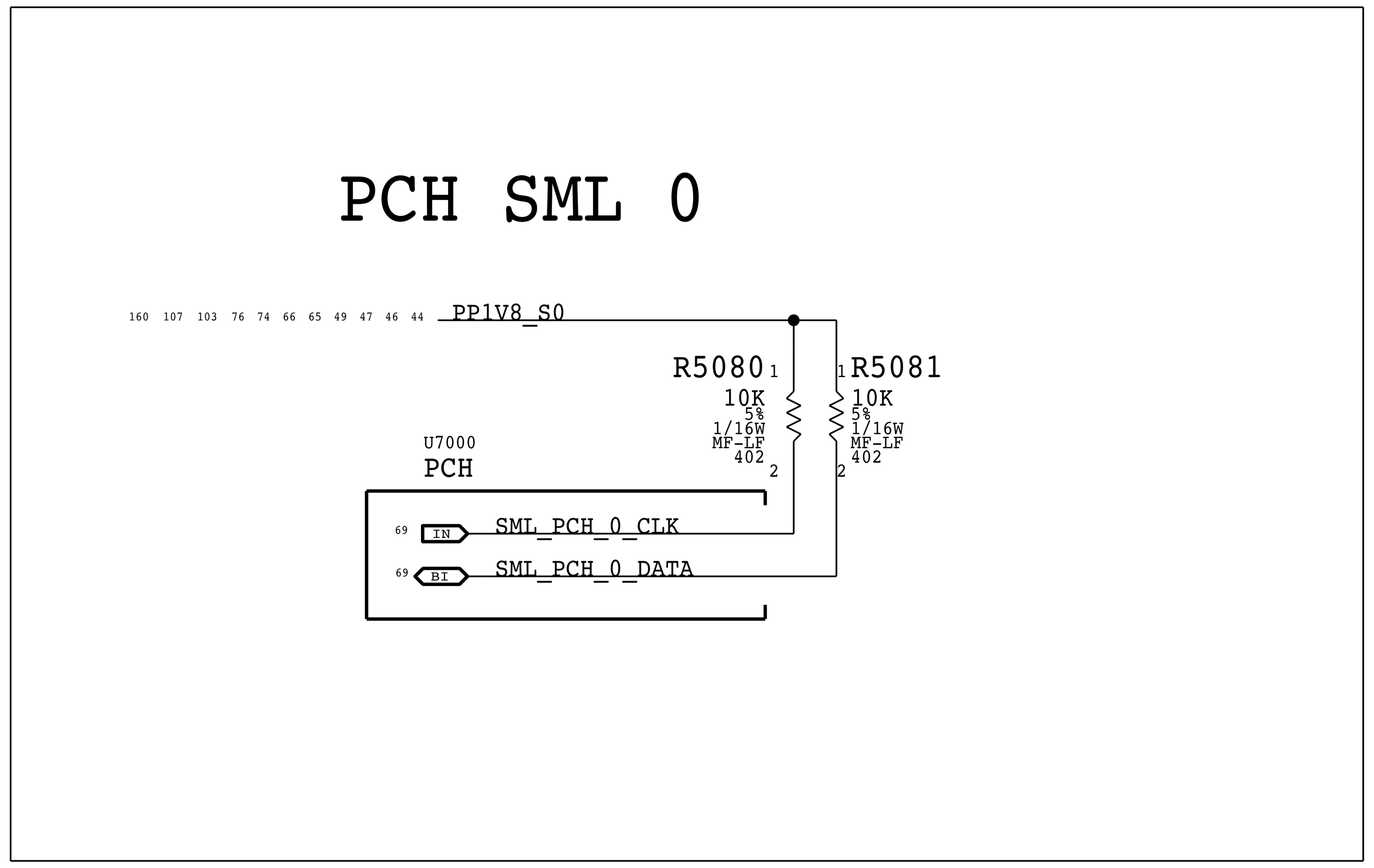
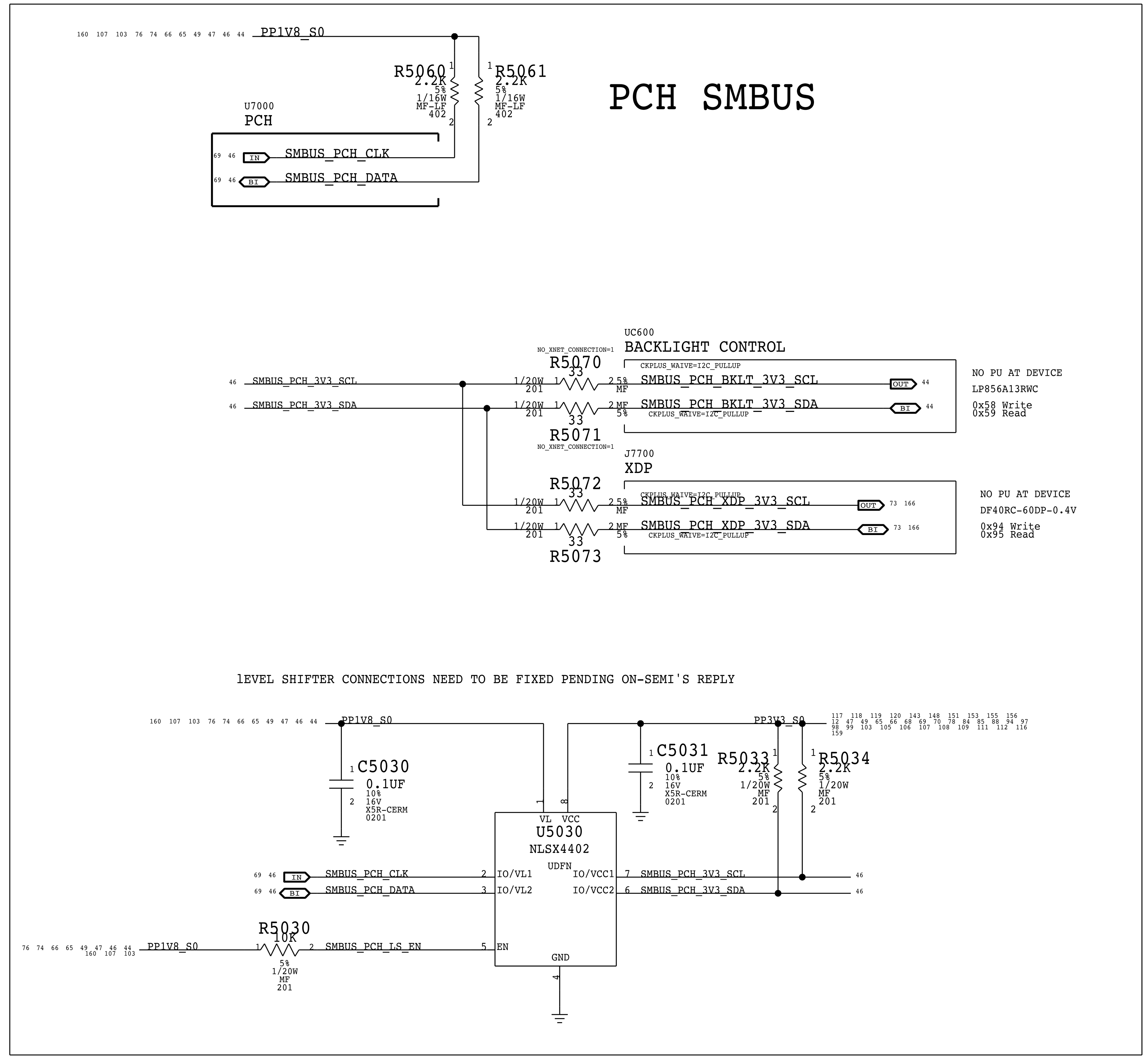
	SHEET 45 OF 171





BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE	I2C PCH
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FOLLOW UP: NEED LEVEL SHIFTER

BOM_COST_GROUP=SOC

PAGE TITLE

I2C SOC 1

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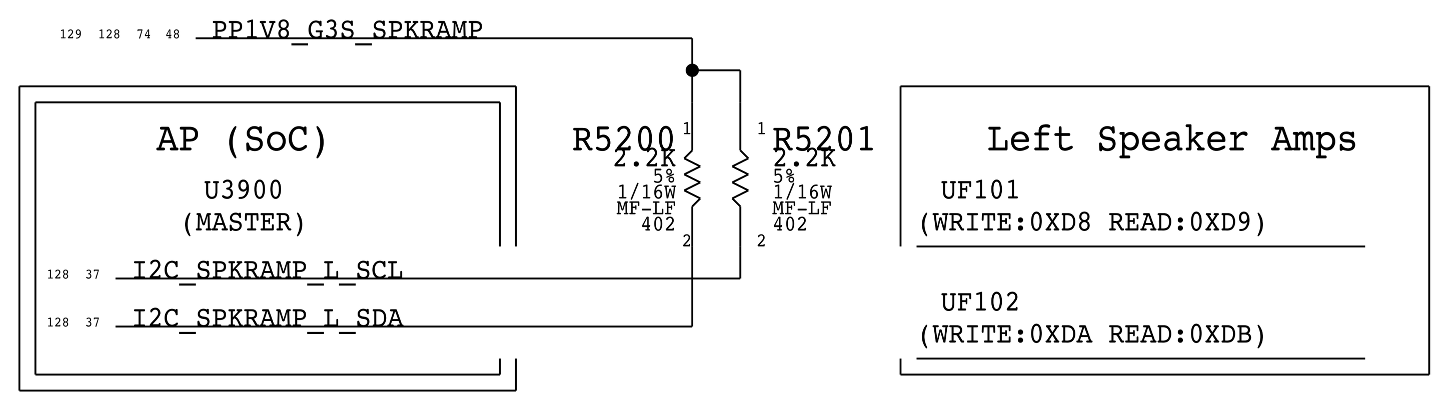
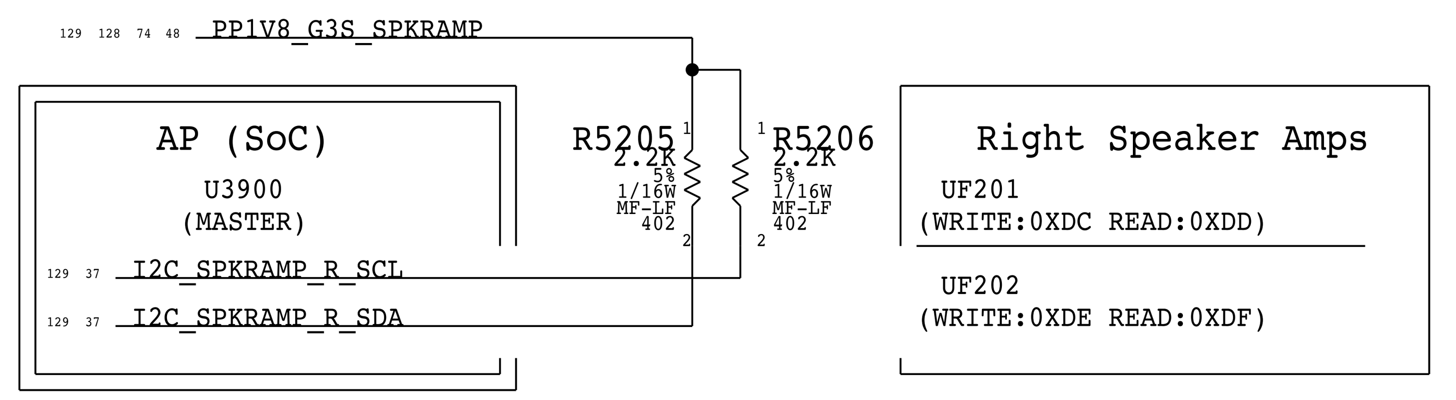
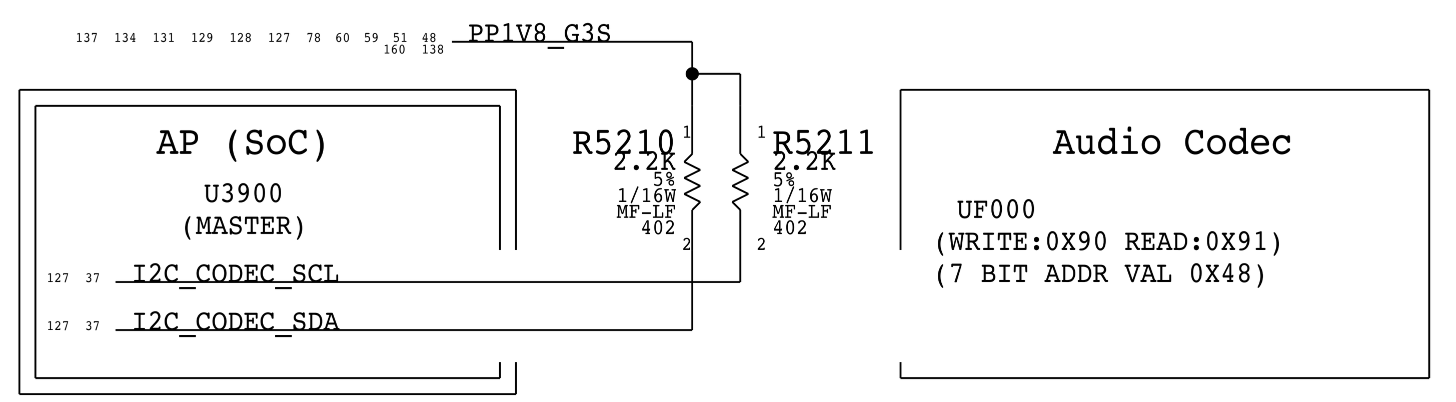
AP I2C "0" G3S Connections



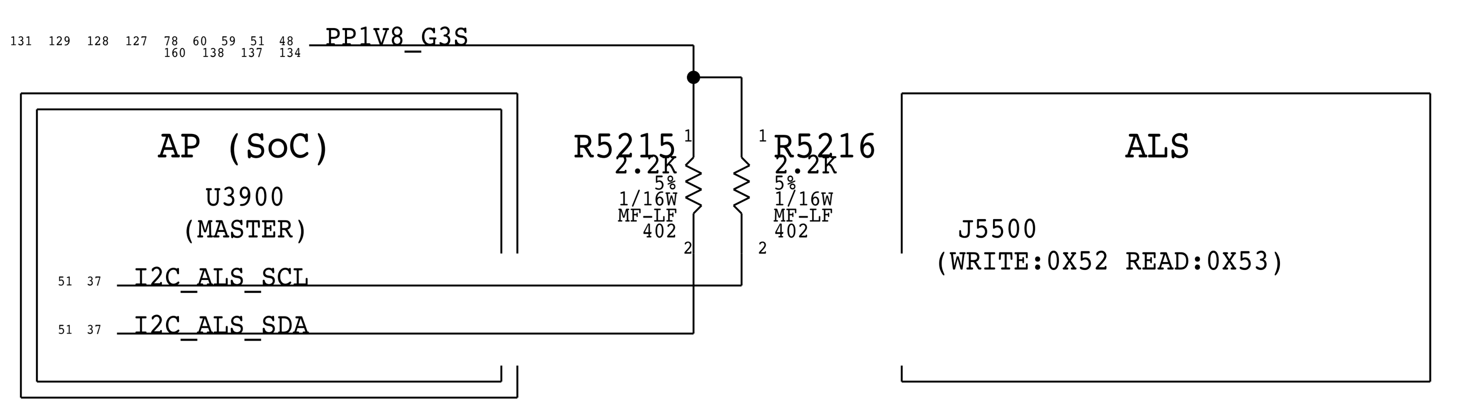
AP I2C "1" G3S Connections



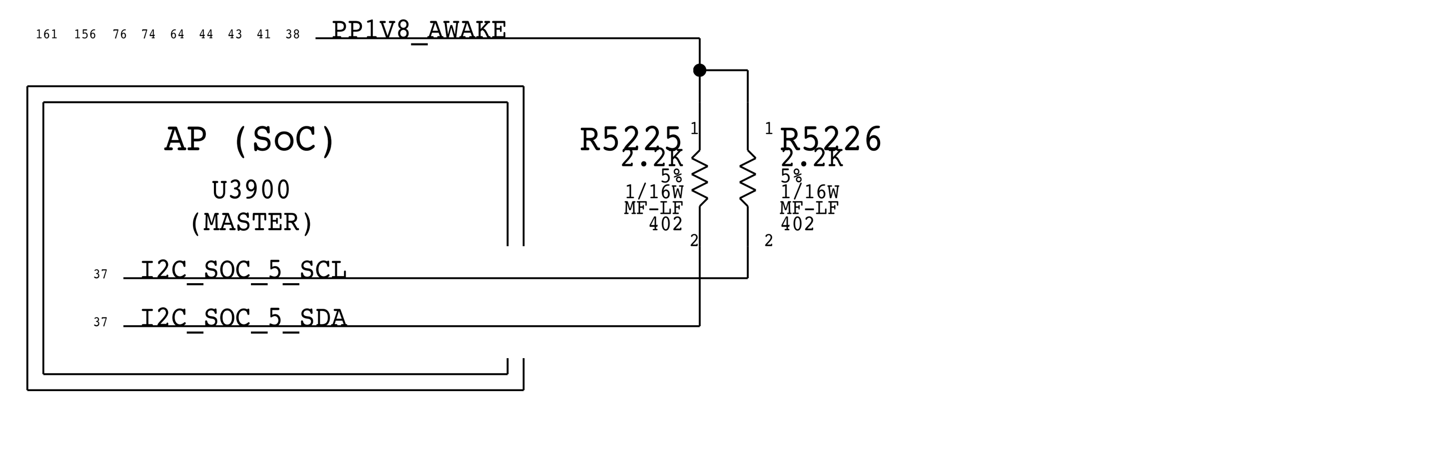
AP I2C "2" Codec Connections



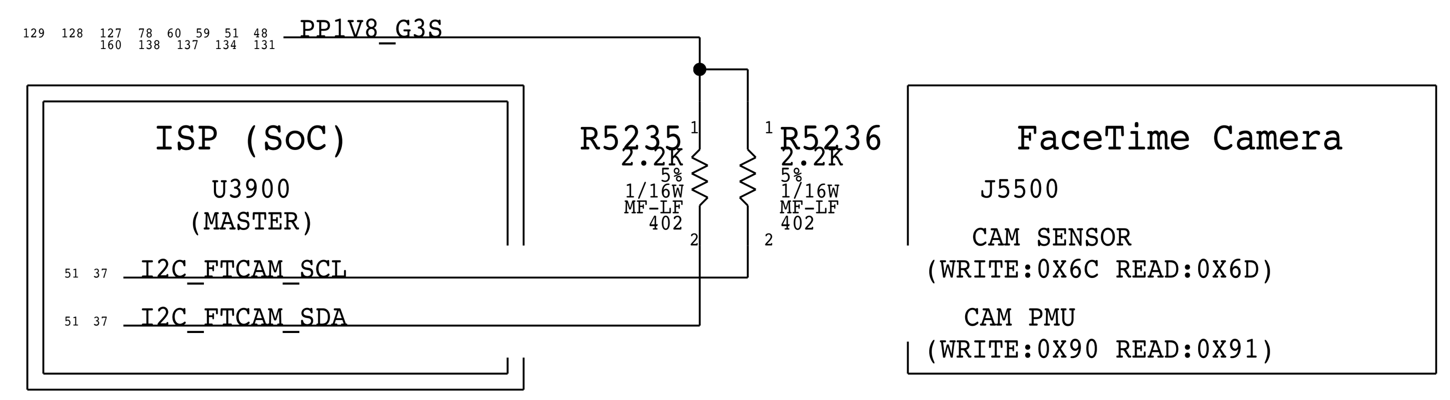
AP I2C "3" G3S CONNECTIONS



AP I2C "5" Awake Connections



ISP I2C "0" G3S CONNECTIONS



BOM_COST_GROUP=SOC

PAGE TITLE
I2C SOC 2

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BOM_COST_GROUP=SOC

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

IO EXPANDER

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PAGE TITLE

IO EXPANDER SUPPORT

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BOM_COST_GROUP=SOC

CAMERA/ALS/DMIC CONNECTOR

ALS I2C = SOC I2C3
2.2K PU TO 1V8_S0

CAM MIPI = SOC MIPI0C

CAM I2C = SOC ISP_I2C0
2.2K PU TO 1V8_S0

DMIC = SOC AOP
1V8 G3S

PP5V0 165 MA MAX
8 MA STANDBY

MODULE = IA CAMERA FLEX
IMAGE SENSOR = CONCORD

PP1V8 10 MA MAX
10 MA STANDBY

MIPI CLK: 766-798 MHZ
MIPI DATA: 1.53-1.59 GBPS

1928 X 1170 RESOLUTION

LAST UPDATE: 2017 MAR 10

BOM_COST_GROUP=CAMERA

SYNC_MASTER=AITKEN SYNC_DATE=03/10/2017

PAGE TITLE: CAMERA/ALS/DMIC CONNECTOR

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DRAWING NUMBER: 051-01649 SIZE: D

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SYNC_MASTER=ANDRES SYNC_DATE=02/13/2017

PAGE TITLE

USB-A EXTERNAL USB PORTS A & B

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BOM_COST_GROUP=USB-A

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

USB-A EXTERNAL USB PORTS C & D

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SYNC_MASTER=BOYLE SYNC_DATE=03/01/2017

PAGE TITLE

ETHERNET CONTROLLER 1

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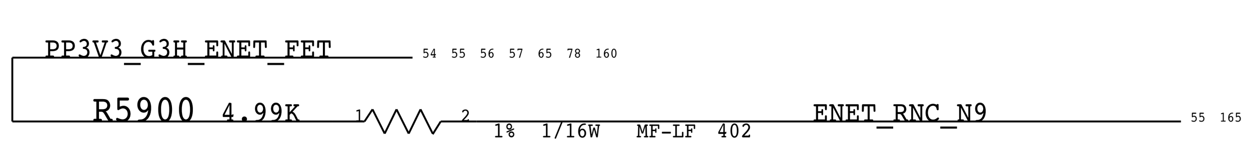
PULL UPS



BOM_COST_GROUP=ETHERNET

SYNC_MASTER=BOYLE
SYNC_DATE=03/01/2017

PAGE TITLE
ETHERNET CONTROLLER 2

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ETHERNET SUPPORT 1

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ETHERNET SUPPORT 2

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3.3V SD CARD LOAD SWITCH

U6250	EDP: 1.05A
Part	SLG5AP1438V
Type	Load Switch
R(on)	15 mOhm Typ 17 mOhm Max
Current	2.5A

NOTE:

HOST PROVIDES DVDD33 AND AVDD33
GL3227 GENERATES 1.2V USING INTERNAL LDO OR DCDC. DCDC IS ENABLED IN FW

FOR USH-I: 3.3V @ 900MA

FOR USH-II: 3.3V @ 700MA

1.2V SVR OPERATES @ 1.6 MHZ

GL3227 PROVIDES POWER TO SD CARD

FOR UHS-I: SD_V33 @ 800MA

FOR UHS-II: SD_V33 @ 400MA
SD_V18 @ 200MA MAX

PIN 29 IS SVR GND
DO NOT CONNECT PIN 29 TO EPAD DIRECTLY
CONNECT PIN 29 TO PIN 27 CAP FIRST
THEN TO MAIN GND

NOTE:

UHS-2 DIFF PAIRS 100 OHM
400 MV PEAK TO PEAK
UP TO 1.56 GBIT/S

UHS-2 DIFF CLK UP TO 52 MHZ

USH-1 SE SIGNALS 50-OHM

SD_CLK MAX FREQ IS 208 MHZ
AT SDR104 MODE

LED IS ON WHEN POWER ON
BLINKING DURING R/W OPERATIONS

BOM_COST_GROUP=SD CARD

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE
SD CONTROLLER UHS-II

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BOM_COST_GROUP=SD CARD

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

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SD CONNECTOR UHS-II

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SYNC_MASTER=AITKEN SYNC_DATE=03/06/2017

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FANS

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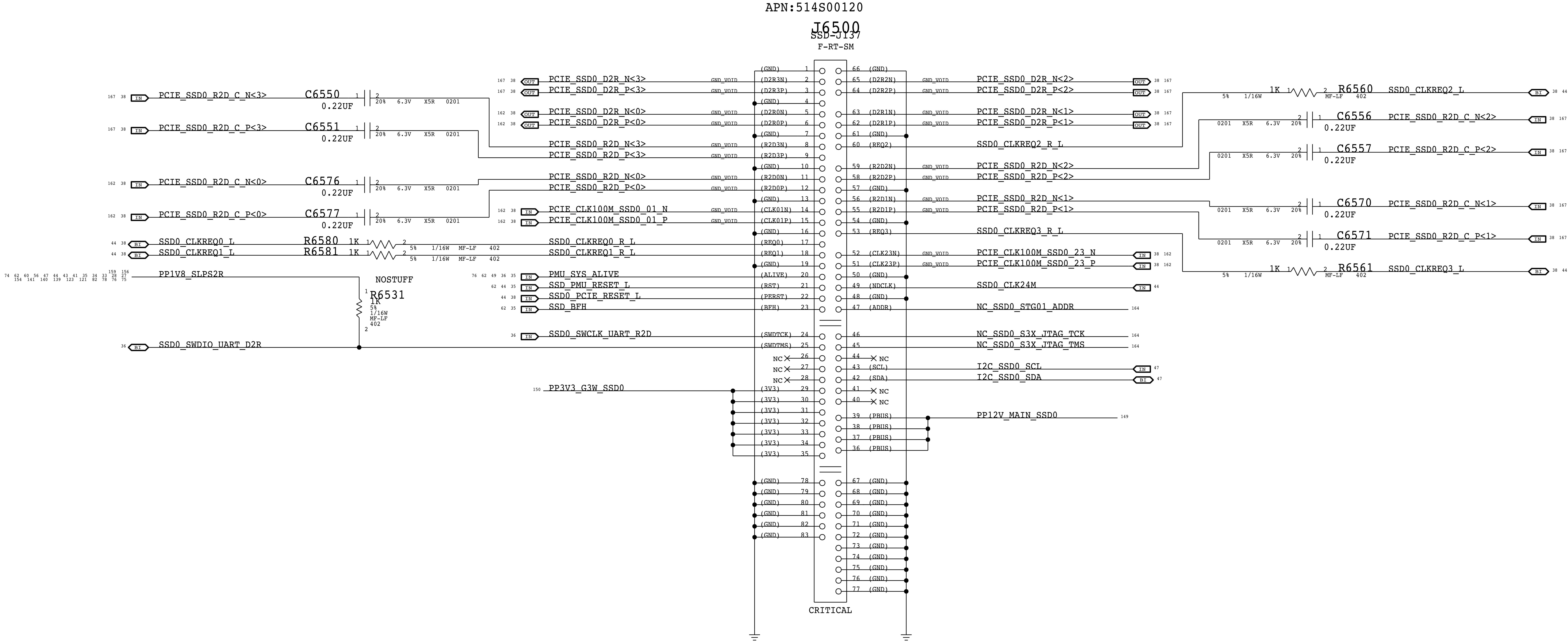
SSD GS5 CONNECTOR A



BOM_COST_GROUP=SSD

PAGE TITLE
SSD CONN A

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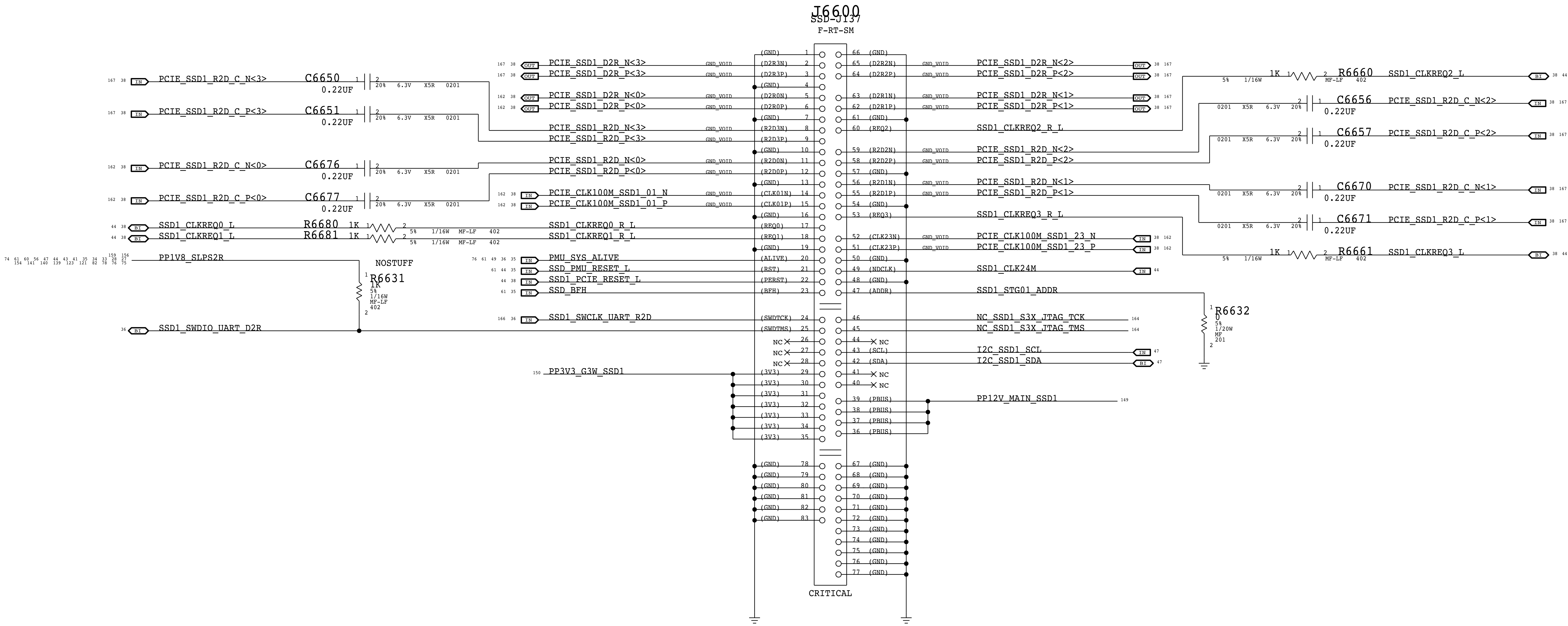
SSD GS5 CONNECTOR B



BOM_COST_GROUP=SSD

PAGE TITLE
SSD CONN B

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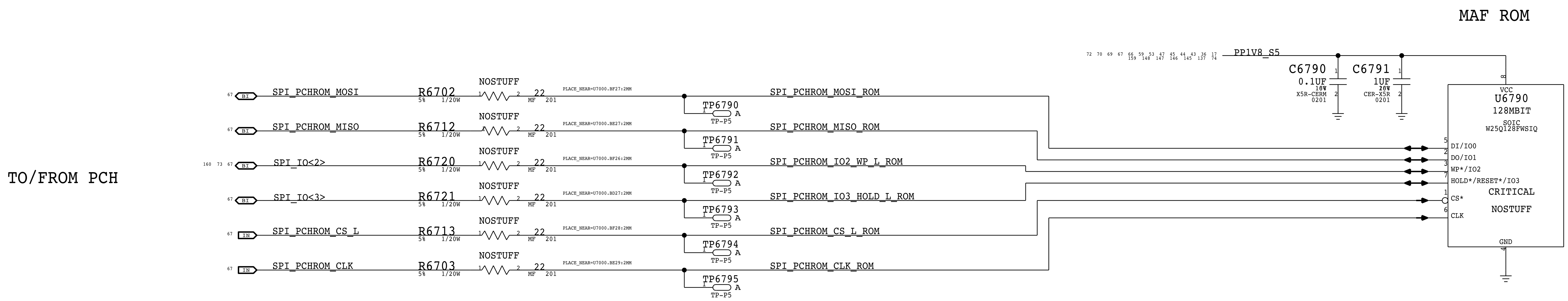

SPI ROM

Quad-IO Mode (Mode 0 & 3) supported.
SPI Frequency: 50MHz for CPU, 40MHz for SMC.

Quad SPI and QPI instructions require the non-volatile Quad Enable bit (QE)
in Status Register-2 to be set. When QE=1, the /WP pin becomes IO2 and /HOLD pin becomes IO3.

NOTE: If HOLD* is asserted
ROM will ignore SPI cycles
in normal and Dual-IO modes.

BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

SPI ROM

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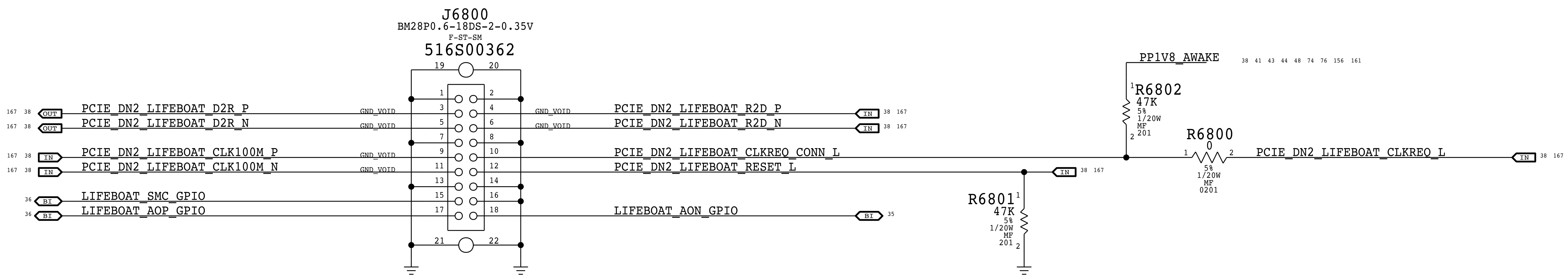

LIFEBOAT GPIOS ASSIGNED PER <RDAR://33517854>

BOM_COST_GROUP=SSD

SYNC_MASTER=AMAN SYNC_DATE=07/25/2017

PAGE TITLE

LIFEBOAT

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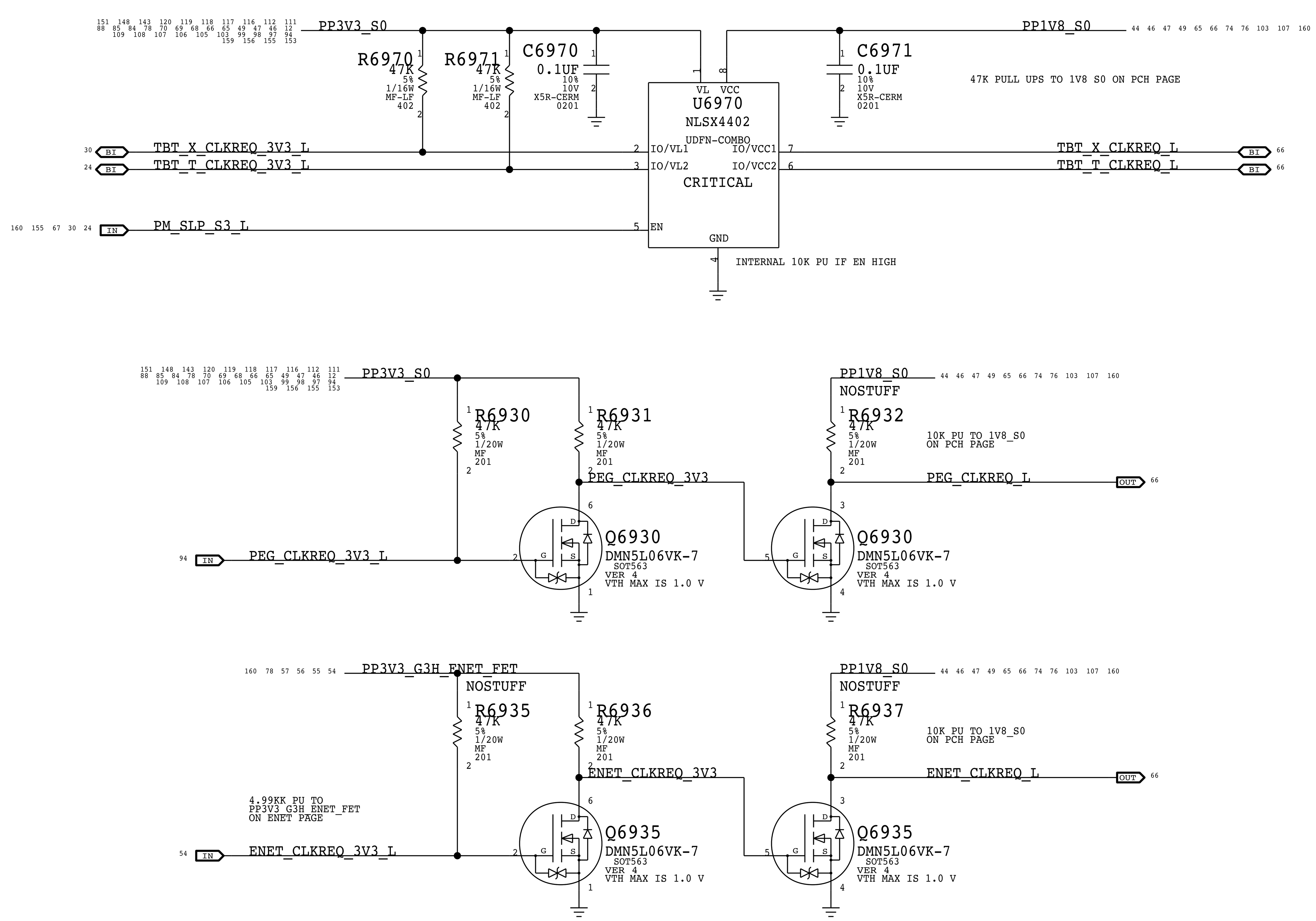
CLKREQ LEVEL SHIFTERS



BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

PCH LEVEL SHIFTERS

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D

REVISION 6.21.0

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BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=AITKEN SYNC_DATE=03/06/2017

PAGE TITLE

PCH CLOCKS, JTAG

Apple Inc.

DRAWING NUMBER: 051-01649 SIZE: D

REVISION: 6.21.0

BRANCH

SYNC_MASTER=AITKEN SYNC_DATE=03/10/2017

PAGE TITLE

PCH DMI, PM, MISC

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D

REVISION 6.21.0

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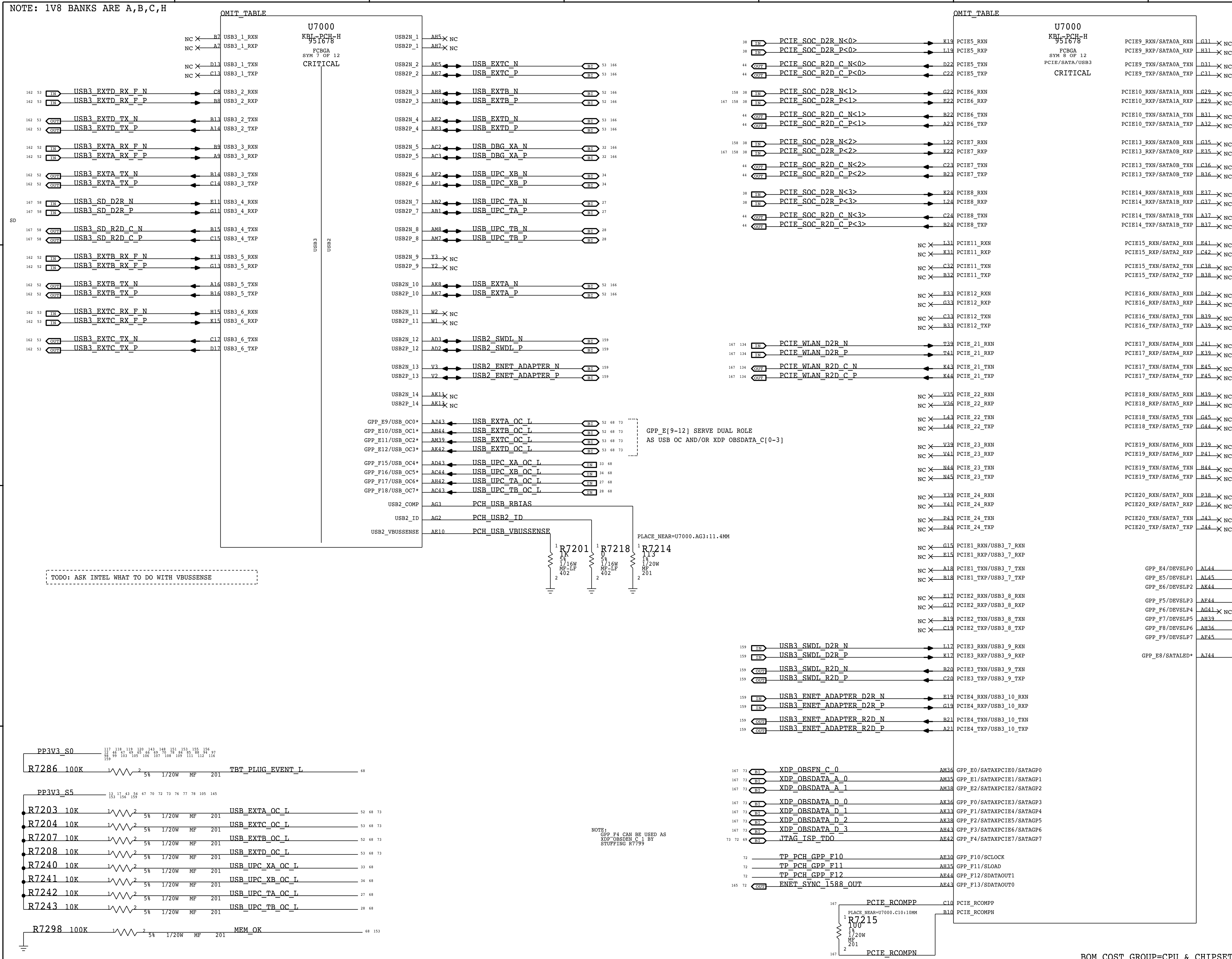

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

PCH PCIE, USB

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D

REVISION 6.21.0

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SYNC_MASTER=AITKEN SYNC_DATE=03/10/2017

PAGE TITLE

PCH GPIO

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REVISION 6.21.0

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NOTE: 1V8 BANKS ARE A,B,C,H

PAGE TITLE

PCH POWER, DECOUPLING

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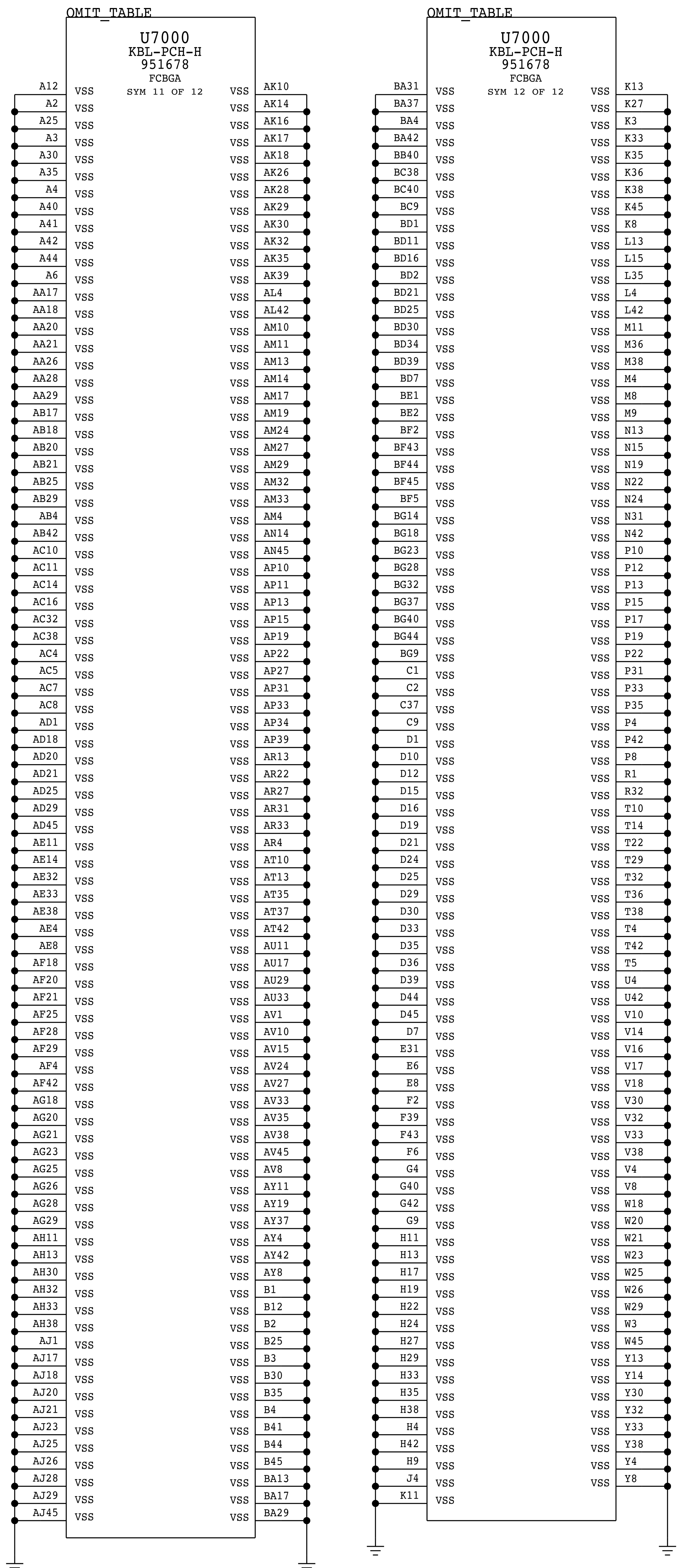
REVISION 6.21.0

BRANCH

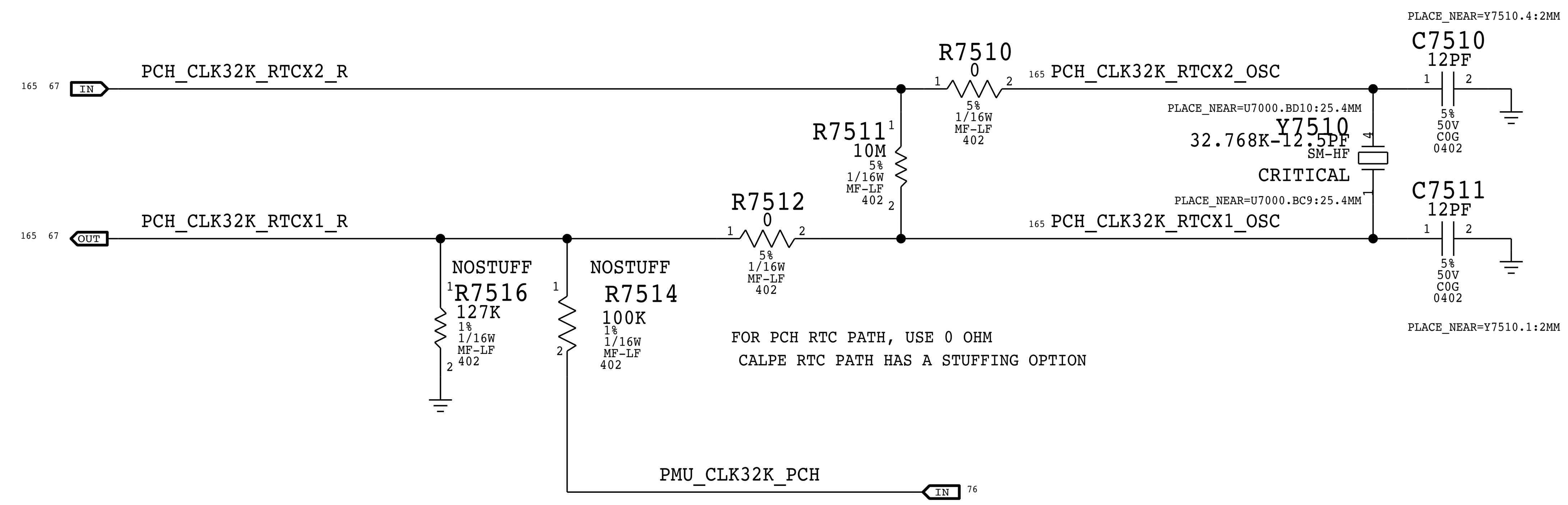
BOM_COST_GROUP=CPU & CHIPSET

8 7 6 5 4 3 2 1

D C B A



PCH RTC Crystal



BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=AITKEN SYNC_DATE=02/07/2017

PAGE TITLE

PCH GND, RTC, ME

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D

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PCH TEST POINTS
TP7600 TP_PCH_GPP_B2
TP7601 TP_PCH_GPD7
TP7602 TP_HDA_SYNC
TP7603 TP_PM_SLP_A_L
TP7605 TP_PCH_GPP_B19
TP7606 TP_PCH_GPP_H20
TP7607 TP_PCH_GPP_H13
TP7608 TP_PCH_GPP_H10
TP7609 TP_PCH_GPP_H5
TP7610 TP_PCH_GPP_H8
TP7612 TP_PCH_GPP_H4
TP7613 TP_PCH_GPP_A6
TP7614 TP_PCH_GPD11
TP7615 TP_HDA_RST_L
TP7616 TP_HDA_SDI1
TP7617 TP_HDA_BCLK
TP7618 TP_HDA_SDIO
TP7619 TP_PCH_GPD9
TP7620 TP_GPP_A10
TP7621 TP_PCH_GPP_H19
TP7622 TP_PCH_GPP_D10
TP7623 TP_PCH_GPP_H14
TP7624 TP_PCH_GPP_H7
TP7625 TP_PCH_GPD8
TP7626 TP_PCH_SLP_LAN_L
TP7627 TP_CLINK_RESET_L
TP7628 TP_CLINK_CLK
TP7629 TP_CLINK_DATA
TP7630 TP_FPF_VREF
TP7631 TP_PCH_GPP_H11
TP7632 TP_PCH_GPP_D9
TP7634 TP_FPF_MON
TP7635 TP_PCH_2_CPU_TRIGGER
TP7636 TP_CPU_2_PCH_TRIGGER
TP7637 PCH_PM_DOWN
TP7638 TP_PCH_GPP_F11
TP7639 TP_PCH_GPP_F8
TP7640 TP_PCH_GPP_F9
TP7641 TP_PCH_GPP_F12
TP7644 TP_PCH_GPP_F23
TP7645 TP_PCH_GPP_F10
TP7646 TP_KBLX_CLK24M
TP7648 TP_PCH_GPP_A22
TP7649 TP_PCH_GPP_B23
PM_SLP_S0_L OPTION
NOSTUFF
R7690 0
PCH_PM_SLP_S0_L
PM_SLP_S0_L
R7691 100K 5% 1/20W MF 201
ETHERNET TIMESYNC
R7693 0
ENET_SYNC_1588_OUT
(FROM PCH)
ENET_SYNC_1588
(AQC107)
R7694 0
ENET_SYNC_1588_IN
(TO PCH)
XDP HOOK[2] LEVEL SHIFTER
PP3V3_S5
R7652 10K 5% 1/20W MF 201
(150 PU TO VCCIO ON XDP PAGE)
PCH_GPD_0_FET_L
XDP_CPU_PWR_DEBUG_FET_L
R7654 0
XDP_CPU_PWR_DEBUG_L
(CPU, XDP)
(1K PU TO 3V3 S5 ON PCH PAGE)
PCH_GPD_0
(PCH)
R7650 0
PCH_GPD_0_FET
Q7650 DMN5L06VK-7 SOT563 VER 4 VTH MAX IS 1.0 V
NOSTUFF
R7653 0 5% 1/16W MF-LF 402
RIDGE JTAG ISOLATION
JTAG_TBT_T_TMS_PCH R7610 0 JTAG_TBT_T_TMS
JTAG_TBT_X_TMS_PCH R7607 0 JTAG_TBT_X_TMS
JTAG_ISP_TDO R7608 0 JTAG_TBT_TDO
JTAG_ISP_TDI R7609 0 JTAG_TBT_TDI
JTAG_ISP_TCK R7623 0 JTAG_TBT_TCK
TBT_X_PWR_EN_PCH R7624 0 TBT_X_CIO_PWR_EN
TBT_T_PWR_EN_PCH R7620 0 TBT_T_CIO_PWR_EN
TBT_PWR_EN GOES HIGH FOR JTAG PROGRAMMING
PCH DEBUG UART HEADER
PP3V3_G3S
DEBUG_UART R7633 100K 5% 1/16W MF-LF 402
DEBUG_UART C7630 0.1UF 10% 10V X5R-CERM 0201
PP1V8_S5
DEBUG_UART C7631 0.1UF 10% 10V X5R-CERM 0201
DEBUG_UART R7632 100K 5% 1/16W MF-LF 402
Q7632 DMN32D2LFB4 DFN1006H4-3 SYM_VER_3
DEBUG_UART
PCH_DEBUGUARTCON_EN_L
U7630 74AVC2T245-S500 XQFN-1
VCCA VCCB OE* DIR1 A1 A2 DIR2 B1 B2 GND
DEBUG_UART
TP7690 PCH_DEBUGUARTCON_EN
TP7691 PCH_DEBUG_UART_D2R_3V3
TP7692 PCH_DEBUG_UART_R2D_3V3
TP7693
PCH_DEBUG_UART_D2R_R R7698 0 PCH_DEBUG_UART_D2R
PCH_DEBUG_UART_R2D_R R7699 0 PCH_DEBUG_UART_R2D
DEBUG_UART PLACE_NEAR=U7000.AY44:10MM
PLACE_NEAR=U7000.BA33:10MM DEBUG_UART
NOTE:
R7698,R7699 SHOULD BE PLACED CLOSE TO PCH PINS AND ALSO CLOSE TO RESISTORS FOR ACE PATH ALLOWING US TO KEEP PCH UART WITHOUT A STUB
BOM_COST_GROUP=CPU & CHIPSET
SYNC_MASTER=AITKEN
SYNC_DATE=03/10/2017
PAGE TITLE
PCH SUPPORT
Apple Inc.
DRAWING NUMBER 051-01649
SIZE D
REVISION 6.21.0
BRANCH
PAGE 76 OF 255
SHEET 72 OF 171

NOTE: JTAG PINS CAN BE LEFT FLOATING ON PCH AND CPU

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE	XDP
DRAWING NUMBER	051-01649
SIZE	D
REVISION	6.21.0
BRANCH	
PAGE	77 OF 255
SHEET	73 OF 171

Apple Inc.

Note : Design based on Calpe ERS - D2449-A0-110-00_0v3.pdf (Radar# 24696002)

Optimize components for individual projects based on EDP(A)

PAGE TITLE

PMIC BUCKS AND SWS

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DRAWING NUMBER 051-01649 SIZE D

REVISION 6.21.0

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BOM_COST_GROUP=SOC

PAGE TITLE

PMIC LDOS

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PBUS	Rising Vth	Falling Vth
12V	9.076V	8.443V

BOM_COST_GROUP=SOC

PAGE TITLE
PMIC GPIOS & CONTROL

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PAGE TITLE
PMIC SUPPORT

Apple Inc.

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SIZE D
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BRANCH
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BOM_COST_GROUP=PLATFORM POWER

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

POWER FETS

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BRANCH

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SHEET 78 OF 171

3V3 G3W REGULATOR	
VOUT	3.32 V
PEAK (4 US)	7.6 A
OC	12.1 A
AVG	6.1 A
FREQ	500 KHZ

5V G3S USBC REGULATOR	
VOUT	5.1 V
PEAK (4 US)	13.2 A
OC (TYP)	21.2 A
AVG	10.2 A
FREQ	500 KHZ

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE: VR P3V3 G3W, P5V G3S USBC

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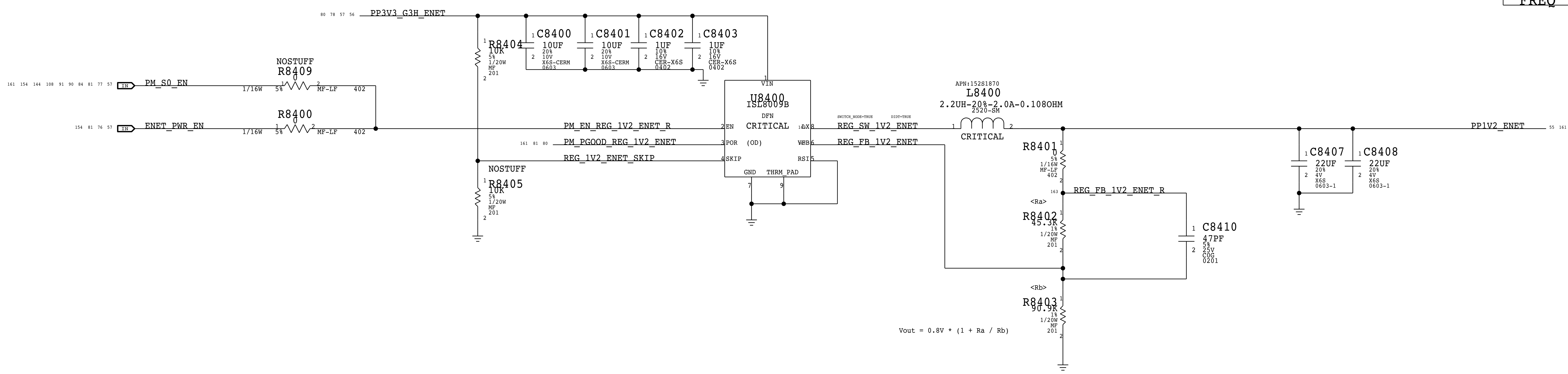
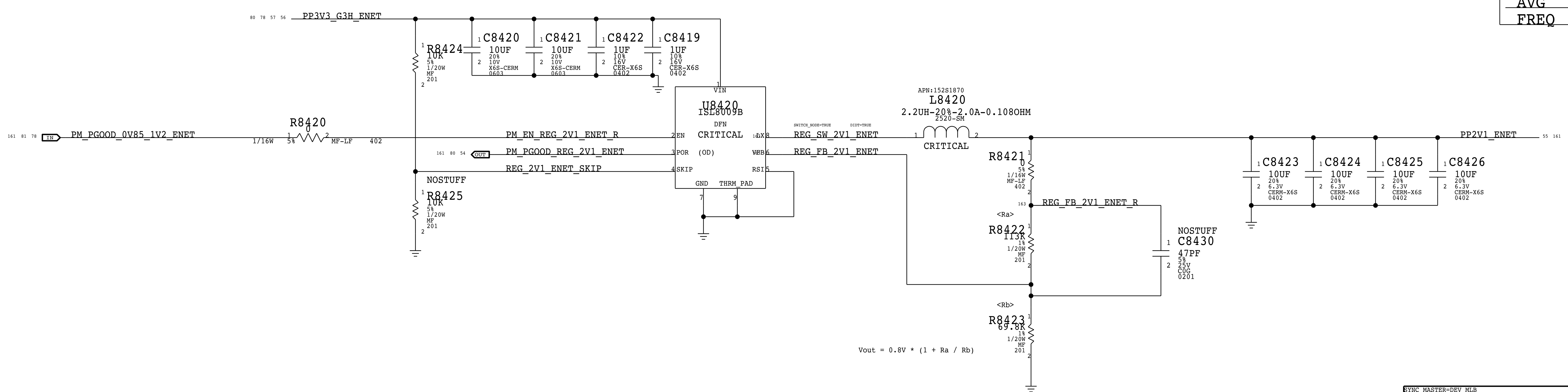
ENET 1V2 REGULATOR	
VOUT	1.20 V
PEAK (4 US)	0.9 A
OC (TYP)	2.2 A
AVG	0.8 A
FREQ	1.6 MHZ

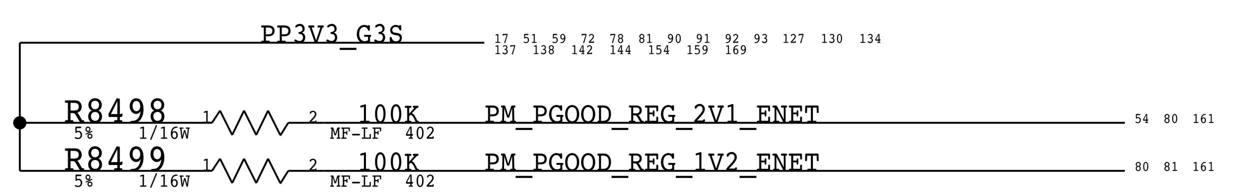
1.2V ENET REGULATOR



ENET 2V1 REGULATOR	
VOUT	2.10 V
PEAK (4 US)	0.5 A
OC (TYP)	2.2 A
AVG	0.5 A
FREQ	1.6 MHZ

2.1V ENET REGULATOR





SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

VR ENET 1V2, 2V1

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D

REVISION 6.21.0

BRANCH

BOM_COST_GROUP=PLATFORM POWER

ENET 0V85 REGULATOR

VOUT		0.85 V
PEAK	(4 US)	6.5 A
OC	(TYP)	12 A
AVG		6.5 A
FREQ		600 KHZ

FSEL STRAP	SW FREQ
GND	300 kHz
VCC	1 MHz
100k to GND	600 kHz
FLOAT	500 kHz

BOM_COST_GROUP=PLATFORM POWER

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

VR ENET 0V85

Apple Inc.

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3V3 G3H REGULATOR

VOUT	3.3 V
PEAK (4 US)	16.9 A
OC (TYP)	25.7 A
AVG	8 A
FREQ	500 KHZ

5V G3S REGULATOR

VOUT	5.1 V
PEAK (4 US)	10.8 A
OC (TYP)	21.2 A
AVG	6.5 A
FREQ	500 KHZ

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

VR P3V3 G3H, P5V G3S

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D

REVISION 6.21.0

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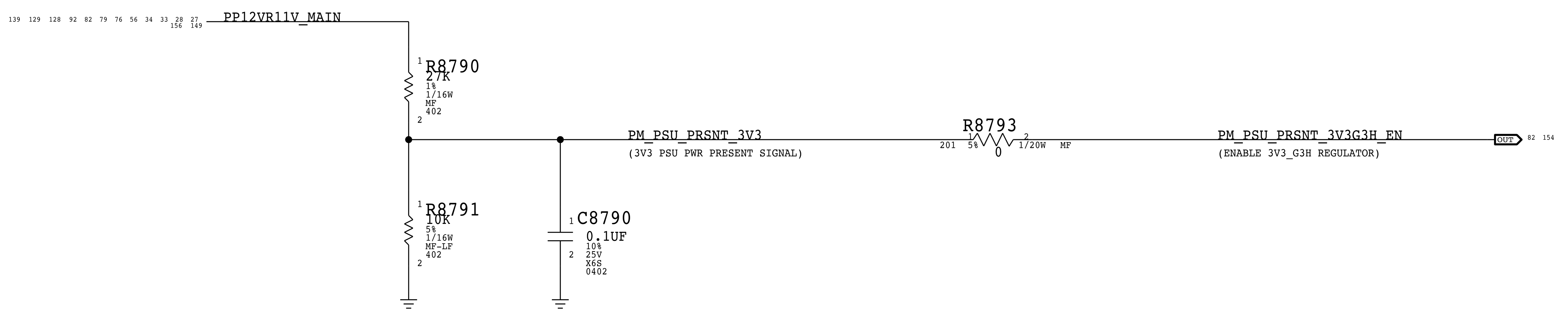

BOM_COST_GROUP=PLATFORM POWER

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

PSU PRESENCE G3H ENABLE

Apple Inc.

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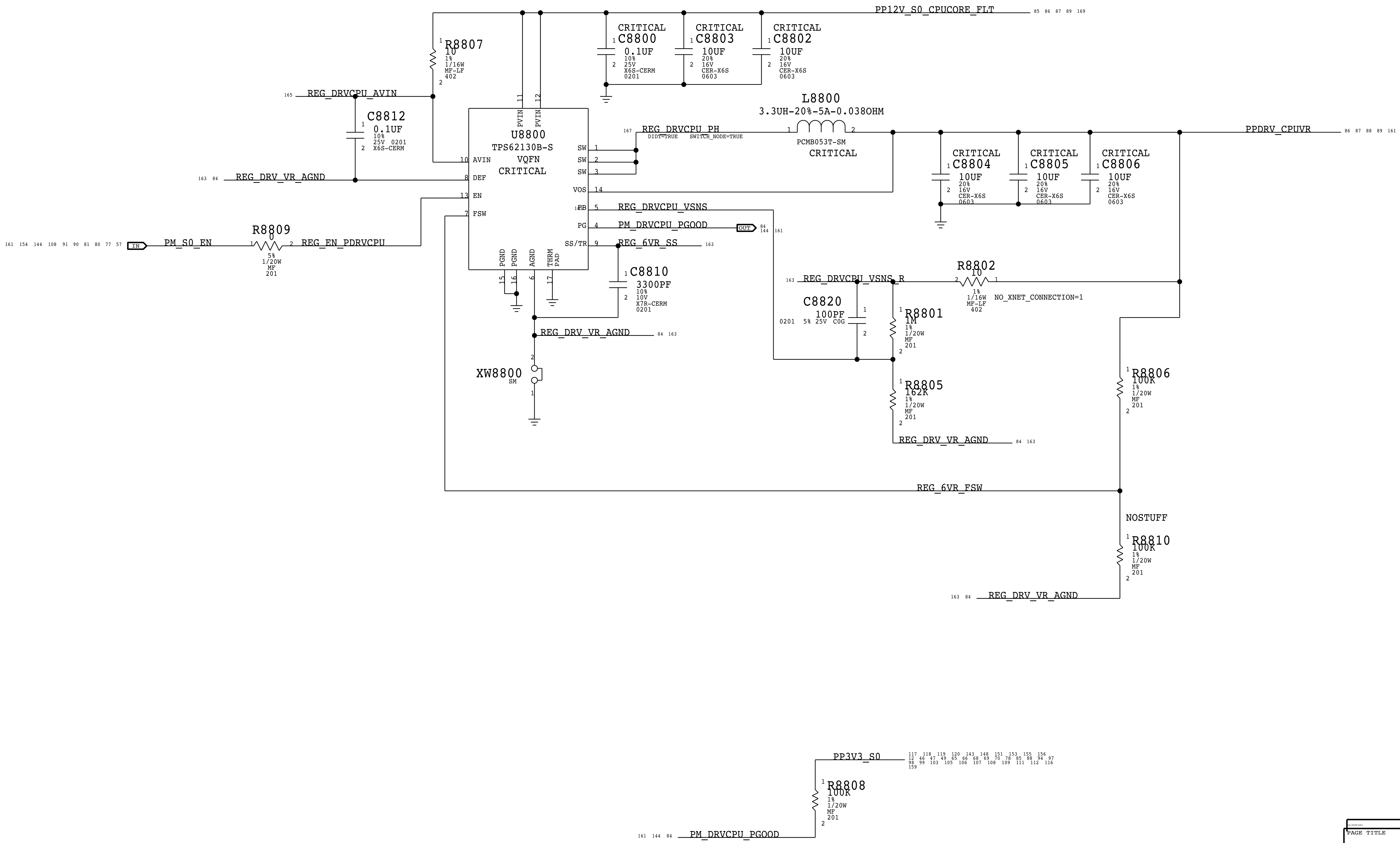
CPU DRV S0 REGULATOR	
VOUT	5.75 V
PEAK	0.5 A
OC (TYP)	4.2 A
AVG	0.5 A
FREQ	1.25 MHZ



BOM_COST_GROUP=PLATFORM POWER

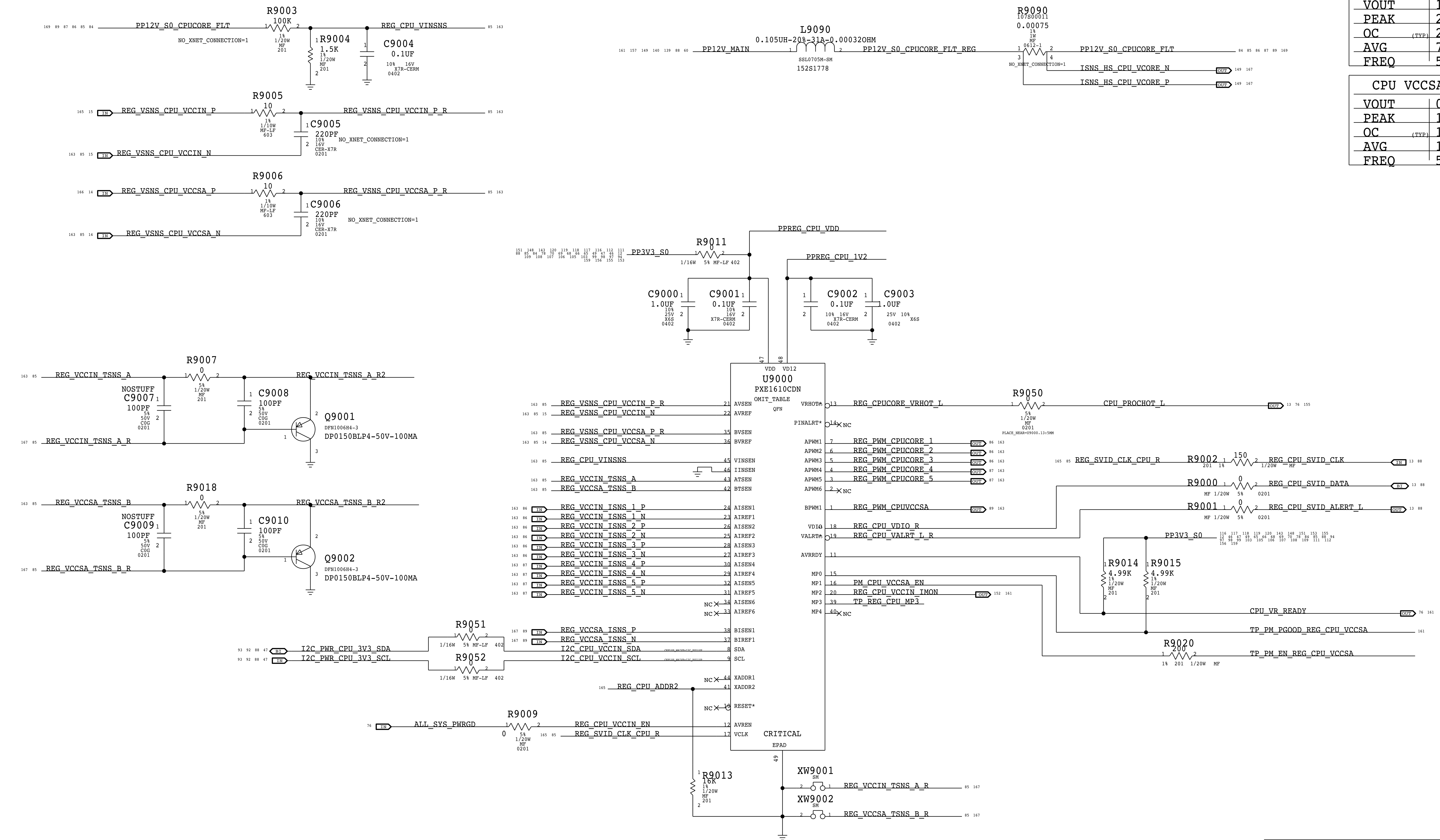
PAGE TITLE
VR 5.75V CPUVR

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D
REVISION 6.21.0
BRANCH
PAGE 88 OF 255
SHEET 84 OF 171

CPU VCCIN REGULATOR	
VOUT	1.8 V
PEAK	200 A
OC (TYP)	240 A
AVG	77 A
FREQ	500 KHZ

CPU VCCSA REGULATOR	
VOUT	0.85 V
PEAK	15 A
OC (TYP)	18 A
AVG	14 A
FREQ	500 KHZ



ACCESS POINT FOR I2C
DEBUG HOOKUP
PLACE GND TP NEARBY

	I2C	PMBUS1	PMBUS2
SMBUS ADDR:	0XA0/A1	0X80/81	0X82/83

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
VR CPU CORE & VCCSA CTRL

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D
REVISION 6.21.0
BRANCH

PAGE TITLE
VR CPU CORE PHASES 1-3

Apple Inc.

DRAWING NUMBER: 051-01649
SIZE: D
REVISION: 6.21.0
BRANCH

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
VR CPU CORE PHASES 4-5

Apple Inc.

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REVISION 6.21.0

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CPU VCCIO REGULATOR	
VOUT	1 V
PEAK (4 US)	12 A
OC (TYP)	15 A
AVG	11 A
FREQ	500 KHZ

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
VR CPU VCCIO

Apple Inc.

DRAWING NUMBER 051-01649
SIZE D
REVISION 6.21.0
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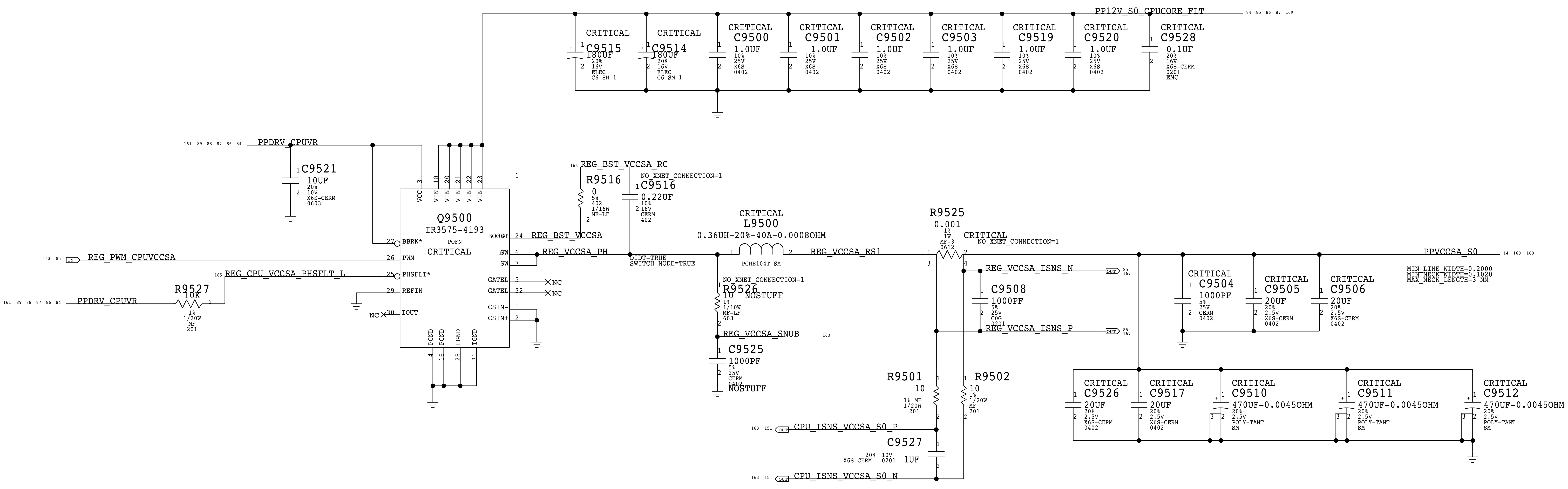

CPU VCCSA REGULATOR
SHARES CONTROLLER ON CSA90

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE: VR CPU VCCSA

Apple Inc.

DRAWING NUMBER: 051-01649 SIZE: D

REVISION: 6.21.0

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CPU VPP01 REGULATOR

VOUT	2.57 V
PEAK	5.92 A
OC (TYP)	18.75 A
AVG	4.4 A
FREQ	500 KHZ

CPU VTT01 REGULATOR

VOUT	0.606 V
PEAK	1.5 A
OC (TYP)	3 A
AVG	1.5 A

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE	VR CPU VPP & VTT CH 0,1
DRAWING NUMBER	051-01649
SIZE	D
Apple Inc.	
REVISION	6.21.0
BRANCH	
PAGE	96 OF 255
SHEET	90 OF 171

CPU VPP23 REGULATOR	
VOUT	2.57 V
PEAK	5.92 A
OC (TYP)	18.75 A
AVG	4.4 A
FREQ	500 KHZ

CPU VTT23 REGULATOR	
VOUT	0.606 V
PEAK	1.5 A
OC (TYP)	3 A
AVG	1.5A A

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE	VR CPU VPP & VTT CH 2,3
DRAWING NUMBER	051-01649
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Apple Inc.	
REVISION	6.21.0
BRANCH	
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SHEET	91 OF 171

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
VR CPU VDDQ CH 0,1

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D

REVISION 6.21.0

BRANCH

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE

VR CPU VDDQ CH 2,3

Apple Inc.

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SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

GPU PCIE, STRAPS, JTAG

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D

REVISION 6.21.0

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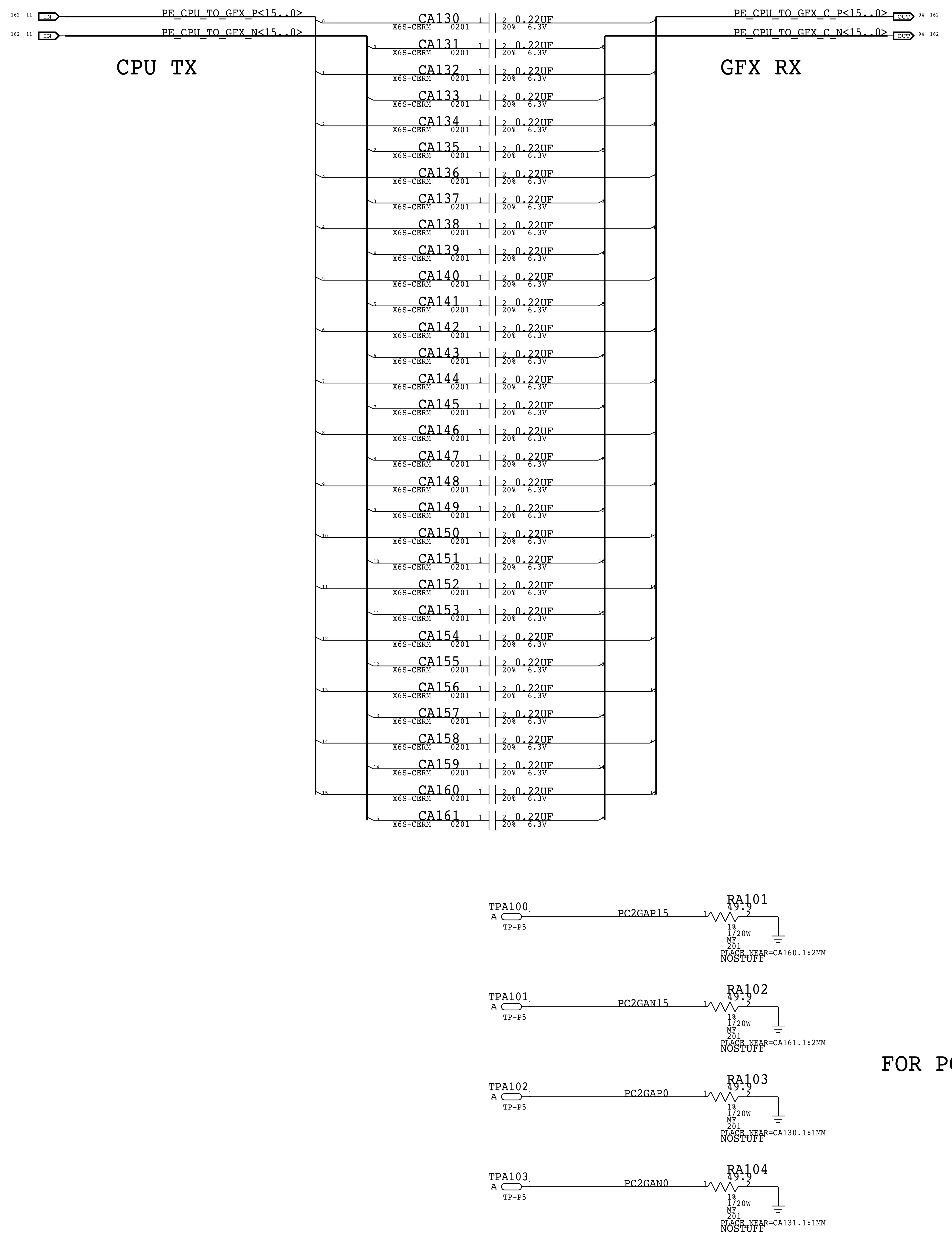
PAGE 100 OF 255

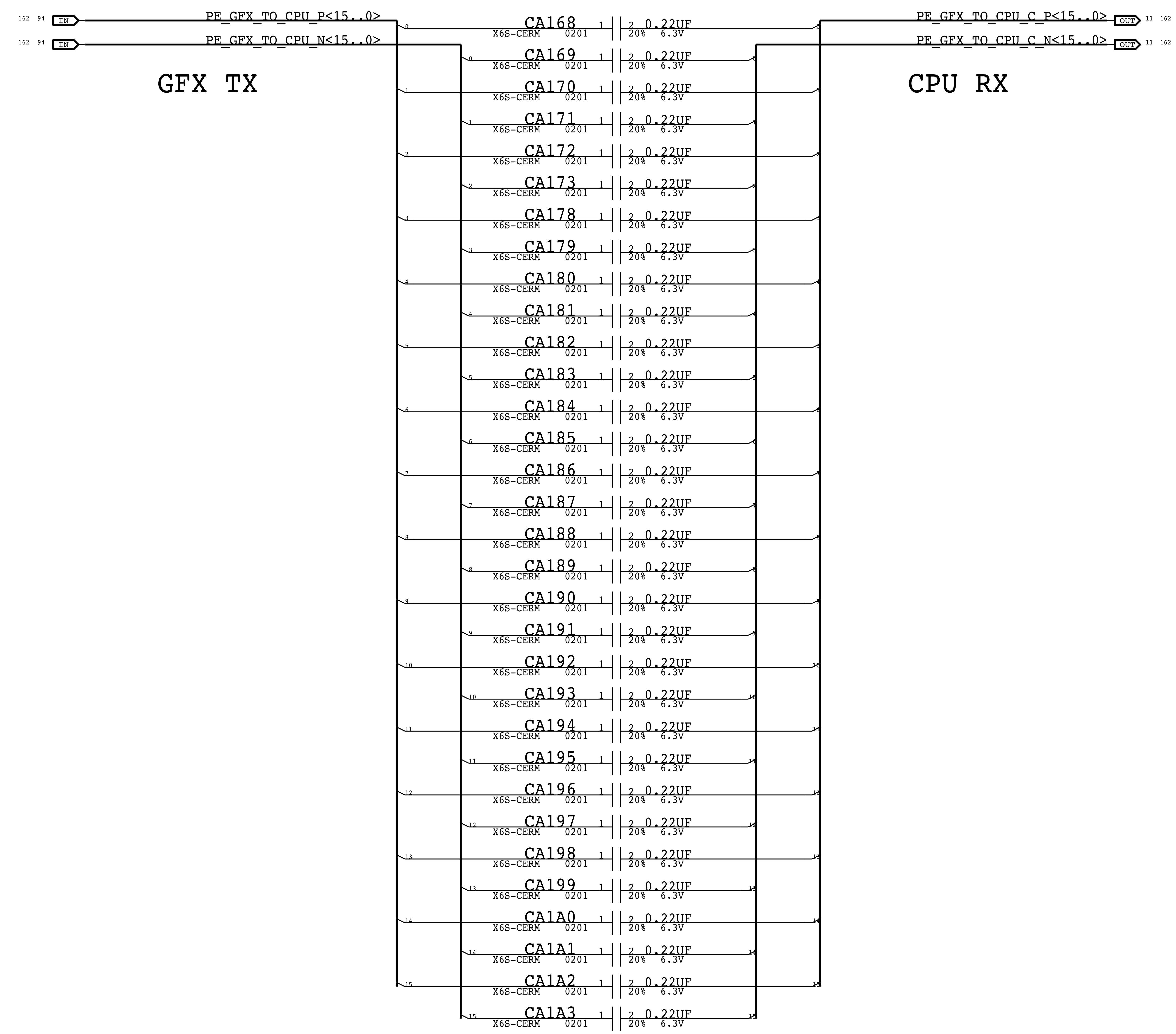
SHEET 94 OF 171

BOM_COST_GROUP=GRAPHICS

PCIE AC COUPLING CAPS CPU TO GFX
PLACE NEAR GFX



PCIE AC COUPLING CAPS GFX TO CPU
PLACE NEAR CPU



FOR PCIE MEASUREMENT

BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE
GPU PCIE AC CAPS

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DRAWING NUMBER 051-01649 SIZE D
REVISION 6.21.0
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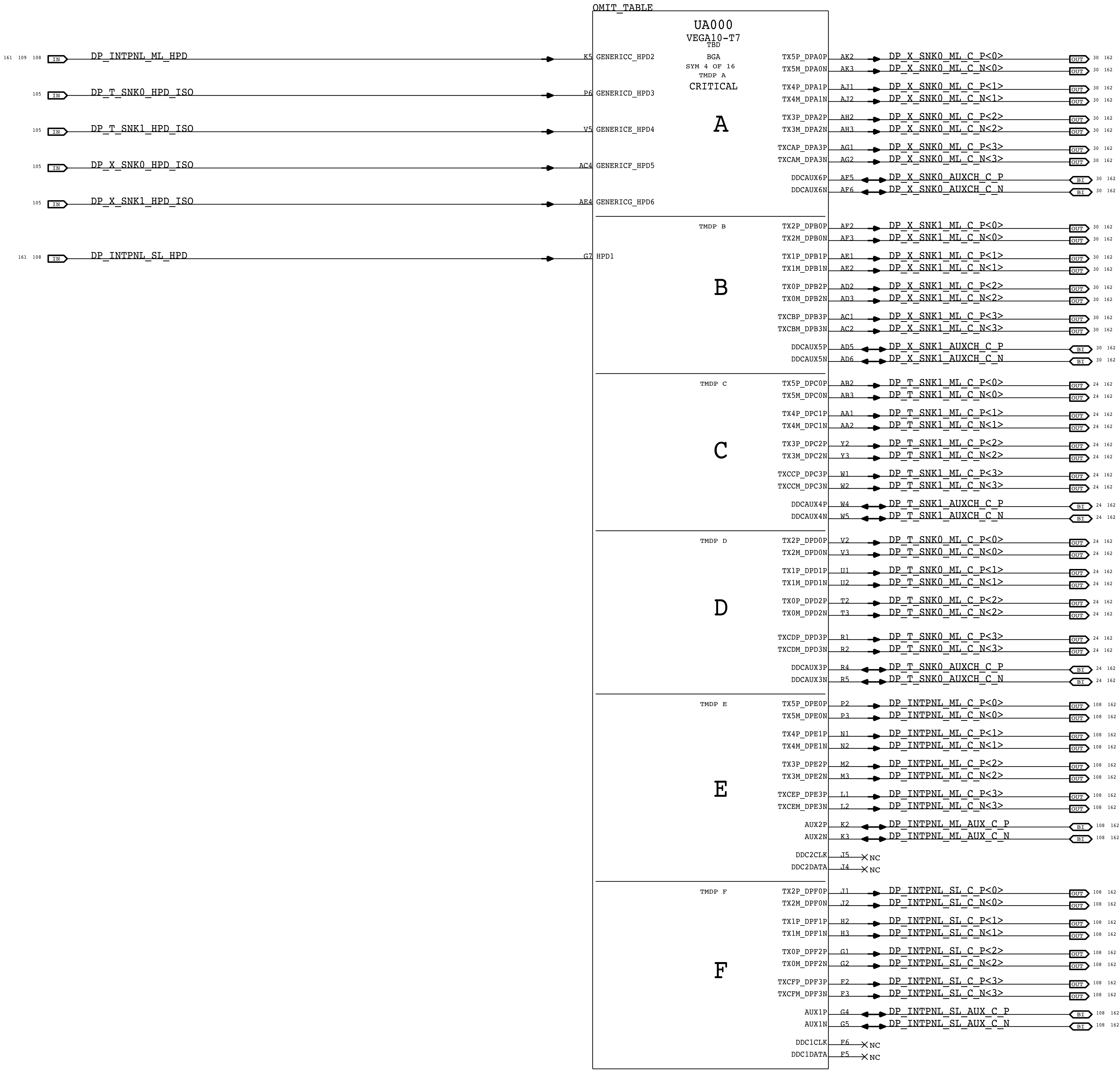

AC CAPS ON RIDGE PAGES

AC CAPS ON PANEL CONNECTOR PAGE

NOTE:

USB-C DOES NOTE SUPPORT DP++
VIDEO PORTS ARE CONFIG TO DP IN VBIOS

BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=DEV_MLB	SYNC_DATE=11/06/2016
PAGE TITLE	GPU DISPLAYPORT
Apple Inc.	DRAWING NUMBER 051-01649 SIZE D
	REVISION 6.21.0
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SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

GPU GPIOS

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REVISION 6.21.0

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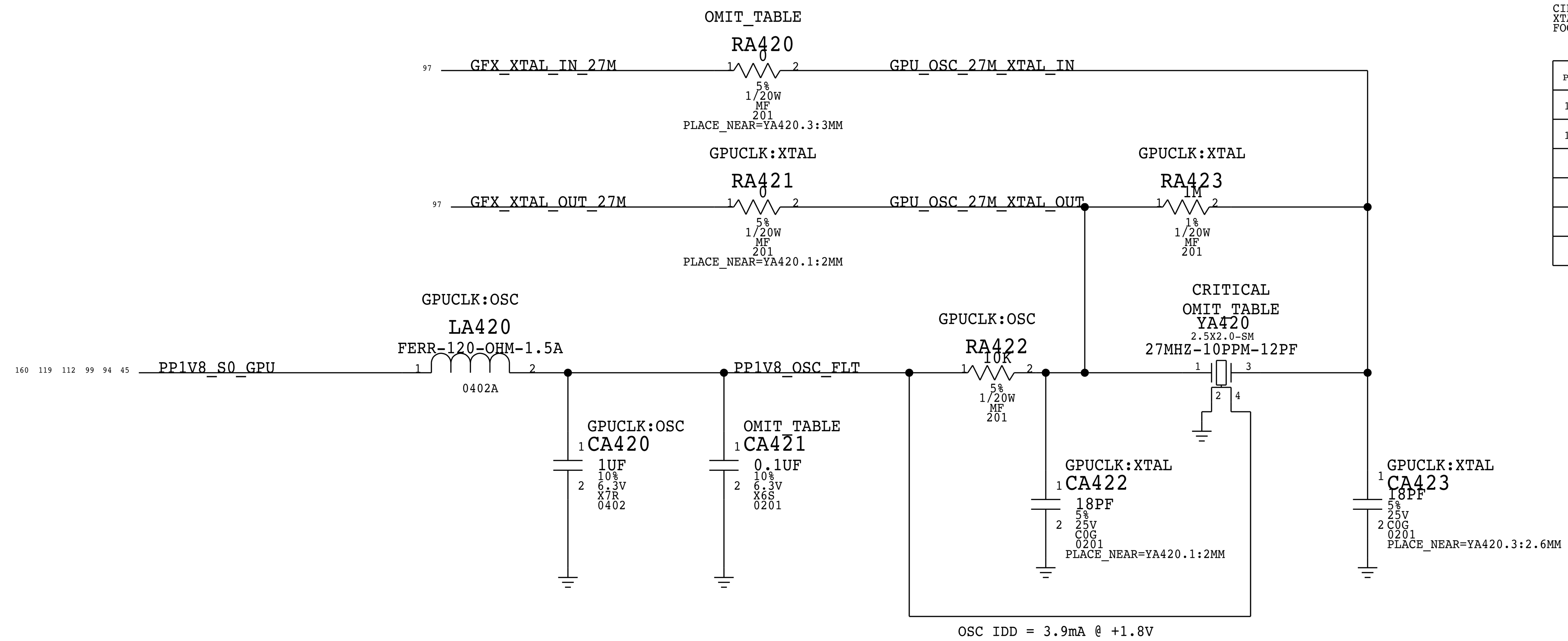
BOM_COST_GROUP=GRAPHICS

GPU 27 MHZ CLK

PIN	OSC	XTAL
1	OE/ST*	XTAL PIN 1
2	GND	NC
3	VOUT	XTAL PIN 2
4	VDD	NC

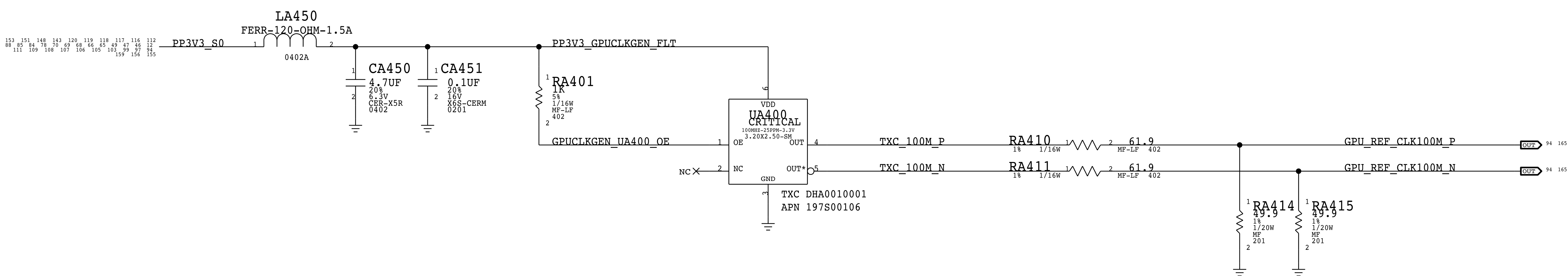


CIRCUIT FOR EITHER OSC OR XTAL, BOTH SHARE SAME THE FOOTPRINT

XTALIN IS 1.8V LEVEL
CLK GEN: 100 MHZ, NON SPREAD SPECTRUM, 1.8V

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
197S00056	1	OSC,MEMS,100MHZ,+/-20%,1V8,2520	YA420	GPUCLK:OSC
197S00142	1	XTAL,27MHZ,10PPM,12PF,40OHM,2520	YA420	GPUCLK:XTAL
117S0080	1	RES,33OHM,5%,0201	RA420	GPUCLK:OSC
117S0201	1	RES,0 OHM,5%,0201	RA420	GPUCLK:XTAL
117S0201	1	RES,0 OHM,5%,0201	CA421	GPUCLK:XTAL
132S0395	1	CAP,0.1UF,10%,0201	CA421	GPUCLK:OSC

GPU 100 MHZ CLK
(NO SPREAD)



BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE
GPU CLK GEN

Apple Inc.

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SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

GPU MEMORY AND IO PWR

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DRAWING NUMBER 051-01649 SIZE D

REVISION 6.21.0

BRANCH

PAGE 105 OF 255

SHEET 99 OF 171

BOM_COST_GROUP=GRAPHICS

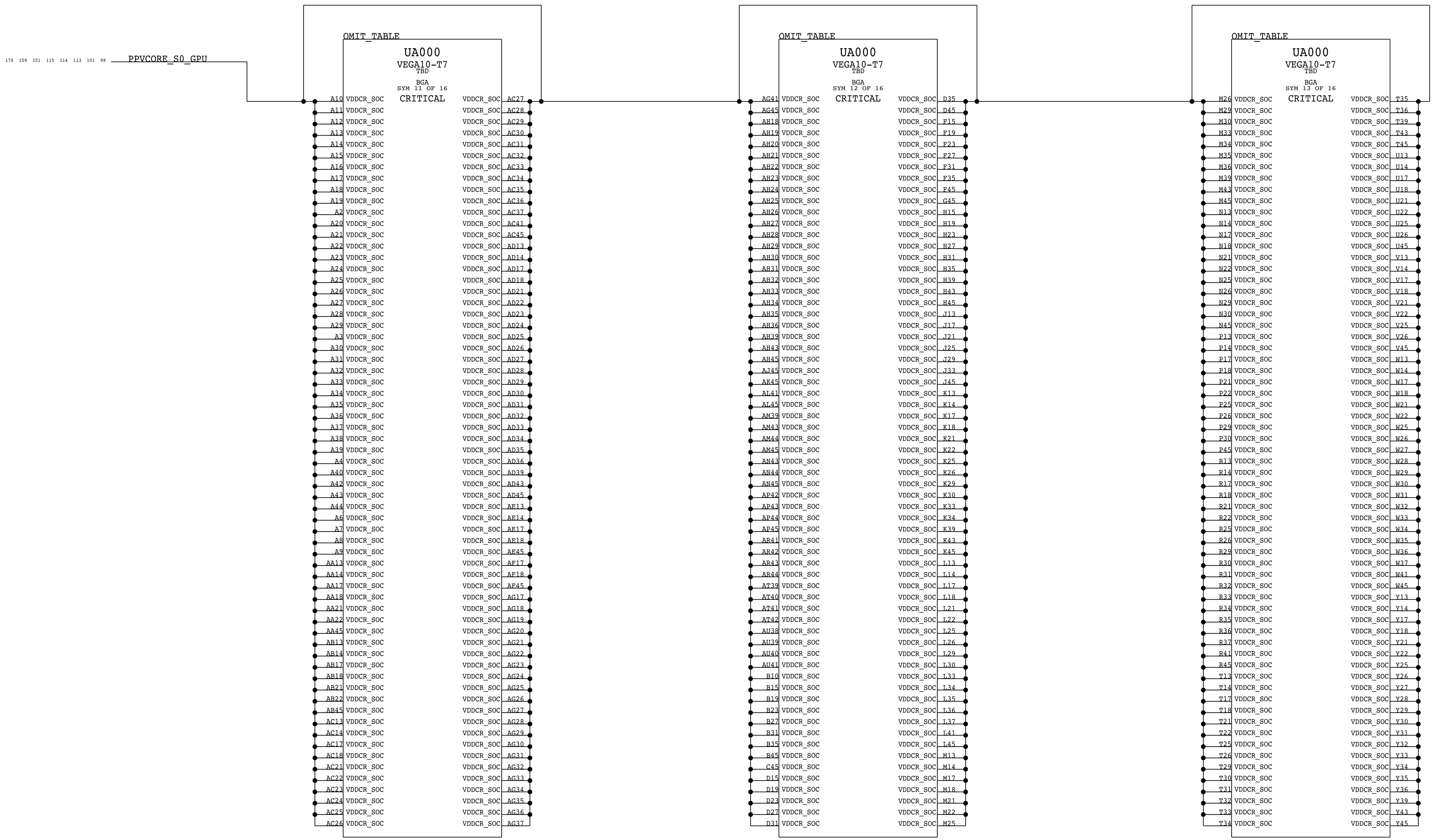
8	7	6	5	4	3	2	1

D

C

B

A



BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

GPU CORE PWR

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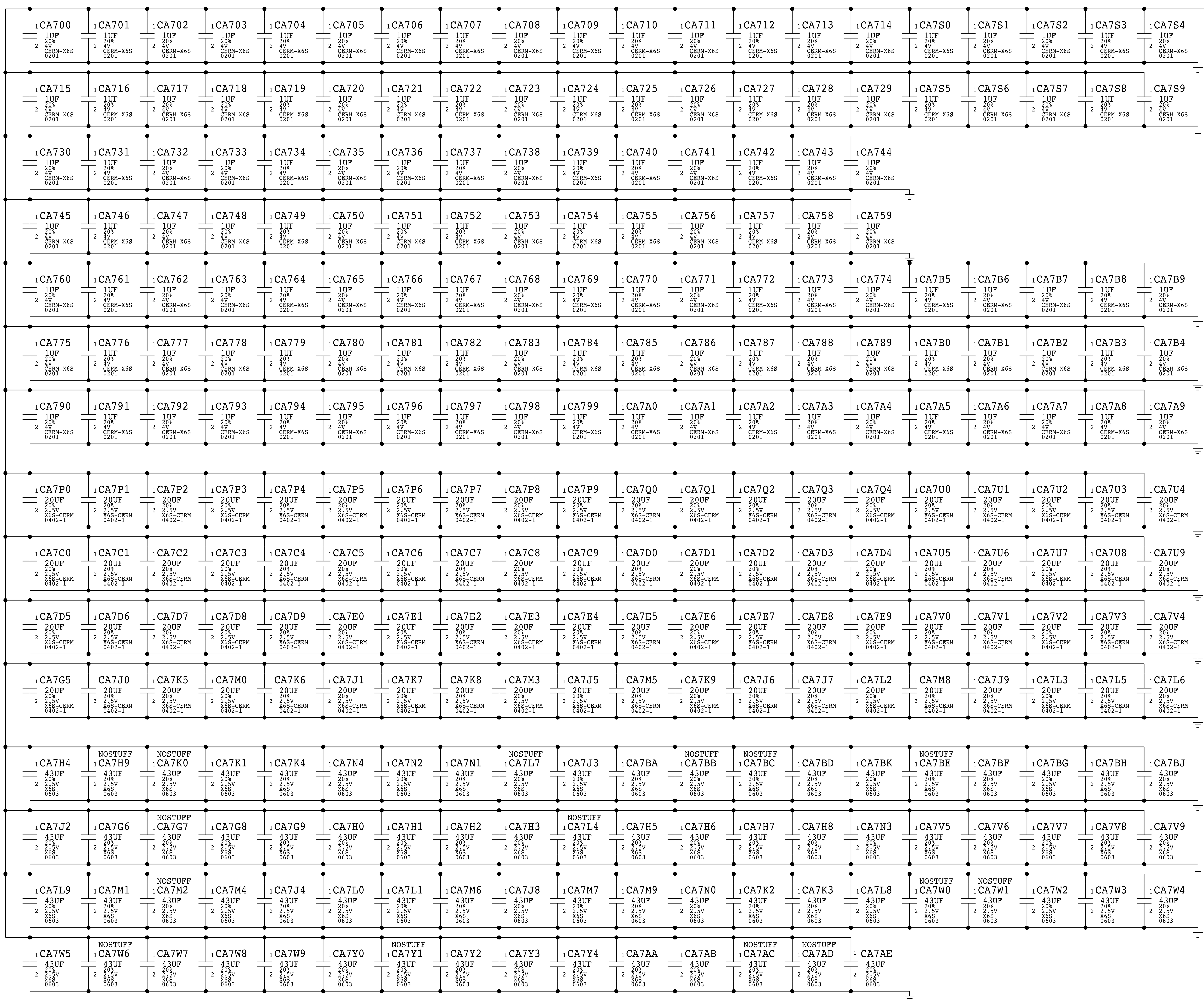

GPU VDDC DECOUPLING
PLACE ALL CAPS UNDER GPU

1UF X130 138S0775
20UF X80 138S00036
43UF X60 138S00168
43UF X15 138S00168 NOSTUFF

BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=DEV_MLB SYNC_DATE=11/09/2016

PAGE TITLE
GPU CORE PWR DECOUPLING

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D

REVISION 6.21.0

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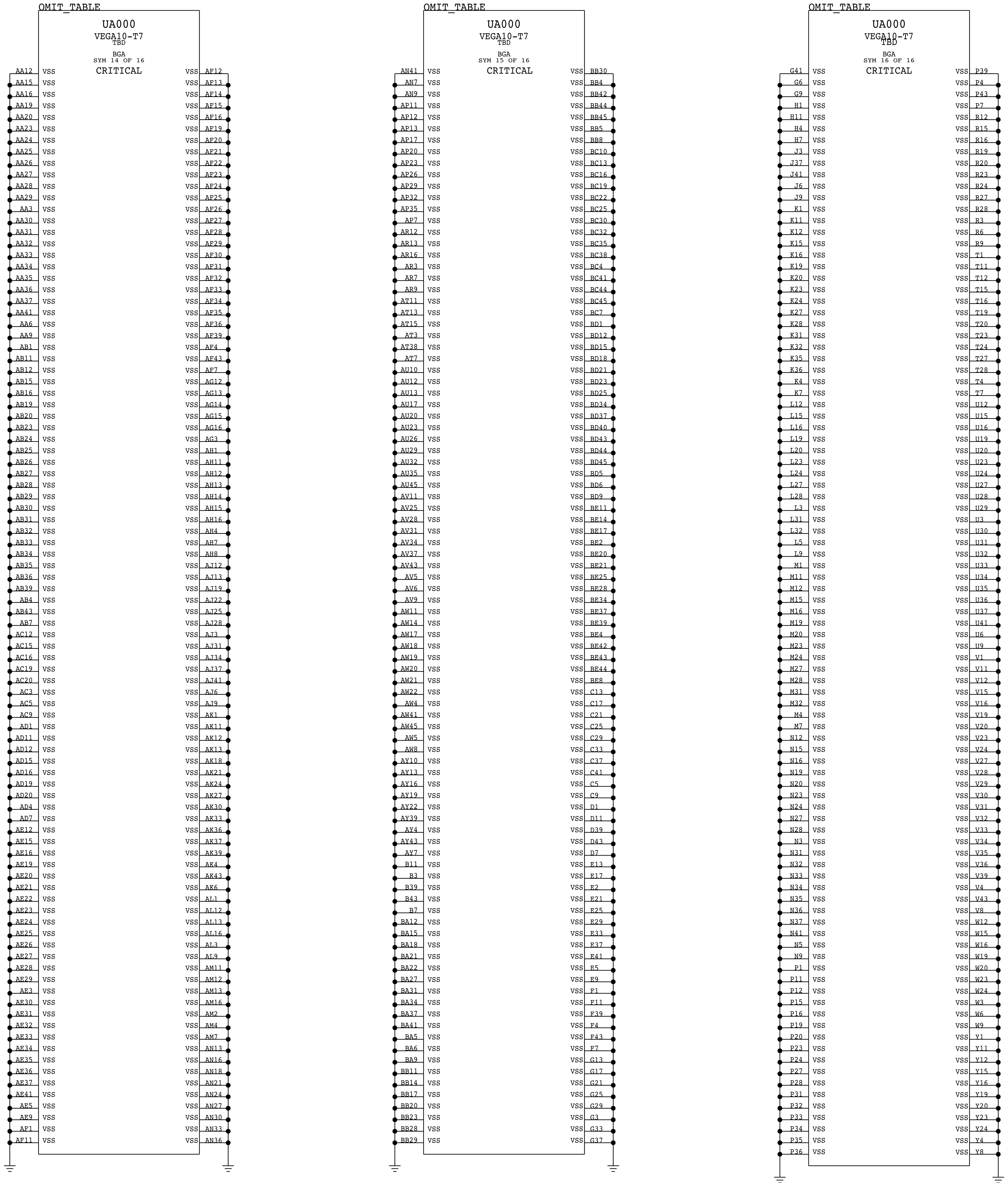
PAGE 107 OF 255

SHEET 101 OF 171



BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

GPU GND

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D

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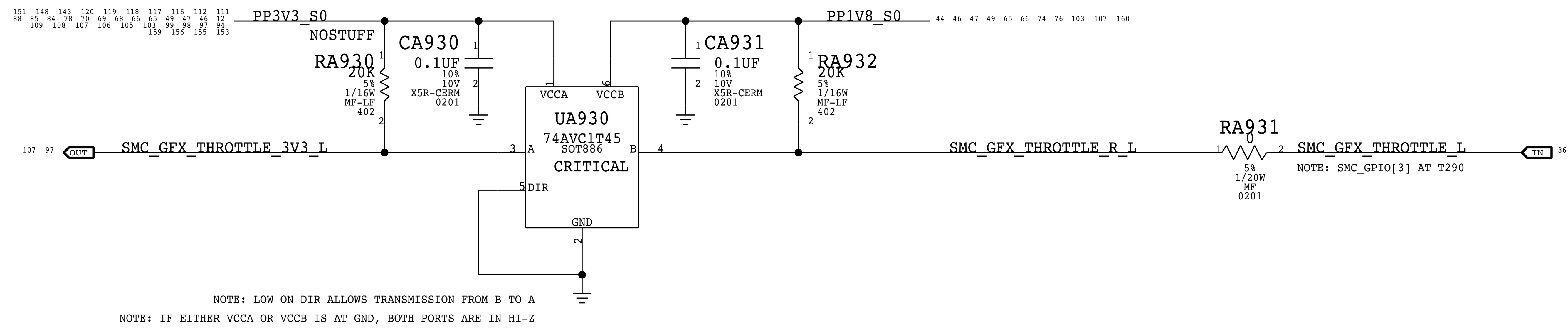
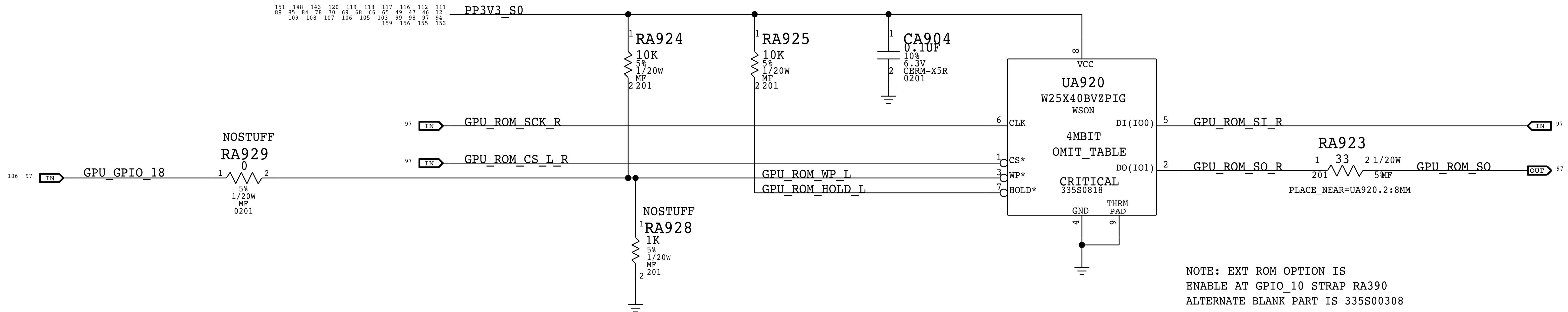
PAGE 108 OF 255

SHEET 102 OF 171

GFX THROTTLE LEVEL SHIFTING

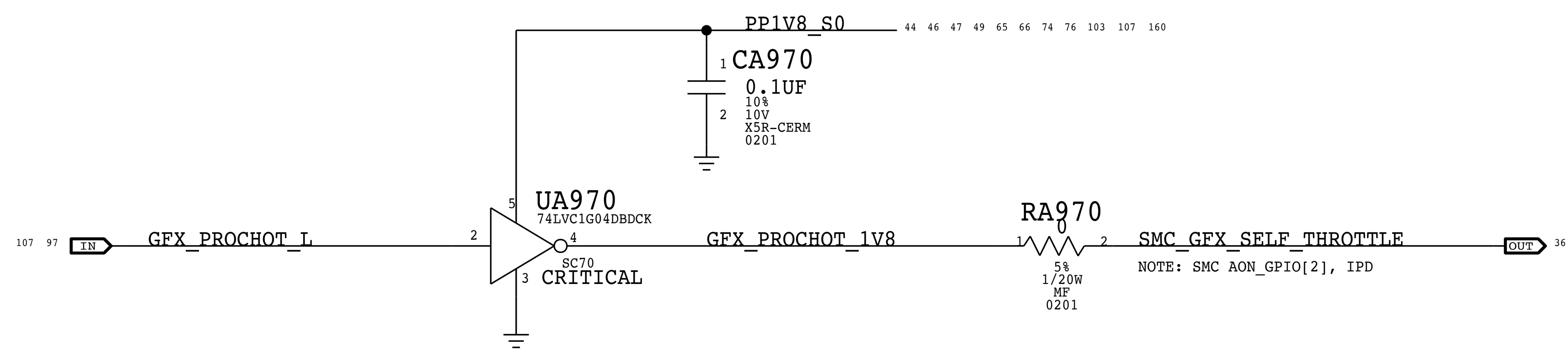


GPU BOOT ROM



NOTE: EXT ROM OPTION IS
ENABLE AT GPIO_10 STRAP RA390
ALTERNATE BLANK PART IS 335S00308

PROCHOT INVERTER AND LEVEL SHIFTER



BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE
GPU ROM, TEMP, LVL SHIFT

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D
REVISION 6.21.0
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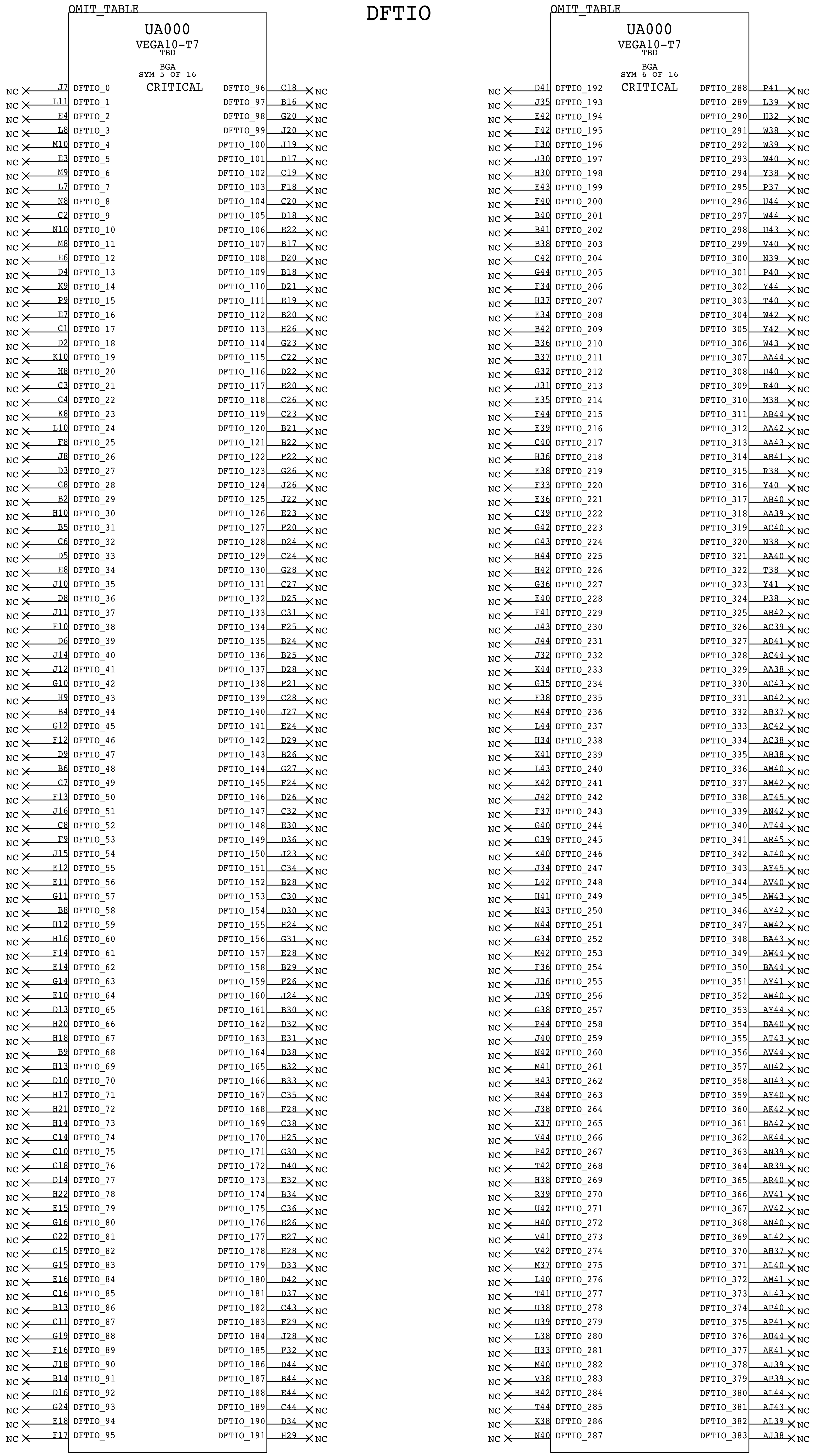

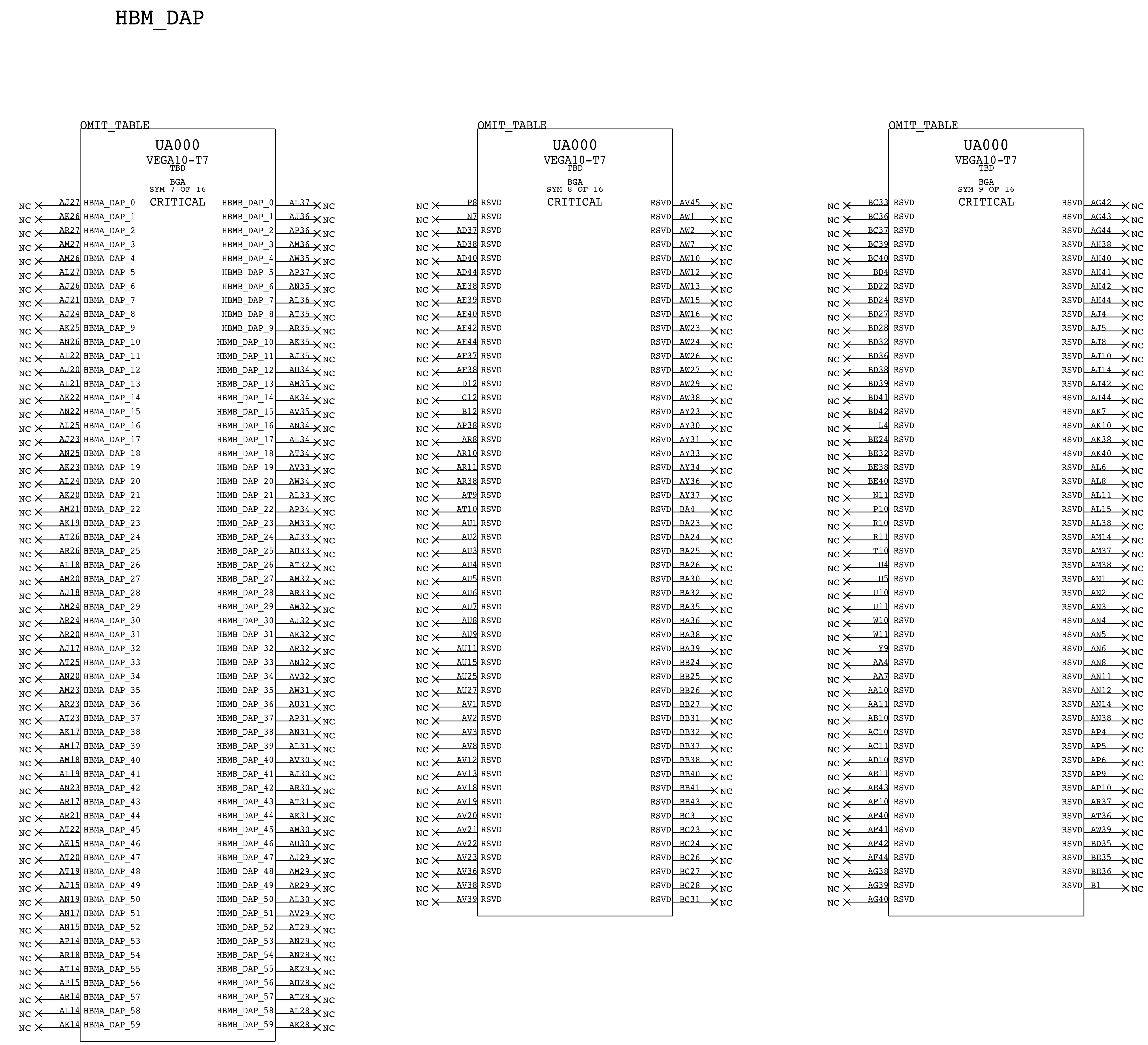

NOTE:

AJ18 AND AR33 ARE NC FOR HBM2
HBM2 DOES NOT NEED THE CONNECTION
TO GND FOR NORMAL OPERATION
HBM2 USES A MRS SETTING TO LOCK OUT
THE DIRECT ACCESS (DA) BUS

BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

GPU DA, DFTIO

Apple Inc.

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051-01649	D

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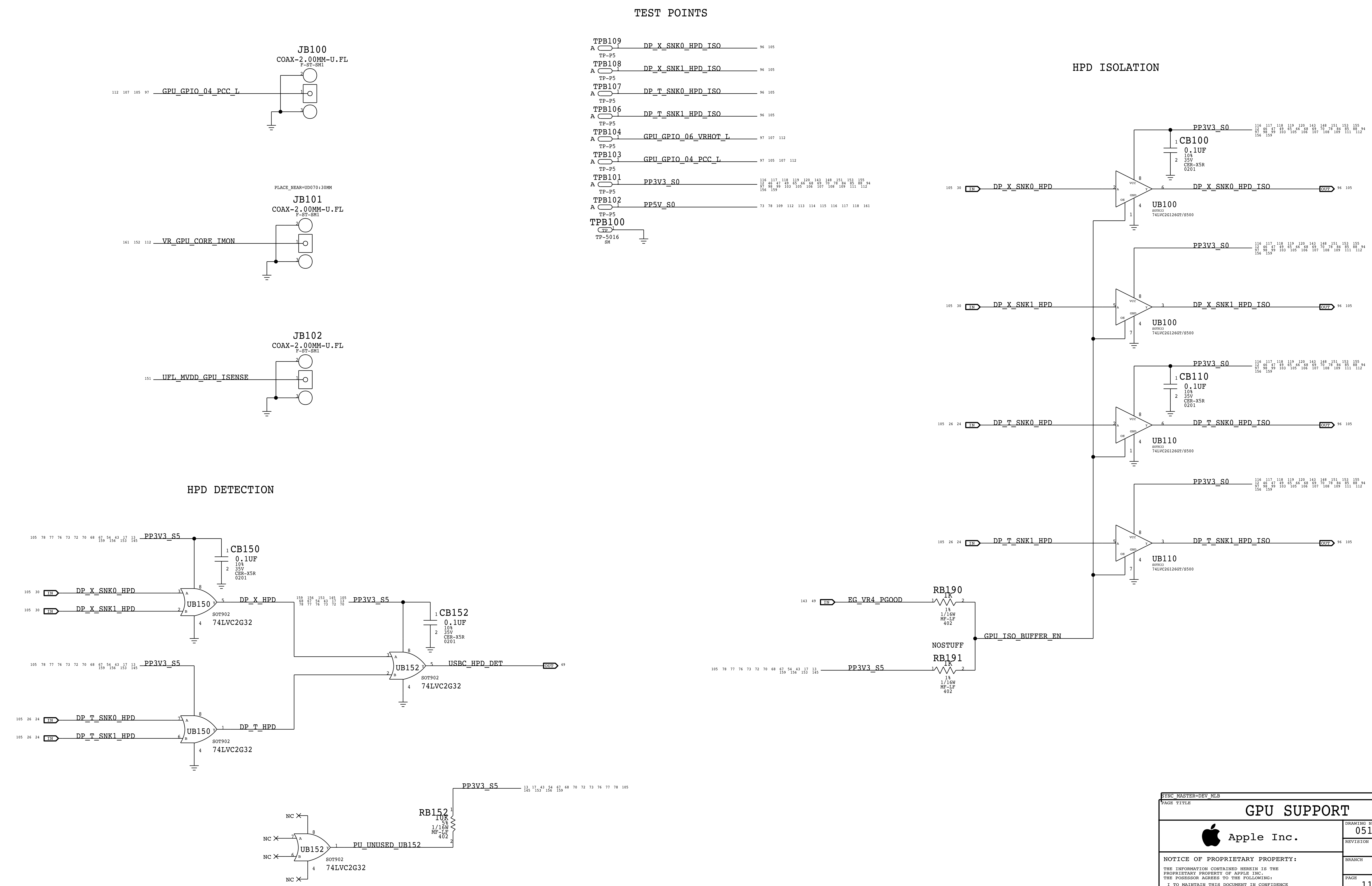

BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

GPU SUPPORT

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D

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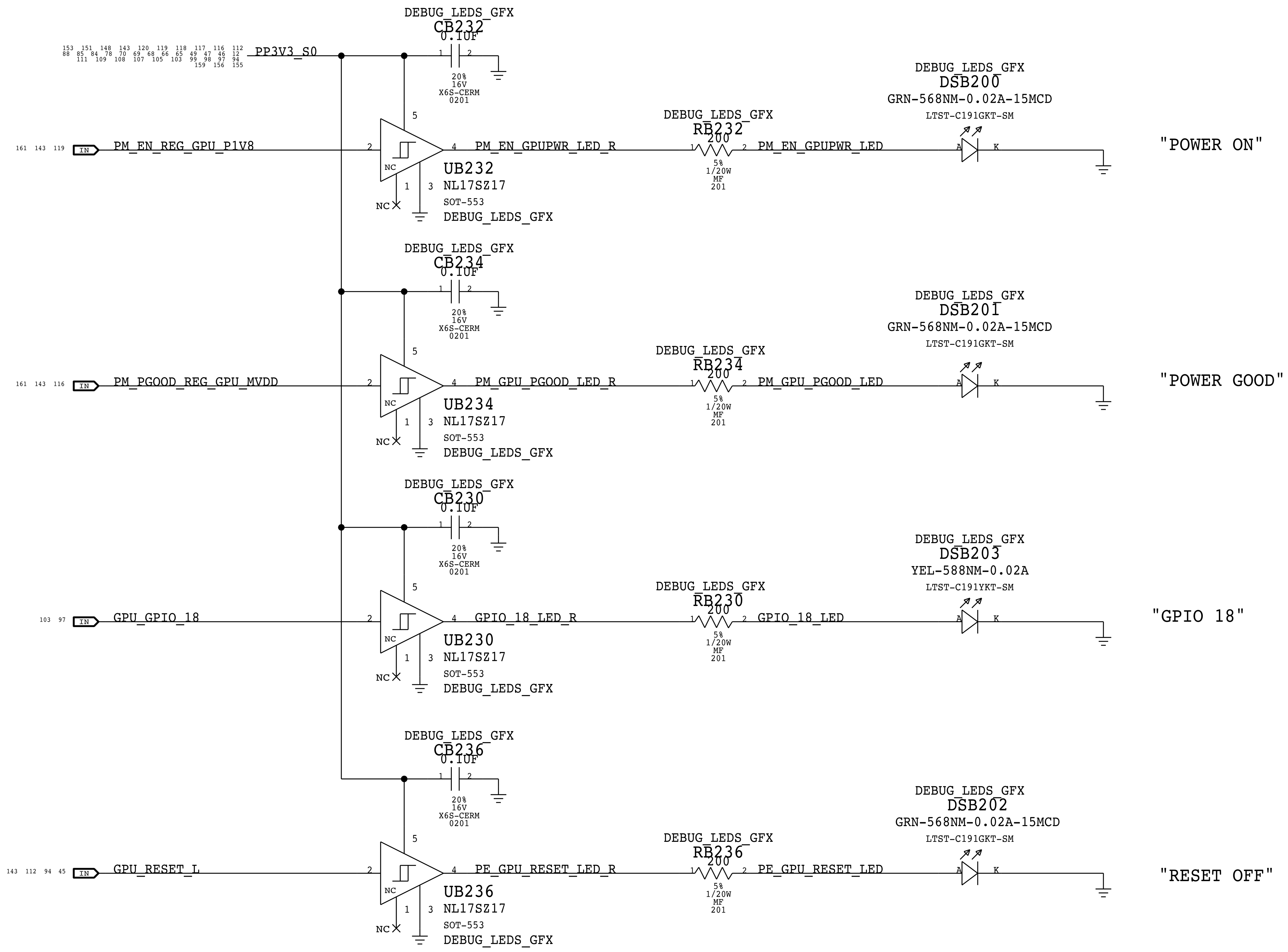
GPU DEBUG LEDS



BOM_COST_GROUP=DEBUG

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE: GPU DEBUG LEDS

Apple Inc.

DRAWING NUMBER: 051-01649 SIZE: D

REVISION: 6.21.0

BRANCH

PAGE: 112 OF 255

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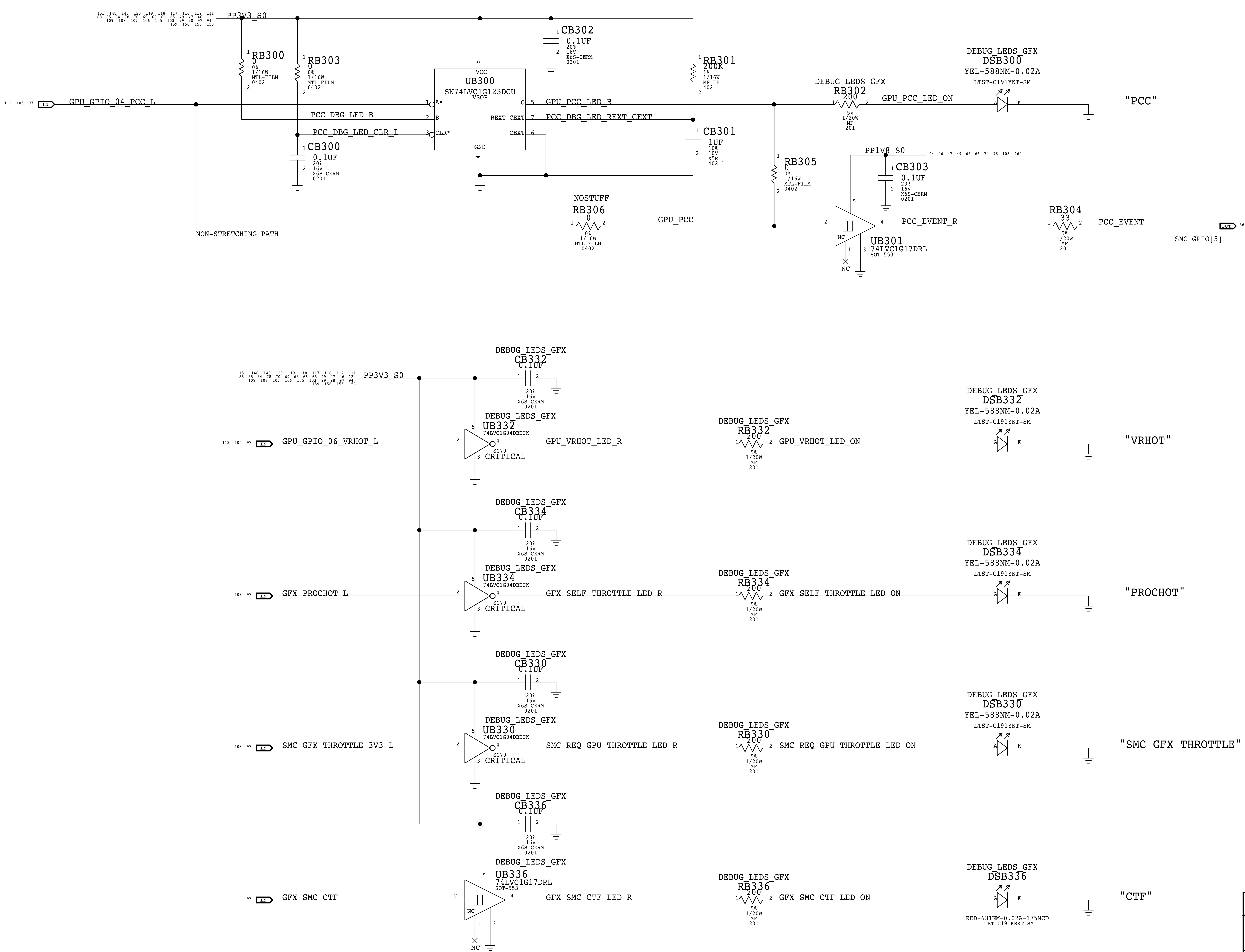
GPU POWER MANAGEMENT LEDS



BOM_COST_GROUP=DEBUG

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE
GPU POWER MGMT LEDS

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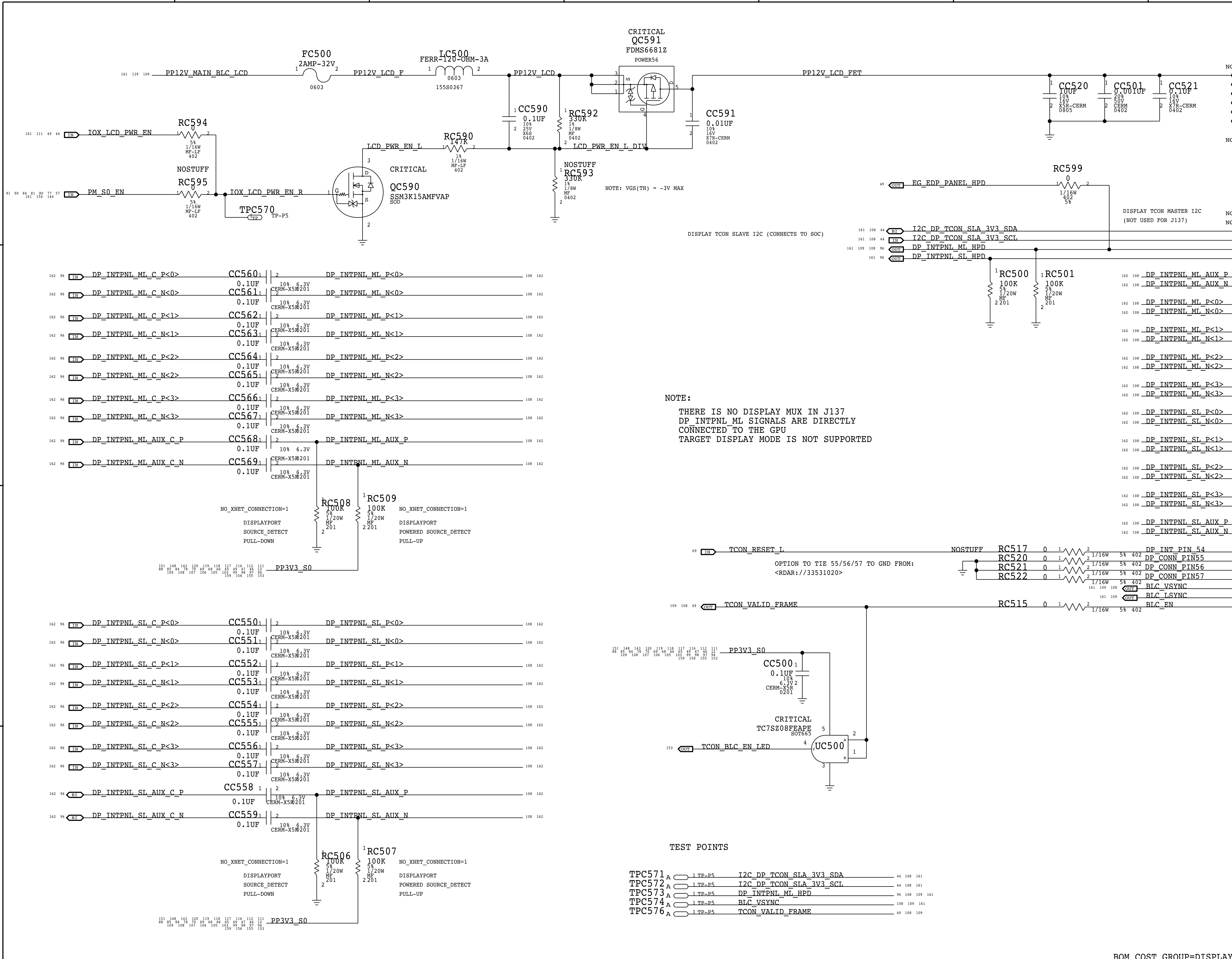

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

DISPLAY INTERNAL DP CONN

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PAGE TITLE

DISPLAY BACKLIGHT DRIVER

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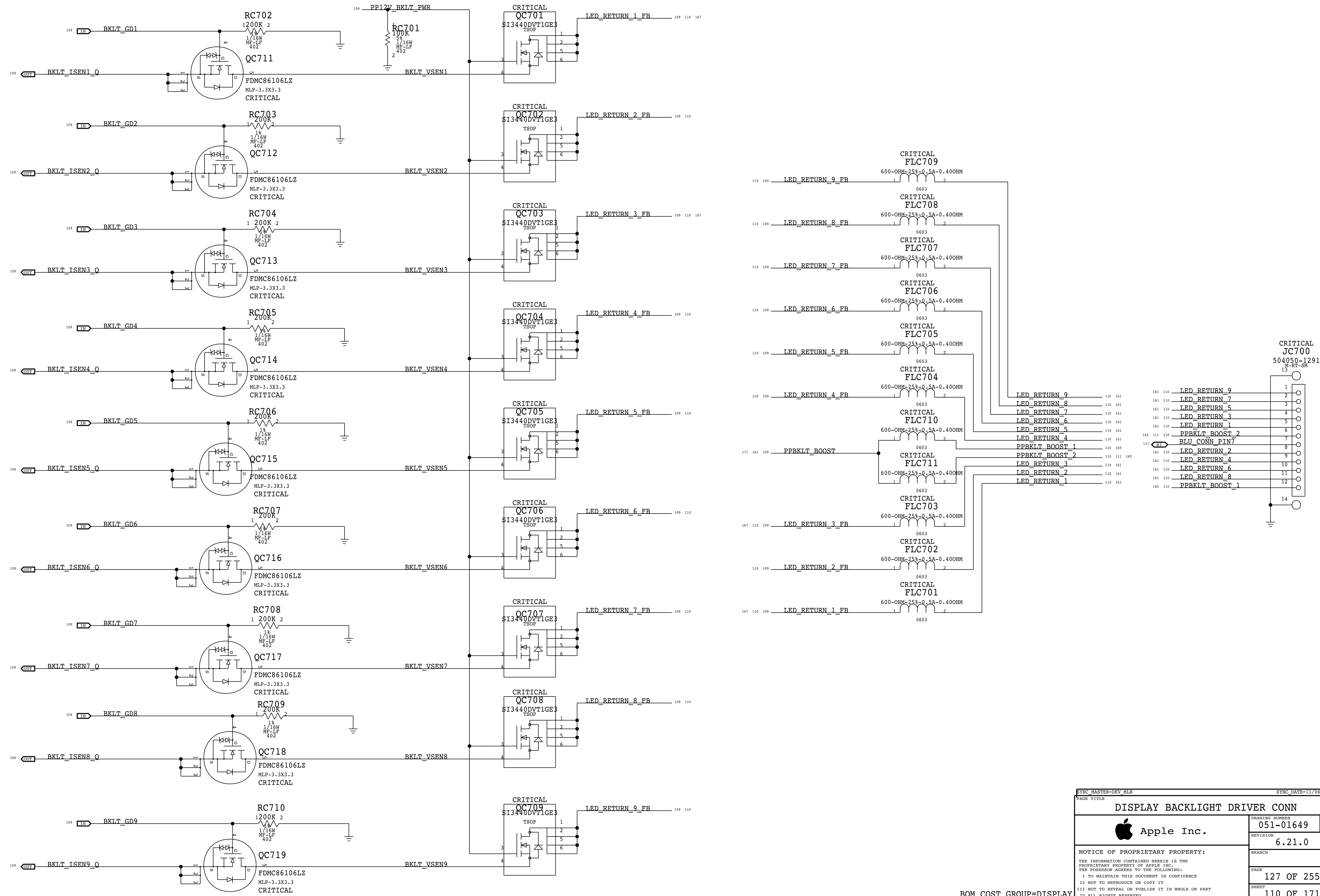

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

DISPLAY BACKLIGHT DRIVER CONN

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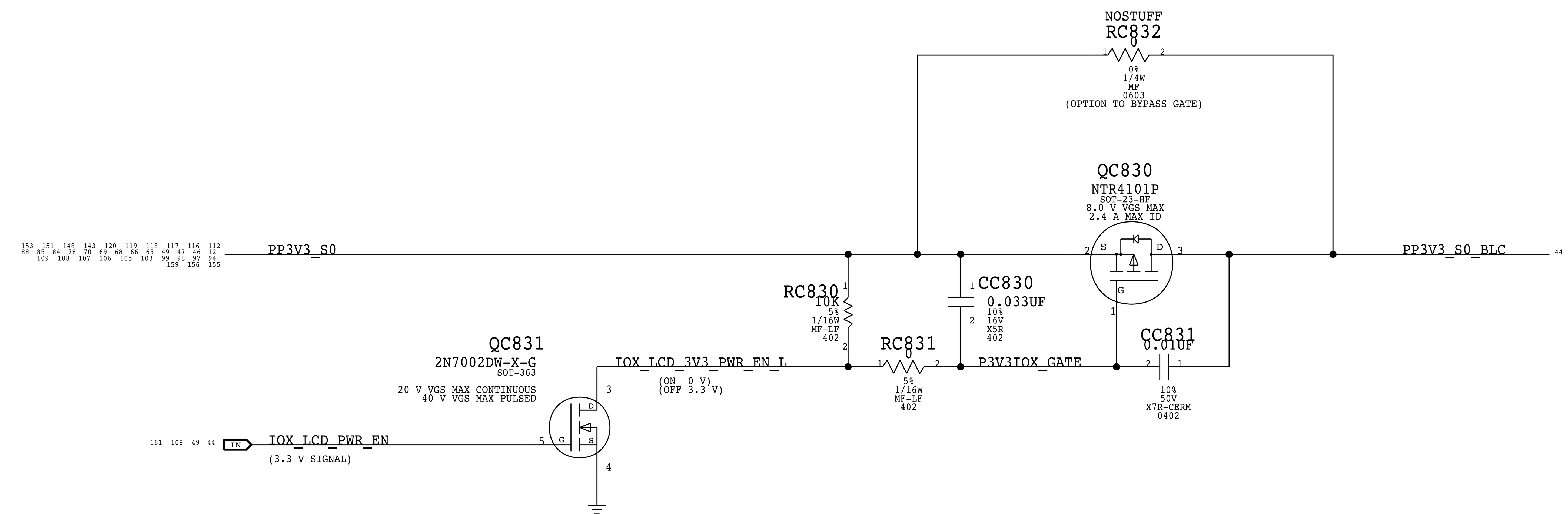
D

DISPLAY 3V3 FET

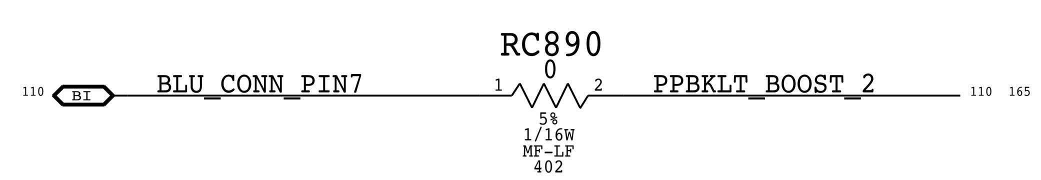
GATED POWER BECAUSE THE DISPLAY AND 3V3 S0 MAY NOT ALWAYS BE ON TOGETHER



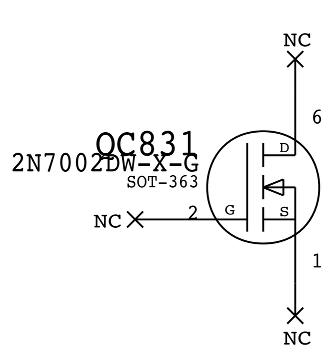
C

B

OPTION TO TIE BLUE CONN PIN 7 TO PIN 6 FROM:
<RDAR://33531020>



A



BOM_COST_GROUP=DISPLAY

PAGE TITLE	DISPLAY BACKLIGHT SUPPORT
Apple Inc.	DRAWING NUMBER 051-01649, SIZE D
	REVISION 6.21.0
	BRANCH
NOTICE OF PROPRIETARY PROPERTY:	PAGE 128 OF 255
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SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

VR GFX CORE CTRL

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MAX TEMP LIMITS

IR3575M	125 C (TJ)	353S4458
470 UF	95 C	128S0381
0.15 UH	125 C	152S00302

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

VR GFX CORE PHASES 1-3

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BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

VR GFX CORE PHASES 4-6

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SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

VR GFX CORE PHASES 7-8

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GFX MVDD REGULATOR	
VOUT	1.2 V
PEAK (4 US)	35 A
OC (TYP)	51.2 A
AVG	25 A
FREQ	500 KHZ

NOTE:
A0 SILICON DEFAULT SETTING IS 1.35V

NOTE:
REGULATOR REQUIRES A MINIMUM LOAD TO PREVENT NOISE IN THE AUDIO FREQUENCIES

VID 1	VID 0	P1 GPU MVDD	EVT GPU MVDD
0	0	1.35 V	1.30 V
0	1	1.30 V	1.25 V
1	0	1.25 V	1.20 V
1	1	1.20 V	1.15 V

A0 GPU: 1.35V
A1 GPU: 1.20V (P1C & EVT)

	RD480	RD481	RD482	RD483
P1A/B	S	NS	S	NS
P1C	S	NS	NS	S
EVT	S	NS	NS	S

BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE
VR GFX MVDD

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GFX VDDCI REGULATOR	
VOUT	0.900 V
PEAK (4 US)	6 A
OC (TYP)	12.7 A
AVG	4 A
FREQ	500 KHZ

NOTE:
A0 SILICON DEFAULT SETTING IS 0.985V

NOTE:
REGULATOR REQUIRES
A MINIMUM LOAD TO
PREVENT NOISE IN THE
AUDIO FREQUENCIES

	VID 1	VID 0	GPU VDDCI
	1	1	0.865 V
DEFAULT ->	1	0	0.901 V
	0	1	0.951 V
	0	0	1.001 V

P1A/B/C: 0.985 V DEFAULT
EVT: 0.900 V DEFAULT

BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE
VR GFX VDDCI

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GFX 0V8 REGULATOR	
VOUT	0.9 V
PEAK (4 US)	5 A
OC (TYP)	14.2 A
AVG	4.5 A
FREQ	300 KHZ

VID 1	VID 0	GPU V0V8
1	1	0.802 V
DEFAULT -> 1	0	0.900 V
0	1	0.951 V
0	0	1.000 V

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

VR GFX 0V8

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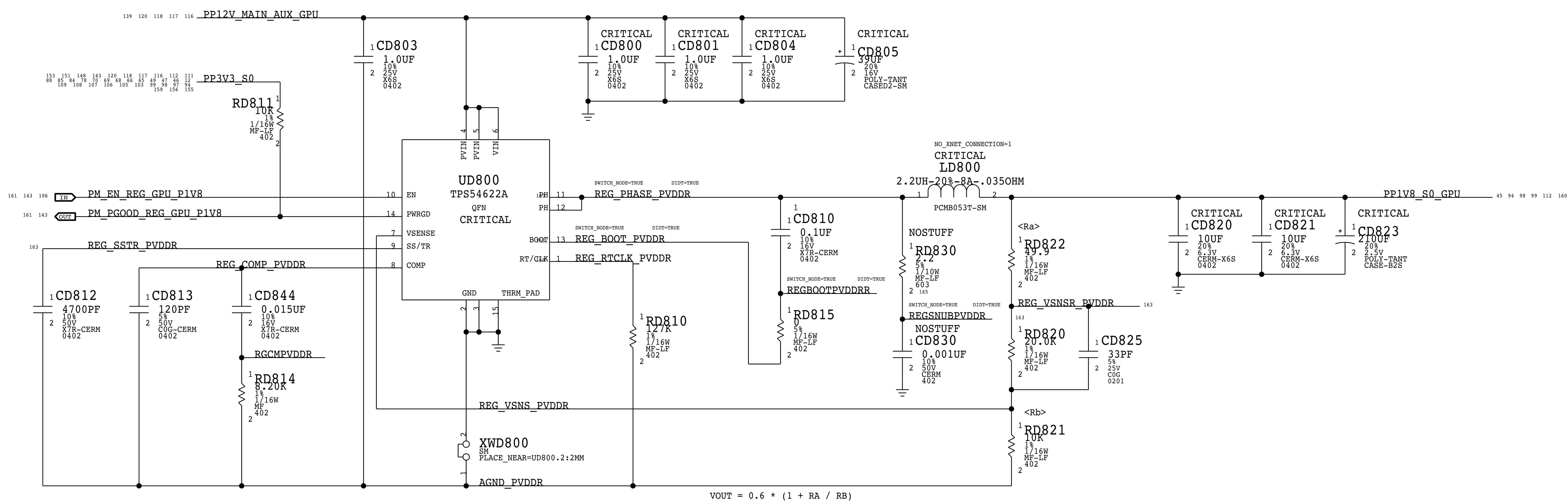
BOM_COST_GROUP=GRAPHICS

GFX 1V8 REGULATOR	
VOUT	1.803 V
PEAK (4 US)	4 A
OC (TYP)	11 A
AVG	3.2 A
FREQ	379 KHZ



BOM_COST_GROUP=GRAPHICS

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

VR GFX 1V8

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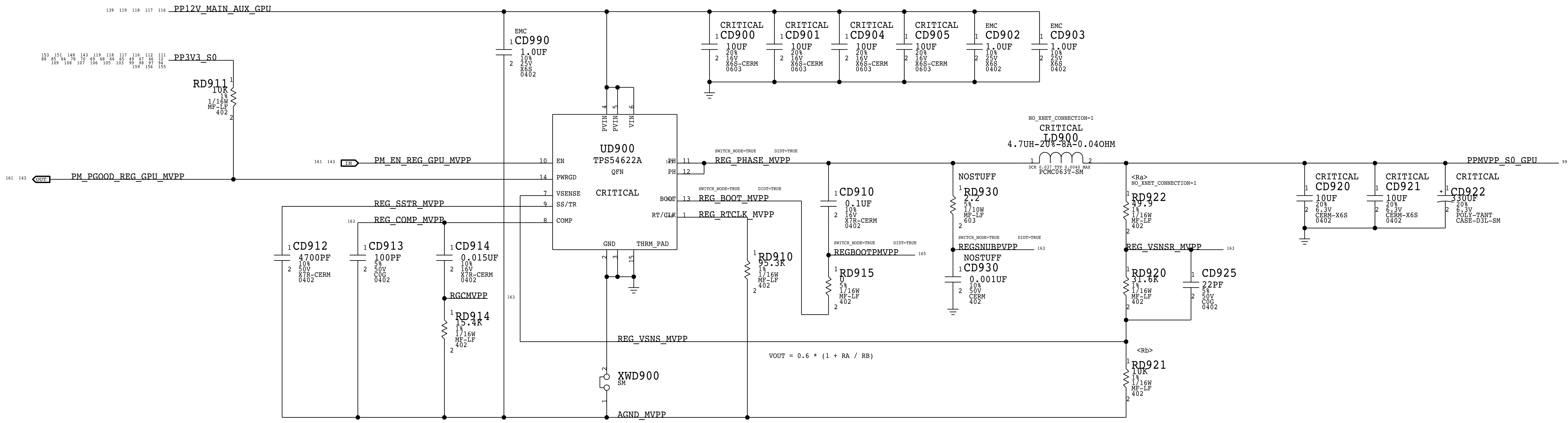
BRANCH

GFX MVPP REGULATOR		
VOUT		2.499 V
PEAK	(4 US)	2 A
OC	(TYP)	11 A
AVG		1.5 A
FREQ		500 KHZ



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SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

VR GFX MVPP

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DEV SUPPORT

Apple Inc.

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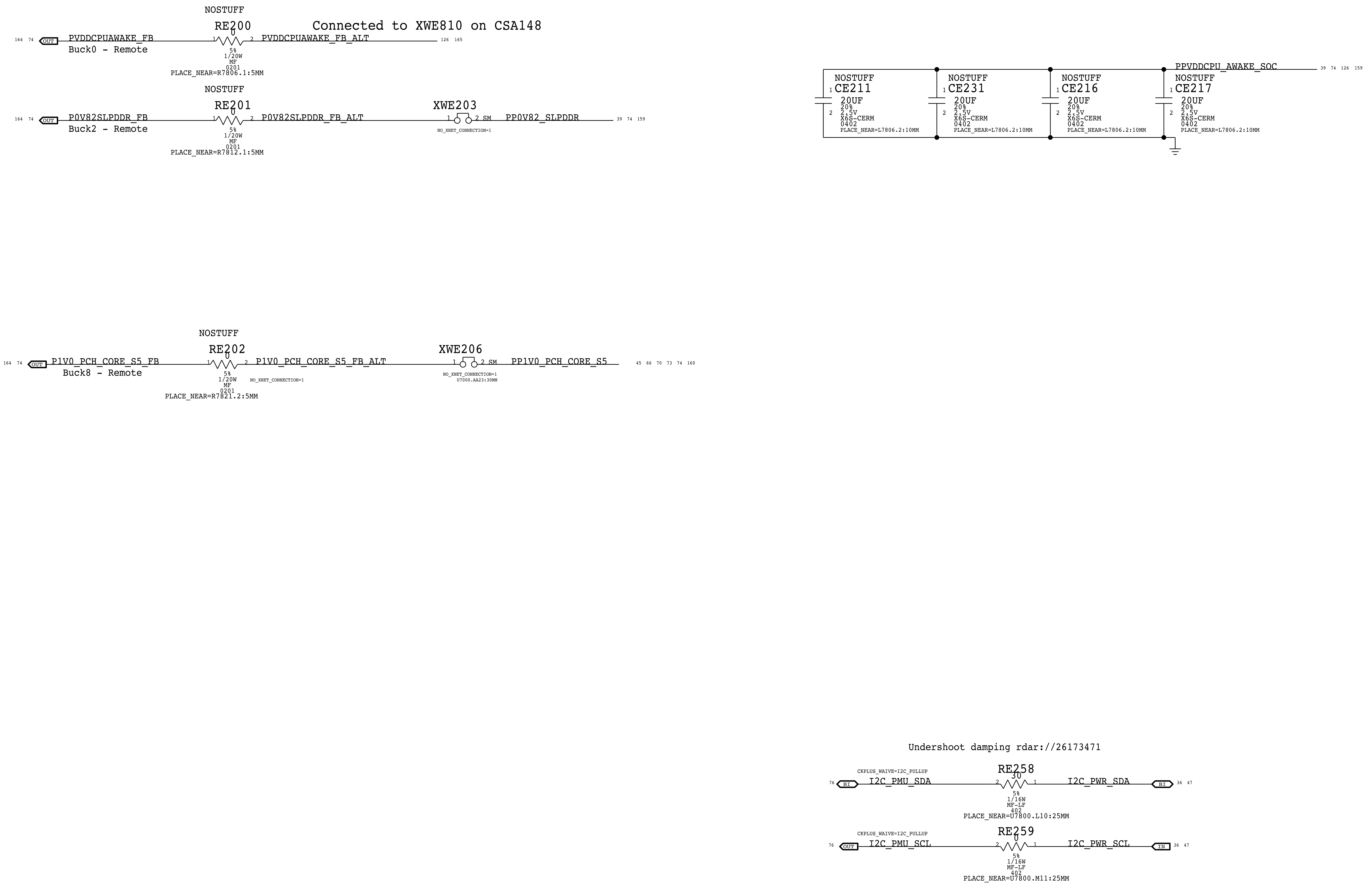
Alternate Feedback Sense



BOM_COST_GROUP=DEBUG

PAGE TITLE
PMU DEV SUPPORT

Apple Inc.

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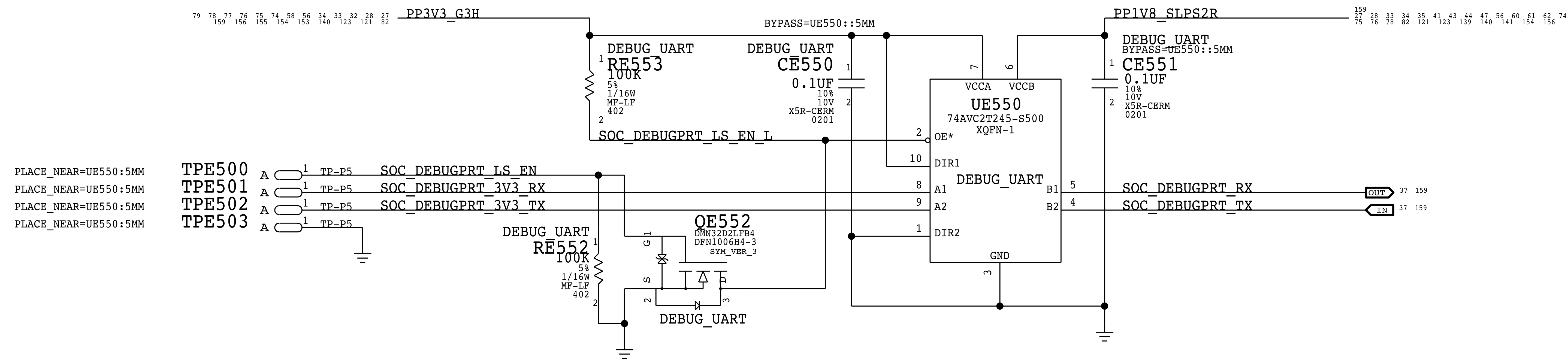
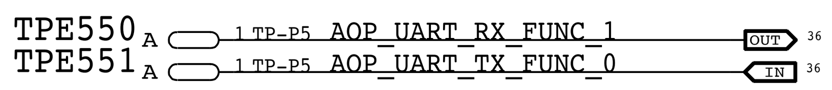
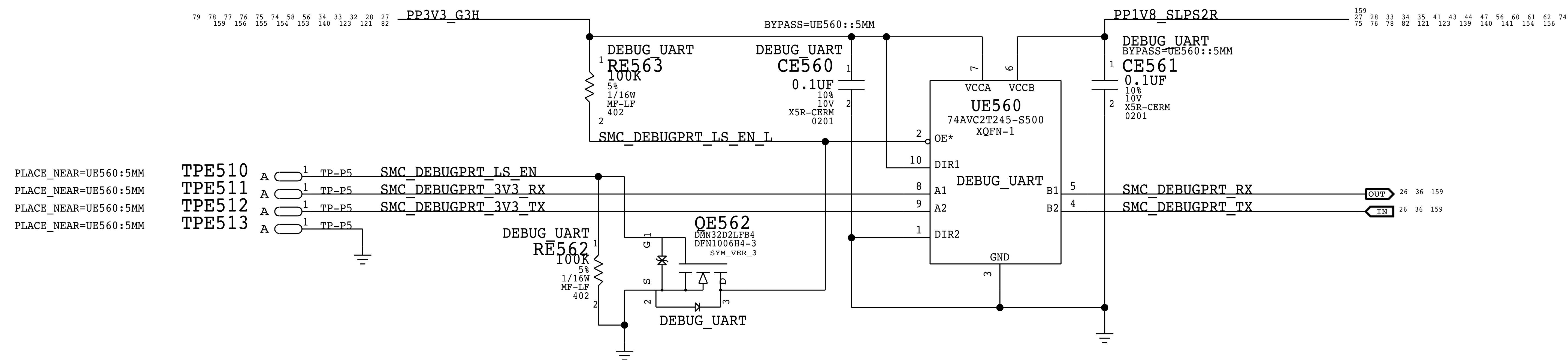
SOC UART0 Header

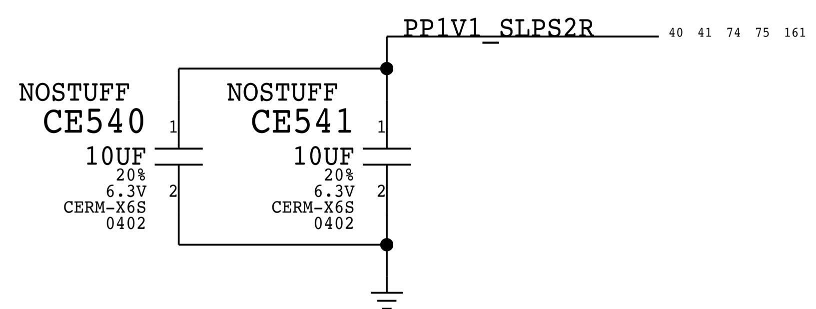


Debug header for SOC AOP UART



SMC UART0 Header





XDP_PRESENT and PCIE_UP_CLKREQ PU are moved to their pages

BOM_COST_GROUP=DEBUG

PAGE TITLE	SOC DEV SUPPORT
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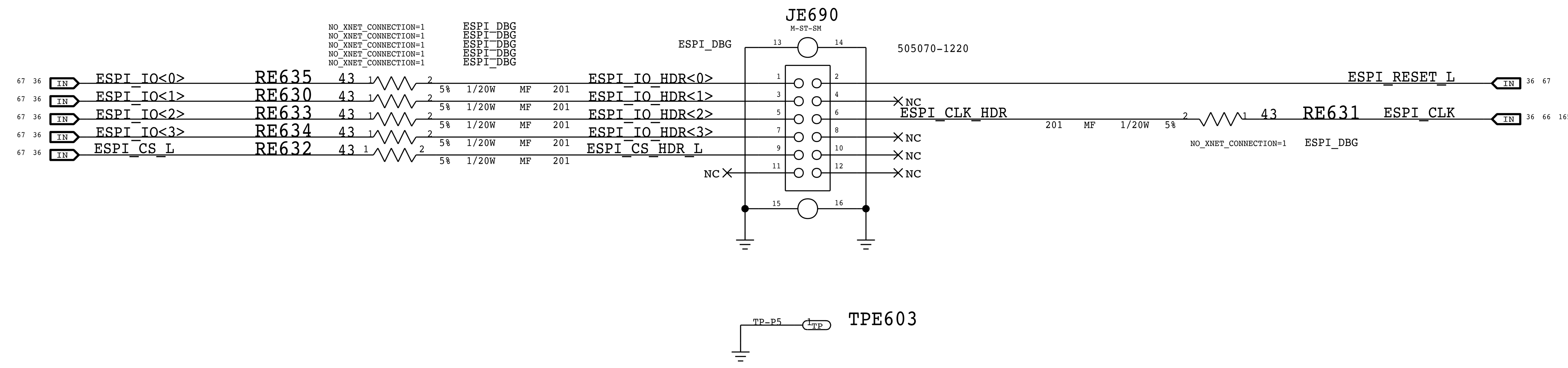
IV ALL RIGHTS RESERVED	SHEET 123 OF 171

eSPI Analyzer

Placement study first



BOM_COST_GROUP=DEBUG

PAGE TITLE	DEBUG HEADERS 1
Apple Inc.	DRAWING NUMBER 051-01649 SIZE D
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8	7	6	5	4	3	2	1

D

C

B

A

BOM_COST_GROUP=DEBUG

PAGE TITLE

DEBUG HEADERS 2

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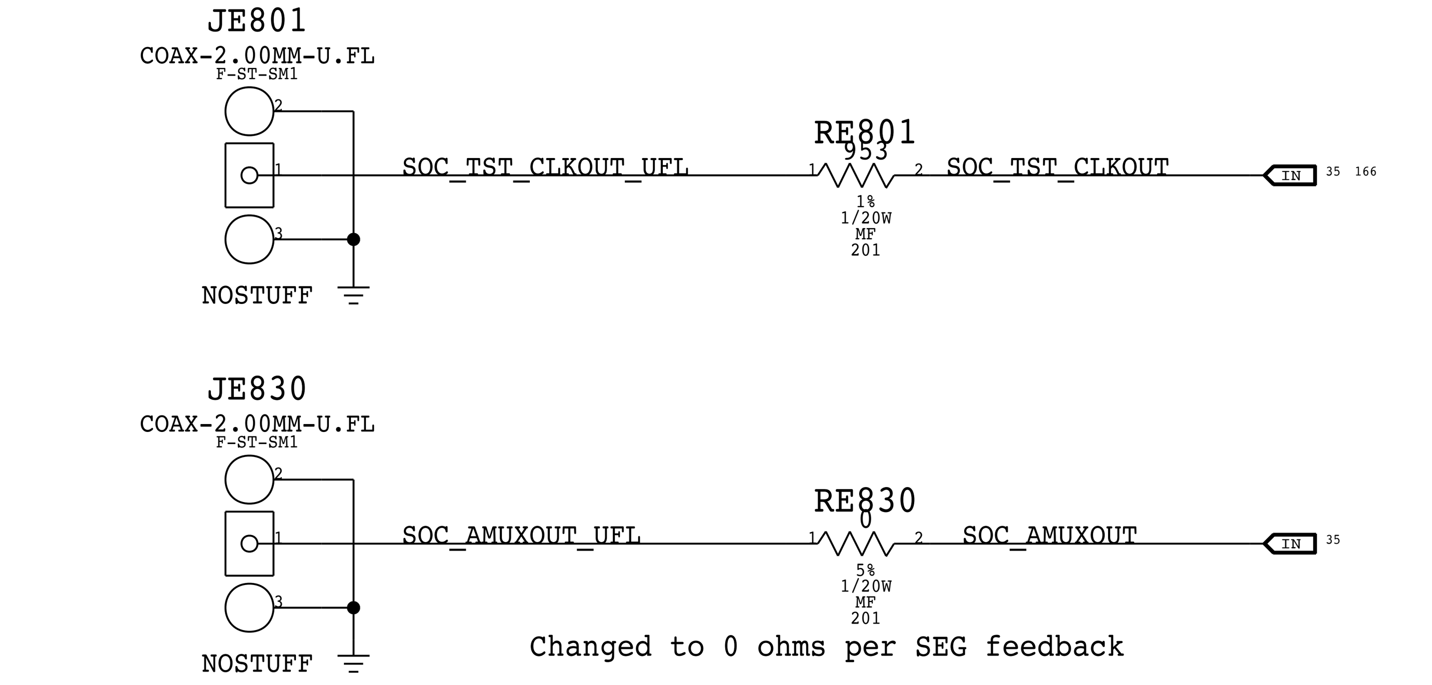
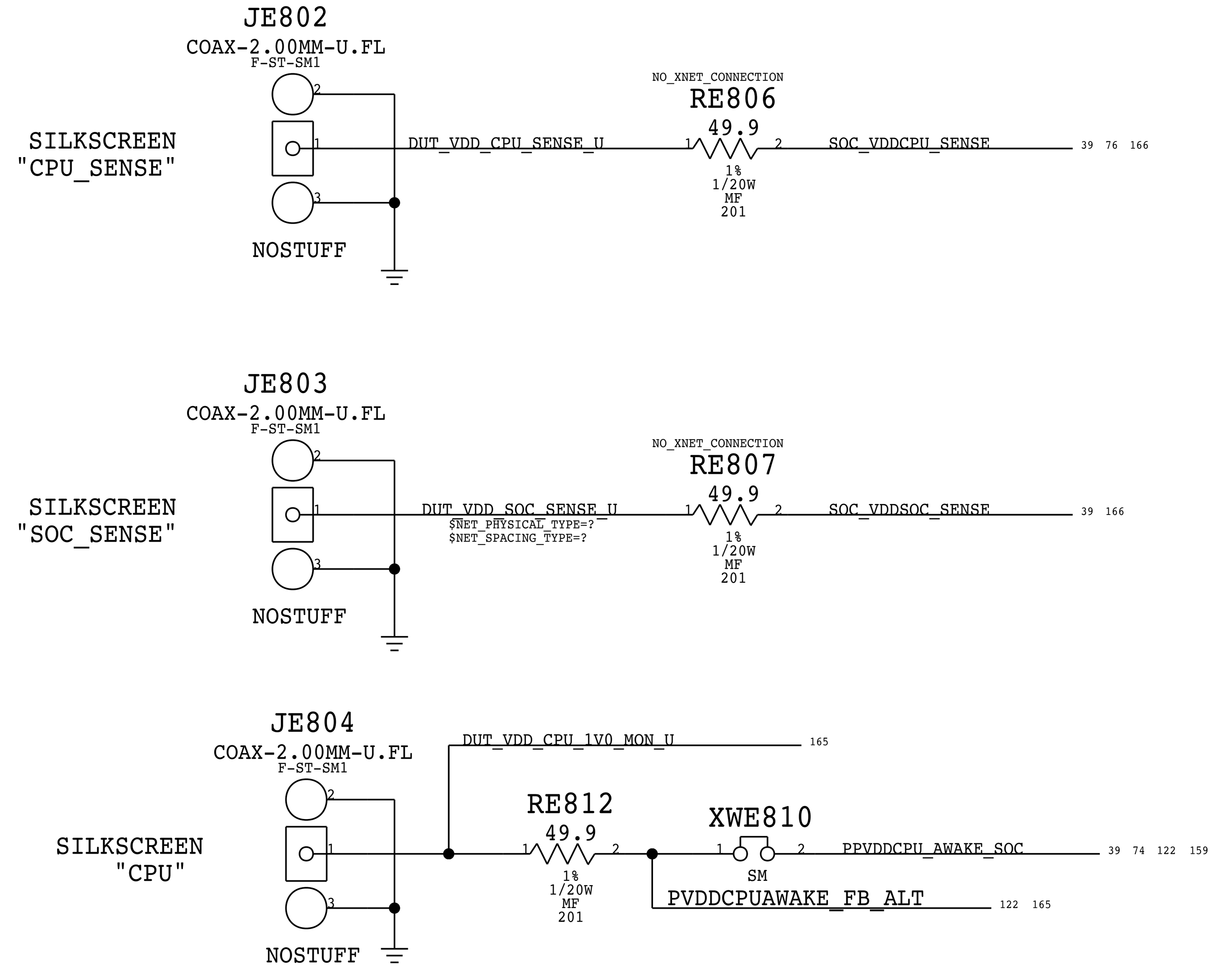
U.FL CONNECTOR FOR SOC CLOCKS



U.FL CONNECTOR FOR SOC POWER



BOM_COST_GROUP=DEBUG

PAGE TITLE	DEBUG SOC U.FL CONNECTORS
Apple Inc.	DRAWING NUMBER 051-01649 SIZE D
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AD1	AD0	I2C ADDR
GND	GND	0X48 <--
GND	1.8V	0X49
1.8V	GND	0X4A
1.8V	1.8V	0X4B

BOM_COST_GROUP=AUDIO

PAGE TITLE

AUDIO JACK CODEC

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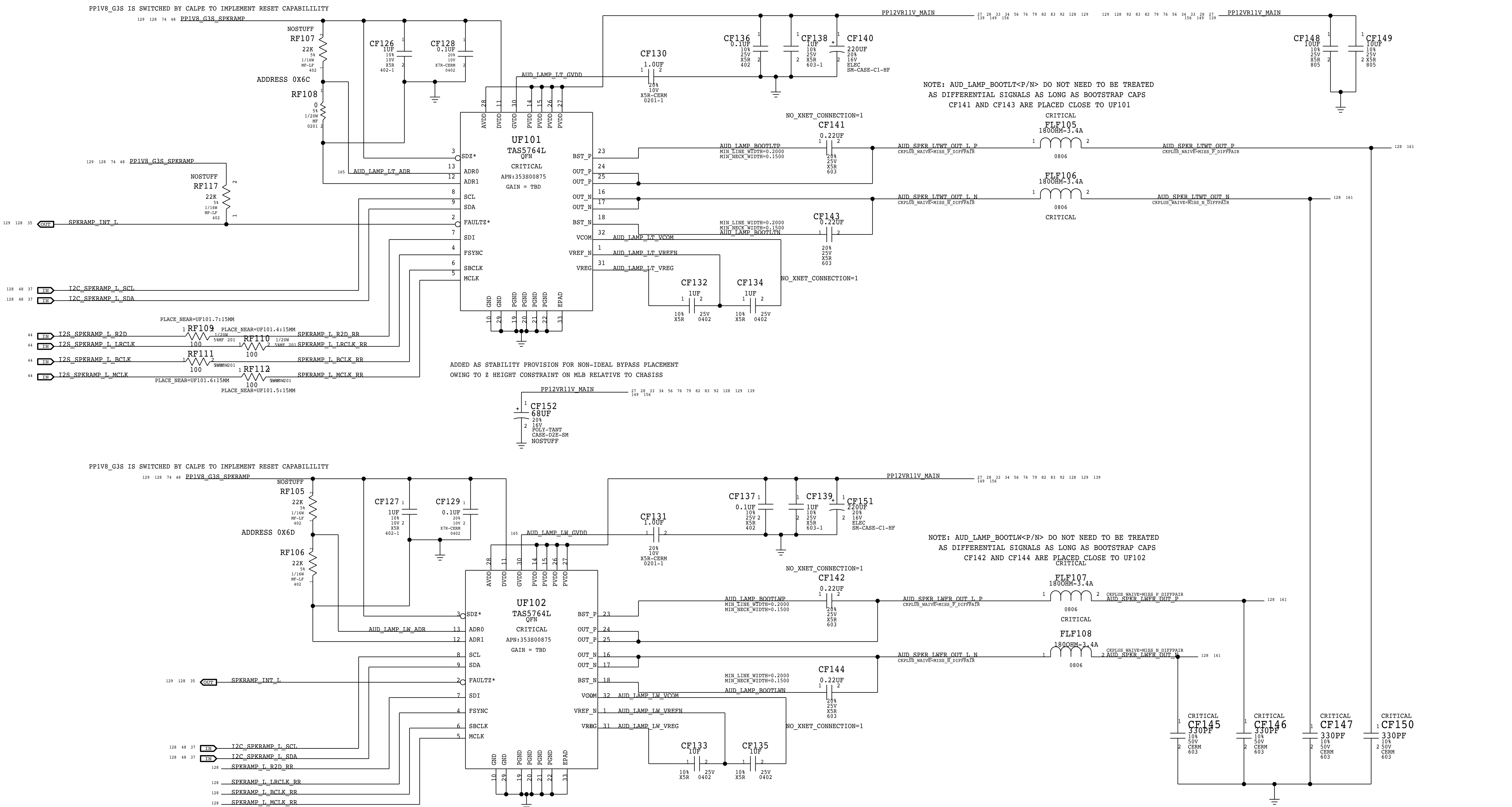
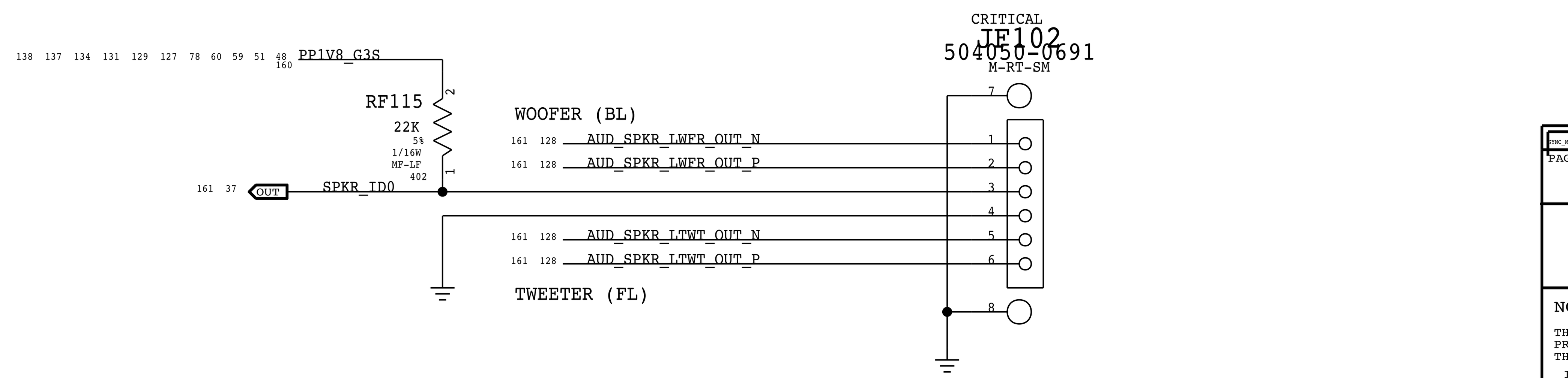
SPEAKER AMPLIFIERS - LEFT CHANNEL

2X MONO SPEAKER AMPLIFIERS (TAS5764L)



SPEAKER CABLE CONNECTORS

APPLE P/N 518S0862



BOM_COST_GROUP=AUDIO

PAGE TITLE

AUDIO LEFT SPKR AMP

Apple Inc.

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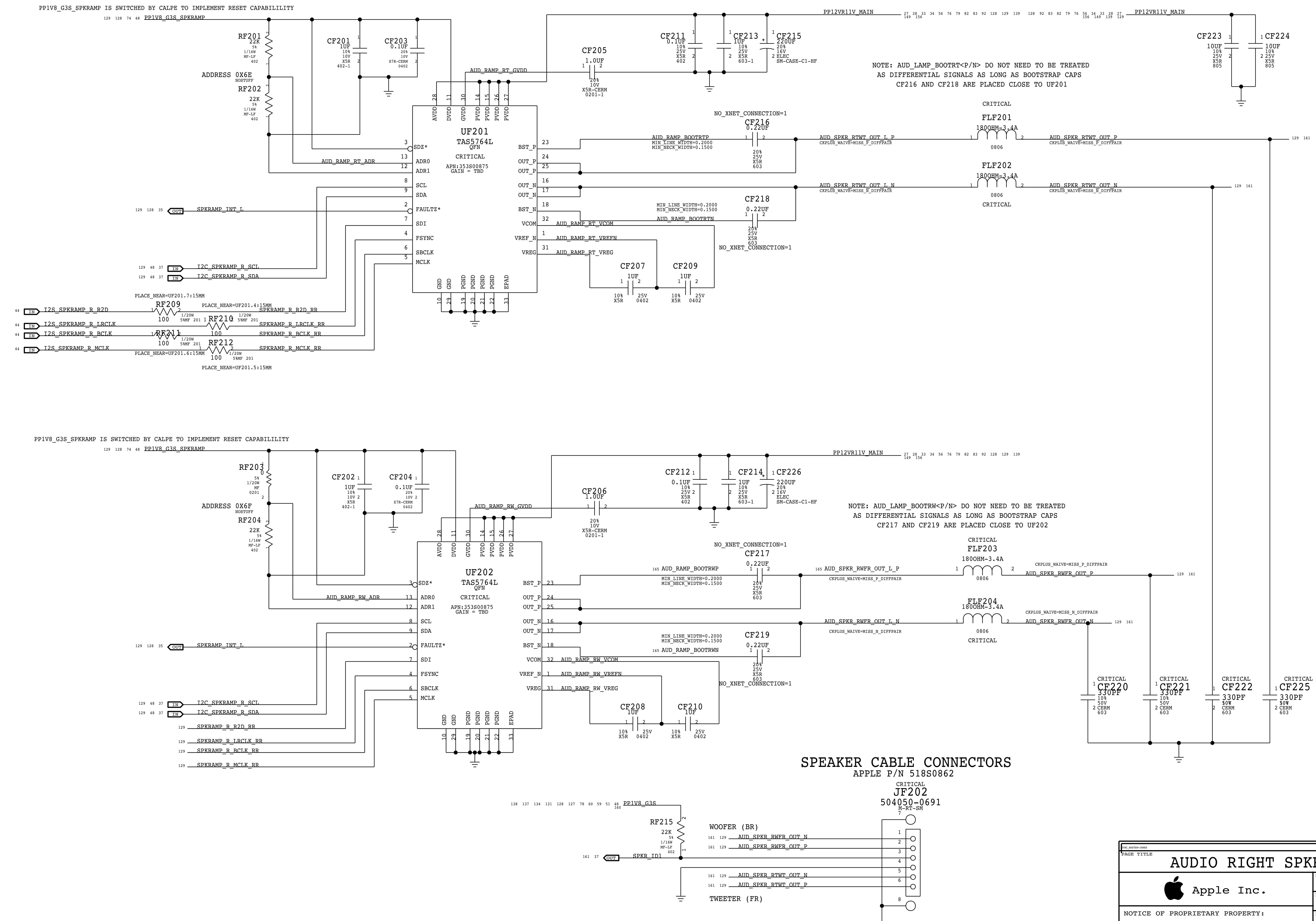
BRANCH

SPEAKER AMPLIFIERS - RIGHT CHANNEL

2X MONO SPEAKER AMPLIFIERS (TAS5764L)



BOM_COST_GROUP=AUDIO

PAGE TITLE

AUDIO RIGHT SPKR AMP

Apple Inc.

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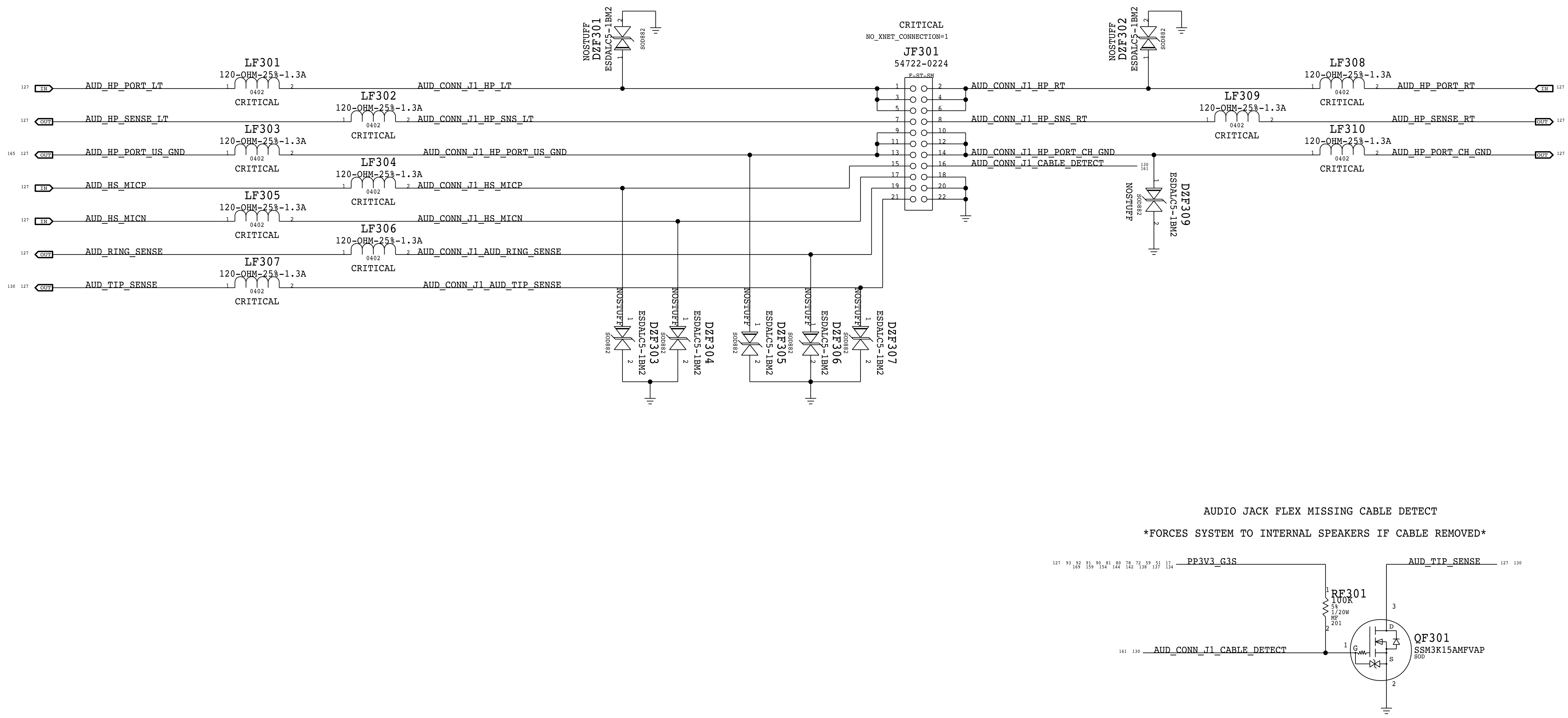

BOM_COST_GROUP=AUDIO

PAGE TITLE
AUDIO JACK CONNECTOR

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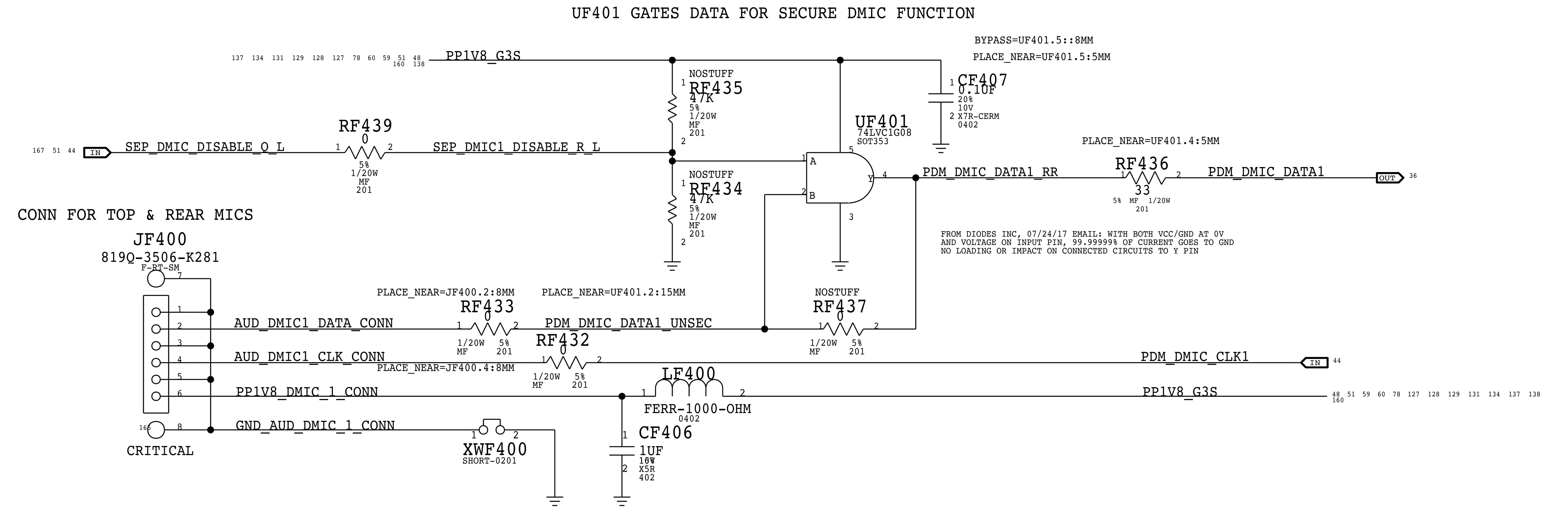

BOM_COST_GROUP=AUDIO

PAGE TITLE

AUDIO MIC CONNECTOR

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BOM_COST_GROUP=AUDIO

PAGE TITLE

AUDIO MISC 1

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SYNC_MASTER=JAMES SYNC_DATE=05/12/2017

PAGE TITLE

AUDIO MISC 2

Apple Inc.

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PAGE TITLE

WIFI/BT MODULE 1

Apple Inc.

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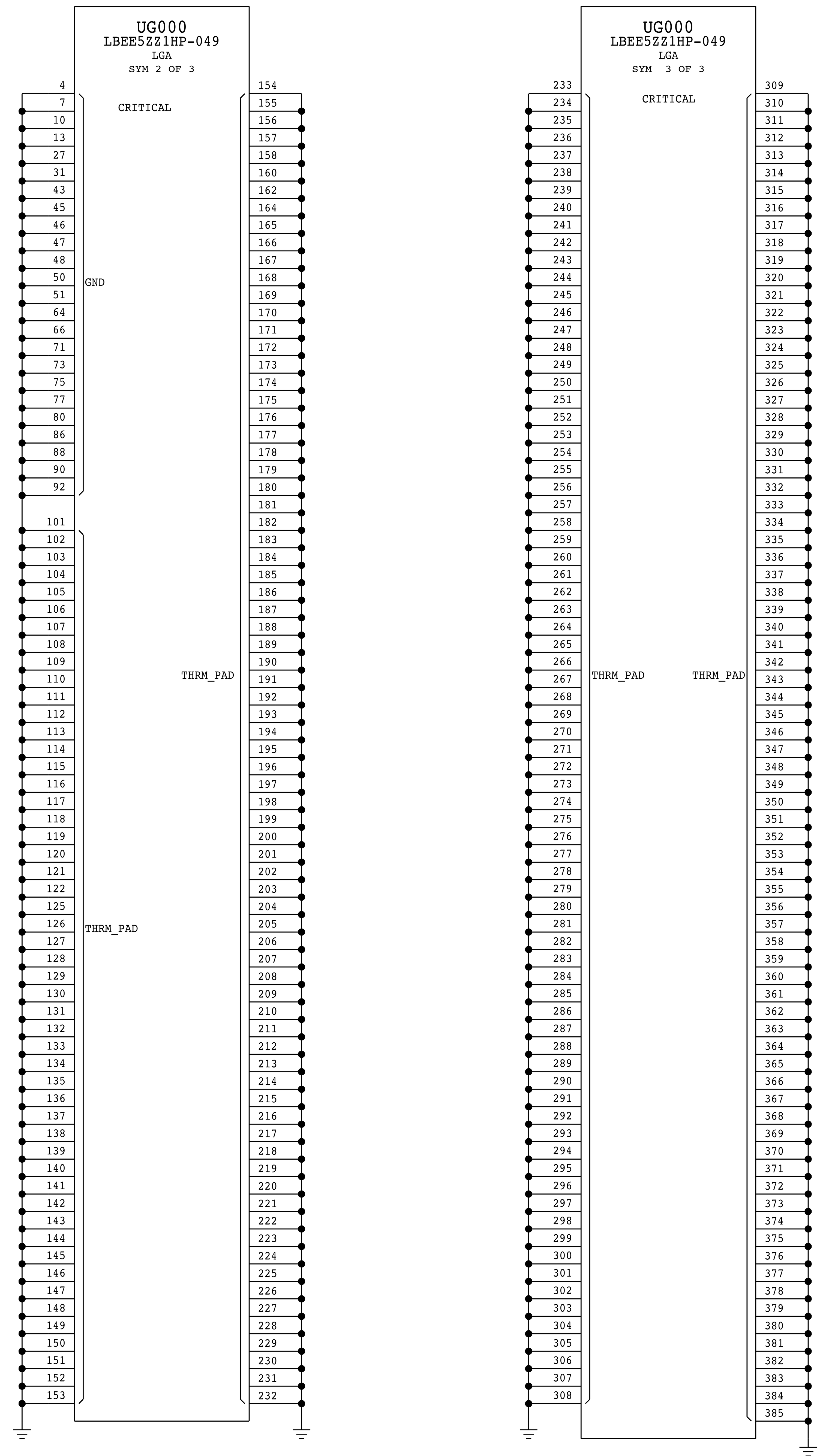

BOM_COST_GROUP=WIRELESS

PAGE TITLE

WIFI/BT MODULE 2

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BOM_COST_GROUP=WIRELESS

PAGE TITLE

WIFI/BT MODULE 3

Apple Inc.

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PCH - BT UART BUFFERS
NOTE: CTS/RTS Swapped
PCH - WLAN UART BUFFERS
PCH ROM BOOT ISOLATION
(2.4K PU ON HARPOON PAGE)
SW REQUESTS PCH TO DRIVE PER <RDAR://32857700>
BT SERIAL FLASH
OMIT_TABLE
UG450
W25Q20EWUXIE
CRITICAL
AIRPORT DEBUG HEADER
JG401
AA25D-S038VA1
516S1074
WLAN SERIAL EEPROM
UG410
CAT93C86B
CRITICAL
WIRELESS TEST POINTS
TPG454 WLAN_SPROM_MOSI
TPG455 WLAN_SPROM_MISO
TPG456 WLAN_SPROM_CLK
TPG457 WLAN_SPROM_CS
TPG458 SWD_WLAN_SWDIO
TPG459 WLAN_THROTTLE
TPG460 SWD_WLAN_SWCLK
TPG461 WLAN_JTAG_TRST_L
TPG462 WLAN_JTAG_TDO
TPG463 PCIE_CLK100M_WLAN_P
TPG464 PCIE_CLK100M_WLAN_N
WLAN AUDIO SYNC BUFFER
CLK GENERATOR
UG420
32.768KHZ-+/-100PPM
CRITICAL
BOM_COST_GROUP=WIRELESS
PAGE TITLE
WIFI/BT SUPPORT 1
Apple Inc.
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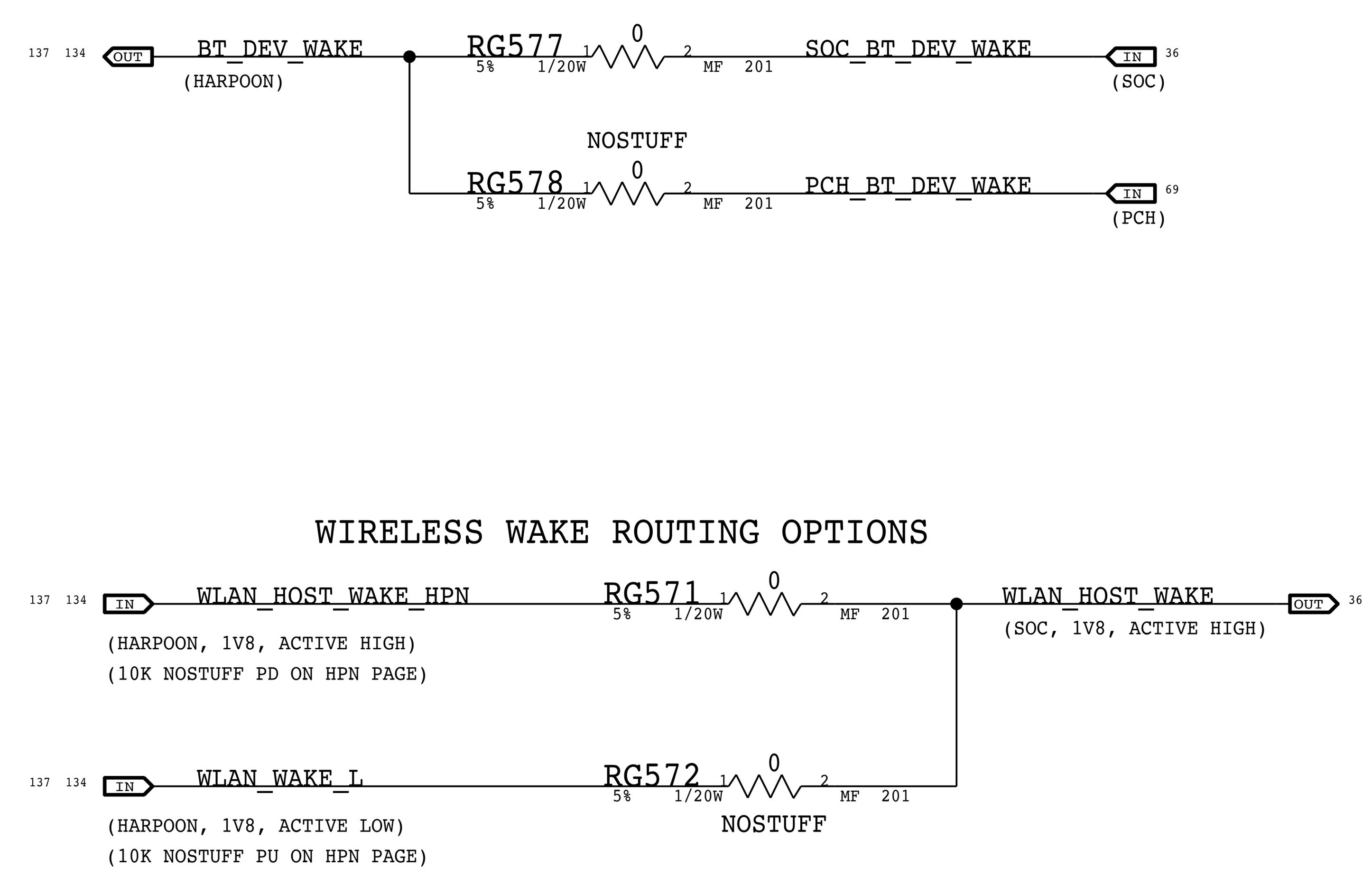
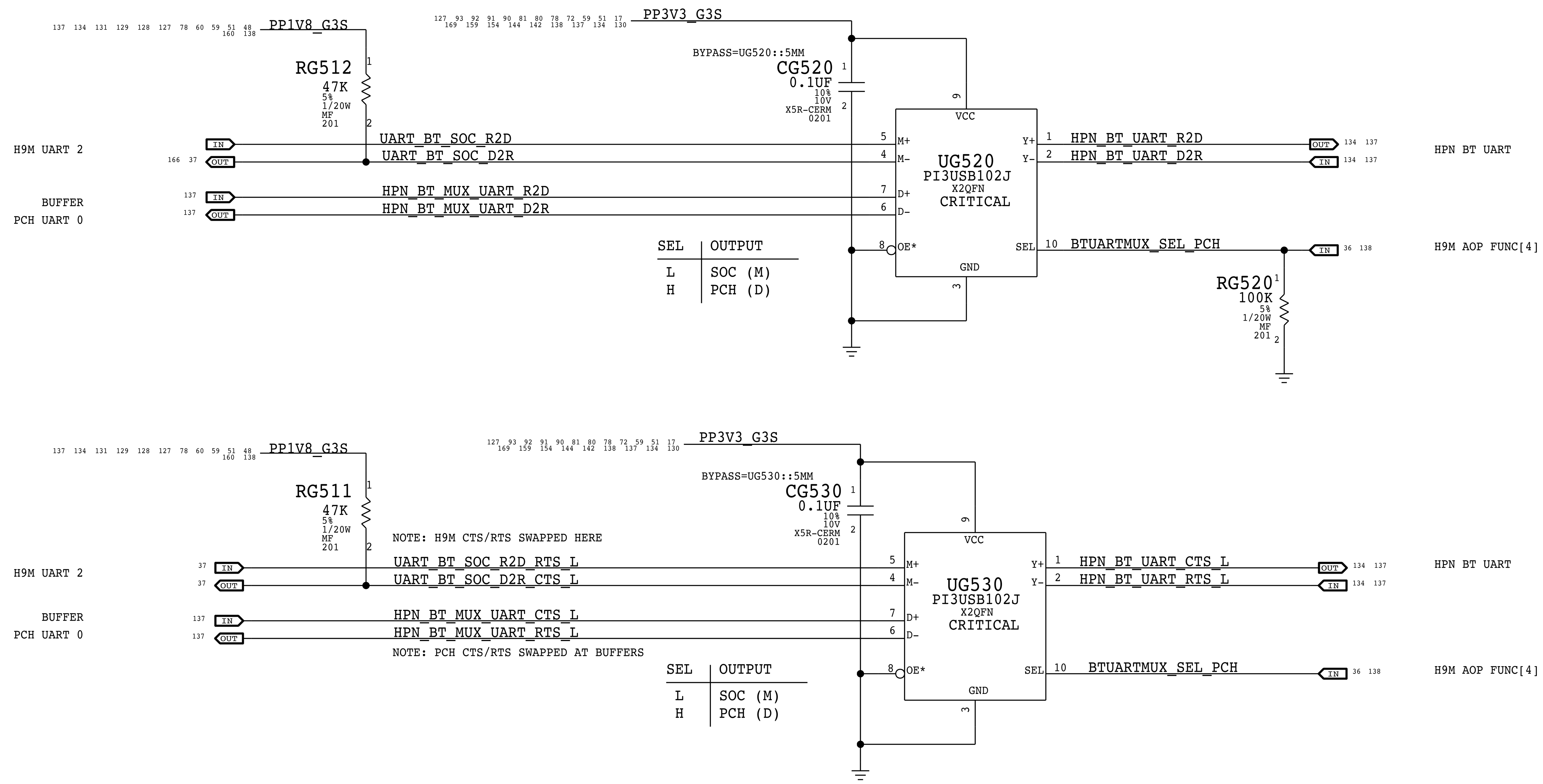
Software	I2S_SEL	UART_SEL	I2S I/F	UART I/F
Gen1 (macOS)	0	1	UART (PCH)	UART (PCH)
Gen1 (Bootcamp)	1	1	I2S (PCH)	UART (PCH)
Gen2	X	0	X	UART (SOC)

BT DEV WAKE ROUTING OPTION
<RDAR://33783520>

WIRELESS WAKE ROUTING OPTIONS





BOM_COST_GROUP=WIRELESS

PAGE TITLE
WIFI/BT SUPPORT 2

Apple Inc.

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BOTH 12V INPUTS FROM THE ACDC ARE THE SAME
SECONDARY SIDE OUTPUT FROM THE ACDC

THERE ARE TWO OUTPUTS BECAUSE THE BOARD IS LARGE
AND HAS A CHOKEPOINT ABOVE THE GPU
HAVING TWO OUTPUTS ALLOWS ONE TO GO TO EACH SIDE
OF THE BOARD

BOM_COST_GROUP=PLATFORM POWER

SYNC_MASTER=AMAN SYNC_DATE=06/16/2017

PAGE TITLE
POWER INPUT FROM AC/DC 1

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DRAWING NUMBER 051-01649 SIZE D

REVISION 6.21.0

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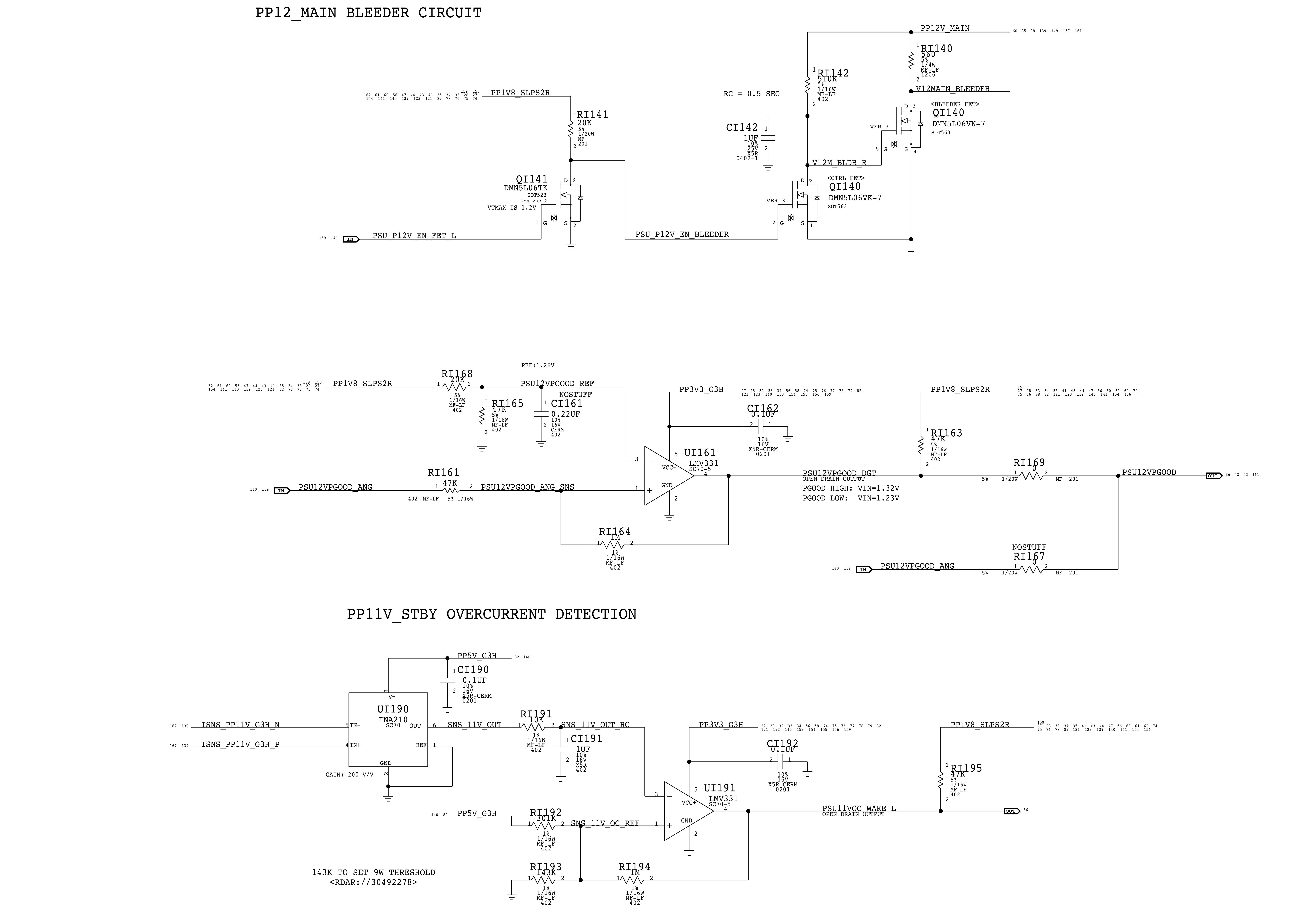

BOM_COST_GROUP=PLATFORM POWER

SYNC_MASTER=AMAN SYNC_DATE=06/16/2017

PAGE TITLE: POWER INPUT FROM AC/DC 2

Apple Inc.

DRAWING NUMBER: 051-01649 SIZE: D

REVISION: 6.21.0

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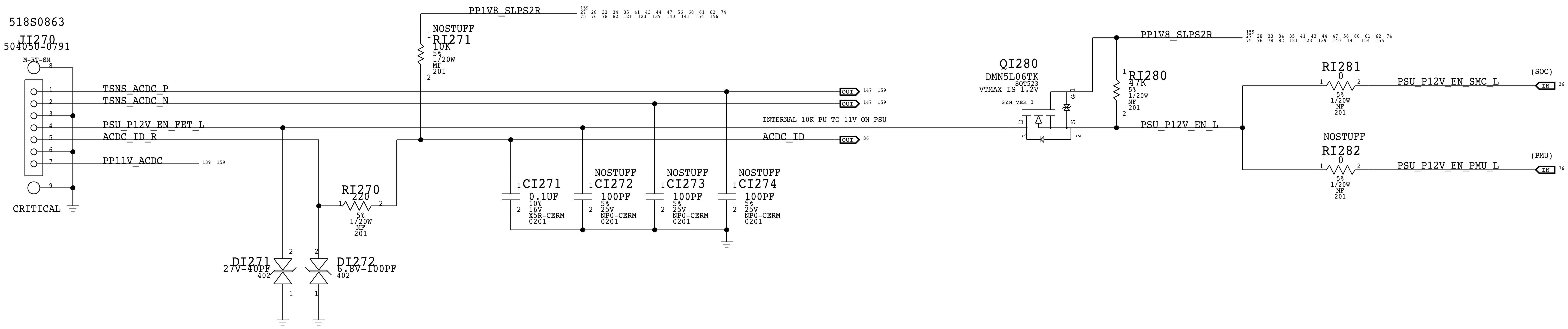
POR: 7 PIN SIGNAL + AUX POWER CONNECTOR



SYNC_MASTER=AMAN SYNC_DATE=06/22/2017

PAGE TITLE

POWER INPUT FROM AC/DC 3

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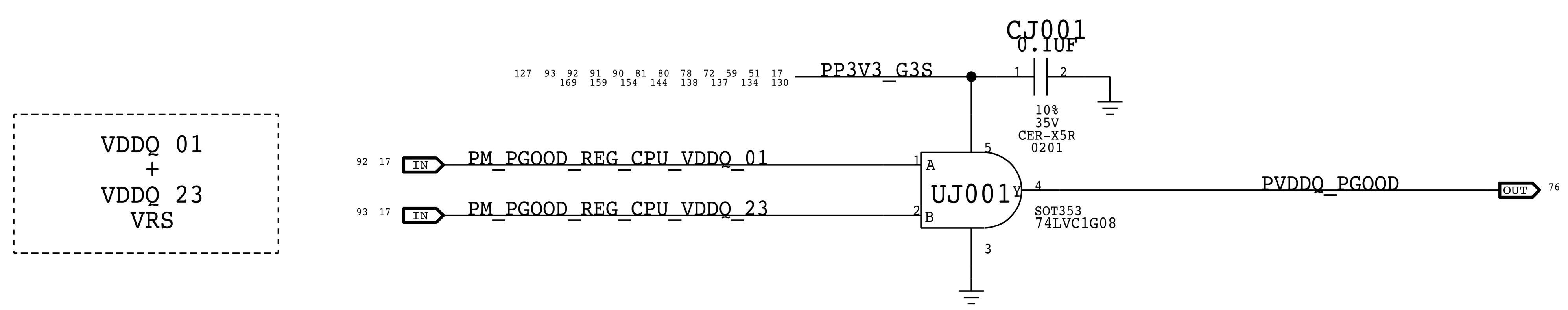
BOM_COST_GROUP=PLATFORM POWER

MEMORY POWER PGOOD



Power Sequencing requirements

System:

1. 12V_ACDC -> 3V42G3H -> 12V_S5/12V_S5_SSD -> 3V3_S5 -> 1V0_S5 -> 5V_S4/3V3_S4 -> 1V0_S3

Intel CPU:

2. 1V0_S3 -> 2V5_S3 -> VDDQ_S3 -> 12V_S0 -> 5V_S0 -> 3V3_S0 -> GPU Rails -> VCCIO_S0-> VCC_CPU_S0 (VCCGT & VCCSA)

BOM_COST_GROUP=PLATFORM POWER

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

POWER SEQUENCING S3

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ALL POWER ENABLE AND POWER GOOD SIGNALS ARE 3V3 SIGNALS
POWER SEQUENCING REQUIREMENTS
AMD GPU:
12V_S0 -> 5V_S0 ->3V3_S0 -> 1V8(VDD_CT) -> 0V8 -> GPUCORE (VDDC) -> VDDCI -> MVPP -> MVDD
POWER SEQUENCE FROM 1V8 TO MVDD IN LESS THAN 20MS
MAX SLEW RATE ON ALL POWER RAILS IS 50 MV / US
1V8 RAIL MUST REACH STEADY STATE AT LEAST 10 US BEFORE OTHER RAILS START TO RAMP
MVPP MUST REACH STEADY STATE BEFORE MVDD STARTS TO RAMP
DISABLING ALL POWER RAILS AT THE SAME TIME IS OK PER <RDAR://27812223>

BOM_COST_GROUP=PLATFORM POWER

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE
POWER SEQUENCING S0 1

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BOM_COST_GROUP=PLATFORM POWER

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

POWER SEQUENCING S0 2

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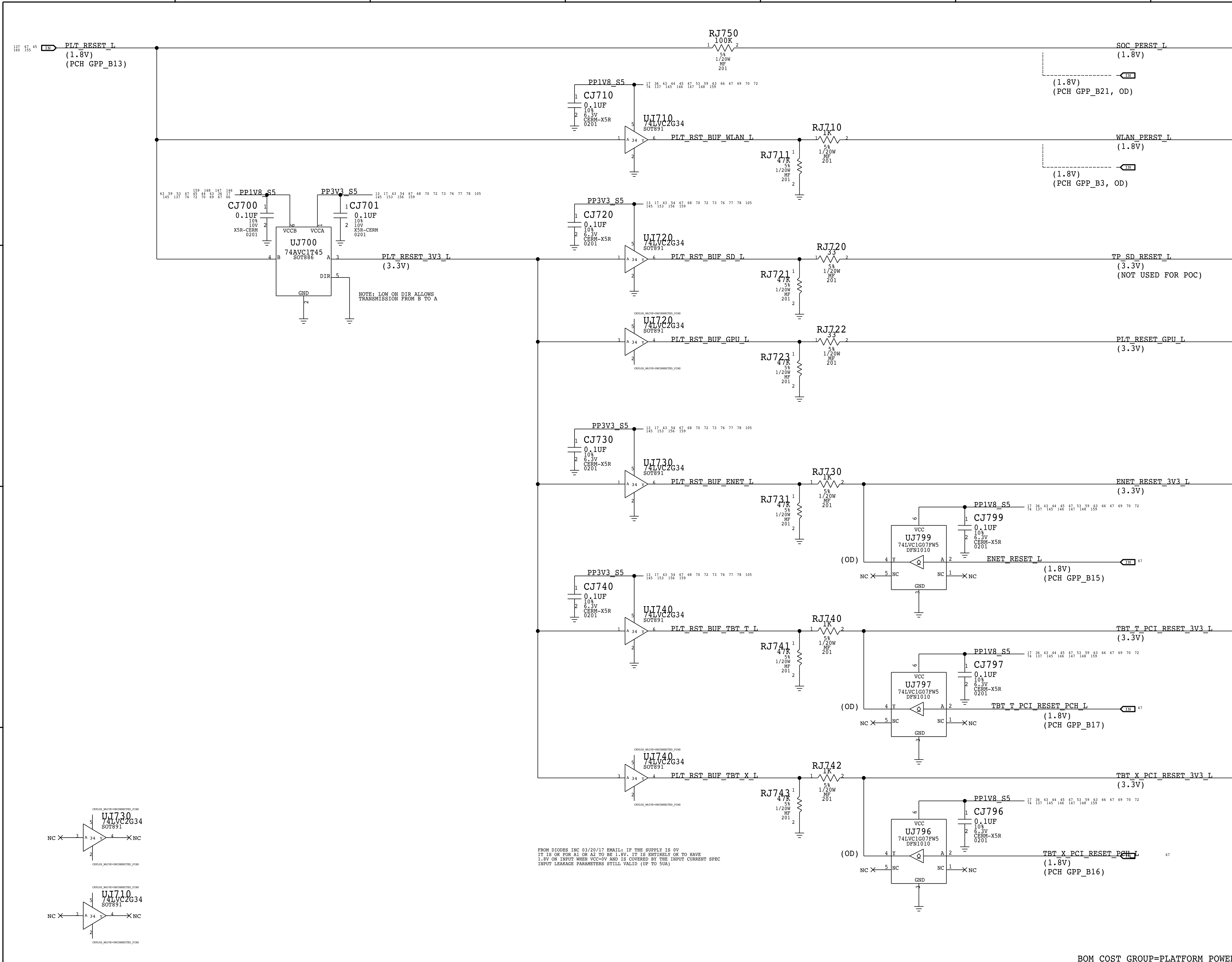

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

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RESETS

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J135 SCHEMATC SAYS NEED DIFFERENT APN 353S2908??

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

BOM_COST_GROUP=SENSORS

PAGE TITLE
SENSORS THERMAL 1

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BOM_COST_GROUP=SENSORS

PAGE TITLE
SENSORS THERMAL 2

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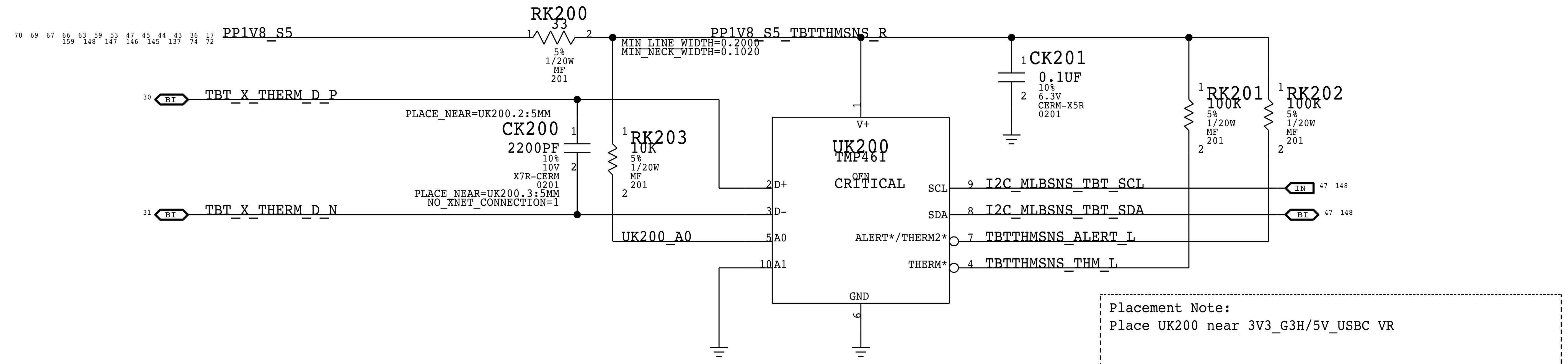
Thermal Sensor A: Thunderbolt Die

I2C Write: 0xD8, I2C Read: 0xD9

THERMAL DIODE: TBT DIE (TBT1 - KEY TI0D)

Placement Note:
The P leg connects to THERMDA pin of the TBT chip, the N leg connect to TitanRidge pin V9.

I2C Device (UK200) Address 0X4A:
I2C Write 0x94, I2C Read 0x95



Placement Note:
Place UK200 near 3V3_G3H/5V_USBC VR

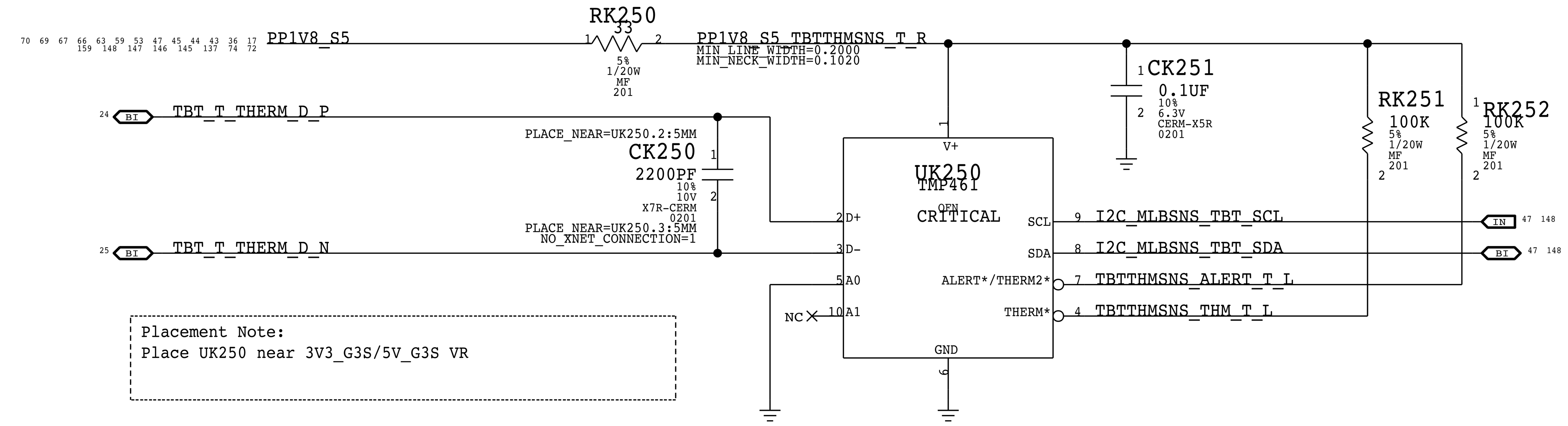
Thermal Sensor B: Thunderbolt Die

I2C Write: 0xB8, I2C Read: 0xB9

THERMAL DIODE: TBT DIE (TBT2 - KEY TI1D)

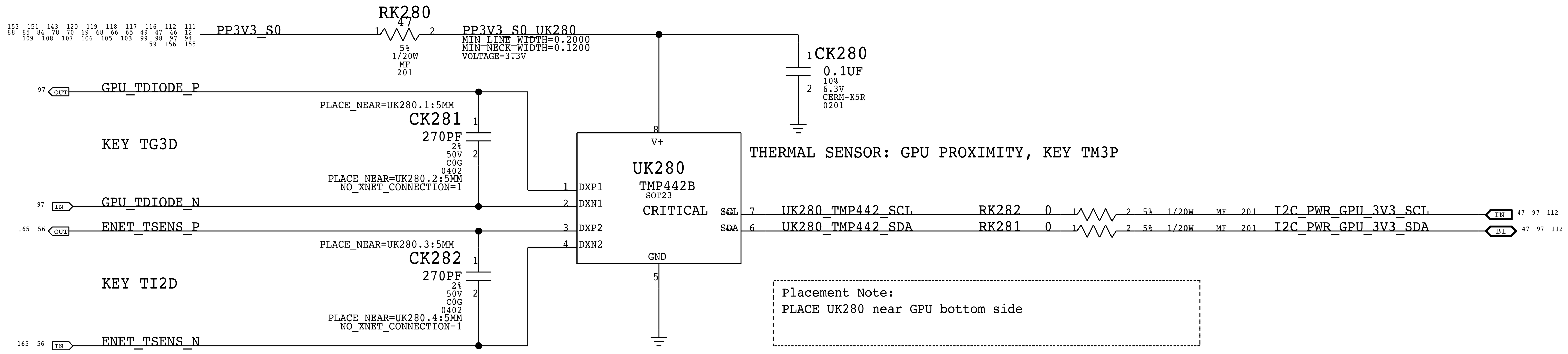
Placement Note:
The P leg connects to THERMDA pin of the TBT chip, the N leg connect to pin AC22.

I2C Device (UK250) Address 0x4B:
I2C Write 0x96, I2C Read 0x97



Placement Note:
Place UK250 near 3V3_G3S/5V_G3S VR

GPU THERMAL - KEY TM3P



Placement Note:
PLACE UK280 near GPU bottom side

I2C WRITE: 0X9A, I2C READ: 0X9B
I2C DEVICE (UK280) ADDRESS 0X4D

BOM_COST_GROUP=SENSORS

PAGE TITLE
SENSORS THERMAL 3

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SENSORS POWER HIGHSIDE 1

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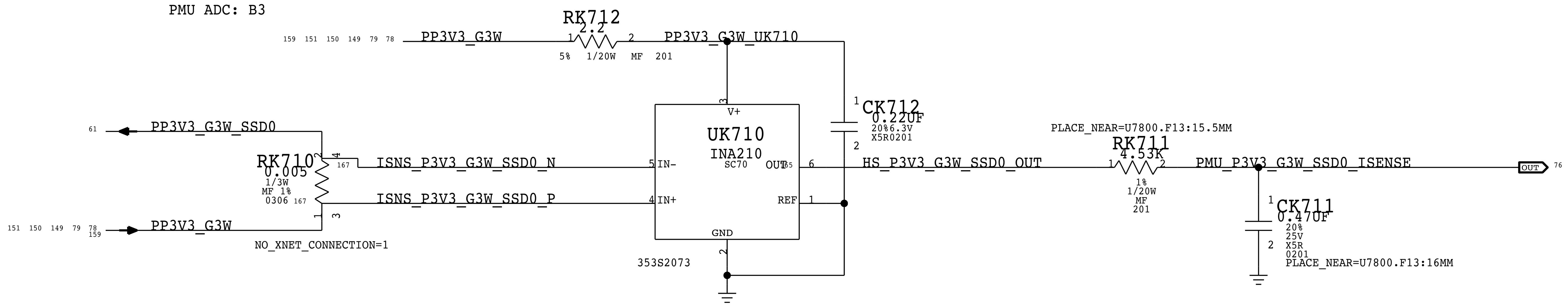
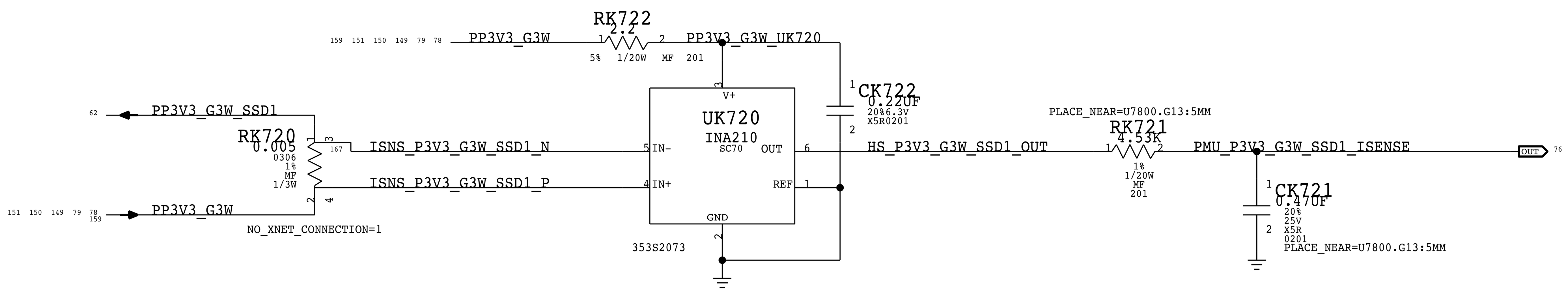
PP3V3_G3W_SSD0 HIGH SIDE CURRENT SENSE (KEY IH0R)

GAIN: 200X, EDP: 1.1 A
RSENSE: 0.005 (RK710) OR RSENSE SHORT
VSENSE: 13 MV, RANGE: 2.5 A
PMU ADC: B3



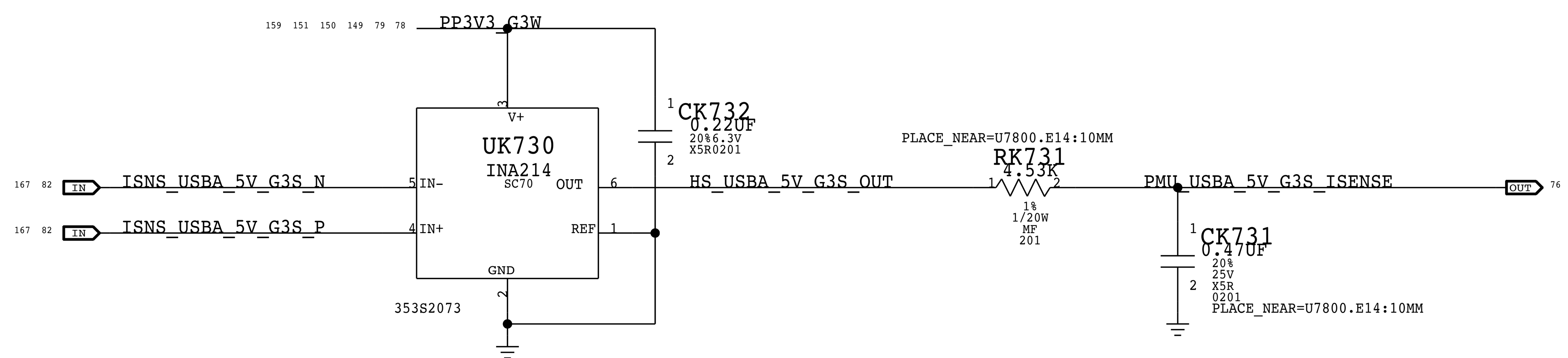
PP3V3_G3W_SSD1 HIGH SIDE CURRENT SENSE (KEY IH1R)

GAIN: 200X, EDP: 1.1 A
RSENSE: 0.005 (RK760) OR RSENSE SHORT
VSENSE: 13 MV, RANGE: 2.5 A
PMU ADC: B4



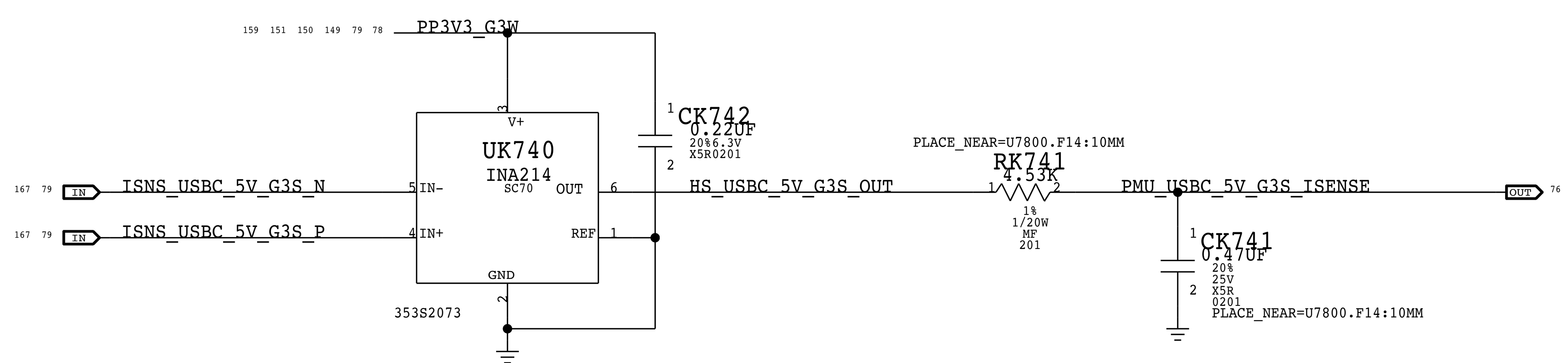
PP5V_G3S_USBA HIGH SIDE CURRENT SENSE (KEY II0R)

GAIN: 100X, EDP: 5.5 A
RSENSE: 0.003 (R8690) OR RSENSE SHORT
VSENSE: 25 MV, RANGE: 8.3 A
PMU ADC: A6



PP5V_G3S_USBC HIGH SIDE CURRENT SENSE (KEY II1R)

GAIN: 100X, EDP: 13 A
RSENSE: 0.001 (R8390) OR RSENSE SHORT
VSENSE: 25 MV, RANGE: 25 A
PMU ADC: A7



BOM_COST_GROUP=SENSORS

PAGE TITLE: SENSORS POWER HIGHSIDE 2

Apple Inc.

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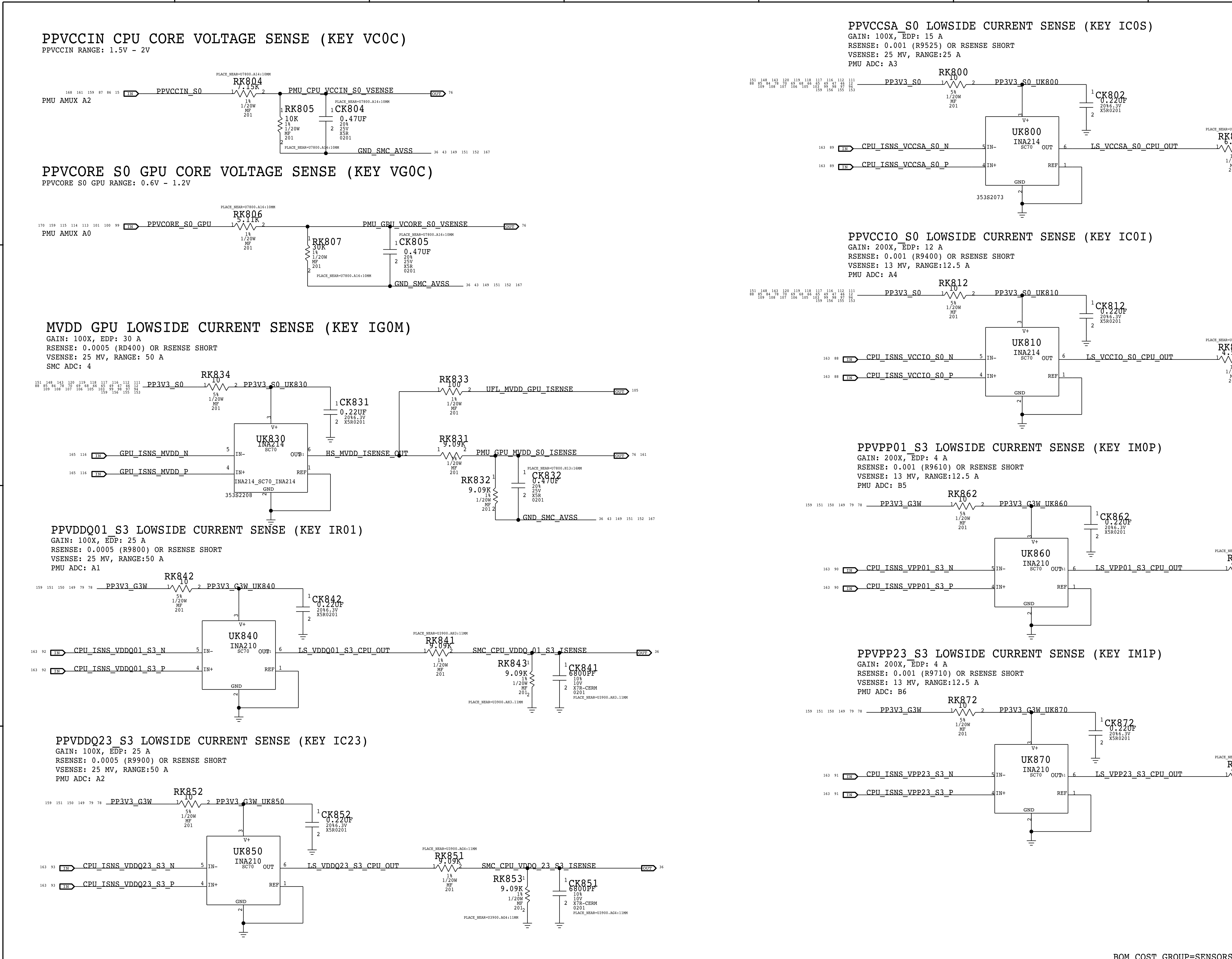

PAGE TITLE

SENSORS POWER LOWSIDE 1

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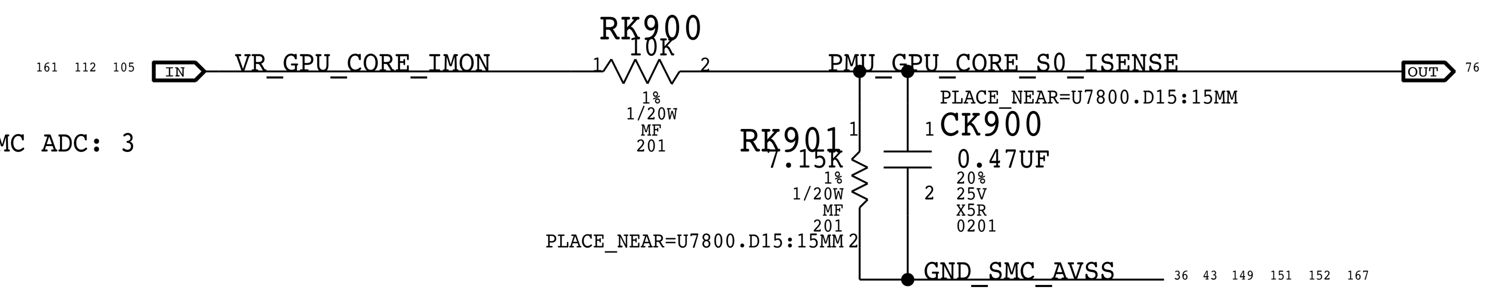
GPU CORE CURRENT SENSE VIA IMON (KEY IG0C)

CPU CORE CURRENT RANGE: 432A

VOLTAGE RANGE: INPUT 2.16V/432A - OUTPUT 2.16 *7.15/(10+7.15) = 2.2mV

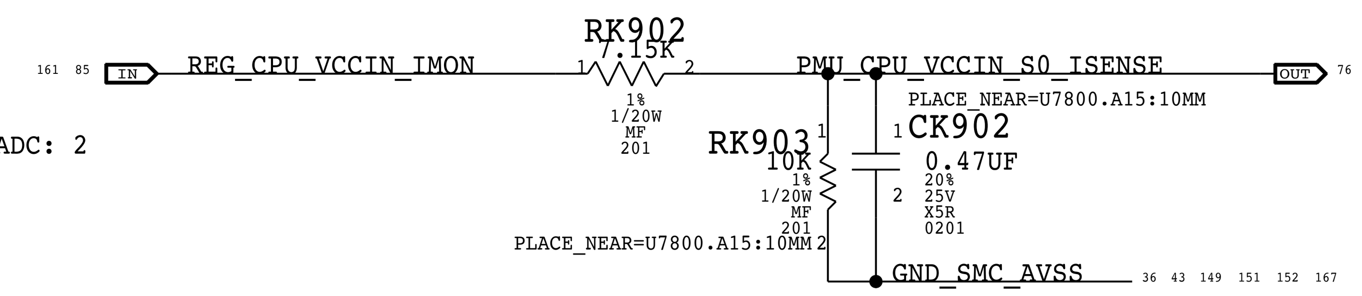
VSENSE: 2.2mV/A



CPU CORE CURRENT SENSE VIA IMON (KEY IC0C)

CPU CORE CURRENT RANGE: 200A

VSENSE: 1.25 MV/A



BOM_COST_GROUP=SENSORS

PAGE TITLE
SENSORS POWER LOWSIDE 2

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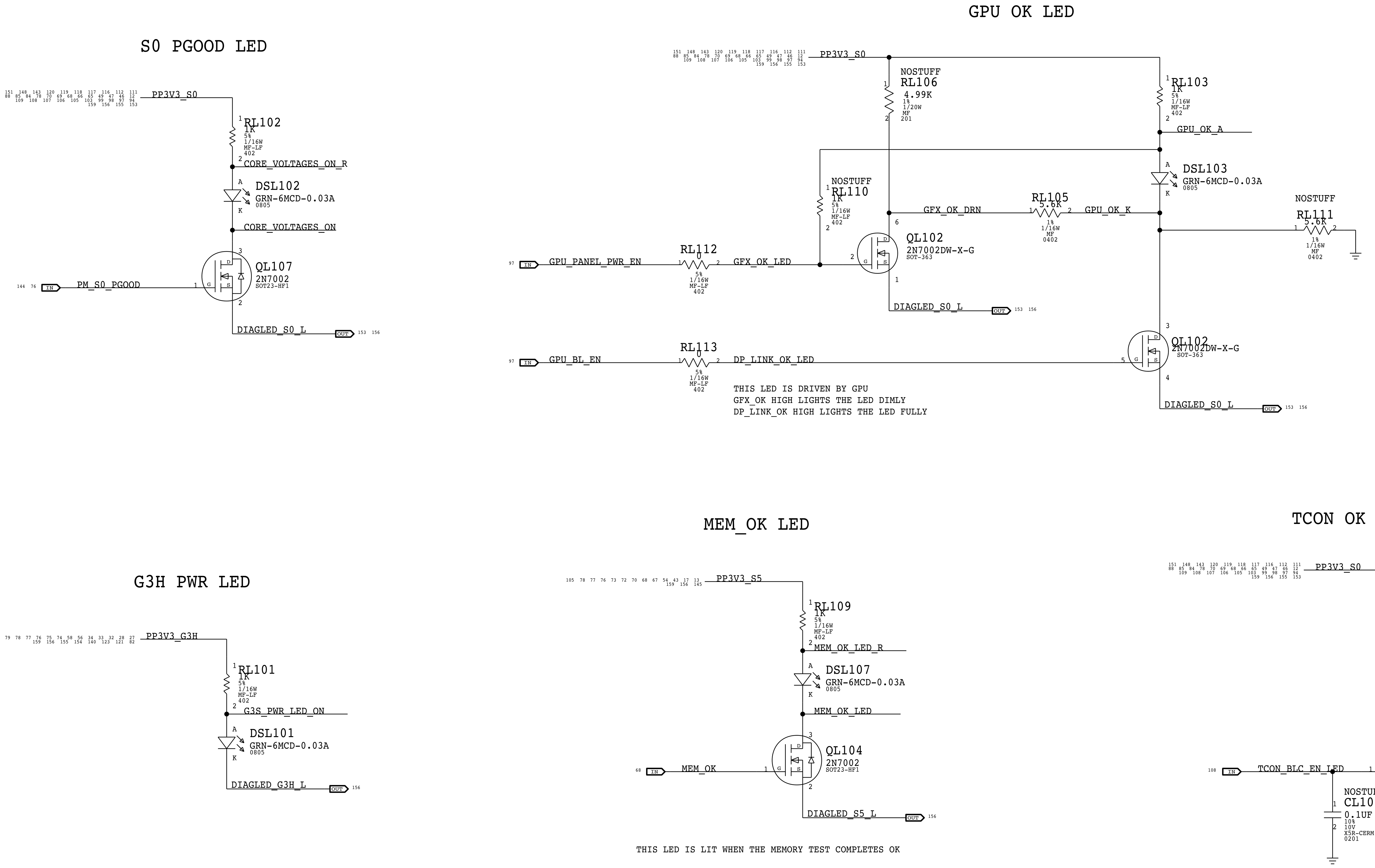
POR LED LIST TRACKED AT <RDAR://30685511>
NOTE, NONE OF THESE WILL TURN ON BY DEFAULT UNLESS SWITCH SL550 IS PUSHED



BOM_COST_GROUP=DEBUG

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE
LEDS POR

Apple Inc.

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LEDS DEBUG 1

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SYNC_MASTER=AMAN SYNC_DATE=01/20/2017

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LEDS DEBUG 2

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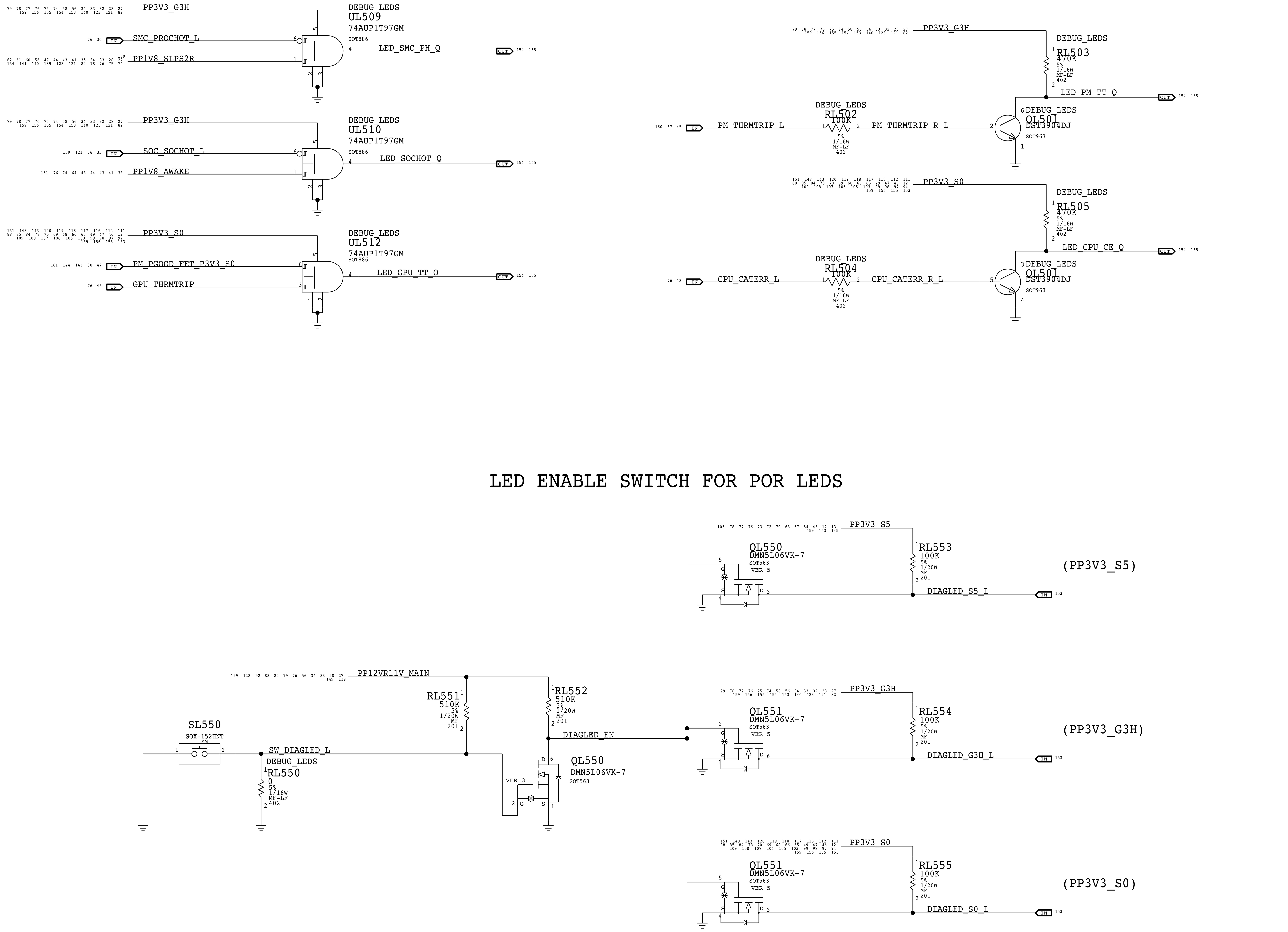

BOM_COST_GROUP=DEBUG

PAGE TITLE
LEDS SUPPORT

Apple Inc.

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BRANCH
PAGE 215 OF 255
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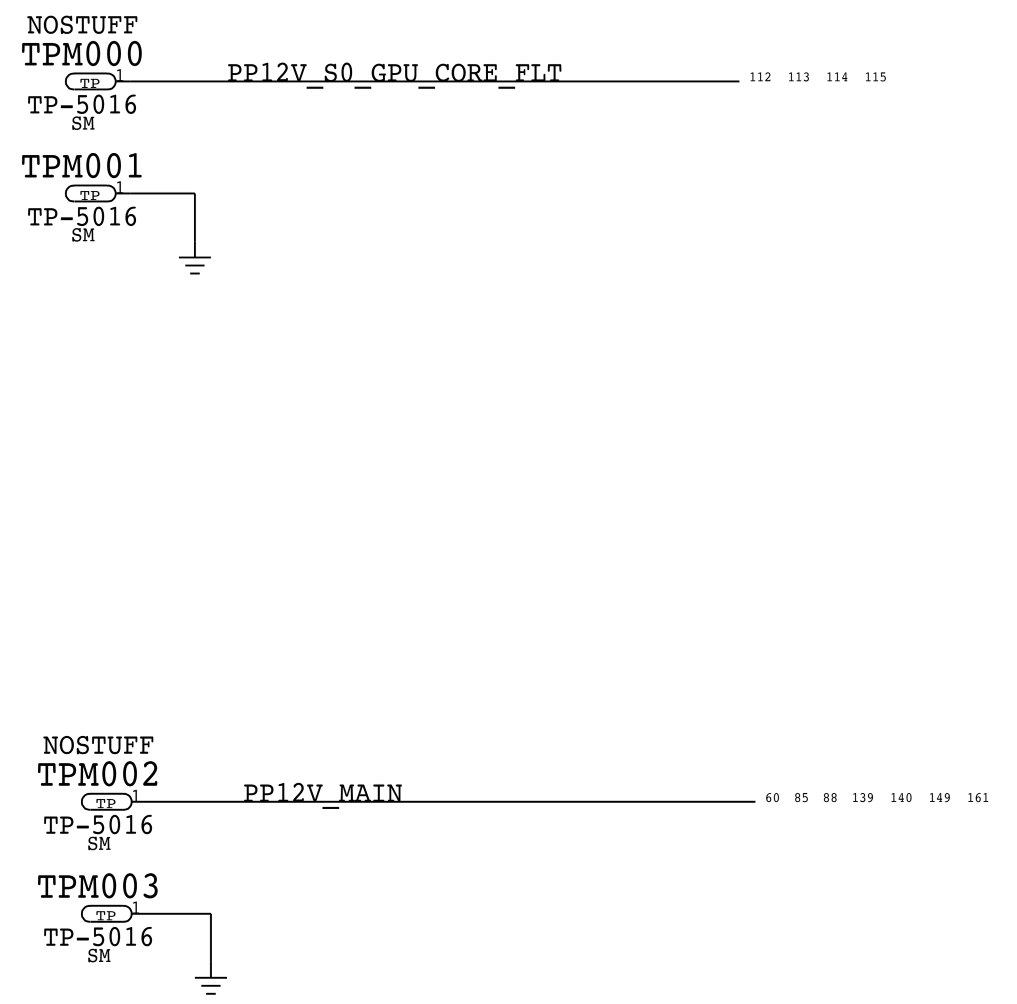

BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE
PSU RESISTANCE MEASUREMENT

Apple Inc.

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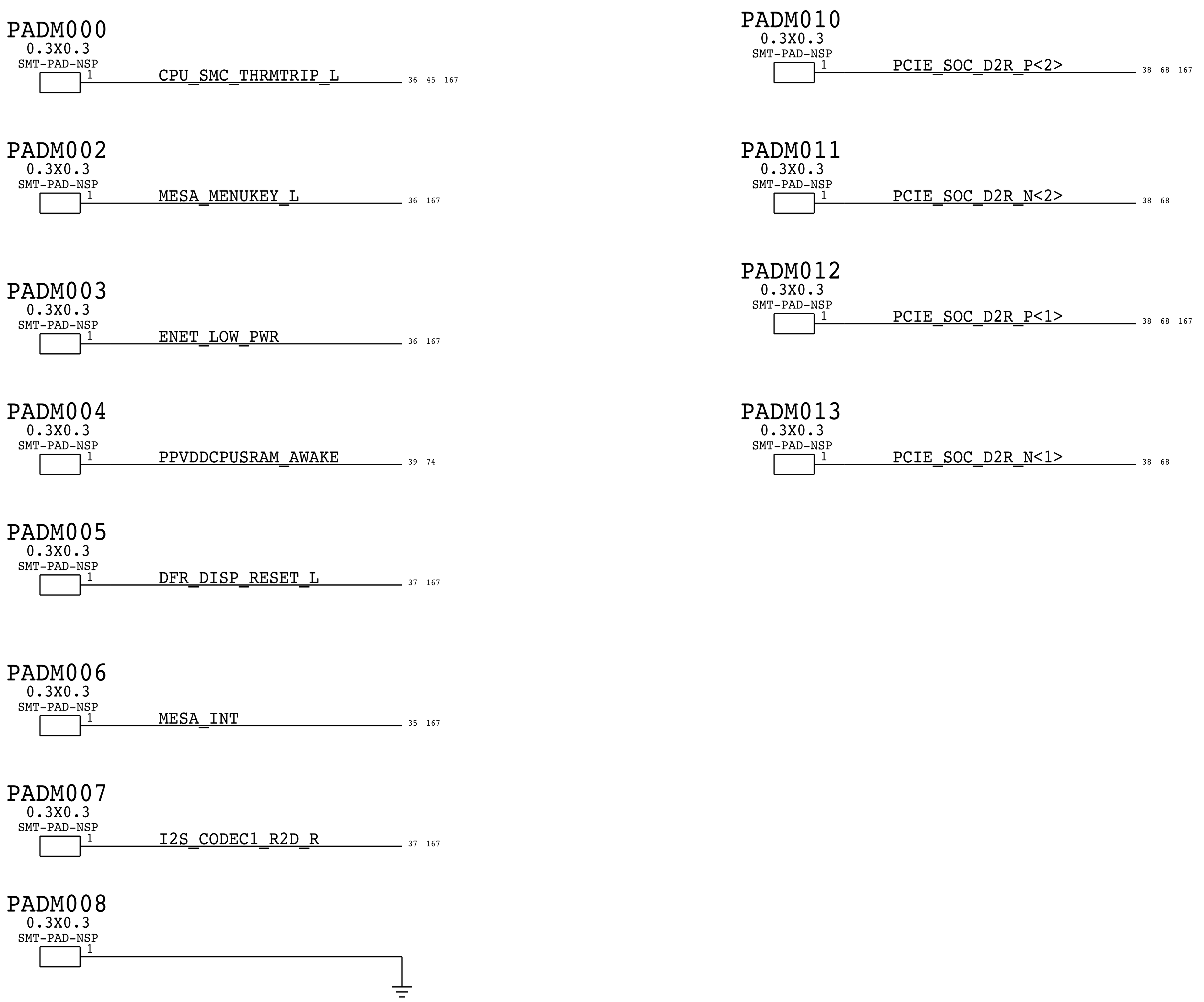
SMALL PADS TO EXPOSE NETS IN SOC AREA WHERE TP WON'T FIT



BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE
TEST PADS FOR SOC AREA

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D

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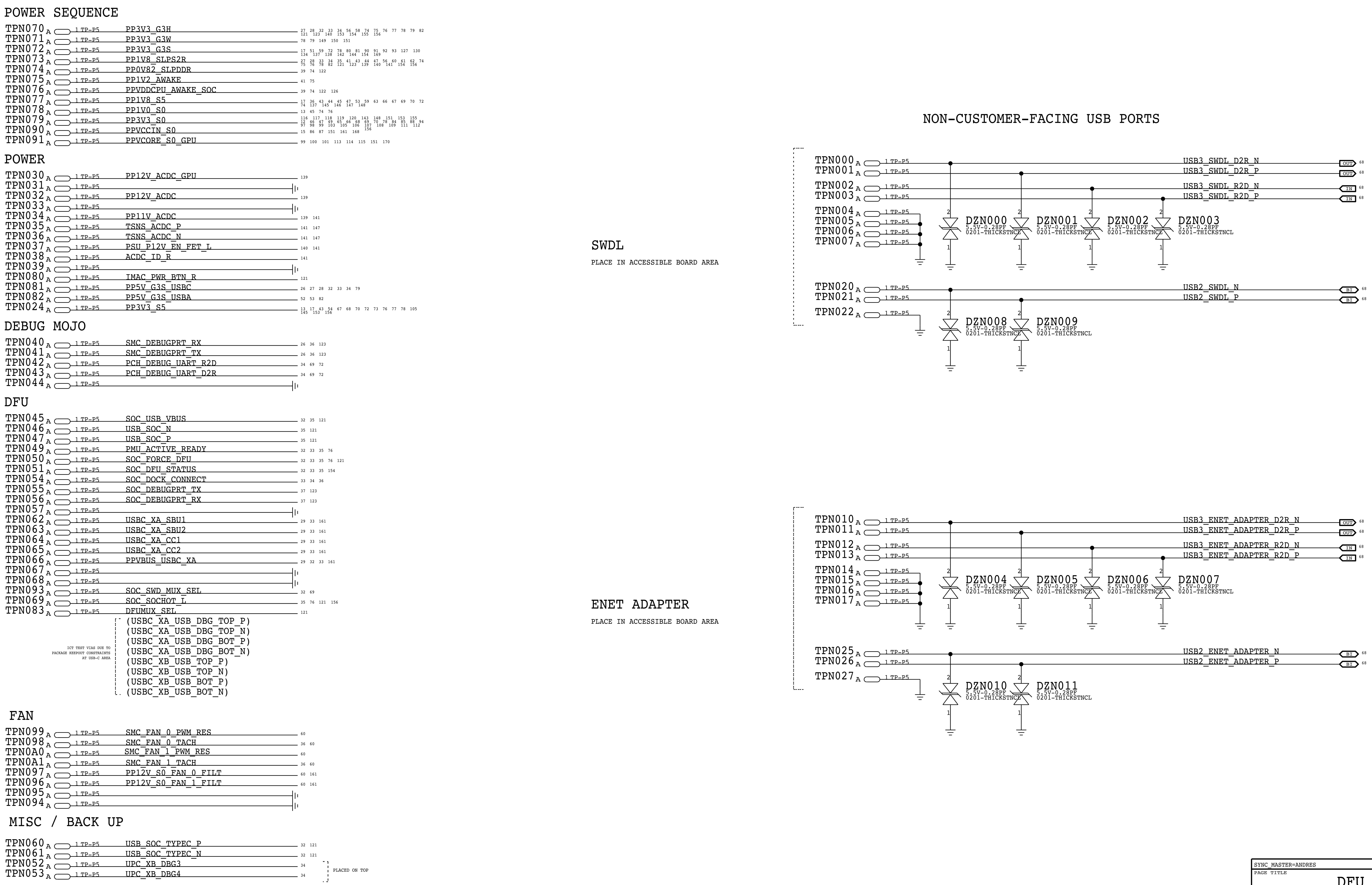
DFU TEST POINTS <RDAR://30758642>



BOM_COST_GROUP=PCB

SYNC_MASTER=ANDRES SYNC_DATE=02/13/2017

PAGE TITLE: DFU TEST POINTS

Apple Inc.

DRAWING NUMBER: 051-01649 SIZE: D

REVISION: 6.21.0

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SVT TEST POINTS <RDAR://30758642>

ITP-XDP

73 13	BI	XDP_CPU_PREQ_L	FUNC_TEST=TRUE
73 13	BI	XDP_CPU_PRDY_L	FUNC_TEST=TRUE
73 66 13	BI	XDP_CPU_TCK	FUNC_TEST=TRUE
73 13	BI	XDP_CPU_TDI	FUNC_TEST=TRUE
73 13	BI	XDP_CPU_TDO	FUNC_TEST=TRUE
73 13	BI	XDP_CPU_TMS	FUNC_TEST=TRUE
73 66	BI	PCH_ITP_PMODE	FUNC_TEST=TRUE
74 73 70 66 45 122	BI	PP1V0_PCH_CORE_S5	FUNC_TEST=TRUE
73 13	BI	XDP_CPU_TRST_L	FUNC_TEST=TRUE
79 73 67 53 43	BI	PM_RSMRST_PCH_L	FUNC_TEST=TRUE
76 73 67	BI	PM_PWRBTN_L	FUNC_TEST=TRUE
73 67 63	BI	SPI_IO<2>	FUNC_TEST=TRUE
73 67 43	BI	PM_PCH_SYS_PWROK	FUNC_TEST=TRUE
73 72 12	BI	XDP_CPU_PWR_DEBUG_L	FUNC_TEST=TRUE
73 67 43	BI	PM_SYSRST_L	FUNC_TEST=TRUE
11	BI	CPU_DEBUG_EN_L	FUNC_TEST=TRUE

BSCAN

73 66	BI	XDP_PCH_TMS	FUNC_TEST=TRUE
73 66	BI	XDP_PCH_TDI	FUNC_TEST=TRUE
73 66	BI	XDP_PCH_TDO	FUNC_TEST=TRUE
73	BI	XDP_JTAG_CPU_ISOL_L	FUNC_TEST=TRUE
72 30 24	BI	JTAG_TBT_TDI	FUNC_TEST=TRUE
72 30 24	BI	JTAG_TBT_TDO	FUNC_TEST=TRUE
72 30 24	BI	JTAG_TBT_TCK	FUNC_TEST=TRUE
72 24	BI	JTAG_TBT_T_TMS	FUNC_TEST=TRUE
72 30	BI	JTAG_TBT_X_TMS	FUNC_TEST=TRUE
121 36	BI	JTAG_SOC_TDO	FUNC_TEST=TRUE
121 36	BI	JTAG_SOC_TDI	FUNC_TEST=TRUE
36	BI	TP_JTAG_SOC_TRST_L	FUNC_TEST=TRUE
94	BI	GPU_JTAG_TCK	FUNC_TEST=TRUE
94	BI	GPU_JTAG_TDI	FUNC_TEST=TRUE
94	BI	GPU_JTAG_TDO	FUNC_TEST=TRUE
94	BI	GPU_JTAG_TMS	FUNC_TEST=TRUE
94	BI	GPU_JTAG_TRST_L	FUNC_TEST=TRUE
161 56 54	BI	ENET_JTAG_TCK	FUNC_TEST=TRUE
161 56 54	BI	ENET_JTAG_TDI	FUNC_TEST=TRUE
161 56 54	BI	ENET_JTAG_TDO	FUNC_TEST=TRUE
161 56 54	BI	ENET_JTAG_TMS	FUNC_TEST=TRUE
161 56 54	BI	ENET_JTAG_TRST_L	FUNC_TEST=TRUE
137 134 36	BI	SWD_WLAN_SWDIO	FUNC_TEST=TRUE
137 134 36	BI	SWD_WLAN_SWCLK	FUNC_TEST=TRUE
137 134 37	BI	WLAN_THROTTLE	FUNC_TEST=TRUE
137 134	BI	WLAN_JTAG_TDO	FUNC_TEST=TRUE
137 134	BI	WLAN_JTAG_TRST_L	FUNC_TEST=TRUE

BSCAN COMPLIANCE

67	BI	PM_SLP_SUS_L	FUNC_TEST=TRUE
76 72 36	BI	PM_SLP_S0_L	FUNC_TEST=TRUE
155 67 65 30 24	BI	PM_SLP_S3_L	FUNC_TEST=TRUE
155 67	BI	PM_SLP_S4_L	FUNC_TEST=TRUE
155 67	BI	PM_SLP_S5_L	FUNC_TEST=TRUE
72 67	BI	PCH_PM_SLP_S0_L	FUNC_TEST=TRUE
156 67 45	BI	PM_THRMTRIP_L	FUNC_TEST=TRUE
155 145 137 67 45	BI	PLT_RESET_L	FUNC_TEST=TRUE
43 36	BI	SMC_PCH_SYS_PWROK	FUNC_TEST=TRUE
43 36	BI	SMC_PCH_PWROK	FUNC_TEST=TRUE
43 36	BI	SMC_SYSRST_L	FUNC_TEST=TRUE
72 67 13	BI	PCH_PM_DOWN	FUNC_TEST=TRUE
67	BI	PCH_SRTCRST_PCH_L	FUNC_TEST=TRUE
67	BI	PCH_PROCPWRGD	FUNC_TEST=TRUE
17 12	BI	VDDQ_AB_PWRGD_BUFFER	FUNC_TEST=TRUE
17 12	BI	VDDQ_CD_PWRGD_BUFFER	FUNC_TEST=TRUE
17	BI	PGOOD_VDDQ_01_R_INV	FUNC_TEST=TRUE
17	BI	PGOOD_VDDQ_23_R_INV	FUNC_TEST=TRUE
67 13	BI	CPU_PWRGD	FUNC_TEST=TRUE
121 76 44 36	BI	PMU_COLD_RESET_L	FUNC_TEST=TRUE
36	BI	SOC_JTAG_SEL	FUNC_TEST=TRUE
121 35	BI	SOC_TESTMODE	FUNC_TEST=TRUE
94	BI	GPU_JTAG_TESTEN	FUNC_TEST=TRUE
137 134	BI	WLAN_JTAG_SEL	FUNC_TEST=TRUE
24	BI	TBT_T_TEST_EN	FUNC_TEST=TRUE
30	BI	TBT_X_TEST_EN	FUNC_TEST=TRUE

VOLTAGE MONITORING

88 76 73	BI	PVCCIO_PGOOD	FUNC_TEST=TRUE
168 89 14	BI	PPVCCSA_S0	FUNC_TEST=TRUE
107 103 76 74 66 65 49 47 46 44	BI	PP1V8_S0	FUNC_TEST=TRUE
168 92 90 22 19 18 17 14	BI	PPVDDQ01_S3	FUNC_TEST=TRUE
168 93 91 67 22 21 20 17 14	BI	PPVDDQ23_S3	FUNC_TEST=TRUE
119 112 99 98 94 45	BI	PP1V8_S0_GPU	FUNC_TEST=TRUE
78 65 57 56 55 54	BI	PP3V3_G3H_ENET_FET	FUNC_TEST=TRUE
168 155 88 73 43 17 14 13 12 11	BI	PPVCCIO_S0	FUNC_TEST=TRUE
137 134 131 129 128 127 78 60 59 51 48 138	BI	PP1V8_G3S	FUNC_TEST=TRUE

BOM_COST_GROUP=PCB

SYNC_MASTER=DEV_MLB SYNC_DATE=11/06/2016

PAGE TITLE

SVT TEST PROPERTIES

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FCT TEST POINTS <RDAR://30758642>

VOLTAGE MONITORING

PP3V3_S0SW_SD	FUNC_TEST=TRUE
PP1V2_SD	FUNC_TEST=TRUE
PP3V3_S0_TBT	FUNC_TEST=TRUE
PP5V_S0	FUNC_TEST=TRUE
PP5V_G3S_USBC_REG	FUNC_TEST=TRUE
PP1V2_ENET	FUNC_TEST=TRUE
PP2V1_ENET	FUNC_TEST=TRUE
PP0V85_ENET	FUNC_TEST=TRUE
PP5V_G3S	FUNC_TEST=TRUE
PPDRV_CPUVR	FUNC_TEST=TRUE
PPDDRVTT01_S0	FUNC_TEST=TRUE
PPDDRVTT23_S0	FUNC_TEST=TRUE
PPVPP01_S3	FUNC_TEST=TRUE
PPVPP23_S3	FUNC_TEST=TRUE
PP12V_LCD_FET	FUNC_TEST=TRUE
PPBKLT_BOOST	FUNC_TEST=TRUE
PPMVDD_S0_GPU	FUNC_TEST=TRUE
PP12V_MAIN	FUNC_TEST=TRUE
PP12V_MAIN_BLC_LCD	FUNC_TEST=TRUE
PP0V9_TBT_T_SVR	FUNC_TEST=TRUE
PP3V3_TBT_T_S0	FUNC_TEST=TRUE
PP3V3_TBT_T_F	FUNC_TEST=TRUE
PP3V3_TBT_T_SX	FUNC_TEST=TRUE
PP0V9_TBT_X_SVR	FUNC_TEST=TRUE
PP0V9_TBT_X_LVR	FUNC_TEST=TRUE
PP3V3_TBT_X_S0	FUNC_TEST=TRUE
PP3V3_TBT_X_SX	FUNC_TEST=TRUE
PP3V3_TBT_X_LC	FUNC_TEST=TRUE
PP1V1_SLPDDR	FUNC_TEST=TRUE
PP0V9_SLPDDR	FUNC_TEST=TRUE
PP1V8_AWAKE	FUNC_TEST=TRUE
PP1V1_SLPS2R	FUNC_TEST=TRUE
PP3V3_AWAKE	FUNC_TEST=TRUE

MIKEY

AUD_CONN_J1_HP_LT	FUNC_TEST=TRUE
AUD_CONN_J1_HP_RT	FUNC_TEST=TRUE
AUD_CONN_J1_HP_SNS_LT	FUNC_TEST=TRUE
AUD_CONN_J1_HP_SNS_RT	FUNC_TEST=TRUE
AUD_CONN_J1_HP_PORT_US_GND	FUNC_TEST=TRUE
AUD_CONN_J1_HP_PORT_CH_GND	FUNC_TEST=TRUE
AUD_CONN_J1_HS_MICP	FUNC_TEST=TRUE
AUD_CONN_J1_CABLE_DETECT	FUNC_TEST=TRUE
AUD_CONN_J1_HS_MICN	FUNC_TEST=TRUE
AUD_CONN_J1_AUD_RING_SENSE	FUNC_TEST=TRUE
AUD_CONN_J1_AUD_TIP_SENSE	FUNC_TEST=TRUE

WIFI AND BT

VDD1P5	FUNC_TEST=TRUE
VIN_RFLDO	FUNC_TEST=TRUE
VDD1P2	FUNC_TEST=TRUE

SD

PP1V8_GL_SDCONN	FUNC_TEST=TRUE
SDCONN_R_D3	FUNC_TEST=TRUE
SDCONN_CMD_R	FUNC_TEST=TRUE
SDCONN_R_CLK	FUNC_TEST=TRUE
SDCONN_R_D2	FUNC_TEST=TRUE
SDCONN_DETECT_R_L	FUNC_TEST=TRUE

POWER ENABLES AND PGOODS

SD_PWR_EN	FUNC_TEST=TRUE
PM_EN_FET_P3V3_S0	FUNC_TEST=TRUE
PM_G3S_EN	FUNC_TEST=TRUE
TBT_PWR_EN	FUNC_TEST=TRUE
PM_PGOOD_0V85_1V2_ENET	FUNC_TEST=TRUE
PM_EN_FET_P5V_S0	FUNC_TEST=TRUE
P3V3G3W_EN	FUNC_TEST=TRUE
PM_EN_REG_P5V_G3S_USBC	FUNC_TEST=TRUE
PM_EN_REG_1V2_ENET_R	FUNC_TEST=TRUE
PM_EN_REG_2V1_ENET_R	FUNC_TEST=TRUE
PM_EN_REG_0V85_ENET_R	FUNC_TEST=TRUE
PM_EN_REG_P3V3_G3H_R	FUNC_TEST=TRUE
PM_EN_REG_P5V_G3S_R	FUNC_TEST=TRUE
PM_S0_EN	FUNC_TEST=TRUE
REG_CPU_VCCIN_EN	FUNC_TEST=TRUE
PM_VCCIO_EN_R	FUNC_TEST=TRUE
REG_CPU_VDDQ23_EN	FUNC_TEST=TRUE
REG_CPU_VDDQ01_EN	FUNC_TEST=TRUE
PM_EN_REG_VPP01_R	FUNC_TEST=TRUE
PM_EN_REG_VPP23_R	FUNC_TEST=TRUE
IOX_LCD_PWR_EN	FUNC_TEST=TRUE
BKLT_BL_EN	FUNC_TEST=TRUE
VR_GPU_ENABLE	FUNC_TEST=TRUE
PM_EN_REG_GPU_MVDD	FUNC_TEST=TRUE
PM_EN_REG_GPU_VDDCI	FUNC_TEST=TRUE
PM_EN_REG_GPU_P0V8	FUNC_TEST=TRUE
PM_EN_REG_GPU_P1V8	FUNC_TEST=TRUE
PM_EN_REG_GPU_MVPP	FUNC_TEST=TRUE
PM_PGOOD_FET_P3V3_S0	FUNC_TEST=TRUE
PM_PGOOD_FET_P3V3G3S	FUNC_TEST=TRUE
PM_PGOOD_FET_P5V_S0	FUNC_TEST=TRUE
P3V3G3W_PGOOD	FUNC_TEST=TRUE
REG_P5VG3S_USBC_PGOOD	FUNC_TEST=TRUE
PM_PGOOD_REG_1V2_ENET	FUNC_TEST=TRUE
PM_PGOOD_REG_2V1_ENET	FUNC_TEST=TRUE
PM_PGOOD_REG_0V85_ENET	FUNC_TEST=TRUE
P3V3G3H_PGOOD	FUNC_TEST=TRUE
P5VG3S_PGOOD	FUNC_TEST=TRUE
PM_DRVCPU_PGOOD	FUNC_TEST=TRUE
CPU_VR_READY	FUNC_TEST=TRUE
TP_PM_PGOOD_REG_CPU_VCCSA	FUNC_TEST=TRUE
PM_PGOOD_VTT01_S0	FUNC_TEST=TRUE
PM_PGOOD_VTT23_S0	FUNC_TEST=TRUE
PM_PGOOD_REG_VPP01	FUNC_TEST=TRUE
PM_PGOOD_REG_VPP23	FUNC_TEST=TRUE
PM_PGOOD_REG_GPUVCORE	FUNC_TEST=TRUE
PM_PGOOD_REG_GPU_MVDD	FUNC_TEST=TRUE
PM_PGOOD_REG_GPU_VDDCI	FUNC_TEST=TRUE
PM_PGOOD_REG_GPU_P0V8	FUNC_TEST=TRUE
PM_PGOOD_REG_GPU_P1V8	FUNC_TEST=TRUE
PM_PGOOD_REG_GPU_MVPP	FUNC_TEST=TRUE
PSU12VPGOOD	FUNC_TEST=TRUE
PSU_P12V_EN_L	FUNC_TEST=TRUE

FAN

FAN_1_PWM_FILT	FUNC_TEST=TRUE
FAN_1_TACH_FILT	FUNC_TEST=TRUE
PP12V_S0_FAN_1_FILT	FUNC_TEST=TRUE
FAN_0_PWM_FILT	FUNC_TEST=TRUE
FAN_0_TACH_FILT	FUNC_TEST=TRUE
PP12V_S0_FAN_0_FILT	FUNC_TEST=TRUE

USB-A

PP5V_S4_EXTA_F	FUNC_TEST=TRUE
PP5V_S4_EXTB_F	FUNC_TEST=TRUE
PP5V_S4_EXTC_F	FUNC_TEST=TRUE
PP5V_S4_EXTD_F	FUNC_TEST=TRUE

SENSE RESISTOR MEASUREMENTS

GND_AUDIO_DMIC_1	FUNC_TEST=TRUE
PDM_DMIC_DATA0_CONN	FUNC_TEST=TRUE
PDM_DMIC_CLK0_CONN	FUNC_TEST=TRUE
AUD_SPKR_LWFR_OUT_N	FUNC_TEST=TRUE
AUD_SPKR_LWFR_OUT_P	FUNC_TEST=TRUE
SPKR_ID0	FUNC_TEST=TRUE
AUD_SPKR_LTWT_OUT_N	FUNC_TEST=TRUE
AUD_SPKR_LTWT_OUT_P	FUNC_TEST=TRUE
AUD_SPKR_RWFR_OUT_N	FUNC_TEST=TRUE
AUD_SPKR_RWFR_OUT_P	FUNC_TEST=TRUE
SPKR_ID1	FUNC_TEST=TRUE
AUD_SPKR_RTWT_OUT_N	FUNC_TEST=TRUE
AUD_SPKR_RTWT_OUT_P	FUNC_TEST=TRUE

USB-C

PPVBUS_USBC_TA	FUNC_TEST=TRUE
PPVBUS_USBC_TB	FUNC_TEST=TRUE
PPVBUS_USBC_XA	FUNC_TEST=TRUE
USBC_XA_SBU1	FUNC_TEST=TRUE
USBC_XA_CC1	FUNC_TEST=TRUE
USBC_XA_CC2	FUNC_TEST=TRUE
USBC_XA_SBU2	FUNC_TEST=TRUE
USBC_XA_USB_DBG_BOT_P	FUNC_TEST=TRUE
USBC_XA_USB_DBG_BOT_N	FUNC_TEST=TRUE
USBC_XA_USB_DBG_TOP_P	FUNC_TEST=TRUE
USBC_XA_USB_DBG_TOP_N	FUNC_TEST=TRUE
USBC_XB_USB_BOT_P	FUNC_TEST=TRUE
USBC_XB_USB_BOT_N	FUNC_TEST=TRUE
USBC_XB_USB_TOP_P	FUNC_TEST=TRUE
USBC_XB_USB_TOP_N	FUNC_TEST=TRUE
PPVBUS_USBC_XB	FUNC_TEST=TRUE

MAC-1 DISPLAY

DP_INTPNL_ML_HPD	FUNC_TEST=TRUE
DP_INTPNL_SL_HPD	FUNC_TEST=TRUE

ADCS

SMC_P12V_MAIN_ISENSE	FUNC_TEST=TRUE
SMC_P12V_GPU_ISENSE	FUNC_TEST=TRUE
REG_CPU_VCCIN_IMON	FUNC_TEST=TRUE
VR_GPU_CORE_IMON	FUNC_TEST=TRUE
HS_MVDD_ISENSE_OUT	FUNC_TEST=TRUE
HS_P12V_GPU_ISENSE_OUT	FUNC_TEST=TRUE
HS_P12V_AUX_GPU_IOUT	FUNC_TEST=TRUE
LS_VDDQ01_S3_CPU_OUT	FUNC_TEST=TRUE
PPVCCIN_S0	FUNC_TEST=TRUE
LS_VPP01_S3_CPU_OUT	FUNC_TEST=TRUE
LS_VPP23_S3_CPU_OUT	FUNC_TEST=TRUE
PMU_GPU_MVDD_S0_ISENSE	FUNC_TEST=TRUE

DISPLAY

I2C_DP_TCON_SLA_3V3_SDA	FUNC_TEST=TRUE
I2C_DP_TCON_SLA_3V3_SCL	FUNC_TEST=TRUE
DP_INTPNL_ML_HPD	FUNC_TEST=TRUE
DP_INTPNL_SL_HPD	FUNC_TEST=TRUE
DP_INT_PIN_54	FUNC_TEST=TRUE
BLC_VSYNC	FUNC_TEST=TRUE
BLC_LSYNC	FUNC_TEST=TRUE
BLC_EN	FUNC_TEST=TRUE

BACKLIGHT TEST

LED_RETURN_9	FUNC_TEST=TRUE
LED_RETURN_7	FUNC_TEST=TRUE
LED_RETURN_5	FUNC_TEST=TRUE
LED_RETURN_3	FUNC_TEST=TRUE
LED_RETURN_1	FUNC_TEST=TRUE
LED_RETURN_2	FUNC_TEST=TRUE
LED_RETURN_4	FUNC_TEST=TRUE
LED_RETURN_6	FUNC_TEST=TRUE
LED_RETURN_8	FUNC_TEST=TRUE
BLC_LSYNC	FUNC_TEST=TRUE
PP12V_BKLT_FUSED	FUNC_TEST=TRUE
PP12V_BKLT_SNS	FUNC_TEST=TRUE

CAMERA

I2C_FTCAM_SCL_CONN	FUNC_TEST=TRUE
I2C_FTCAM_SDA_CONN	FUNC_TEST=TRUE
PP1V8_G3S_DMIC_F	FUNC_TEST=TRUE

ALS

I2C_ALS_SDA_CONN	FUNC_TEST=TRUE
I2C_ALS_SCL_CONN	FUNC_TEST=TRUE

ETHERNET

ENET_JTAG_TRST_L	FUNC_TEST=TRUE
ENET_JTAG_TDI	FUNC_TEST=TRUE
ENET_JTAG_TCK	FUNC_TEST=TRUE
ENET_JTAG_TMS	FUNC_TEST=TRUE
ENET_JTAG_TDO	FUNC_TEST=TRUE

BOM_COST_GROUP=PCB

ICT NO TEST <RDAR://30758642>

MEMORY

MEM_0_DQ<63..0>	NO_TEST=1
MEM_0_CS_L<3..0>	NO_TEST=1
MEM_0_CLK_P<1..0>	NO_TEST=1
MEM_0_CLK_N<1..0>	NO_TEST=1
MEM_0_DQS_P<17..0>	NO_TEST=1
MEM_0_DQS_N<17..0>	NO_TEST=1
MEM_0_MA<17..0>	NO_TEST=1
MEM_0_ECC<7..0>	NO_TEST=1
MEM_0_CKE<1..0>	NO_TEST=1
MEM_0_BA<1..0>	NO_TEST=1
MEM_0_BG<1..0>	NO_TEST=1
MEM_0_ODT<1..0>	NO_TEST=1
MEM_0_PARITY	NO_TEST=1
MEM_0_ACT_L	NO_TEST=1
MEM_0_ALERT_L	NO_TEST=1
MEM_1_DQ<63..0>	NO_TEST=1
MEM_1_CS_L<3..0>	NO_TEST=1
MEM_1_CLK_P<1..0>	NO_TEST=1
MEM_1_CLK_N<1..0>	NO_TEST=1
MEM_1_DQS_P<17..0>	NO_TEST=1
MEM_1_DQS_N<17..0>	NO_TEST=1
MEM_1_MA<17..0>	NO_TEST=1
MEM_1_ECC<7..0>	NO_TEST=1
MEM_1_CKE<1..0>	NO_TEST=1
MEM_1_BA<1..0>	NO_TEST=1
MEM_1_BG<1..0>	NO_TEST=1
MEM_1_ODT<1..0>	NO_TEST=1
MEM_1_PARITY	NO_TEST=1
MEM_1_ACT_L	NO_TEST=1
MEM_1_ALERT_L	NO_TEST=1
MEM_2_DQ<63..0>	NO_TEST=1
MEM_2_CS_L<3..0>	NO_TEST=1
MEM_2_CLK_P<1..0>	NO_TEST=1
MEM_2_CLK_N<1..0>	NO_TEST=1
MEM_2_DQS_P<17..0>	NO_TEST=1
MEM_2_DQS_N<17..0>	NO_TEST=1
MEM_2_MA<17..0>	NO_TEST=1
MEM_2_ECC<7..0>	NO_TEST=1
MEM_2_CKE<1..0>	NO_TEST=1
MEM_2_BA<1..0>	NO_TEST=1
MEM_2_BG<1..0>	NO_TEST=1
MEM_2_ODT<1..0>	NO_TEST=1
MEM_2_PARITY	NO_TEST=1
MEM_2_ACT_L	NO_TEST=1
MEM_2_ALERT_L	NO_TEST=1
MEM_3_DQ<63..0>	NO_TEST=1
MEM_3_CS_L<3..0>	NO_TEST=1
MEM_3_CLK_P<1..0>	NO_TEST=1
MEM_3_CLK_N<1..0>	NO_TEST=1
MEM_3_DQS_P<17..0>	NO_TEST=1
MEM_3_DQS_N<17..0>	NO_TEST=1
MEM_3_MA<17..0>	NO_TEST=1
MEM_3_ECC<7..0>	NO_TEST=1
MEM_3_CKE<1..0>	NO_TEST=1
MEM_3_BA<1..0>	NO_TEST=1
MEM_3_BG<1..0>	NO_TEST=1
MEM_3_ODT<1..0>	NO_TEST=1
MEM_3_PARITY	NO_TEST=1
MEM_3_ACT_L	NO_TEST=1
MEM_3_ALERT_L	NO_TEST=1

PCIE

PCIE_ENET_D2R_P<3..0>	NO_TEST=1
PCIE_ENET_D2R_N<3..0>	NO_TEST=1
PCIE_ENET_R2D_P<3..0>	NO_TEST=1
PCIE_ENET_R2D_N<3..0>	NO_TEST=1
PCIE_ENET_D2R_C_P<3..0>	NO_TEST=1
PCIE_ENET_D2R_C_N<3..0>	NO_TEST=1
PCIE_ENET_R2D_C_P<3..0>	NO_TEST=1
PCIE_ENET_R2D_C_N<3..0>	NO_TEST=1
PCIE_TBT_T_D2R_P<3..0>	NO_TEST=1
PCIE_TBT_T_D2R_N<3..0>	NO_TEST=1
PCIE_TBT_T_R2D_P<3..0>	NO_TEST=1
PCIE_TBT_T_R2D_N<3..0>	NO_TEST=1
PCIE_TBT_T_D2R_C_P<3..0>	NO_TEST=1
PCIE_TBT_T_D2R_C_N<3..0>	NO_TEST=1
PCIE_TBT_T_R2D_C_P<3..0>	NO_TEST=1
PCIE_TBT_T_R2D_C_N<3..0>	NO_TEST=1
PCIE_TBT_X_D2R_P<3..0>	NO_TEST=1
PCIE_TBT_X_D2R_N<3..0>	NO_TEST=1
PCIE_TBT_X_R2D_P<3..0>	NO_TEST=1
PCIE_TBT_X_R2D_N<3..0>	NO_TEST=1
PCIE_TBT_X_D2R_C_P<3..0>	NO_TEST=1
PCIE_TBT_X_D2R_C_N<3..0>	NO_TEST=1
PCIE_TBT_X_R2D_C_P<3..0>	NO_TEST=1
PCIE_TBT_X_R2D_C_N<3..0>	NO_TEST=1
PE_GFX_TO_CPU_P<15..0>	NO_TEST=1
PE_GFX_TO_CPU_N<15..0>	NO_TEST=1
PE_CPU_TO_GFX_P<15..0>	NO_TEST=1
PE_CPU_TO_GFX_N<15..0>	NO_TEST=1
PE_GFX_TO_CPU_C_P<15..0>	NO_TEST=1
PE_GFX_TO_CPU_C_N<15..0>	NO_TEST=1
PE_CPU_TO_GFX_C_P<15..0>	NO_TEST=1
PE_CPU_TO_GFX_C_N<15..0>	NO_TEST=1
DMI_SB_TO_CPU_P<3..0>	NO_TEST=1
DMI_SB_TO_CPU_N<3..0>	NO_TEST=1
DMI_CPU_TO_SB_P<3..0>	NO_TEST=1
DMI_CPU_TO_SB_N<3..0>	NO_TEST=1
DMI_SB_TO_CPU_C_P<3..0>	NO_TEST=1
DMI_SB_TO_CPU_C_N<3..0>	NO_TEST=1
DMI_CPU_TO_SB_C_P<3..0>	NO_TEST=1
DMI_CPU_TO_SB_C_N<3..0>	NO_TEST=1
PCIE_SOC_R2D_P<3..0>	NO_TEST=1
PCIE_SOC_R2D_N<3..0>	NO_TEST=1
PCIE_SOC_D2R_C_P<3..0>	NO_TEST=1
PCIE_SOC_D2R_C_N<3..0>	NO_TEST=1
PCIE_CLK100M_SOC_P	NO_TEST=1
PCIE_CLK100M_SOC_N	NO_TEST=1
PCIE_SSD0_R2D_P<0>	NO_TEST=1
PCIE_SSD0_R2D_N<0>	NO_TEST=1
PCIE_SSD0_R2D_C_P<0>	NO_TEST=1
PCIE_SSD0_R2D_C_N<0>	NO_TEST=1
PCIE_SSD0_D2R_P<0>	NO_TEST=1
PCIE_SSD0_D2R_N<0>	NO_TEST=1
PCIE_CLK100M_SSD0_01_P	NO_TEST=1
PCIE_CLK100M_SSD0_01_N	NO_TEST=1
PCIE_CLK100M_SSD0_23_P	NO_TEST=1
PCIE_CLK100M_SSD0_23_N	NO_TEST=1
PCIE_SSD1_R2D_P<0>	NO_TEST=1
PCIE_SSD1_R2D_N<0>	NO_TEST=1
PCIE_SSD1_R2D_C_P<0>	NO_TEST=1
PCIE_SSD1_R2D_C_N<0>	NO_TEST=1
PCIE_SSD1_D2R_P<0>	NO_TEST=1
PCIE_SSD1_D2R_N<0>	NO_TEST=1
PCIE_CLK100M_SSD1_01_P	NO_TEST=1
PCIE_CLK100M_SSD1_01_N	NO_TEST=1
PCIE_CLK100M_SSD1_23_P	NO_TEST=1
PCIE_CLK100M_SSD1_23_N	NO_TEST=1

DISPLAY PORT

DP_X_SNK0_ML_C_P<3..0>	NO_TEST=1
DP_X_SNK0_ML_C_N<3..0>	NO_TEST=1
DP_X_SNK0_AUXCH_C_P	NO_TEST=1
DP_X_SNK0_AUXCH_C_N	NO_TEST=1
DP_X_SNK0_ML_P<3..0>	NO_TEST=1
DP_X_SNK0_ML_N<3..0>	NO_TEST=1
DP_X_SNK0_AUXCH_P	NO_TEST=1
DP_X_SNK0_AUXCH_N	NO_TEST=1
DP_X_SNK1_ML_C_P<3..0>	NO_TEST=1
DP_X_SNK1_ML_C_N<3..0>	NO_TEST=1
DP_X_SNK1_AUXCH_C_P	NO_TEST=1
DP_X_SNK1_AUXCH_C_N	NO_TEST=1
DP_X_SNK1_ML_P<3..0>	NO_TEST=1
DP_X_SNK1_ML_N<3..0>	NO_TEST=1
DP_X_SNK1_AUXCH_P	NO_TEST=1
DP_X_SNK1_AUXCH_N	NO_TEST=1
DP_T_SNK0_ML_C_P<3..0>	NO_TEST=1
DP_T_SNK0_ML_C_N<3..0>	NO_TEST=1
DP_T_SNK0_AUXCH_C_P	NO_TEST=1
DP_T_SNK0_AUXCH_C_N	NO_TEST=1
DP_T_SNK0_ML_P<3..0>	NO_TEST=1
DP_T_SNK0_ML_N<3..0>	NO_TEST=1
DP_T_SNK0_AUXCH_P	NO_TEST=1
DP_T_SNK0_AUXCH_N	NO_TEST=1
DP_T_SNK1_ML_C_P<3..0>	NO_TEST=1
DP_T_SNK1_ML_C_N<3..0>	NO_TEST=1
DP_T_SNK1_AUXCH_C_P	NO_TEST=1
DP_T_SNK1_AUXCH_C_N	NO_TEST=1
DP_T_SNK1_ML_P<3..0>	NO_TEST=1
DP_T_SNK1_ML_N<3..0>	NO_TEST=1
DP_T_SNK1_AUXCH_P	NO_TEST=1
DP_T_SNK1_AUXCH_N	NO_TEST=1
DP_INTPNL_ML_C_P<3..0>	NO_TEST=1
DP_INTPNL_ML_C_N<3..0>	NO_TEST=1
DP_INTPNL_ML_AUX_C_P	NO_TEST=1
DP_INTPNL_ML_AUX_C_N	NO_TEST=1
DP_INTPNL_ML_P<3..0>	NO_TEST=1
DP_INTPNL_ML_N<3..0>	NO_TEST=1
DP_INTPNL_ML_AUX_P	NO_TEST=1
DP_INTPNL_ML_AUX_N	NO_TEST=1
DP_INTPNL_SL_C_P<3..0>	NO_TEST=1
DP_INTPNL_SL_C_N<3..0>	NO_TEST=1
DP_INTPNL_SL_AUX_C_P	NO_TEST=1
DP_INTPNL_SL_AUX_C_N	NO_TEST=1
DP_INTPNL_SL_P<3..0>	NO_TEST=1
DP_INTPNL_SL_N<3..0>	NO_TEST=1
DP_INTPNL_SL_AUX_P	NO_TEST=1
DP_INTPNL_SL_AUX_N	NO_TEST=1

ETHERNET

ENET_CONN_A_P	NO_TEST=1
ENET_CONN_A_N	NO_TEST=1
ENET_CONN_B_P	NO_TEST=1
ENET_CONN_B_N	NO_TEST=1
ENET_CONN_C_P	NO_TEST=1
ENET_CONN_C_N	NO_TEST=1
ENET_CONN_D_P	NO_TEST=1
ENET_CONN_D_N	NO_TEST=1
ENET_A_P	NO_TEST=1
ENET_A_N	NO_TEST=1
ENET_B_P	NO_TEST=1
ENET_B_N	NO_TEST=1
ENET_C_P	NO_TEST=1
ENET_C_N	NO_TEST=1
ENET_D_P	NO_TEST=1
ENET_D_N	NO_TEST=1

MIPI

NC_MIPI_PLCAM_DATA_P<0>	NO_TEST=1
NC_MIPI_PLCAM_DATA_N<0>	NO_TEST=1
NC_MIPI_PLCAM_DATA_P<1>	NO_TEST=1
NC_MIPI_PLCAM_DATA_N<1>	NO_TEST=1
NC_MIPI_PLCAM_CLK_P	NO_TEST=1
NC_MIPI_PLCAM_CLK_N	NO_TEST=1
NC_MIPI_DFR_DATA_P	NO_TEST=1
NC_MIPI_DFR_DATA_N	NO_TEST=1
NC_MIPI_DFR_CLK_P	NO_TEST=1
NC_MIPI_DFR_CLK_N	NO_TEST=1

I2C

I2C_SNS0_S0_SCL	NO_TEST=1

USB-A 3.0

USB3_EXTD_RX_F_P	NO_TEST=1
USB3_EXTD_RX_F_N	NO_TEST=1
USB3_EXTD_RX_P	NO_TEST=1
USB3_EXTD_RX_N	NO_TEST=1
USB3_EXTD_TX_P	NO_TEST=1
USB3_EXTD_TX_N	NO_TEST=1
USB3_EXTD_TX_C_P	NO_TEST=1
USB3_EXTD_TX_C_N	NO_TEST=1
USB3_EXTD_TX_F_P	NO_TEST=1
USB3_EXTD_TX_F_N	NO_TEST=1
USB3_EXTC_RX_F_P	NO_TEST=1
USB3_EXTC_RX_F_N	NO_TEST=1
USB3_EXTC_RX_P	NO_TEST=1
USB3_EXTC_RX_N	NO_TEST=1
USB3_EXTC_TX_P	NO_TEST=1
USB3_EXTC_TX_N	NO_TEST=1
USB3_EXTC_TX_C_P	NO_TEST=1
USB3_EXTC_TX_C_N	NO_TEST=1
USB3_EXTC_TX_F_P	NO_TEST=1
USB3_EXTC_TX_F_N	NO_TEST=1
USB3_EXTB_RX_F_P	NO_TEST=1
USB3_EXTB_RX_F_N	NO_TEST=1
USB3_EXTB_RX_P	NO_TEST=1
USB3_EXTB_RX_N	NO_TEST=1
USB3_EXTB_TX_P	NO_TEST=1
USB3_EXTB_TX_N	NO_TEST=1
USB3_EXTB_TX_C_P	NO_TEST=1
USB3_EXTB_TX_C_N	NO_TEST=1
USB3_EXTB_TX_F_P	NO_TEST=1
USB3_EXTB_TX_F_N	NO_TEST=1
USB3_EXTA_RX_F_P	NO_TEST=1
USB3_EXTA_RX_F_N	NO_TEST=1
USB3_EXTA_RX_P	NO_TEST=1
USB3_EXTA_RX_N	NO_TEST=1
USB3_EXTA_TX_P	NO_TEST=1
USB3_EXTA_TX_N	NO_TEST=1
USB3_EXTA_TX_C_P	NO_TEST=1
USB3_EXTA_TX_C_N	NO_TEST=1
USB3_EXTA_TX_F_P	NO_TEST=1
USB3_EXTA_TX_F_N	NO_TEST=1

BOM_COST_GROUP=PCB

ICT NO TEST <RDAR://30758642>

GRAPHICS VOLTAGE REGULATORS

(each net: NO_TEST=1)

REG_BOOST_GPUCORE_1
REG_BOOST_GPUCORE_1_RC
REG_BOOST_GPUCORE_2
REG_BOOST_GPUCORE_2_RC
REG_BOOST_GPUCORE_3
REG_BOOST_GPUCORE_3_RC
REG_BOOST_GPUCORE_4
REG_BOOST_GPUCORE_4_RC
REG_BOOST_GPUCORE_5
REG_BOOST_GPUCORE_5_RC
REG_BOOST_GPUCORE_6
REG_BOOST_GPUCORE_6_RC
REG_BOOST_GPUCORE_7
REG_BOOST_GPUCORE_7_RC
REG_BOOST_GPUCORE_8
REG_BOOST_GPUCORE_8_RC
VR_PHASE_GPUCORE_1
VR_PHASE_GPUCORE_2
VR_PHASE_GPUCORE_3
VR_PHASE_GPUCORE_4
VR_PHASE_GPUCORE_5
VR_PHASE_GPUCORE_6
VR_PHASE_GPUCORE_7
VR_PHASE_GPUCORE_8
REGLGP0V8
REGSNUBPVDDR
REGSNUBPVPP
REG_BOOT_MVDD
REG_BOOT_MVPP
REG_BOOT_P0V8
REG_BOOT_P0V8_RC
REG_COMP_MVPP
REG_COMP_PVDDR
REG_GPUCORE_ISNS_1_N
REG_GPUCORE_ISNS_1_P
REG_GPUCORE_ISNS_1_R_N
REG_GPUCORE_ISNS_1_R_P
REG_GPUCORE_ISNS_2_N
REG_GPUCORE_ISNS_2_P
REG_GPUCORE_ISNS_2_R_N
REG_GPUCORE_ISNS_2_R_P
REG_GPUCORE_ISNS_3_N
REG_GPUCORE_ISNS_3_P
REG_GPUCORE_ISNS_3_R_N
REG_GPUCORE_ISNS_3_R_P
REG_GPUCORE_ISNS_4_N
REG_GPUCORE_ISNS_4_P
REG_GPUCORE_ISNS_4_R_N
REG_GPUCORE_ISNS_4_R_P
REG_GPUCORE_ISNS_5_N
REG_GPUCORE_ISNS_5_P
REG_GPUCORE_ISNS_5_R_N
REG_GPUCORE_ISNS_5_R_P
REG_GPUCORE_ISNS_6_N
REG_GPUCORE_ISNS_6_P
REG_GPUCORE_ISNS_6_R_N
REG_GPUCORE_ISNS_6_R_P
REG_GPUCORE_ISNS_7_N
REG_GPUCORE_ISNS_7_P
REG_GPUCORE_ISNS_7_R_N
REG_GPUCORE_ISNS_7_R_P
REG_GPUCORE_ISNS_8_N
REG_GPUCORE_ISNS_8_P
REG_GPUCORE_ISNS_8_R_N
REG_GPUCORE_ISNS_8_R_P
REG_LGATE_MVDD
REG_LGATE_P0V8
REG_MVDD_FB
REG_MVDD_FB_R
REG_MVDD_FSEL
REG_MVDD_OCSET
REG_MVDD_RTN
REG_MVDD_RTN_R
REG_MVDD_SET0
REG_MVDD_SET1
REG_MVDD_SET1_R
REG_MVDD_SREF

GRAPHICS VOLTAGE REGULATORS

(each net: NO_TEST=1)

REG_P0V8_FB
REG_P0V8_FB_R
REG_P0V8_FSEL
REG_P0V8_OCSET
REG_P0V8_RTN
REG_P0V8_RTN_R
REG_P0V8_SET0
REG_P0V8_SET1
REG_P0V8_SREF
REG_P0V8_VO
REG_PHASE_MVDD
REG_PHASE_MVPP
REG_PHASE_P0V8
REG_PHASE_SNS_P0V8
REG_PWM_GPUCORE_1
REG_PWM_GPUCORE_2
REG_PWM_GPUCORE_3
REG_PWM_GPUCORE_4
REG_PWM_GPUCORE_5
REG_PWM_GPUCORE_6
REG_PWM_GPUCORE_7
REG_PWM_GPUCORE_8
REG_RTCLK_MVPP
REG_SNUBBER_GPUCORE_1
REG_SNUBBER_GPUCORE_2
REG_SNUBBER_GPUCORE_3
REG_SNUBBER_GPUCORE_4
REG_SNUBBER_GPUCORE_5
REG_SNUBBER_GPUCORE_6
REG_SNUBBER_GPUCORE_7
REG_SNUBBER_GPUCORE_8
REG_SNUBBER_MVDD
REG_SNUBBER_P0V8
REG_SSTR_MVPP
REG_UGATE_MVDD
REG_VSNSR_MVPP
REG_VSNS_MVPP
REG_COMP_MVPP
RGCMVPP
VR_PHASE_GPUCORE_1
VR_PHASE_GPUCORE_2
VR_PHASE_GPUCORE_3
VR_PHASE_GPUCORE_4
VR_PHASE_GPUCORE_5
VR_PHASE_GPUCORE_6
VR_PHASE_GPUCORE_7
VR_PHASE_GPUCORE_8
VR_PHASE_MVDD

ETHERNET VOLTAGE REGULATORS

(each net: NO_TEST=1)

P3V3ENET_SS
REG_0V85_ENET_FB
REG_0V85_ENET_OCSET
REG_0V85_ENET_RTN
REG_0V85_ENET_FB
REG_0V85_ENET_FB_N
REG_0V85_ENET_FB_P
REG_0V85_ENET_FSEL
REG_0V85_ENET_OCSET
REG_0V85_ENET_RTN
REG_0V85_ENET_SENSE_N
REG_0V85_ENET_SENSE_P
REG_0V85_ENET_SREF
REG_BOOT_0V85_ENET
REG_BOOT_0V85_ENET_RC
REG_LGATE_0V85_ENET
REG_PHASE_0V85_ENET
REG_PHASE_0V85_ENET_TGR
REG_PVCC_U8500
REG_SNUBBER_0V85_ENET
REG_UGATE_0V85_ENET
REG_UGATE_0V85_ENET_R
REG_VCC_U8500
REG_FB_1V2_ENET
REG_FB_1V2_ENET_R
REG_FB_2V1_ENET
REG_FB_2V1_ENET_R

CPU VOLTAGE REGULATORS

(each net: NO_TEST=1)

REG_BOOST_CPUCORE_1
REG_BOOST_CPUCORE_1_RC
REG_BOOST_CPUCORE_2
REG_BOOST_CPUCORE_2_RC
REG_BOOST_CPUCORE_3
REG_BOOST_CPUCORE_3_RC
REG_BOOST_CPUCORE_4
REG_BOOST_CPUCORE_4_RC
REG_BOOST_CPUCORE_5
REG_BOOST_CPUCORE_5_RC
REGSNUBPVDDR
REGSNUBPVPP
REG_6VR_FSW
REG_6VR_SS
CPU_ISNS_VCCIO_S0_N
CPU_ISNS_VCCIO_S0_P
CPU_ISNS_VCCSA_S0_N
CPU_ISNS_VCCSA_S0_P
CPU_ISNS_VDDQ01_S3_N
CPU_ISNS_VDDQ01_S3_P
CPU_ISNS_VDDQ23_S3_N
CPU_ISNS_VDDQ23_S3_P
CPU_ISNS_VPP01_S3_N
CPU_ISNS_VPP01_S3_P
CPU_ISNS_VPP23_S3_N
CPU_ISNS_VPP23_S3_P
REG_6VR_FSW
REG_6VR_SS
REG_BOOST_VCCIO
REG_BOOT_PVDDR
REG_BOOT_VDDCI
REG_BOOT_VDDCI_RC
REG_BOOT_VPP01
REG_BOOT_VPP23
REG_BST_CPU_VDDQ01
REG_BST_CPU_VDDQ23
REG_BST_VCCSA
REG_CPU_VDDQ01_SW
REG_CPU_VDDQ23_SW
REG_CPU_VINSNS
REG_DRVCPU_VSNS
REG_DRVCPU_VSNS_R
REG_DRV_VR_AGND
REG_FSEL_VPP23
REG_ISNS_CPU_VDDQ01_N
REG_ISNS_CPU_VDDQ01_P
REG_ISNS_CPU_VDDQ23_N
REG_ISNS_CPU_VDDQ23_P
REG_ISNS_REF_VCCIO
REG_ISNS_VCCIO
REG_LGATE_VDDCI
REG_LG_VPP01
REG_LG_VPP23
REG_PHASE_CPUCORE_1
REG_PHASE_CPUCORE_2
REG_PHASE_CPUCORE_3
REG_PHASE_CPUCORE_4
REG_PHASE_CPUCORE_5
REG_PHASE_PVDDR
REG_PHASE_VDDCI
REG_PHASE_VDDCI_L
REG_PHASE_VPP01
REG_PHASE_VPP01_L
REG_PHASE_VPP23
REG_PHASE_VPP23_L
REG_PWM_CPUCORE_1
REG_PWM_CPUCORE_2
REG_PWM_CPUCORE_3
REG_PWM_CPUCORE_4
REG_PWM_CPUCORE_5
REG_PWM_CPUVCCSA
REG_PWM_CPU_VDDQ01
REG_PWM_CPU_VDDQ23
REG_PWM_VCCIO
REG_RTCLK_PVDDR
REG_SNUBBER_CPUCORE_1
REG_SNUBBER_CPUCORE_2
REG_SNUBBER_CPUCORE_3
REG_SNUBBER_CPUCORE_4
REG_SNUBBER_CPUCORE_5
REG_SNUBBER_VDDCI
REG_SNUB_VPP01_RC
REG_SNUB_VPP23_RC
REG_SSTR_PVDDR

CPU VOLTAGE REGULATORS

(each net: NO_TEST=1)

REG_UGATE_VDDCI
REG_UGATE_VDDCI_R
REG_UG_VPP01
REG_UG_VPP01_R
REG_UG_VPP23
REG_UG_VPP23_R
REG_VCCIN_ISNS_1_N
REG_VCCIN_ISNS_1_P
REG_VCCIN_ISNS_2_N
REG_VCCIN_ISNS_2_P
REG_VCCIN_ISNS_3_N
REG_VCCIN_ISNS_3_P
REG_VCCIN_ISNS_4_N
REG_VCCIN_ISNS_4_P
REG_VCCIN_ISNS_5_N
REG_VCCIN_ISNS_5_P
REG_VCCIN_ISNS_A
REG_VCCIO_SW
REG_VCCSA_PH
REG_VCCSA_PH
REG_VCCSA_SNUB
REG_VCCSA_TSNS_B
REG_VDDCI_FB
REG_VDDCI_FB_P
REG_VDDCI_FSEL
REG_VDDCI_OCSET
REG_VDDCI_RTN
REG_VDDCI_SET0
REG_VDDCI_SET1
REG_VDDCI_SREF
REG_VIN_SNS_CPU_VDDQ01
REG_VIN_SNS_VCCIO
REG_VIN_SNS_VDDQ23
REG_VIN_SNS_VDDQ23
REG_VPP01
REG_VPP01_FB
REG_VPP01_FB_XW_N
REG_VPP01_FB_XW_P
REG_VPP01_OCSET
REG_VPP01_RTN
REG_VPP01_SREF
REG_VPP01_VO
REG_VPP23_FB
REG_VPP23_FB_XW_N
REG_VPP23_FB_XW_P
REG_VPP23_OCSET
REG_VPP23_RTN
REG_VPP23_SREF
REG_VPP23_VO
REG_VSNSR_PVDDR
REG_VSNS_CPU_VCCIN_N
REG_VSNS_CPU_VCCIN_P_R
REG_VSNS_CPU_VCCIO_N
REG_VSNS_CPU_VCCIO_P_R
REG_VSNS_CPU_VCCSA_N
REG_VSNS_CPU_VCCSA_P_R
REG_VSNS_CPU_VDDQ01_N
REG_VSNS_CPU_VDDQ01_R_P
REG_VSNS_CPU_VDDQ23_R_P
REG_VSNS_PVDDR
REG_VTT01_REFIN
REG_VTT23_REFIN
SNUB_CPU_VDDQ23_RC
SNUB_CPU_VDDQ_01

WIRELESS ANTENNAS

(each net: NO_TEST=1)

50_0_ANT
50_0_COM
50_A_0_DIPLEXER
50_G_0_DIPLEXER
50_A_0_MATCH
50_G_0_MATCH
50_1_ANT
50_1_COM
50_A_1_DIPLEXER
50_G_1_DIPLEXER
50_A_1_MATCH
50_G_1_MATCH
50_2_ANT
50_2_COM
50_A_2_DIPLEXER
50_G_2_DIPLEXER
50_A_2_MATCH
50_G_2_MATCH
50_3_ANT
50_3_ANT_IND
BT_ONLY_MATCH_IND
BT_ONLY_MATCH

SYSTEM VOLTAGE REGULATORS

(each net: NO_TEST=1)

P5V_G3S_SNS_P
REG_BOOT_P3V3G3H
REG_BOOT_P3V3G3H_RC
REG_BOOT_P5VG3S
REG_BOOT_P5VG3S_RC
REG_LGATE_P3V3G3H
REG_LGATE_P5VG3S
REG_P3V3G3H_DCR_ISNS_N
REG_P3V3G3H_DCR_ISNS_P
REG_P3V3G3H_FSET
REG_P3V3G3H_ISEN
REG_P3V3G3H_OCSET
REG_P3V3G3H_VOUT
REG_P3V3G3H_VOUT_R
REG_P5VG3S_FB
REG_P5VG3S_FSET
REG_P5VG3S_ISEN
REG_P5VG3S_OCSET
REG_P5VG3S_VOUT_R
REG_PHASE_P3V3G3H
REG_PHASE_P3V3G3H_TGR
REG_PHASE_P5VG3S
REG_SNUBBER_P3V3G3H
REG_SNUBBER_P5VG3S
REG_UGATE_P3V3G3H
REG_UGATE_P5VG3S
REG_VIN_U8600
REG_SNUBBER_P3V3G3W
REG_P3V3G3W_VOUT_R
REG_VIN_U8300
REG_VCC1_U8300
REG_UGATE_P3V3G3W
REG_LGATE_P3V3G3W
REG_BOOT_P3V3G3W
REG_P3V3G3W_ISEN
REG_P3V3G3W_VOUT
REG_P3V3G3W_FSET
REG_P3V3G3W_VOUT
REG_P5VG3S_FCCM
REG_UGATE_P5VG3S_USBC
REG_BOOT_P5VG3S_USBC
REG_PHASE_P5VG3S_USBC_TGR
REG_LGATE_P5VG3S_USBC
REG_P5VG3S_USBC_ISEN
REG_P5VG3S_USBC_OCSET
REG_P5VG3S_USBC_VOUT
REG_P5VG3S_USBC_FSET
REG_P5VG3S_USBC_FB
REG_BOOT_P5VG3S_USBC_RC
REG_P5VG3S_USBC_VOUT_R
REG_SNUBBER_P5VG3S_USBC

BOM_COST_GROUP=PCB

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USB-C

- DP_TA_AUXCH_P
- DP_TA_AUXCH_N
- DP_TA_AUXCH_C_P
- DP_TA_AUXCH_C_N
- DP_TB_AUXCH_P
- DP_TB_AUXCH_N
- DP_TB_AUXCH_C_P
- DP_TB_AUXCH_C_N
- DP_XA_AUXCH_P
- DP_XA_AUXCH_N
- DP_XA_AUXCH_C_P
- DP_XA_AUXCH_C_N
- DP_XB_AUXCH_P
- DP_XB_AUXCH_N
- DP_XB_AUXCH_C_P
- DP_XB_AUXCH_C_N
- USBC_TA_D2R_P<2..1>
- USBC_TA_D2R_N<2..1>
- USBC_TA_D2R_CONN_P<2..1>
- USBC_TA_D2R_CONN_N<2..1>
- USBC_TA_R2D_C_P<2..1>
- USBC_TA_R2D_C_N<2..1>
- USBC_TB_D2R_P<2..1>
- USBC_TB_D2R_N<2..1>
- USBC_TB_D2R_CONN_P<2..1>
- USBC_TB_D2R_CONN_N<2..1>
- USBC_TB_R2D_C_P<2..1>
- USBC_TB_R2D_C_N<2..1>
- USBC_XA_D2R_P<2..1>
- USBC_XA_D2R_N<2..1>
- USBC_XA_D2R_CONN_P<2..1>
- USBC_XA_D2R_CONN_N<2..1>
- USBC_XA_R2D_C_P<2..1>
- USBC_XA_R2D_C_N<2..1>
- USBC_XB_D2R_P<2..1>
- USBC_XB_D2R_N<2..1>
- USBC_XB_D2R_CONN_P<2..1>
- USBC_XB_D2R_CONN_N<2..1>
- USBC_XB_R2D_C_P<2..1>
- USBC_XB_R2D_C_N<2..1>
- USBC_TA_D2R_RC_N<2>
- USBC_TA_D2R_RC_P<2>
- USBC_TA_D2R_RC_N<1>
- USBC_TA_D2R_RC_P<1>
- USBC_TA_R2D_CONN_N<1>
- USBC_TA_R2D_CONN_N<2>
- USBC_TA_R2D_CONN_P<1>
- USBC_TA_R2D_CONN_P<2>
- USBC_TA_R2D_RC_N<1>
- USBC_TA_R2D_RC_P<1>
- USBC_TA_R2D_RC_N<2>
- USBC_TA_R2D_RC_P<2>
- USBC_TB_D2R_RC_N<2>
- USBC_TB_D2R_RC_P<2>
- USBC_TB_D2R_RC_N<1>
- USBC_TB_D2R_RC_P<1>
- USBC_TB_R2D_CONN_N<1>
- USBC_TB_R2D_CONN_N<2>
- USBC_TB_R2D_CONN_P<1>
- USBC_TB_R2D_CONN_P<2>
- USBC_TB_R2D_RC_N<1>
- USBC_TB_R2D_RC_P<1>
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- USBC_TB_R2D_RC_P<2>
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- USBC_XA_D2R_RC_P<2>
- USBC_XA_D2R_RC_N<1>
- USBC_XA_D2R_RC_P<1>
- USBC_XA_R2D_CONN_N<1>
- USBC_XA_R2D_CONN_N<2>
- USBC_XA_R2D_CONN_P<1>
- USBC_XA_R2D_CONN_P<2>
- USBC_XA_R2D_RC_N<1>
- USBC_XA_R2D_RC_P<1>
- USBC_XA_R2D_RC_N<2>
- USBC_XA_R2D_RC_P<2>
- USBC_XB_D2R_RC_N<2>
- USBC_XB_D2R_RC_P<2>
- USBC_XB_D2R_RC_N<1>
- USBC_XB_D2R_RC_P<1>
- USBC_XB_R2D_CONN_N<1>
- USBC_XB_R2D_CONN_N<2>
- USBC_XB_R2D_CONN_P<1>
- USBC_XB_R2D_CONN_P<2>
- USBC_XB_R2D_RC_N<1>
- USBC_XB_R2D_RC_P<1>
- USBC_XB_R2D_RC_N<2>
- USBC_XB_R2D_RC_P<2>

NO CONNECTS

- NC_UPC_TA_HV_GATE1
- NC_UPC_TA_HV_GATE2
- NC_UPC_TB_HV_GATE1
- NC_UPC_TB_HV_GATE2
- NC_UPC_TB_I2C_ADDR
- NC_UPC_XA_HV_GATE1
- NC_UPC_XA_HV_GATE2
- NC_USBC_XA_USB_BOT_P
- NC_USBC_XA_USB_BOT_N
- NC_UPC_XB_HV_GATE1
- NC_UPC_XB_HV_GATE2
- NC_UPC_XB_I2C_ADDR
- NC_SSD0_S3X_JTAG_TCK
- NC_SSD0_S3X_JTAG_TMS
- NC_SSD0_STG01_ADDR
- NC_SSD1_S3X_JTAG_TCK
- NC_SSD1_S3X_JTAG_TMS
- NC_TSNS_T2_DX7N

PMIC VOLTAGE REGULATORS

- P0V9SLPDDR_FB
- P0V9SLPDDR_SW0
- P0V9SLPDDR_SW1
- P1V1SLPS2R_FB
- P1V1SLPS2R_SW0
- P1V1SLPS2R_SW1
- P1V8SLPS2R_FB
- P1V8SLPS2R_SW0
- P1V8_SLPS2R_CT
- P1V0_PCH_CORE_S5_FB
- P0V82SLPDDR_FB
- P0V82SLPDDR_SW0
- P0V82SLPDDR_SW1
- PVDDCPUAWAKE_FB
- PVDDCPUAWAKE_SW0
- PVDDCPUAWAKE_SW1
- PVDDCPUAWAKE_SW2
- PVDDCPUAWAKE_SW3
- PVDDCPUSRAMAWAKE_FB
- PVDDCPUSRAMAWAKE_SW0

BLC VOLTAGE REGULATORS

- BKLT_SNUBBER
- BKLT_BST_FB
- BKLT_BST_FB_R

BOM_COST_GROUP=PCB

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AUD_HP_PORT_US_GND
AUD_LAMP_BOOTLTN
AUD_LAMP_BOOTLTP
AUD_LAMP_BOOTLWN
AUD_LAMP_BOOTLWP
AUD_LAMP_LT_ADR
AUD_LAMP_LW_GVDD
AUD_LAMP_LW_VCOM
AUD_LAMP_LW_VREG
AUD_PWR_EN
AUD_RAMP_BOOTRTN
AUD_RAMP_BOOTRTP
AUD_RAMP_BOOTRWN
AUD_RAMP_BOOTRWP
AUD_SPKR_RTWT_OUT_L_N
AUD_SPKR_RTWT_OUT_L_P
AUD_SPKR_RWFR_OUT_L_N
AUD_SPKR_RWFR_OUT_L_P
BKLT_SNSN
BKLT_SNSP
BOARD_REV0
BOARD_REV1
BOARD_REV2
BOOT_CONFIG0
BOOT_CONFIG1
BOOT_CONFIG2
BT_ONLY_MATCH
BT_SF_CLK
BT_SF_CS_L
BT_SF_CS_L_R
BT_SF_HOLD_L
BT_SF_MISO
BT_SF_MOSI
BT_SF_WP_L
CPU_CATERR_R_L
CPU_ERROR_0_LED
CPU_ERROR_0_LED_R
CPU_ERROR_0_Q
CPU_ERROR_1_LED
CPU_ERROR_1_LED_R
CPU_ERROR_1_Q
CPU_ERROR_2_LED
CPU_ERROR_2_LED_R
CPU_ERROR_2_Q
CPU_ERROR_L_0_R
CPU_ERROR_L_1_R
CPU_ERROR_L_2_R
CPU_FIVR_FAULT_LED
CPU_FIVR_FAULT_LED_R
CPU_FIVR_FAULT_R
CPU_PROCHOT_LED
CPU_PROCHOT_LED_R
CPU_PROCHOT_Q
CPU_PROCHOT_R1_L
CPU_THERMTRIP_LED
CPU_THERMTRIP_LED_R
CPU_THERMTRIP_L_R
CPU_THERMTRIP_Q
DP_XA_AUXCH_C_N
DP_XA_AUXCH_C_P
DP_XA_AUXCH_N
DP_XA_AUXCH_P
DP_XB_AUXCH_C_N
DP_XB_AUXCH_C_P
DP_XB_AUXCH_N
DP_XB_AUXCH_P
DP_X_SNK_RBIAS
DUT_VDD_CPU_1V0_MON_U
DUT_VDD_CPU_SENSE_U
DUT_VDD_SOC_SENSE_U
ENET_A_CMC_N

ENET_A_CMC_P
ENET_A_CT_CONN
ENET_A_CT_PHY
ENET_BG_REXT1_R
ENET_B_CMC_N
ENET_B_CMC_P
ENET_B_CT_CONN
ENET_B_CT_PHY
ENET_CLKREQ_PCH_L
ENET_C_CMC_N
ENET_C_CMC_P
ENET_C_CT_CONN
ENET_C_CT_PHY
ENET_D_CMC_N
ENET_D_CMC_P
ENET_D_CT_CONN
ENET_D_CT_PHY
ENET_EFUSE_EN
ENET_MISO_R
ENET_RG_K12
ENET_RNC_N9
ENET_SMB_CLK_R
ENET_SMB_DAT_R
ENET_SYNC_1588
ENET_SYNC_1588_IN
ENET_SYNC_1588_OUT
ENET_TSENS_A_N
ENET_TSENS_A_P
ENET_TSENS_E_N
ENET_TSENS_N
ENET_TSENS_P
ENET_XTALI
ENET_XTALO
ENET_XTALO_R
ENET_XTAL_SELECT_L
ESPI_CLK
ESPI_CLK60M_R
ESPI_IO_HDR<0>
ESPI_IO_HDR<1>
ESPI_IO_HDR<2>
ESPI_IO_HDR<3>
ESPI_IO_R<0>
ESPI_IO_R<1>
ESPI_IO_R<2>
ESPI_IO_R<3>
FBMVDD_ALTV0_R
FBMVDD_ALTV1_R
FBP0V8_ALTV0_R
FBP0V8_ALTV1_R
FBVDDCI_ALTV0_R
FBVDDCI_ALTV1_R
FB_VDDCI_MEM
FB_VDDCR_HBM
FB_VDDCR_SOC
FB_VDDIO_MEM_GPU
FB_VDDIO_MEM_HBM
FB_VSS_A
GND_AUD_DMIC_1_CONN
GPU_GPIO_14
GPU_ISNS_MVDD_N
GPU_ISNS_MVDD_P
GPU_REF_CLK100M_N
GPU_REF_CLK100M_P
HS_MICN
HS_MICP
HS_P3V3_G3W_SSD0_OUT
I2C_CPU_VCCIN_SCL
I2C_CPU_VCCIN_SDA
I2C_SEP_SCL
I2C_SEP_SDA
I2C_TBT_T_RIDGE_SCL
I2C_TBT_X_RIDGE_SCL
I2C_TBT_X_RIDGE_SDA
I2C_VCCIO_SCL
I2C_VCCIO_SDA
I2C_VDDQ01_SCL
I2C_VDDQ01_SDA
I2C_VDDQ23_SCL
I2C_VDDQ23_SDA

I2S_CODEC_BCLK_R
I2S_CODEC_BCLK_RR
I2S_CODEC_D2R_R
I2S_CODEC_LRCLK_RR
I2S_CODEC_R2D_R
I2S_CODEC_R2D_RR
I2S_SPKRAMP_L_BCLK_R
I2S_SPKRAMP_L_MCLK_R
I2S_SPKRAMP_L_R2D_R
I2S_SPKRAMP_R_BCLK_R
I2S_SPKRAMP_R_R2D_R
J5500_PIN_5
L83_FLYC
L83_FLYN
L83_FLYP
L83_HSBIAS_FILT
L83_HSBIAS_FILT_REF
L83_VCP_FILTN
LDO_CORE
LDO_RTC
LED_CPU_CE
LED_CPU_CE_Q
LED_CPU_CE_R
LED_DDR
LED_DDR_R
LED_ENET_EN_R
LED_G3S
LED_G3S_R
LED_G3W
LED_G3W_R
LED_GPU_TT
LED_GPU_TT_Q
LED_GPU_TT_R
LED_MAIN
LED_MAIN_R
LED_PM_TT
LED_PM_TT_Q
LED_PM_TT_R
LED_S0_RUN
LED_S0_RUN_R
LED_S2R
LED_S2R_R
LED_SMC_PH
LED_SMC_PH_Q
LED_SMC_PH_R
LED_SOCHOT
LED_SOCHOT_Q
LED_SOCHOT_R
LED_SOC_DFU
LED_SOC_DFU_R
LED_TBT_EN_R
LED_VCCIO
LED_VCCIO_R
MIPI_A1N
MIPI_A1P
MIPI_B1N
MIPI_B1P
MIPI_FTCAM_CLK_N
MIPI_FTCAM_CLK_P
MIPI_FTCAM_C_CLK_N
MIPI_FTCAM_C_CLK_P
MIPI_FTCAM_C_DATA_N<0>
MIPI_FTCAM_C_DATA_N<1>
MIPI_FTCAM_C_DATA_P<0>
MIPI_FTCAM_C_DATA_P<1>
MIPI_FTCAM_DATA_N<0>
MIPI_FTCAM_DATA_N<1>
MIPI_FTCAM_DATA_P<0>
MIPI_FTCAM_DATA_P<1>
MIPI_LED_A
MIPI_LED_K
MIPI_SW_BYP0
MIPI_SW_BYP1
MIPI_SW_BYP2
MIPI_SW_BYP3
MIPI_SW_BYP4
MIPI_SW_BYP5
MIPI_SW_G0
MIPI_SW_G1

MIPI_SW_G2
MIPI_SW_G3
MIPI_SW_G4
MIPI_SW_G5
P0V82SLPDDR_FB_ALT
P1V0_PCH_CORE_S5_FB_ALT
P1V1SLPDDR_RAMP
P3V3S5_RAMP
P3V3_G3S_CT
PCH_BT_ROM_BOOT_ISO
PCH_CLK32K_RTCX1_OSC
PCH_CLK32K_RTCX1_R
PCH_CLK32K_RTCX2_OSC
PCH_CLK32K_RTCX2_R
PCH_DIFFCLK_BIASREF
PCH_GPD_0_FET
PCH_INTRUDER_L
PCH_SUSACK_L
PCH_UART2_CTS_L
PCH_USB2_ID
PCH_USB_VBUSSENSE
PCH_XTAL24_IN
PCH_XTAL24_IN_R
PCH_XTAL24_OUT
PCH_XTAL24_OUT_R
PDM_DMIC_CLK0_R
PDM_DMIC_CLK1_R
PD_PCH_GPP_I0
PD_PCH_GPP_I1
PD_PCH_GPP_I2
PD_PCH_GPP_I4
PEG_CLKREQ_PCH_L
PLT_RESET_L_LED
PLT_RESET_L_LED_R
PLT_RST_BUF_ENET_L
PLT_RST_BUF_GPU_L
PLT_RST_BUF_SD_L
PLT_RST_BUF_TBT_T_L
PLT_RST_BUF_TBT_X_L
PLT_RST_BUF_WLAN_L
PMU_CLK32K_PCH_R
PMU_CLK32K_SOC_R
PMU_INT_L
PMU_IREF
PMU_LDO3_OUT
PMU_VOUT_RTC
PMU_VPUMP
PMU_VREF
PMU_XTAL1
PMU_XTAL1_R
PMU_XTAL2
PM_SLP_S3_L_LED
PM_SLP_S3_L_LED_R
PM_SLP_S4_L_LED
PM_SLP_S4_L_LED_R
PM_SLP_S5_L_LED
PM_SLP_S5_L_LED_R
PM_THRMTRIP_R_L
PM_VTT01_EN
PM_VTT23_EN
PP1V8_DMIC_1_CONN
PP3V3_G3H_ENET_EFUSE
PPBKLT_BOOST_1
PPBKLT_BOOST_2
PU_CLKREQ_ANALYZER_L
PU_CLKREQ_ANA_PCH_L
PVDDCPUAWAKE_FB_ALT
REGBOOTPMVPP
REGBOOTPVDDRR

REG_1V2_ENET_SKIP
REG_2V1_ENET_SKIP
REG_BOOT_MVDD_RC
REG_BOOT_VPP23_RC
REG_BST_CPU_VDDQ23_RC
REG_BST_VCCSA_RC
REG_BST_VPP01_RC
REG_CPUCORE_VRHOT_L
REG_CPU_ADDR2
REG_CPU_SVID_ALERT_SKL_L
REG_CPU_SVID_CLK_SKL
REG_CPU_SVID_DATA_SKL
REG_CPU_VALRT_L_R
REG_CPU_VCCSA_PHSFLT_L
REG_CPU_VDDQ01_ADDR2
REG_CPU_VDDQ01_ISNS_R
REG_CPU_VDDQ01_RC
REG_CPU_VDDQ01_SVID_ALERT_R
REG_CPU_VDDQ01_SVID_DATA_R
REG_CPU_VDDQ23_ADDR2
REG_CPU_VDDQ23_IL_R
REG_CPU_VDDQ23_PHSFLT_L
REG_CPU_VDDQ23_SVID_ALERT_L_R
REG_CPU_VDDQ23_SVID_DATA_R
REG_CPU_VDDQ_SVID_ALERT_SKL_L
REG_CPU_VDDQ_SVID_CLK_SKL
REG_CPU_VDDQ_SVID_DATA_SKL
REG_CPU_VDIO_R
REG_DRVCPU_AVIN
REG_GPUCORE_CFILT
REG_GPUCORE_ISNS_3_R_P
REG_MVDD_OCSET_R
REG_MVDD_VO_R
REG_P0V8_SET1_R
REG_P3V3G3H_FB
REG_P3V3G3W_FB
REG_P3V3G3W_OCSET
REG_PHSFLT_GPUCORE_1_L
REG_PHSFLT_GPUCORE_2_L
REG_PHSFLT_GPUCORE_3_L
REG_PHSFLT_GPUCORE_4_L
REG_PHSFLT_GPUCORE_5_L
REG_PHSFLT_GPUCORE_6_L
REG_PHSFLT_GPUCORE_7_L
REG_PHSFLT_GPUCORE_8_L
REG_PPDRV_CPUVR
REG_PVCC_UD450
REG_PVCC_UD650
REG_PVCC_UD750
REG_PVCC_VPP01
REG_PVCC_VPP23
REG_RS_VPP01
REG_RS_VPP23
REG_SNUBBER_VCCIO_RC
REG_SVID_ALERT_CPU_VCCIO_R
REG_SVID_CLK_CPU_R
REG_SVID_DATA_CPU_VCCIO_R
REG_SVID_VCLK_CPU_VDDQ01_R
REG_SVID_VCLK_CPU_VDDQ23_R
REG_SVID_VCLK_VCCIO_R
REG_VCC1_U8600
REG_VCCIO_ADR2
REG_VCCIO_BOOST_RC
REG_VCCIO_ISNS_R
REG_VCCSA_RS1
REG_VCC_UD450
REG_VCC_UD650
REG_VCC_UD750
REG_VCC_VPP01
REG_VCC_VPP23
REG_VDDCI_FB_N
REG_VDDCI_SET1_R
REG_VPP01_ISNS_N
REG_VPP01_ISNS_P
REG_VPP23_ISNS_N
REG_VPP23_ISNS_P
REG_VSNS_CPU_VCCIN_P
REG_VSNS_CPU_VCCIO_P

(All nets: NO_TEST=1)

BOM_COST_GROUP=PCB

PAGE TITLE: ICT NO TEST 4

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D

REVISION 6.21.0

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ICT "NICE TO HAVE" LIST <RDAR://30758642>

Net	Property
REG_VSNS_CPU_VCCSA_P	NO_TEST=1
SDCONN_CLK	NO_TEST=1
SDCONN_CMD	NO_TEST=1
SDCONN_D0	NO_TEST=1
SDCONN_D1	NO_TEST=1
SDCONN_R_WP	NO_TEST=1
SDCONN_UHS2_CLK_P	NO_TEST=1
SDCONN_UHS2_D0_N	NO_TEST=1
SDCONN_UHS2_D0_P	NO_TEST=1
SDCONN_UHS2_D1_N	NO_TEST=1
SDCONN_UHS2_D1_P	NO_TEST=1
SD_SPI_CLK_R	NO_TEST=1
SD_SPI_CS_L_R	NO_TEST=1
SD_SPI_HOLD	NO_TEST=1
SD_SPI_MISO_R	NO_TEST=1
SD_SPI_MOSI_R	NO_TEST=1
SEP_DMIC0_DISABLE_R_L	NO_TEST=1
SEP_DMIC1_DISABLE_R_L	NO_TEST=1
SMBUS_PCH_XDP_3V3_SCL	NO_TEST=1
SMBUS_PCH_XDP_3V3_SDA	NO_TEST=1
SOC_AMUXOUT_UFL	NO_TEST=1
SOC_CLKREQ_PCH_L	NO_TEST=1
SOC_COLD_RESET_L	NO_TEST=1
SOC_MIPI0C_REXT	NO_TEST=1
SOC_MIPI1C_REXT	NO_TEST=1
SOC_MIPID_REXT	NO_TEST=1
SOC_TST_CLKOUT	NO_TEST=1
SOC_TST_CLKOUT_UFL	NO_TEST=1
SOC_VDDCPU_SENSE	NO_TEST=1
SOC_VDDSOC_SENSE	NO_TEST=1
SOC_XTAL24M_IN	NO_TEST=1
SOC_XTAL24M_OUT	NO_TEST=1
SOC_XTAL24M_OUT_R	NO_TEST=1
SPI_SOCROM_CLK_R	NO_TEST=1
SPI_SOCROM_MOSI_R	NO_TEST=1
SPI_TPAD_CLK	NO_TEST=1
SPI_TPAD_MISO	NO_TEST=1
SPI_TPAD_MOSI	NO_TEST=1
SR1P2_VLX	NO_TEST=1
SR1P4_VLX	NO_TEST=1
SR1P8_VLX	NO_TEST=1
SSD0_CLK24M_R	NO_TEST=1
SSD0_CLKREQ0_R_L	NO_TEST=1
SSD0_CLKREQ1_R_L	NO_TEST=1
SSD1_CLK24M_R	NO_TEST=1
SSD1_CLKREQ0_R_L	NO_TEST=1
SSD1_CLKREQ1_R_L	NO_TEST=1
SSD1_SWCLK_UART_R2D	NO_TEST=1
TBT_X_RSENSE	NO_TEST=1
TXC_100M_N	NO_TEST=1
TXC_100M_P	NO_TEST=1
UART_BT_SOC_D2R	NO_TEST=1
UART_BT_SOC_R2D	NO_TEST=1
UK280_TMP442_SCL	NO_TEST=1
UK280_TMP442_SDA	NO_TEST=1
UPC_XA_R_OSC	NO_TEST=1
UPC_XA_USB_TOP_P	NO_TEST=1
USB2_EXTA_N	NO_TEST=1
USB2_EXTA_P	NO_TEST=1
USB2_EXTB_N	NO_TEST=1
USB2_EXTB_P	NO_TEST=1
PP1V1_SLPDDR_SOC_XTAL_F	NO_TEST=1

Net	Property
USB2_EXTC_N	NO_TEST=1
USB2_EXTC_P	NO_TEST=1
USB2_EXTD_N	NO_TEST=1
USB2_EXTD_P	NO_TEST=1
USB_DBG_XA_F_N	NO_TEST=1
USB_DBG_XA_F_P	NO_TEST=1
USB_DBG_XA_N	NO_TEST=1
USB_DBG_XA_P	NO_TEST=1
USB_EXTA_N	NO_TEST=1
USB_EXTA_P	NO_TEST=1
USB_EXTB_N	NO_TEST=1
USB_EXTB_P	NO_TEST=1
USB_EXTC_N	NO_TEST=1
USB_EXTC_P	NO_TEST=1
USB_EXTD_N	NO_TEST=1
USB_EXTD_P	NO_TEST=1
USB_SD_DM	NO_TEST=1
USB_SD_DP	NO_TEST=1
USB_UPC_TA_F_N	NO_TEST=1
USB_UPC_TA_F_P	NO_TEST=1
USB_UPC_TB_F_N	NO_TEST=1
USB_UPC_TB_F_P	NO_TEST=1
USB_UPC_XB_F_N	NO_TEST=1
USB_UPC_XB_F_P	NO_TEST=1
V10_SMBCLK	NO_TEST=1
V10_SMBDAT	NO_TEST=1
VDD1P2_C	NO_TEST=1
VDD1P5_C	NO_TEST=1
VR_GPU_ADDR_OFFSET	NO_TEST=1
VR_GPU_IMON_DIGITAL_FILTER	NO_TEST=1
VR_GPU_IMON_EN	NO_TEST=1
VR_GPU_IMON_RG	NO_TEST=1
VR_GPU_IOUT_ALERT_L	NO_TEST=1
VR_GPU_PHSFLT_L	NO_TEST=1
VR_GPU_RCSM	NO_TEST=1
VR_GPU_RCSN_R	NO_TEST=1
VR_GPU_RCSP	NO_TEST=1
VR_GPU_RCSP_R	NO_TEST=1
VR_GPU_SM_ALERT_L	NO_TEST=1
VR_GPU_SVC_R	NO_TEST=1
VR_GPU_SVD_R	NO_TEST=1
VR_GPU_SVT_R	NO_TEST=1
VR_GPU_SYS_PWROK_L	NO_TEST=1
VR_GPU_VAUXSENSE	NO_TEST=1
VR_GPU_VINSEN	NO_TEST=1
VR_GPU_VINSEN	NO_TEST=1
VR_GPU_VRDY1	NO_TEST=1
VR_GPU_VRHOT_CRIT_L	NO_TEST=1
VSENSE_PMAX	NO_TEST=1
VSNS_GPU_MVDD_N	NO_TEST=1
VSNS_GPU_MVDD_P	NO_TEST=1
VSNS_GPU_VDDC_N	NO_TEST=1
VSNS_GPU_VDDC_P	NO_TEST=1
VSNS_GPU_VDDC_P_R	NO_TEST=1
WLAN_PMU_TEST_O	NO_TEST=1
XDP_CPU_PWRBTN_L	NO_TEST=1
UPC_TA_DEBUGUART_TX	NO_TEST=1
UPC_TA_GPIO1	NO_TEST=1
UPC_TA_SWD_CLK	NO_TEST=1
UPC_TA_SWD_DATA	NO_TEST=1
UPC_TB_DEBUGUART_TX	NO_TEST=1
UPC_TB_GPIO1	NO_TEST=1
UPC_XA_DEBUGUART_TX	NO_TEST=1
UPC_XB_DEBUGUART_TX	NO_TEST=1
UPC_XB_GPIO1	NO_TEST=1

BOM_COST_GROUP=PCB

PAGE TITLE
ICT NO TEST 5

Apple Inc.

DRAWING NUMBER 051-01649 SIZE D
REVISION 6.21.0
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MISC NETS NOT REQUESETED BY ICT OR FCT <RDAR://30758642>

TBT_X_RBIAS
TBT_X_PCIE_BIAS
USBC_TA_USB_TOP_N
USBC_TA_USB_TOP_P
USBC_TB_USB_BOT_N
USBC_TB_USB_BOT_P
USBC_TB_USB_TOP_N
USBC_TB_USB_TOP_P
CLKTEST_C_N
CLKTEST_C_P
DP_T_SNK_RBIAS
DP_T_SRC_RBIAS
TBT_TA_USB2_RBIAS
TBT_TB_USB2_RBIAS
TBT_T_HDMI_DDC_CLK
TBT_T_HDMI_DDC_DATA
TBT_T_PCIE_BIAS
TBT_T_RBIAS
TBT_T_XTAL25M_OUT_R
LED_RETURN_1_FB
LED_RETURN_3_FB
USB3_SD_D2R_N
USB3_SD_D2R_P
USB3_SD_D2R_C_N
USB3_SD_D2R_C_P
USB3_SD_R2D_N
USB3_SD_R2D_P
REG_PHSFLT_CPUCORE_1_L
REG_PHSFLT_CPUCORE_2_L
REG_PHSFLT_CPUCORE_3_L
REG_PHSFLT_CPUCORE_4_L
REG_PHSFLT_CPUCORE_5_L
WLAN_CLKREQ_PCH_L
PCH_SUSWARN_L
PCH_USB_RBIAS
PP1V0_S5_PCH_VCC_MPHYPCIE3_PLL
PCIE_WLAN_R2D_C_N
PCIE_WLAN_R2D_C_P
PCIE_RCOMPN
PCIE_RCOMPP
XDP_VR_READY
TBT_X_TMU_CLK_OUT
TBT_T_TMU_CLK_OUT
P5V_G3S_SNS_N
REG_MVDD_VO
REG_P5VG3S_VOUT
REG_VCC2_U8600
REG_VDDCI_VO
ARKANOID_T_P5
ARKANOID_T_P7
FAN_0_TACH_FILT
PCIE_WLAN_R2D_N
PCIE_WLAN_R2D_P
MEM_0_SA<0>
MEM_0_SA<1>
MEM_0_SA<2>
MEM_1_SA<0>
MEM_1_SA<1>
MEM_1_SA<2>
MEM_2_SA<0>
MEM_2_SA<1>
MEM_2_SA<2>
MEM_3_SA<0>
MEM_3_SA<1>
MEM_3_SA<2>
ISNS_PP11V_G3H_N
ISNS_PP11V_G3H_P
GND_SMC_AVSS
PCIE_DN2_LIFEBOAT_D2R_P
PCIE_DN2_LIFEBOAT_D2R_N
PCIE_DN2_LIFEBOAT_R2D_P
PCIE_DN2_LIFEBOAT_R2D_N
PCIE_DN2_LIFEBOAT_CLK100M_P
PCIE_DN2_LIFEBOAT_CLK100M_N
PCIE_DN2_LIFEBOAT_CLKREQ_L
PCIE_DN2_LIFEBOAT_RESET_L
PCIE_DN2_LIFEBOAT_R2D_C_P
PCIE_DN2_LIFEBOAT_R2D_C_N

PP1V0_PCH_CORE_S5_SW0
PP1V0_PCH_CORE_S5_SW1
PCIE_SSD0_D2R_N<1>
PCIE_SSD0_D2R_N<2>
PCIE_SSD0_D2R_N<3>
PCIE_SSD0_D2R_P<1>
PCIE_SSD0_D2R_P<2>
PCIE_SSD0_D2R_P<3>
PCIE_SSD0_R2D_C_N<1>
PCIE_SSD0_R2D_C_N<2>
PCIE_SSD0_R2D_C_N<3>
PCIE_SSD0_R2D_C_P<1>
PCIE_SSD0_R2D_C_P<2>
PCIE_SSD0_R2D_C_P<3>
PCIE_SSD0_R2D_N<1>
PCIE_SSD0_R2D_N<2>
PCIE_SSD0_R2D_N<3>
PCIE_SSD0_R2D_P<1>
PCIE_SSD0_R2D_P<2>
PCIE_SSD0_R2D_P<3>
PCIE_SSD1_D2R_N<1>
PCIE_SSD1_D2R_N<2>
PCIE_SSD1_D2R_N<3>
PCIE_SSD1_D2R_P<1>
PCIE_SSD1_D2R_P<2>
PCIE_SSD1_D2R_P<3>
PCIE_SSD1_R2D_C_N<1>
PCIE_SSD1_R2D_C_N<2>
PCIE_SSD1_R2D_C_N<3>
PCIE_SSD1_R2D_C_P<1>
PCIE_SSD1_R2D_C_P<2>
PCIE_SSD1_R2D_C_P<3>
PCIE_SSD1_R2D_N<1>
PCIE_SSD1_R2D_N<2>
PCIE_SSD1_R2D_N<3>
PCIE_SSD1_R2D_P<1>
PCIE_SSD1_R2D_P<2>
PCIE_SSD1_R2D_P<3>
PP0V9_SLPDDR_SOC_PCIEREFBUF
PP1V2_AWAKE_SOC_PCIEREFBUF_F
PP1V2_AWAKE_SOC_PLLCPU_F
PP1V2_AWAKE_SOC_PLLSOC_F
SMC_FAN_0_PWM
SMC_PECI_TX_R
SPMI_CLK_R
SPMI_DATA_R
PMU_CPU_VCCIO_S0_ISENSE
TBT_XA_USB2_RBIAS
TBT_XB_USB2_RBIAS
USB3_SD_R2D_C_N
USB3_SD_R2D_C_P
PPHV_USBC_XB
USBC_TA_USB_BOT_N
USBC_TA_USB_BOT_P
REG_0V85_ENET_VO
SDCONN_CLK_F
REG_P5VG3S_USBC_N
NC_PMU_RSLOC_RST_L
REG_BOOT_P3V3G3W_RC
REG_P3V3G3W_DCR_P
REG_P5VG3S_USBC_P
REG_VCC2_U8300
REG_P3V3G3W_DCR_N
PP1V2_SD_SW
REG_SW_1V2_ENET
REG_SW_2V1_ENET
REG_UGATE_P0V8
REG_VCCSA_ISNS_N
REG_VCCSA_ISNS_P
PCIE_WLAN_D2R_P
PCIE_WLAN_D2R_N
REG_VSNS_CPU_VDDQ23_N
PCIE_HOT_PLUG_SCL

PCIE_HOT_PLUG_SDA
REG_PHASE_P5VG3S_TGR
PCIE_DN_CLKREQ1_L
ISNS_P12V_MAIN_SSD0_N
ISNS_P12V_MAIN_SSD0_P
ISNS_P12V_MAIN_SSD1_N
ISNS_P12V_MAIN_SSD1_P
REG_DRVCPU_PH
ISNS_HS_CPU_VCORE_N
ISNS_HS_CPU_VCORE_P
ISNS_USBA_5V_G3S_N
ISNS_USBA_5V_G3S_P
ISNS_USBC_5V_G3S_N
ISNS_USBC_5V_G3S_P
PCIE_CLK100M_ANALYZER_N
PCIE_CLK100M_ANALYZER_P
XDP_OBSFN_C_0
XDP_OBSFN_C_1
ISNS_P12V_GPU_N
ISNS_P12V_GPU_P
ISNS_P12V_MAIN_AUX_GPU_N
ISNS_P12V_MAIN_AUX_GPU_P
ISNS_P12V_MAIN_BLC_LCD_N
ISNS_P12V_MAIN_BLC_LCD_P
ISNS_P12V_MAIN_N
ISNS_P12V_MAIN_P
ISNS_P3V3_G3W_SSD0_N
ISNS_P3V3_G3W_SSD0_P
ISNS_P3V3_G3W_SSD1_N
ISNS_P3V3_G3W_SSD1_P
REG_VSNS_CPU_VDDQ01_P
REG_VSNS_CPU_VDDQ23_P
XDP_PCH_PRDY_L
XDP_PCH_PREQ_L
REG_VCCIN_TSNS_A_R
REG_VCCSA_TSNS_B_R
XDP_OBSDATA_A_0
XDP_OBSDATA_A_1
XDP_OBSDATA_A_2
XDP_OBSDATA_A_3
XDP_OBSDATA_B_0
XDP_OBSDATA_B_1
XDP_OBSDATA_B_2
XDP_OBSDATA_B_3
XDP_OBSDATA_C_0
XDP_OBSDATA_C_1
XDP_OBSDATA_C_2
XDP_OBSDATA_C_3
XDP_OBSDATA_D_0
XDP_OBSDATA_D_1
XDP_OBSDATA_D_2
XDP_OBSDATA_D_3
XDP_PCH_TCK_CON
XDP_PIN2
PCIE_WLAN_D2R_C_N
PCIE_WLAN_D2R_C_P
PCH_GPD_0_FET_L
XDP_CPU_PWR_DEBUG_FET_L
IOX_RESET_MASK_L
P1V8S0_VALID_IOX
ENETCONN_MCT_BS
PCH_STRP_SPIROM_SAF
CLK100M_CPU_BCLK_0_P
CLK100M_CPU_BCLK_0_N
CLK100M_CPU_BCLK_1_P
CLK100M_CPU_BCLK_1_N
CLK100M_CPU_BCLK_2_P
CLK100M_CPU_BCLK_2_N

DP_X_SNK0_DDC_CLK
DP_X_SNK0_DDC_DATA
DP_X_SNK1_DDC_CLK
DP_X_SNK1_DDC_DATA
DP_X_SRC_RBIAS
PP0V9_TBT_X_CIO
PP0V9_TBT_X_DP
PP0V9_TBT_X_USB
TBT_XB_LSRX
TBT_XB_LSTX
TBT_X_HDMI_DDC_CLK
TBT_X_HDMI_DDC_DATA
TBT_X_THERM_D_POS
ENET_TSENS_E_P
SEP_DISABLE_STROBE
SEP_DMIC_DISABLE_Q_L
NC_DFR_TOUCH_INT_L
NC_Q6080_S
NC_Q6080_G
NC_Q6080_D
NC_SECI_OUT_D2R
NC_SECI_IN_R2D
I2S_SPKRAMP_L_LRCLK_R
I2S_SPKRAMP_R_LRCLK_R
I2S_CODEC_LRCLK_R
NC_I2S_HAWKING_BCLK_R
NC_SOC_VSSSOC_SENSE
NC_SOC_VSSCPU_SENSE
NC_SOC_AOP_PDM_CLK4
NC_SE_PWR_EN
NC_PMU_CLK32K_GNSS
NC_PMU_CLK32K
NC_PMU_AMUX_BY
NC_PEARL_PWR_EN
NC_LIDINT_L_UARTHDD_D2R
NC_GNSS_HOST_WAKE
NC_BUCK8_SW1
NC_BUCK4_SW1
NC_ALTIMETER_INT
CPU_SMC_THRMTRIP_L
MESA_MENUKEY_L
ENET_LOW_PWR
DFR_DISP_RESET_L
MESA_INT
I2S_CODEC1_R2D_R
TBT_X_XTAL25M_OUT
SOC_DDR0_ZQ
SWD_SOC_SWCLK
SWD_SOC_SWDIO
KBLX_PM_DOWN
PCIE_SOC_R2D_P<3..0>
PCIE_SOC_R2D_N<3..0>
PCIE_SOC_D2R_P<2>
PCIE_SOC_D2R_P<1>
USB3_DM2_D2RP
USB3_DM2_D2RN
USB3_DM2_R2DP
USB3_DM2_R2DN
OSC_1588_50MHZ_P
OSC_1588_50MHZ_N
ENET_OSC_1588_50MHZ_P
ENET_OSC_1588_50MHZ_N
REG_UGATE_P3V3G3H_R
REG_UGATE_P5VG3S_R
REG_UGATE_P5VG3S_USBC_R
ENET_OSC_REFCLK_50MHZ_N
ENET_OSC_REFCLK_50MHZ_P
ENET_REFCLK_50MHZ_N
ENET_REFCLK_50MHZ_P
OSC_REFCLK_OE
PP3V3_G3H_ENET_REFCLK

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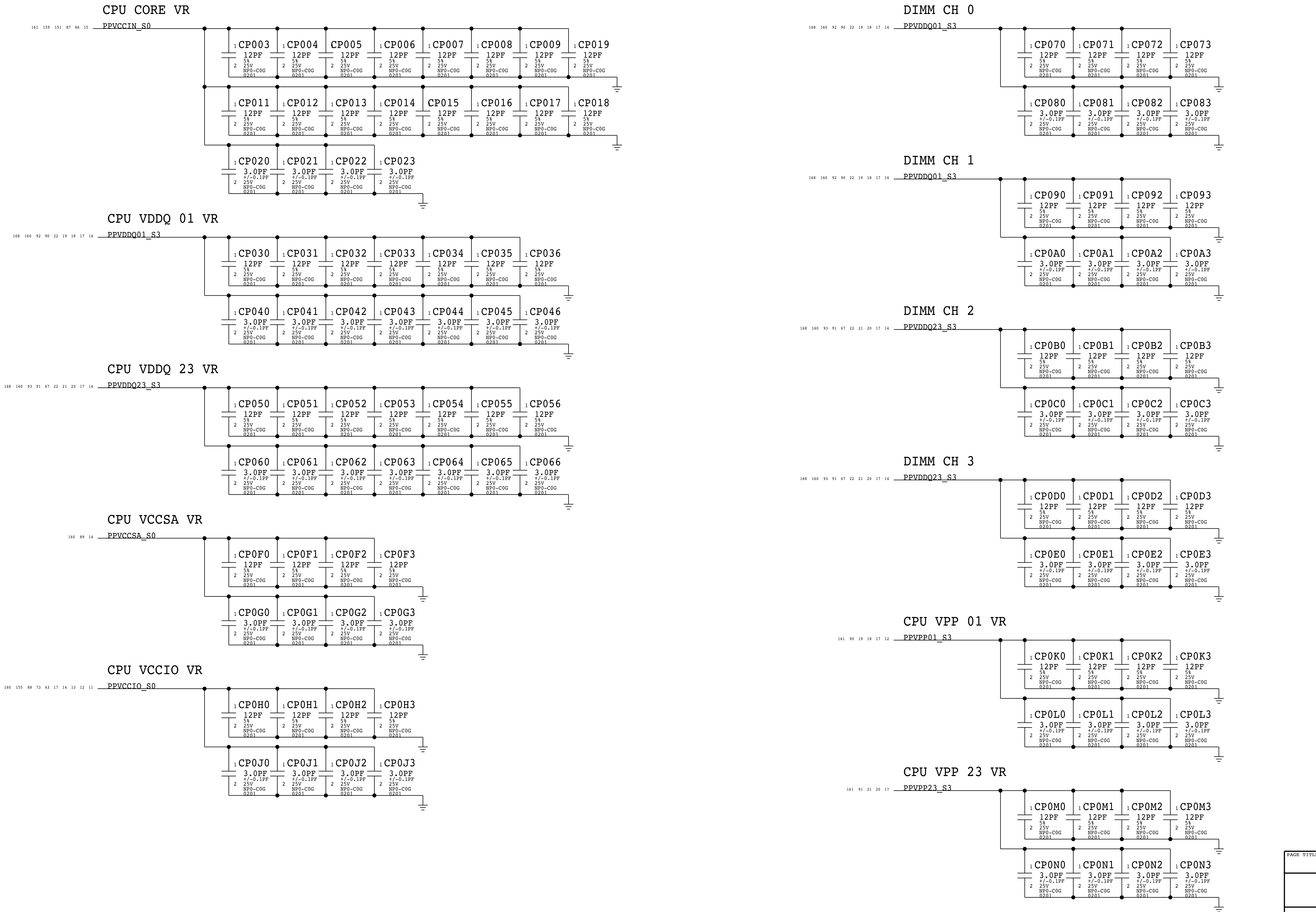

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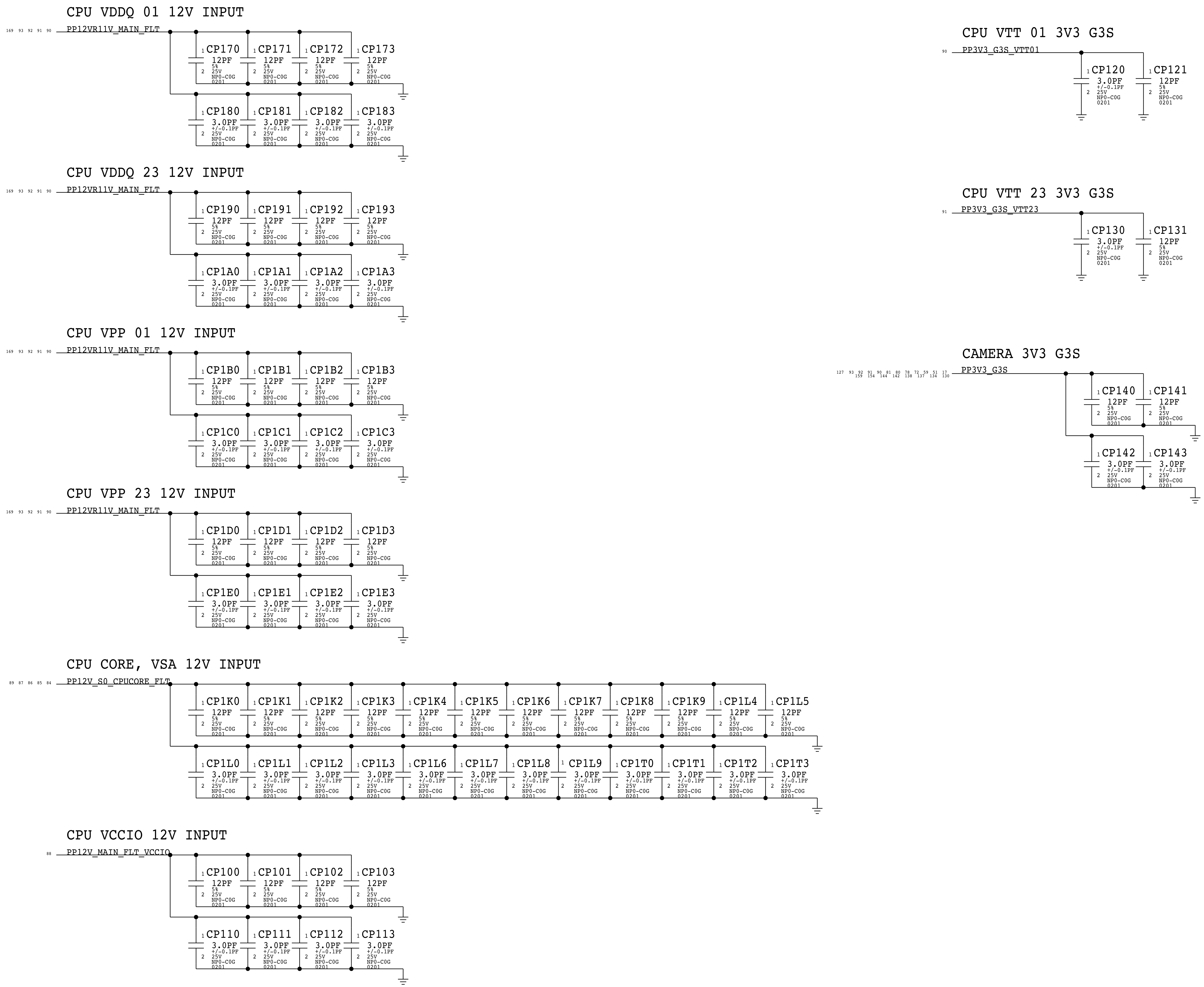

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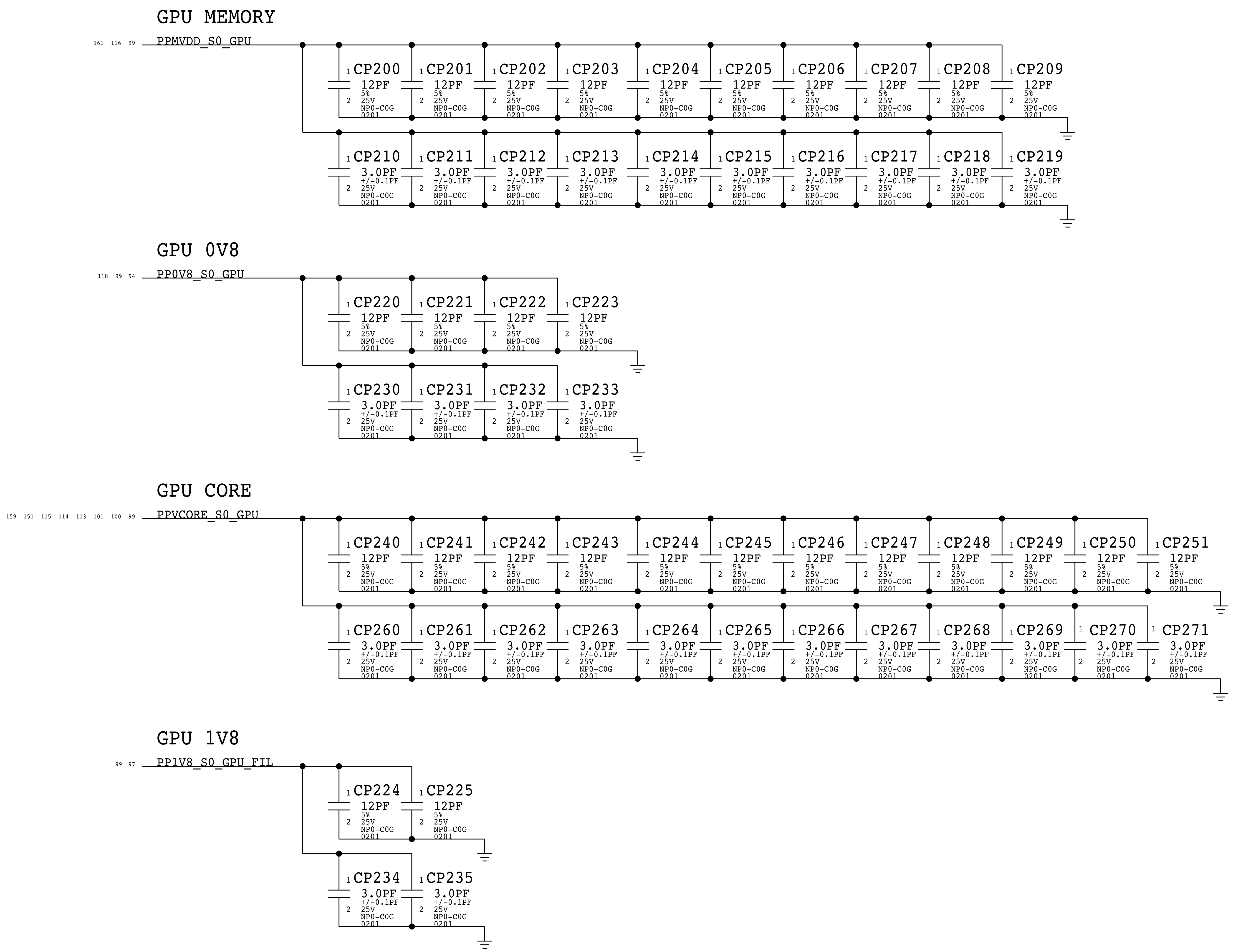

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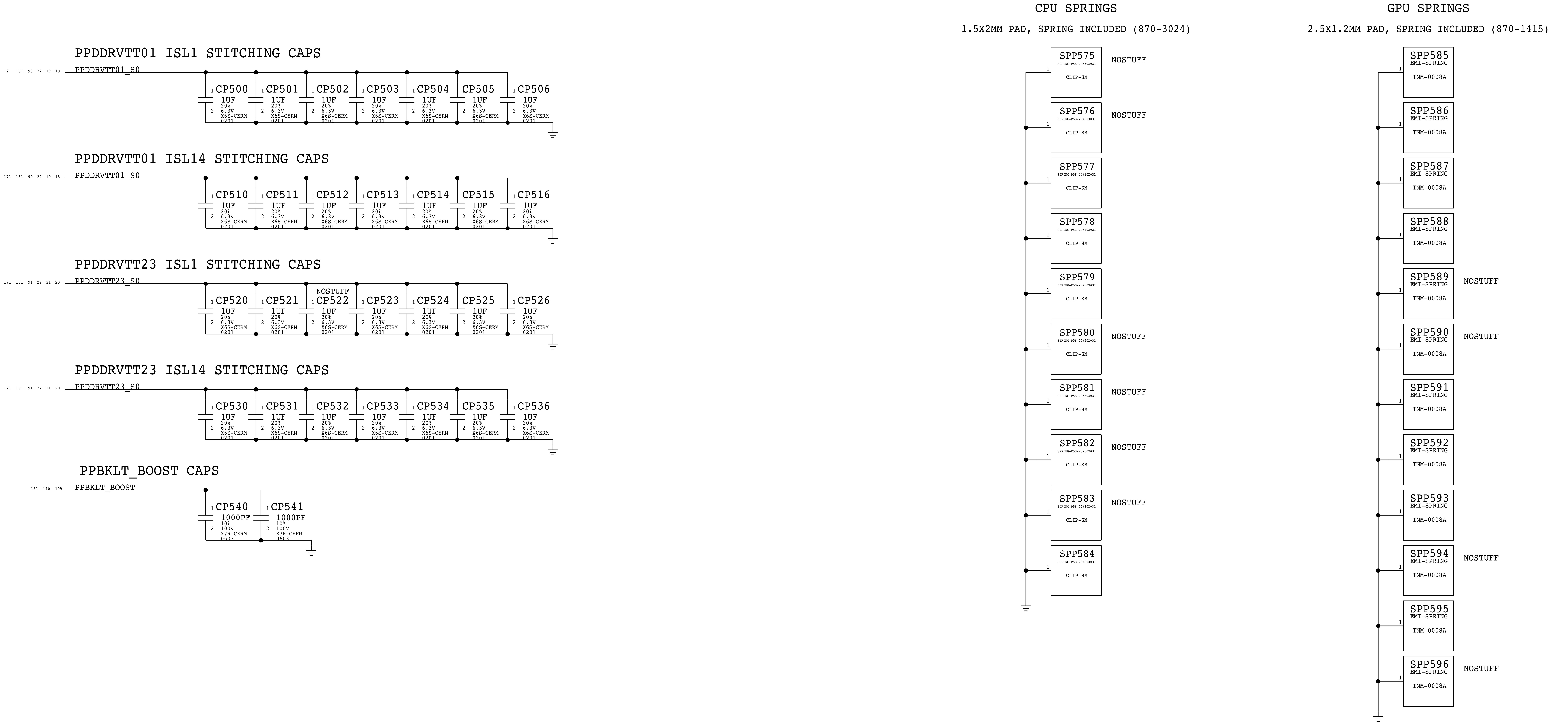

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